

CLOSTRIDIODES DIFFICILE

Awareness of the drivers of C. diff has improved dramatically over the past decade and it is increasingly uncommon to see a true case of primary community C. diff. thanks to our switch away from the broadest spectrum antibiotics. Nevertheless, when we do see a case it can be severe and having a clear management strategy is important. [NICE has now published evidence-based guidance](#) on the best management of C. diff.

There are two interesting recommendations in this guideline:

- **Consider faecal microbiota transplant for recurrent episodes**
 - FMT has been shown to be effective whether delivered via [colonoscopy or oral capsules](#)
 - Presumably, this will be specialist-led rather than the GP sorting it out themselves...
- **Do not advise people taking antibiotics to take pre- or probiotics to prevent C. difficile infection**
 - Many of us might tell patients to consider probiotics to reduce the risk of adverse effects from antibiotics.
 - [Cochrane](#) found high dose probiotics (≥ 5 billion CFUs per day) ARE effective at reducing antibiotic-associated diarrhoea in children, NNT = 9, using *Lactobacillus rhamnosus* or *Saccharomyces boulardii* - there is uncertainty about other types of probiotic
 - [Meta-analysis published in J Clin Gastroenterology 2021](#) showed a relative risk reduction in antibiotic-associated diarrhoea of 38% when probiotics were taken early in antibiotic treatment
 - NICE found [1 meta-analysis](#) showing probiotics statistically significantly reduce the incidence of C. difficile infection in inpatients but not in outpatients or those in mixed settings. General adverse events may be reduced.

So it may still be worthwhile advising probiotics alongside antibiotic treatment to reduce general diarrhoeal adverse events, but prevention of C. difficile is less likely to occur.

KISS: Clostridioides Difficile Infection: Antimicrobial Prescribing

[NICE NG199 July 2021](#)

Assessment

- First episode or recurrence? Severity? Individual risk factors e.g. frailty, age, co-morbidities?
- Review existing antibiotic use - stop unless essential - if essential can it be switched to lower risk abx?
- Review medications - PPIs, laxatives, SADMAN meds (e.g. NSAIDs, diuretics, ACEi, SGLT2i)

Treatment

- Offer an oral antibiotic to adults with C. diff - consider prompt specialist advice prior to Rx
 - 1st line: Vancomycin 125mg orally QDS for 10 days
 - 2nd line: Fidaxomicin 200mg orally BW for 10 days
- In children treatment should be initiated by, or after, advice from a specialist
- Advise on maintaining hydration, preventing spread, safety net
- Reassess if symptoms do not respond as expected or worsen
- Refer to hospital if severely unwell, consider if risk of complications or recurrence is high
- Consider faecal microbiota transplant for recurrence with 2 or more previous episodes

Prevention

- Do not offer antibiotics.
- Do not advise people taking abx to take prebiotics or probiotics to prevent C. diff. infection.

ANTIBIOTIC PRESCRIBING

It should be no surprise, indeed it should be welcomed, that [antibiotic resistance is in the media](#) once again. This year it is *Klebsiella pneumonia* which has developed carbapenem resistance and shows a 6-fold increase in the resistant strain throughout European hospitals ([Nature Microbiology 2019](#)) over the last decade. Carbapenems are considered the last resort antibiotic for multidrug-resistant microbes.

The WHO has named antimicrobial resistance as one of the top 10 threats to the world. Concerns around resistance relate not just to "extremely-resistant bacteria" but at a more grass-roots, practical level. A BMJ paper ([BMJ 2016;352:i939](#)) found that OECD countries demonstrated a pooled prevalence of resistance of 53.4%, and this was substantially higher in non-OECD countries where antibiotics were more freely available to the public.

80% of antibiotic prescribing in the UK occurs in general practice, so we have a central role in making sure we prescribe appropriately. Indeed prescribing has improved, as a recent BMJ editorial ([BMJ 2019;364:l780](#)) explains primary care clinicians have reduced antibiotic prescribing by 13% in the past 5 years, without any increase in serious negative patient outcomes.

But what more can we do to further curb inappropriate antibiotic use? What strategies might prove effective?

1 **Duration of antibiotics** [BMJ 2019;364:l440](#), [BMJ 2019;364:l780](#), [BMJ 2017;358:j3418](#)

Reducing overall exposure to antibiotics may be beneficial. In past, we have often considered resistance is driven by inadequate treatment with courses of antibiotics which are too short, however traditional recommended durations of use far exceed what is usually required to control an infection. Indeed prolonged exposure is more likely to lead to resistance, especially from commensal organisms (e.g. *K pneumonia* in the gut). There has been a clear trend towards shorter duration courses for many diseases in recent years.

But before we can focus on improving appropriate prescribing it is helpful to explore where prescribing may be suboptimal. A recent BMJ cross-sectional study of UK general practices did exactly this.

- The researchers collected data from practices contributing to The Health Improvement Network database between 2013-15 and compared duration of antibiotic prescription to the 2013 PHE guidelines on treating common infections
- 931000 consultations resulted in an antibiotic prescription for a wide variety of common infections
- **Results**
 - Commonest uses: acute cough/bronchitis 41.6%, acute sore throat 25.7%, acute otitis media 8.9%, acute sinusitis 8.2%
 - Certain conditions were frequently associated with prescriptions exceeding the recommended duration:
 - Acute cough/bronchitis (recommended 5 days): 86%
 - Acute sore throat (5d): 84%
 - Acute COPD exacerbation (5d): 89%
 - Acute otitis media (5d): 86%
 - Acute cystitis in females (3d): 55%
 - Duration of abx infrequently exceeded guidelines in pyelonephritis, acute cystitis in men, impetigo and cellulitis
 - Around 75% of antibiotics prescribed were those recommended in the PHE guideline
- **Conclusion:** "*substantial reductions in antibiotic exposure can be accomplished by the aligned duration of antibiotic prescriptions with guidelines*".

The *linked editorial* ([BMJ 2019;364:l780](#)) comments that following the latest national guidance would *reduce antibiotic use by 14 days per 10 prescriptions issued* equating to 65 million fewer antibiotic days each year in the UK, easily exceeding the government's target for Abx reduction by 2020 (half of 2016 rates). It reasonably points out that "prescribers cannot be held responsible for what they were doing before new guidelines were issued" but that we can familiarise ourselves with the current best practice and optimise prescribing.

► *The antibiotic course has had its day* [BMJ 2017;358:j3418](#)

Despite the trend is towards shorter durations of antibiotics, this controversial editorial in the BMJ ([BMJ 2017;358:j3418](#)) goes even further and asks if "the antibiotic course has had its day?".

The author suggests that there is very limited evidence guiding clinicians on the optimal duration of antibiotics for specific infections. Indeed standard courses of 5 or 7 days exist because there are 5 digits on a hand and 7 days in a week rather than any stronger reason.

The paper suggests that perhaps patients simply taking antibiotics "until they feel better" would be as good as fixed doses. Indeed this is an attractive option on the face of it, although given most antibiotics are still 'accidentally' prescribed for viral infections NB can't help but think that would leave many patients expecting a 3-week course...

It certainly has generated debate. This was the intention. It does not have new data showing shorter or flexible courses are actually better than fixed longer prescriptions, but can certainly start to make the medical world challenge established concepts and generate the data we need to provide an answer.

► *Reducing antibiotic side effects* [Cochrane 2019, CD004827](#)

This update of well-trodden ground reconfirms that probiotics DO reduce rates of antibiotic-associated diarrhoea with an NNT = 9. Subgroup analysis shows that higher dose probiotics (≥ 5 billion CFUs or 'colony-forming units', it's a lot of bacteria...) fair even better with an NNT = 6, albeit with slightly less certainty due to smaller sample sizes. Whether the type of probiotic influences outcomes remains unclear.

2 NICE Antimicrobial Guidelines [NICE 2018/19](#)

Concerns around antimicrobial resistance explains the drive from NICE, in conjunction with PHE, to produce evidence-based antimicrobial guidelines for clinicians and disseminate what they consider best practice as widely as possible. Throughout the 10 conditions so far published, there are common themes:

- Encouraging self-care
- Avoiding unnecessary antibiotics
- Back-up antibiotics

These conditions are covered in the coming chapters. Expect more to follow in the near future.

INFECTIOUS DISEASES

KISS: Optimal Duration of Antibiotics

PHE / NICE

Condition	Duration Abx (days)	Condition	Duration Abx (days)
Cough (acute)	5	UTI (lower)	3 (non-pregnant women)
Bronchiectasis	7 - 14		3 (children)
COPD acute exacerbation	5		7 (pregnant women)
Pneumonia (community)	5 (7-10 if poor response)		7 (men)
Otitis media (acute)	5 - 7	UTI (catheter)	7
Sore throat (acute)	5 - 10 (pen V), 5 (clarithro)	Pyelonephritis	7 - 10 (14 if trimethoprim)
Sinusitis (acute)	5	Prostatitis (acute)	14
Scarlet fever	10 (pen V), 5 (clarithro)	Epididymitis	10 - 14
Impetigo	5 (top), 7 (oral)	PID	14
Cellulitis	7 (14 if poor response)	Mastitis	10 - 14
Lyme Disease	21		

NB Learning Points and Quality Improvement Ideas: Antibiotic Prescribing

- We need to be aware of the 'antibiotic iceberg' - the [RCGP Antibiotic Toolkit](#) has patient information and tools which can help tricky consultations and patient education
- The NICE antimicrobial guidelines can be found [here](#)
- Delayed prescribing significantly reduces antibiotic use by patients ([BMJ 2014;348:g1606](#))
- Probiotics reduce the rate of antibiotic-associated diarrhoea (NNT=9) – if patients use them they should take them while receiving the treatment and for 7 days after ([Cochrane CD006095](#))
- For a great insight into the scale of the problem see Dame Sally Davies' book [The Drugs Don't Work](#) or her [TED talk](#) on the subject.



CELLULITIS

Around 1 in 40 people develop cellulitis each year and most are effectively treated in general practice with antibiotics. NICE has recently published guidance on how to optimally manage the condition, and while much is common practice there are some useful tidbits of information. The guideline doesn't touch on diagnosis, which was helpfully covered in this *very helpful BJGP* ([2018;68\(677\):595](#)) paper reviewing some of the uncertainties with the condition.

1 Cellulitis and Erysipelas: antimicrobial prescribing [NICE NG141, Sept 2019](#)

This new guideline applies to both cellulitis (bacterial infection of the lower dermis and subcutaneous tissue, 2/3 due to *Strep pyogenes*, 1/3 *Staph aureus*, then occasional rare causes, e.g. with bites, water exposure) and erysipelas ('superficial cellulitis', the upper dermis is affected +/- subcutaneous lymphatics). Clinically hard to distinguish between the two, they are often considered together and the management set out by NICE is the same for both. Below we will simply use the term cellulitis to cover both conditions.

The guideline is very clear that antibiotics should be used for the treatment of cellulitis, with consideration of the severity, area affected (e.g. periorbital), risk of uncommon pathogens and any microbiology results. Patients should be reassessed if symptoms worsen or fail to improve in 2-3 days.

Interesting points made by the guideline include:

- The skin will take several days to recover after treatment, so the redness from residual inflammation may not have fully resolved by the end of the course of antibiotics.
- Microbial confirmation may be helpful but pick-up rates are very poor. Swabbing is only recommended if the skin is broken and there is an increased risk of an uncommon organism, e.g. after penetrating injury, exposure to water-borne organisms or infection acquired outside the UK.
- In most cases oral antibiotics are as effective as intravenous, so, if the patients can take them orally and not 'seriously unwell', offer this route.
- Flucloxacillin remains the first-line Abx choice in most general practice scenarios - it has a narrow spectrum of activity but is effective against both *Staph aureus* and *Strep pyogenes*. Co-prescription of penicillin and flucloxacillin (historically given to cover both organisms) is not beneficial over flucloxacillin monotherapy. However, flucloxacillin has poor bioavailability so in patients with possible impaired circulation (e.g. those with diabetes or venous insufficiency) NICE suggests up to 1g QDS orally (an off-label dose) may be needed to adequately treat the infection.
- Co-amoxiclav is preferred for treatment of cellulitis around the eyes or nose due to extended cover of *Haemophilus* and anaerobes, which may be more common in presentation. Use 1st line minimises treatment failure. Clarithromycin + metronidazole is 1st line if penicillin allergy in such cases.
- NICE recommend that prophylactic antibiotics should only be used in severe circumstances (2 episodes in 12 months requiring admission or specialist advice) and **BY A SPECIALIST**. Make of that what you will.

2 Diagnosing cellulitis

Almost 1 in 5 antibiotics used in primary care are for cellulitis or soft tissue infection. In an age where we are all trying to reduce antibiotic use, can this be done safely for this condition? Antibiotics effectively treat cellulitis and there is no suggestion that they should be avoided but over-diagnosis and over-treatment is common.

- **Overdiagnosis:** 1 in 3 hospital admissions for "cellulitis" has an alternative cause
 - Mainly venous eczema, lymphoedema or lipodermatosclerosis
 - Also consider DVT, chronic venous insufficiency and superficial thrombophlebitis
- **The BJGP has some helpful 'rules of thumb'**
 - *Bilateral red leg is rarely cellulitis* - probably non-infectious

- **Varicose eczema causes other skin changes** - e.g. haemosiderin deposits, dryness, atrophie blanche, lipodermatosclerosis
 - [CKS](#) has **useful guidance on acute lipodermatosclerosis**, often mistaken for cellulitis, with acute painful inflammation above the ankles - responds to topical steroids ("very potent" strength is advised)
- **The leg raise test can be helpful to distinguish vascular causes** from cellulitis
 - Raise the leg to 45° for 1-2 minutes - redness persists in cellulitis and usually disappears with other causes

Around 1 in 3 patients will have recurrent episodes. We know from RCT data ([NEJM 2013;368:1695](#)) that prophylactic penicillin is effective at reducing recurrence. The study treated patients for 12 months then followed them for a further 2 years. Towards the end of the study, rates were no different between antibiotic and placebo groups, so optimal duration (?indefinite) of antibiotics is uncertain.

Are there non-antibiotic methods to reduce recurrence? One simple strategy is to treat concomitant skin disease, e.g. eczema or fungal foot disease. While again data is lacking, simple emollients or fungal treatments are likely to reduce the risk of cellulitis. If lymphoedema is a problem, exercise and compression hosiery may be effective.

NB Learning Points & Quality Improvement Ideas: Cellulitis

- Cellulitis is common and successfully managed with oral antibiotics. Flucloxacillin monotherapy covers common organisms.
- Consider alternative causes of red legs when assessing - acute lipodermatosclerosis looks similar but requires steroids not antibiotics to treat.
- NICE advises against prophylaxis being initiated from primary care.
- Useful leaflets from the [NHS](#), [British Association of Dermatologists](#) and [Patient.info](#)
- Audit patients with a diagnosis of cellulitis. Which antibiotic have they been prescribed? Does it match the latest recommendations? Has an underlying cause been assessed for? Could this be modified to reduce future risk?



KISS: Acute Cellulitis

[NICE NG 141 Sept 2019](#), [BJGP 2018;68:595](#), [NICE/PHE 2019](#), [CKS 2021](#), [NEJM 2013;368:1695](#)

Diagnosis

- Acutely red, painful, hot, swollen, tender skin with rapid spreading; may have fever, malaise, etc
- Usually leg but can occur anywhere on body including face
- Unilateral – bilateral presentation rare, even in legs
- Assess for cause/predisposing factors (e.g. bites, eczema, fungal infections, lymphoedema), sepsis, abscesses, blistering, necrosis

Differential diagnosis – not an exhaustive list!

- DVT (think risk factors, often less erythema), gout, thrombophlebitis ruptured Baker's cyst
- Necrotising fasciitis – life-threatening, suspect if pain more marked than clinical features suggest
- Lipodermatosclerosis (can become red and painful in acute flares – [CKS](#) suggests treating with very potent topical steroids), lymphoedema, venous insufficiency

Treatment

- Assess and treat for risk factors and co-morbidities (e.g. diabetes, alcohol misuse)
- Urgently admit patients if symptoms or signs of more serious illness:
 - Severely unwell (e.g. tachycardia, vomiting, confusion), unstable comorbidities (e.g. uncontrolled diabetes), vascular compromise (e.g. severe PAD), sepsis, necrotising fasciitis, orbital cellulitis, etc.
- Consider referral or advice if severely unwell, lymphangitis, infection near nose/eyes, risk of uncommon pathogen, spreading infection not responding to oral antibiotics, cannot take oral antibiotics
- Swab only if broken skin and risk of uncommon organisms, e.g. water exposure, bites, facial cellulitis
- Patients with no systemic symptoms and uncontrolled co-morbidities can be managed in primary care
 - Offer oral antibiotics for 5-7 days (up to 14 days in limited cases based on clinical judgement)
 - Flucloxacillin 500mg qds first line, 1g QDS (off-label) if impaired circulation
 - Clarithromycin/erythromycin/doxycycline then if penicillin allergy
 - For doses for adults and children [click here](#) for NICE summary
 - Co-amoxiclav for mild facial cellulitis managed in community
 - If river or seawater exposure PHE recommend d/w microbiology
 - Elevate limb – helps reduce swelling and pain and aids clearance of infection; simple analgesia prn

Follow up

- There may be an increase in redness over the first 24-48 hours, possibly due to toxin release
- Reassess in 2-3 days if no improvement, or at any time if worsening symptoms

Recurrent Cellulitis

- NICE suggest specialist-led only. May be appropriate for patients with 2+ severe episodes in <12/12.
- Oral low-dose penicillin is the treatment of choice. Review every 6 months minimum.



VACCINATIONS

The progress in vaccinations is nothing short of remarkable and it isn't too much of a stretch to think that the developments in mRNA vaccines will revolutionise this field. Nevertheless, new medicines take time and require rigorous trials to access safety and efficacy and all this takes time. But still, there are recent data that show the benefits of existing vaccine programmes, for example, HPV immunisation.

► COVID Vaccinations

This rapidly changing area is a scientific marvel. There is little doubt that vaccinations have altered the course of the pandemic and been highly successful, especially in the elderly and vulnerable groups but also for adults in general allowing returning to more normal day-to-day life. In younger children, the case is less clear-cut. [Data in the BMJ](#) showed in Argentinian children the vaccine had poor efficacy against omicron (16% in younger children - it had been better against earlier variants and was more effective in adolescents at 26%) and that efficacy rapidly waned. Efficacy against death was better at 67% and 98% respectively. A [linked BMJ editorial](#) suggests that non-drug interventions remain central to containing the burden of disease at the community level. With vaccinations for adults being so effective, such measures for other groups are in short supply.

The best way to stay up to date is to consult the [Green Book COVID chapter](#) which contains the latest information.

► HPV Vaccination Efficacy [BMJ2019;364:i1161](#), [BMJ editorial Lancet 2021](#)

The HPV vaccination was introduced in the UK in 2008. Thanks to our comprehensive screening programme there are excellent data for researchers to analyse the effectiveness of the vaccine in the real world. This BMJ study ([BMJ2019;364:i1161](#)) does exactly that, comparing smear test results at age 20 in Scottish cohorts of unvaccinated women born in 1988 to vaccinated women born in 1995-96.

- It shows a *89% reduction in cervical intraepithelial neoplasia (CIN) grade 3+* (from 0.59% to 0.06%). CIN 1 was reduced by 79%.
- Younger age at vaccination was also important - *vaccine effectiveness was 86% for CIN 3+ at age 12-13* vs 51% age 17.
- The data also showed evidence of *herd immunity* in unvaccinated females in the 1995-96 cohort.

In 2021 we had further evidence comprehensively proving the efficacy of the HPV vaccine programme, and **with evidence that it reduces not only CIN3 but also invasive cervical cancer**. This Cancer Research UK-funded trial was a substantial observational trial based on UK data and was published in the [Lancet in November 2021](#) and further [discussed by CRUK here](#):

- Cancer registry data in England was reviewed between 2006 and 2019 for women who had diagnoses of either cervical cancer or CIN3.
- 3 different aged cohorts of vaccinated populations were reviewed against matched controls who had not been vaccinated.
- A massive 13.7 million years of follow-up were included.
- The **relative reductions in cervical cancer rates in the vaccinated vs unvaccinated populations varied depending on the age of vaccination**:
 - Vaccination aged 16-18 years old reduced cervical cancer rates by 34% (95% CI = 25-41).
 - Vaccination aged 14-16 years old reduced cervical cancer rates by 62% (95% CI = 52-71).
 - Vaccination aged 12-13 years old reduced cervical cancer rates by 87% (95% CI = 72-94).
- Likewise, the **risk reductions for CIN3 were also dependent on the age of vaccination**:
 - Vaccination aged 16-18 years old reduced CIN rates by 39% (95% CI = 36-41).
 - Vaccination aged 14-16 years old reduced CIN rates by 75% (95% CI = 72-77).
 - Vaccination aged 12-13 years old reduced CIN rates by 97% (95% CI = 96-98).



These results confirm the effectiveness of the HPV vaccine. They also confirm the vaccines are more effective if given younger thus supporting the current programme of giving the vaccine to 12-13-year-olds, although even in the older cohorts aged 16-18 the vaccine still reduces cervical cancer rates by ~35%. Interestingly, these were results from the use of the bivalent vaccine in both studies (the quadrivalent is now used) but, as the [linked editorial from the BMJ](#) study discusses, even with just HPV 16 and 18 officially covered, there appears to be helpful cross-protection for other invasive strains providing additional benefit above and beyond what would be expected.

This and herd immunity can only strengthen the case for the introduction of the HPV vaccine for males...

► **HPV Vaccination in Males** [DTB 2019;57:85](#), [BMJ2018;362:k3837](#), [JCVI 2018](#)

Following the announcement that the HPV vaccination would be [offered to MSM](#) (men who have sex with men) ≤45yo earlier in the year, in 2018 the [Joint Committee on Vaccination and Immunisation](#) has recommended that HPV vaccination should be offered to boys aged 12 to 13 alongside girls. This reverses the recommendation from last year where it stated this would not be cost-effective.

A recent BMJ editorial ([2018;362:k3837](#)) highlighted some of the benefits of gender-neutral vaccination for HPV. It argues that vaccinating females fails to provide equality, with males missing out on comparable protection, and reinforcing gender stereotypes around sexual health responsibility. Two particular HPV-related diseases are important:

- Oropharyngeal cancer: incidence has increased >100% between 2001 and 2011, and rates are now comparable to cervical cancer (8.9/100000 men vs 9.4/100000 women). 3/4 are in men and half are due to HPV
- Anal cancer: incidence has also increased, 80% is due to HPV; 3.1 v 1.7/100000 women/men.

Reviewed in the DTB ([2019;57:85](#)), the JCVI altered its opinion on HPV vaccination for boys due to "strengthening evidence on the efficacy of HPV vaccination to prevent non-cervical cancers", prevention of anogenital warts, cost-effectiveness review which now includes 30-year benefits, the clear benefits from vaccinating the MSM population pre-sexual debut, and the driving of more robust herd immunity. There would be cascading benefits for females, especially in vaccination rates in girls was to fall.

The vaccination will be administered in schools in England and Northern Ireland from Autumn 2019, in Wales from early 2020 and in Scotland "as soon as practicable".

► **HPV vaccine safety** [DTB 2018;56:88](#)

There was recent media speculation about the quadrivalent HPV vaccine causing autoimmune disorders following a number of case reports. The DTB reviewed recent data exploring these potential issues. A large Canadian cohort study examined >290000 girls aged 12-17y and found no significant risk of developing an autoimmune disorder after vaccination. Condition that did occur within 60 days showed them to be consistent with the background incidence.

NB Learning Points: Vaccinations

- HPV vaccination will be offered to boys aged 12/13 as well as girls in the UK, likely to start in 2019. It is also now offered to MSM ≤45yo.
- Men B vaccination has now been introduced across the UK and Ireland. Parents should administer paracetamol with the vaccination to help control subsequent fever.
- For the most up-to-date information on vaccinations throughout the UK see:
 - [PHE](#), [Immunisation Scotland](#), [Public Health Wales](#), [NI Direct](#), [HSE Ireland](#)

PERTUSSIS

Pertussis remains a very hot topic in primary care even though levels have been suppressed during the pandemic. There was a BMJ clinical update in 2019 ([BMJ 2019;364:l401](#)) on the disease and as always there is good supporting guidance from [Clinical Knowledge Summaries](#).

► What is pertussis?

Caused by the Gram-negative bacterium *Bordetella pertussis*, pertussis is droplet spread and highly contagious - a [useful diagram from Oxford's Vaccination Knowledge Project](#) shows out of 100 non-immunised people who come in to contact with an infected individual, 80 will contract the infection. The disease is cyclical, with peaks every 3-4 years (why this pattern occurs is not known), with the last peak in 2016 and before that in 2012. The latter was a particularly bad year internationally and in the UK 14 infants <3m old died from pertussis infection. In 2018 there was 1 death.

While typically considered an infection of childhood (peak incidence is in infants up to 6m old) this is also the age where we are most vigilant and more likely to notify and investigate. Therefore epidemiological data is likely to be skewed by reporting bias. The condition is in fact seen in all ages, and data suggests that pertussis causes 3% of acute cough in adults. Even this may be inaccurate, as [data also suggests gross under-reporting](#) of the true scale of the condition, by a factor of ~100...

Another cause of whooping cough symptoms is *Bordetella parapertussis*. Immunisation for pertussis provides no cover for the organism, and the usual investigations will fail to identify it. It may account for [up to one-third of whooping cough](#), albeit usually with less severe symptoms.

► How does it present?

Initially, patients have a "catarrhal phase" for 1-2/52, during which is it indistinguishable from other URTIs. After this patients enter the "paroxysmal phase" and may exhibit more typical features. After 1-6/52 patients enter the "convalescent-phase" which may last 3 months, with gradual improvement in cough.

There are specific clinical criteria for diagnosing pertussis:

- Coughing lasting at least 2 weeks with ≥1 of the following:
 - Coughing paroxysms or fits
 - Inspiratory whooping
 - Post-tussive vomiting without other apparent cause
 - Apnoea with or without cyanosis for infants <1yo

Patients with suggestive symptoms <2 /52 duration should be managed as pertussis. Many patients do not have "typical" symptoms, e.g. no whoop is common in adults and very young children. In adults presence of post-tussive vomiting and inspiratory whoop is highly suggestive of pertussis, although vomiting is a less useful feature in children. One useful tip from [a GP who has dedicated his life to understanding pertussis](#) suggests a helpful question is: "do you get frightening attacks of a choking and exhausting cough followed by hours of no coughing at all?" Anecdotally this is highly suspicious for pertussis infection.



► How should we investigate?

The disease is largely a clinical diagnosis, so management does not require investigations. However, pertussis is a notifiable disease and public health will advise on investigations to confirm the diagnosis. Largely, testing is not undertaken in primary care with public health increasingly using oral swabs 3/52 after initial symptoms.

The type of test used depends on the stage of illness and age of the patient. In the first 2 weeks, culture or PCR of nasopharyngeal / pernasal swab or nasopharyngeal aspirate is required. Their sensitivity declines after this, with oral fluid samples (detecting anti-pertussis toxin IgG) or serology (usually reserved for adults) becoming better tests.

► How effective is vaccination?

Primary vaccination has been hugely successful, preventing ~80% of associated mortality, 1.3million deaths worldwide. The UK now uses an acellular vaccine (part of the 6-in-1 given to infants) which has a lower complication rate but with a vaccine efficacy of 84% v 94% for the whole-cell vaccine. Efficacy also falls over time and there is debate as to whether regular boosters throughout life are indicated.

What this means is that it is very possible to be fully vaccinated and still contract pertussis, although the severity and duration of the symptoms will be attenuated and less severe.

► Do I need a booster?

Interestingly, since 2016 the JCVI (Joint Committee on Vaccinations and Immunisations) has recommended that healthcare workers who have contact with pregnant women and infants (e.g. pretty much all of primary care) should have a pertussis booster to reduce the risk of nosocomial spread if they have not had a pertussis vaccination in the past 5 years.

Due to worldwide shortages, this hasn't been implemented but since the second half of 2019 the supply chain has improved and this recommendation is being rolled out in stages. HCW are prioritised into one of 3 groups depending on the level of contact with pregnant women and infants. Unsurprisingly midwives, maternity unit staff, neonatal ICU staff, etc. are prioritised the highest. Those of us working in general practice are in the lower group, 3, and we will be offered the vaccination as stocks allow. Watch this space!

NB Learning Points & Quality Improvement Ideas

- Pertussis is common in all ages but often presents atypically especially in adults and young infants.
- The condition is under-reported and often missed, risking exposure to vulnerable groups.
- Consider pertussis in people with exhausting coughing fits followed by hours of no cough
- Consider antibiotic prophylaxis in vulnerable groups and those living or working with such groups
- Have you had a pertussis vaccination in the past 5 years? If not it may be offered to you soon to help prevent spread to vulnerable groups.

KISS: Pertussis / Whooping Cough [BMJ 2019;364:l401](#), [CKS 2022](#), [PHE guidelines](#)

Presentation – (Clinically suspected or confirmed cases should be Notified to Public Health)

- Incubation period usually 7-10days (range 4-21), infection is highly contagious esp. in catarrhal and first 3 weeks of cough.
- Initial catarrhal stage followed by a cough +/- paroxysms, lasting for 1-6weeks+
- Unvaccinated infants are most vulnerable with highest complications & mortality (3.5% in <6 /12 old)
- Complications include apnoea, pneumonia, seizures, encephalopathy, otitis media, unilateral hearing loss, and then those due to increased intra-thoracic/abdominal pressure, e.g. pneumothorax, hernias, rib fracture, incontinence

Diagnosis

- Check immunisation status. Immunity wanes over time & rising no. of adolescents & adults are being affected.
- Clinical on basis of symptoms: Acute cough for ≥ 14 days without apparent cause plus ≥ 1 of paroxysms of coughing, post-tussive vomiting, inspiratory whoop, apnoea in infants.
- Consider if unwell and recent exposure to a confirmed case of whooping cough.

Public Health Guidance on diagnostic tests ([click here for PHE guidance](#))

- We will be advised by local public health team on appropriate tests once notification made:
- <2 /52 since onset: Culture (nasopharyngeal/pernasal swab) but -ve result does not rule out pertussis.
- <3/52 since onset: PCR on nasopharyngeal or throat swabs can be used.
- >2/52 since onset: IgG serology if >17 y.o. or IgG oral fluid kit if 5-16 years old.
- There are some useful videos on how to take nasopharyngeal [swabs](#) + [aspirate](#)
 - Culture - use nasopharyngeal or pernasal swab in culture medium (ideally charcoal)
 - Oral fluid kit - notify local HPT, they will post one to the patient - not useful if vaccinated in past 1y
 - Serology - for patients not 2-<17yo this is a charged-for test, also not useful if vaccinated in past 1y

Management - the BMJ has a very helpful [infographic](#)

- **Admit if** 1) a child <6 months old and unwell, 2) has significant breathing difficulty, 3) has sig. complication
- **Exclusion from school/place of work** for 48 hours after starting antibiotics or 21 days from the start of symptoms if not taking antibiotics. Illness presenting >21/7 has no exclusions.
- **Antibiotics:** Start as soon as possible after the onset of illness the main benefit is to eradicate the bacteria and prevent transmission...if symptoms have lasted for > 3 weeks, there is no benefit in giving Abx as pt is no longer infectious
 - Use **clarithromycin for 7 days**, or **azithromycin for 3-5days**; erythromycin or co-trimoxazole are alternatives; only erythromycin in pregnancy - [for doses click here](#)
- **Antibiotic prophylaxis for vulnerable contacts** is recommended when the onset of illness was in the preceding 21 days. This should be managed with assistance from the public health team.
 - Includes partially immunised infants, pts with resp disease or immunodeficiency, pregnancy >32/40
 - Also, consider for anyone who lives/works with any of the above

Vaccination

- **Offer to all pregnant women from 16 weeks gestation**
 - For optimal infant protection, it must be administered at least 1 week prior to delivery
- Routine schedule: 2, 3 and 4 months; then a booster at 3 years and 4 months - offer vax if any missed
- Healthcare workers in primary care will be offered pertussis vax to reduce nosocomial spread as stocks allow



IMPETIGO

In February 2020 [NICE published new guidance](#) on impetigo in which they propose a significant change in management, with a move away from topical antibiotics for people with localised non-bullous impetigo, as discussed in [BMJ2019;366;I5162](#).

The change in practice comes following a review by NICE in which they found that a topical antiseptic (such as hydrogen peroxide 1% cream) is just as effective as treating non-bullous localised impetigo as an oral antibiotic. The evidence is for hydrogen peroxide cream, but the committee agreed that other antiseptics are also available and could be used (NB the hydrogen peroxide cream crystacide® is expensive at £8 a tube).

This recommendation to not use either topical or oral antibiotics in non-bullous impetigo initially will certainly be a big change for many of us, and indeed may make us feel a little nervous. But since this NICE guidance was published, we have had some **reassuring data in the form of a systematic review looking at the natural history of non-bullous impetigo** in the [BJGP 2021; 71 \(704\)](#):

- This systematic review identified 7 randomised trials that involved participants with non-bullous impetigo in either the placebo group of randomised trials or in single-group prognostic studies that did not use antibiotics and measured time to resolution or improvement.
- Although the times to resolution were quite wide, the **majority of cases showed resolution of symptoms by day 7 without any antibiotic treatment - with only about ~25% had not resolved within 1 week.**
- The authors conclude that immediate antibiotics for non-bullous impetigo is not mandatory given this data and does **support the NICE recommendation that for localised non-bullous impetigo a non-antibiotic strategy with topical antiseptic is a safe option.**

A crucial part of management is making the **distinction between bullous and non-bullous impetigo**. Bullous impetigo should be treated with oral antibiotics because topical antibiotics will be less able to target the infected area and the patient is at greater risk of spread and systemic illness. All babies under 1 year with bullous impetigo should be referred. For patients with non-bullous impetigo, if it is localised, it can just be treated with topical antiseptic cream. If it is more widespread, then offer a choice of either topical or oral antibiotic. Topical and oral antibiotics appear to be equally effective so patient choice and practicality of application are important. Topical antibiotics are obviously less likely to cause adverse events than oral antibiotics but may increase the risk of resistance. The guidance also contains very useful information on the scenario of difficult-to-treat recurrent impetigo.

The cornerstone of management though is, of course, good hygiene measures...

NB Learning Points & Quality Improvement Ideas: Impetigo

- NICE have produced new guidance for prescribing in impetigo, see KISS summary below
- Good hygiene measures remain paramount
- For localised non-bullous impetigo a topical antiseptic cream can be used and appears to be just as effective as topical antibiotics
- It is important to make the distinction between bullous and non-bullous impetigo
- Bullous impetigo will always need oral antibiotics and should be referred in infants under one
- If treatment fails consider taking a swab and adjusting treatment according to results, and if it is recurrent consider nasal swabs and decolonisation measures if staph carriage is confirmed



KISS: Impetigo

Based on [NICE NG153](#), [NICE-CKS 2020](#)

Background

- Impetigo can affect all ages, is spread by direct contact and the incubation period is 4 to 10 days; good hygiene measures are essential to reduce spread to other parts of the body and other people
- ***Impetigo is either non-bullous or bullous and localised or widespread***
 - non-bullous 75%, can be staph aureus or strep or both
 - bullous 25%, staph aureus
- Serious complications are rare but possible, more likely if immune suppressed e.g. cellulitis, sepsis, 'scalded skin syndrome', post-strep complications (e.g. scarlet fever, and glomerulonephritis)

Management

- ***Refer for urgent assessment*** if serious complications such as suspected sepsis, if impetigo is widespread and patient immune-compromised or bullous impetigo in infants under 1 year
- Do NOT offer a combination of oral and topical antibiotics in impetigo
- Off school until lesions crusted/healed or 48 hours after starting antibiotic
- Hygiene advice e.g. hot wash of clothes & bedding [NHS Choices PIL](#)
- ***Localised non-bullous impetigo***
 - First-line: a topical antiseptic (e.g. hydrogen peroxide 1% cream) tds for 5 to 7 days
 - Second-line a topical antibiotic if antiseptic not suitable e.g. fusidic acid 2% apply ads for 5 to 7 days
- ***Widespread non-bullous impetigo***
 - Offer topical fusidic acid or oral antibiotic
 - Take into account the preference of patient and carers and practicalities of administration
- ***Bullous impetigo, or if the patient is a systemically unwell or high risk of complications***
 - Offer oral antibiotics, oral flucloxacillin (or clarithromycin if allergic or flucloxacillin unsuitable) for 7 days
- ***Safety net***
 - Even though impetigo is generally mild, significant complications from cellulitis to sepsis can develop so advise patients of the need to reassess if symptoms worsen significantly or rapidly
 - If it does not improve after initial treatment with topical or oral antibiotic
 - consider sending a skin swab
 - if have been on topical fusidic acid, switch to oral antibiotic (or to topical mupirocin, if fusidic acid resistance suspected or confirmed on swab)
 - if have been on oral, extend the course pending swab results & adjust according to results
 - if MRSA is identified, get local expert advice
- ***Recurrent impetigo***
 - if frequently recurrent consider a skin swab for testing and consider taking a nasal swab to test for the carriage of staphylococcus and starting treatment for nasal and skin decolonisation if present
 - If staph carriage, eliminate with naseptin qds for 10 days
 - avoid naseptin if peanut or soya allergic, use mupirocin instead
 - decolonise skin with chlorhexidine 4% body wash (hibiscrub) for 5 days
 - apply directly to wet skin, leave for 1 minute before rinsing off
 - use *Dermol®* (emollient containing chlorhexidine) if skin condition or sensitive skin
- **Patient information:** [British Assoc Derm](#)

MUMPS

Mumps has been called "[the forgotten disease](#)", the title being apt - most of the public have no experience of mumps and it is testament to the success of vaccinations that they also have little understanding of the serious complications that can result from prevented infections. While rates have been low throughout the pandemic it was good to have a very useful update from [Clinical Knowledge Summaries](#) on the condition.

► *What is mumps?*

Mumps is caused by the paramyxovirus, leading to an acute infection and the classic "chipmunk face" appearance due to marked parotid swelling. It is droplet, fomite or saliva spread with an incubation period 16-18 days (range 12-25D). It is most infectious 1-2 days pre-symptom development until 9 days after. The virus has a propensity for the nervous system and so meningism is common. The clinical features are highly recognisable but up to 20% of infected individuals may be asymptomatic while still being vectors for the disease.

In 2005 England and Wales saw a huge spike in mumps cases - >40000 - since then the levels have been much lower at around 4000 confirmed cases per year. The start of 2019 has seen an increase of ~50% in cases compared to the same period in the previous year, but despite the media interest, it is certainly not the levels seen a decade ago.

Universities and colleges are the typical places where outbreaks are seen - a combination of close proximity and poor herd immunity. Vaccination (using 2 doses as per the MMR) provides [around 88% protection](#), adequate but well short of other elements of the MMR. As such even those who are fully vaccinated may contract the disease, although the effects are likely to be less severe than in the unvaccinated. Nevertheless, one of the biggest risk factors for mumps remains no vaccination. The good news is that the vast majority of people who actually contract mumps will develop life-long immunity afterwards.

KISS: Clinical Features of Mumps

- **Clinical features**
 - 95% will get parotitis - usually unilateral initially then contralateral side swells, increases for 2-3/7, may cause ear pain and make chewing and talking difficult; 1 in 4 remain unilateral only
 - Non-specific viral symptoms, e.g. mild fever, malaise, muscle aches, often 1/7 prior to parotitis
- **Complications**
 - **Epididymo-orchitis**: up to 38%, mostly unilateral, starts 1-2/52 after parotitis manifests, may cause a transient reduction in sperm count. Infertility rare but sub-fertility common occurs in ~13% - bilateral disease (in 15-30%) associated with worse prognosis, 30-87% develop infertility. Resolution of symptoms generally takes 2/52, abx are unhelpful.
 - **Oophoritis**: up to 7% females, but unlike in males rarely causes fertility issues
 - **Aseptic meningitis**: up to 25%, males > females 3:1, usually benign with full recovery and no risk of mortality
 - **Encephalitis**: affects 0.1%, mortality is 1-5%, long-term morbidity uncommon.
 - **Transient hearing loss**: in ~4%, permanent deafness only in 1:20000, bilateral hearing loss very rare.
 - **Pancreatitis**: ~4% but typically mild and transient
- **Differential diagnosis of parotitis**
 - There are other causes of parotitis including other viruses (EBV, parainfluenza, adenovirus, influenza, coxsackie, parvovirus B19, HIV), bacterial (usually Staph aureus, also TB, usually unilateral), and non-infectious causes, e.g. parotid duct obstruction (e.g. stones), drugs (e.g. thiazides), autoimmune disease (e.g. Sjogren's syndrome), and metabolic disorders (DM, cirrhosis, uraemia)
 - Observational data suggests that the most common causes of parotitis during non-epidemic periods are likely to be EBV, parainfluenza and adenovirus. It appears unclear if these cause as marked swelling as mumps.



► *How should we manage mumps?*

Mumps is a notifiable disease, so Public Health should be informed of any suspected cases. The investigation of choice is salivary mumps IgM, with a response and accuracy effectively as good as serum-based tests. Mouth swabs can be taken from 1-6 weeks after symptoms begin.

Treatment for mumps is supportive only. The virus is self-limiting, resolving in around 1-2 weeks. Beyond simple analgesia for any discomfort, no other medicines are recommended. Human Normal Immunoglobulin does not have evidence to support its role; antibiotics and steroids are not recommended. CKS recommends safety netting RE meningitis and epididymo-orchitis and suggests clinicians "consider" follow-up at 1 week to ensure improvement.

Isolation is recommended, and suspected cases should remain off school or work for 5 days from the development of parotitis. Patients should be given information on the condition ([click here for PILs](#)). There is no specific management for contacts with mumps except checking immunisation status and using it as an opportunity to complete any missed doses. Pregnant women and the immunocompromised are managed no differently to normal, except that vaccination with MMR is contraindicated in the former and not routinely given in the latter.

NB Learning Points & Quality Improvement Ideas

- Mumps is a viral illness, self-limiting in most, but linked with some common complications including epididymo-orchitis, oophoritis, aseptic meningitis and encephalitis.
- Vaccination is only 88% effective, with herd immunity providing the benefit for those without full immunity.
- Treatment is supportive only, no medication has been shown to improve symptoms or prognosis.
- Notify public health of any suspected cases so that they can send the patient an oral swab for confirmation.

MEASLES

100,000 people, predominantly children, die from measles each year around the world - astonishing for such a preventable disease. Yet despite a safe and effective vaccine, the worldwide incidence of measles increased by 300% in 2019.

The UK had 966 confirmed cases in 2018, the highest rates for decades, so it's no surprise measles is a very hot topic. This increase is primarily due to travel within Europe where there were a [staggering 82500 cases](#) and 72 deaths in 2018. Tellingly, around 80% of infected younger people had never been vaccinated.

The BJGP ([2018;68\(674\):404](#)), BMJ ([2017;356:j426](#)) and PHE remind us about identifying and managing measles. While current levels are low due to the pandemic, inevitably they will rise as mixing increases.

► *What is measles?*

Measles is a highly infectious virus transmitted via respiratory droplet. Being in the same room as an infected individual is enough, particularly for 15mins or more. While mortality is rare in the UK, in countries with poor medical access it can be as high as 10% as a result of complications such as pneumonitis, encephalitis, secondary bacterial infections and, in the latter stages, subacute sclerosing panencephalitis.

Vaccination with 2 doses provides excellent immunity (PHE quote nearly 100% effectiveness) which is thought to be lifelong. To achieve control of measles countries need to achieve 95% uptake of the vaccination. While uptake throughout the UK is generally good (~92% - presumably high in 2017 for the WHO to state it was eliminated) in London it is <80%.< /p>

► *How does measles present?*

KISS: Presenting Features of Measles

[BJGP2018;68\(674\):404](#), [BMJ 2017;356:j426](#)

- Incubation period: 7-14 days. Infectious 4 days before and after the rash appears, with peak infectivity in the first 3 days prior to the rash developing
- Initially fever, cough, coryza, conjunctivitis
- Rash follows 2-4 days later - erythematous maculopapular rash, blanching - [click for pics](#)
 - RASH STARTS ON FACE, THEN SPREADS DOWN THE BODY
- Koplik's spots - white spots on the oral mucosa - these are PATHOGNOMONIC
- Risk factors which should prompt suspicion:
 - Unvaccinated/partial vaccination
 - Live in high-risk communities with suboptimal MMR uptake (e.g. travelling communities, new migrants, Steiner communities, Charedi Orthodox communities)
 - Recent travel to measles-endemic countries or areas where there are outbreaks - e.g. Romania and Italy, and in Asia and Africa
 - All ages are affected, but 15-25yo are at particular risk

► *How is measles diagnosed?*

PHE now recommends oral fluid as the optimal sample for measles surveillance. While being more acceptable and minimally invasive than blood tests, it can still be tested for IgM, IgG and measles RNA. Therefore it can: 1) reliably confirm or exclude measles, 2) shows if this is a primary or re-infection, and 3) genotype confirmed cases.



Testing IgM on oral fluid is more sensitive and more specific than serum, especially in the first few days of the illness. IgM antibodies are +ve in >50% of samples on D1, and >90% by D3. However, given this limitation, if taken within 7 days of onset, PCR is also performed for RNA detection, which can be positive even before the onset of the rash.

Serum can also be tested and is the most appropriate way of assessing the immune status of contacts.

► *How is measles managed?*

Measles is a notifiable disease.

Measles is usually self-limiting and coryza settle in around a week. For most patients management is the same as a viral infection: rest, fluids, paracetamol PRN with safety netting for complications (e.g. symptoms of ear infection, breathlessness, persistent fever, mental change, etc.) Patients with such changes or higher-risk patients (pregnant, immunocompromised, etc.) should be admitted.

Patients should avoid contact with pregnant women, children (in particular babies who will not have been vaccinated yet) and unvaccinated people.

Public health may offer **post-exposure prophylaxis with human normal immunoglobulin to non-immune contacts at higher risk** (pregnant, immunocompromised, <1yo). This is most effective when used < 3 days from exposure, but can be used for up to 6 days. **It has an NNT = 2!** (For PHE guidelines on [PEP click here](#) and [full PHE guidance click here](#).)

Unsurprisingly, public health is trying to spread to strong message to the general public about ensuring appropriate immunisation with 2 doses of MMR, and for anyone who is unsure or has missed one or both doses to contact their GP practice to catch up. Opportunistic vaccination in those who have not had 2 doses is recommended.

Of note, measles vaccination can be given from 6 months of age if the child is travelling to a high-risk country. If they are vaccinated before the age of 1y, they will still require a further 2 doses at the usual times.

NB Learning Points and Quality Improvement Ideas: Measles

- Consider measles in patients with non-specific viral features and maculopapular rash
- Be suspicious in patients with fever and new rash returning from Europe - ask about vaccination history
- Notify public health of positive cases as contacts may be given immunoglobulins
- Audit your practice population for MMR immunisation status
 - Contact patients with incomplete MMR courses, check the records are accurate and invite them to complete their vaccinations

LOWER RESPIRATORY TRACT INFECTION

There is no doubt, that if we suspect pneumonia then patients merit antibiotics, and as soon as possible. The *new NICE antibiotic prescribing guidance* ([NICE NG138, 2019](#)) *recommends treatment in the community within 4 hours*. But how long are antibiotics needed to resolve a pneumonia?

► Amoxicillin duration & dose for CAP in children [Health Tech Ass 2021](#)

This UK & Ireland multicentre randomised blinded non-inferiority trial asked whether 3 days of antibiotics for community-acquired pneumonia in children >6 months old between 6 and 24kg. 824 children were recruited from paediatric ED and inpatients.

The **primary outcome was clinically indicated antibacterial treatment for respiratory infection within 28 days other than that given as part of the trial. This occurred in 12.5% of participants in BOTH groups**. There was also no difference between a higher (70-90mg/kg/day) vs lower (35-50mg/kg/day) dose of amoxicillin. In a range of measured secondary outcomes exploring pneumonia symptoms and adverse events, the only difference was in time to resolution of cough - marginally faster with 7 days of treatment, 10d v 12d. This data certainly supports shorter durations of antibiotics than traditionally considered necessary even for serious infections such as pneumonia.

► Antibiotics for LRTI in children in primary care - ARCTIC PC [Lancet Sept 2021](#)

Observational data shows that antibiotics are prescribed for at least 40% of children with a 'chest infection', usually diagnosed due to a cough and largely upper respiratory in origin. In this group antibiotics are ineffective. However, most of us would consider a lower respiratory tract infection to be more serious and prescribing rates would be even higher. But do antibiotics make a difference in LRTI in children?

A new primary care-based double-blind, randomised placebo-controlled trial examined the benefit of antibiotics (amoxicillin 50mg/kg for 7 days) in children aged 6 months to 12 years with acute uncomplicated LRTI for <21 days. Children with likely pneumonia or who were severely unwell were not enrolled, also if a non-infective origin (e.g. hayfever) was suspected. The **primary outcome was duration of moderately bad or worse symptoms**. Secondary outcomes included severity of various symptoms (e.g. cough, SOB, feeling unwell, fever), total symptom duration and side effects. The results showed:

- Of 432 children, median durations of moderately bad or worse symptoms were similar between both treatment and placebo groups.
- There was also no benefit seen in important subgroups including children with chest signs, fever, SOB or chest rattle.
- Rates of side effects were similar between groups and significant complication rates were low.

The authors conclude that: "*amoxicillin for uncomplicated chest infections in children is unlikely to be clinically effective either overall or for key subgroups in whom antibiotics are commonly prescribed. Unless pneumonia is suspected, clinicians should provide safety-netting advice but not prescribe antibiotics for most children presenting with chest infections.*" This provides us with more confidence in our management when assessing children with possible lower respiratory tract infection that holding off antibiotics as we would with clear URTI is unlikely to do harm.

NB Learning Points & Quality Improvement Ideas: Lower Respiratory Tract Infections

- Shorter duration antibiotic courses are equally effective as longer in children with pneumonia.
- Antibiotics are ineffective in children with LRTI (and no signs of pneumonia).
- Audit your patients presenting with lower respiratory tract infection without pneumonia
 - Set a standard (even if this is "relatively" high, e.g. 50%) for maximum antibiotic use
 - What percentage have been prescribed antibiotics? Consider re-audit after 6 months.

KISS: Management of Lower Respiratory Tract Infections

[NICE CG191](#), [NICE 2019 NG138](#) [NICE NG120 2019](#)

Clinically assess patient for pneumonia

- Is there diagnostic uncertainty regarding whether this is pneumonia? Are antibiotics being considered?
 - **If yes, consider a point-of-care CRP test if available**
 - No treatment if CRP 20mg/L or less
 - Delayed script if CRP 20-100mg/L
 - Immediate antibiotics if CRP >100mg/L

If pneumonia is likely:

- Use **CRB65 score to assess risk** of mortality and guide whether admission is required
 - 1 point for 1) confusion, 2) RR >30, 3) SBP <90 or DBP <60mmHg, 4) Age 65+
 - Use clinical judgment, including social situation, alongside CRB65 to guide management:
 - Score 0 = low severity, usually managed in the community
 - Score 2+ = consider hospital assessment as higher risk
- **If managing at home**
 - **Start antibiotics within 4 hours**; consider to local prescribing guidelines
 - **First line for mild or moderate severity:** amoxicillin 500mg TDS, 5 days only
 - **Penicillin allergy:** doxycycline, clarithromycin, erythromycin - all 5 days
 - High severity 1st line: co-amoxiclav + clarithromycin or erythromycin orally 5/7 (levofloxacin if pen allergy)
 - If atypical pathogens suspected use amoxicillin + clarithromycin or erythromycin 5 days
 - **Consider extending the duration for longer than 5 days** where symptoms do not improve as expected after 3 days
 - **Safety net appropriately**, including that patients/families/carers should seek further advice if symptoms do not begin to improve within 3 days of starting an antibiotic, or sooner if they worsen
 - If symptoms have not improved after Abx send a sputum sample
 - **Refer to hospital** if more serious illness, failure to improve as expected with antibiotics, abx-resistant organisms/unable to take oral antibiotics or a child/young person
- **Explain to patients that resolution takes time**
 - 1 week: fever should have resolved
 - 4 weeks: chest pain and sputum production should have substantially reduced
 - 6 weeks: cough and breathlessness should have substantially reduced
 - 3 months: most symptoms should have resolved but fatigue may still be present
 - 6 months: most people will feel back to normal

If pneumonia is clinically excluded and POC CRP not used for guidance:

- The majority of infections are viral, self-limiting and antibiotics are not required ([Lancet Infect. Dis. 2013;13:123-129](#), [BJGP 2014;64:e75](#)); the harms are likely to outweigh benefits
- Oral prednisolone is also ineffective and do not prescribe inhaled corticosteroid either.
- Do not prescribe inhaled B-agonists.

SHINGLES

Herpes zoster virus (HZV) is high on the agenda in the medical community - much of this is sparked by new data on the new, non-live shingles vaccine, Shingrix ([BMJ2018;363:k4029](#)) and on the effectiveness of the UK shingles vaccination programme ([Lancet2018;3:e82](#) & [BMJ Open 2020](#))

Along with this, there have been some useful reviews on the management of shingles and in particular, HZV ophthalmicus in the BMJ ([2019;364:k5095](#) and [2019;364:k5234](#)), summarised in our KISS at the end of the chapter.

► **Evaluation herpes zoster vaccination programme** [Lancet2018;3:e82](#), [DTB2018;56:10](#) [BMJ Open 2020](#)

Since 2013 the UK has used Zostavax, a live attenuated (LA) shingles vaccine, to deliver a shingles vaccination programme in people aged 70-80. This [population-based study](#), reviewed in the [DTB](#), is the first evaluation of how it has performed in England, 3 years into the programme.

- 72% uptake in the 1st routine cohort, 58% for current targeted cohorts
- Incidence of shingles fell by 35% in the routine and 33% in the catch-up groups
- Post-herpetic neuralgia fell by 50% in the routine and 38% in the catch-up groups
- The authors suggest this is equivalent to vaccine effectiveness at 3y of 60% against HZV and 70-88% against neuralgia, which is actually better than seen in the clinical trials and equates to 17000 fewer cases of shingles and 3300 fewer cases of post-herpetic neuralgia in 5.5million people in the first 3 years alone.

There were some limitations of this research, as identified by the DTB but since then we have had further data ([BMJ Open 2020](#)) confirming large and prolonged **reductions in not only the incidence of shingles, but also consultations rates for post-herpetic neuralgia (PHN), and importantly reductions in hospitalisation:**

- Population-based study to assess the impact of the herpes zoster vaccination programme in the 5 years after introduction using both primary care data and hospital episode statistics.
- In the **5 years post-implementation there were reductions in:**
 - Hospitalisations with zoster of 37% (primary vaccinated cohorts aged 70) and 49% (catch-up groups aged 78-79)
 - PHN consultations of 75% (primary vaccinated cohorts aged 70) and 66% (catch-up groups aged 78-79)
- The authors **estimate that over this 5 year follow-up period the vaccination programme saved 40,500 zoster consultations and 1840 fewer hospitalisations.**

► **Efficacy and safety of HZ vaccines in adults aged ≥50** [BMJ2018;363:k4029](#)

Until now Zostavax has been the only shingles vaccination available in the UK on the NHS. One of the limitations of Zostavax, as a live vaccine, is it is contraindicated in immunosuppressed patients. There is now a second licensed option, Shingrix. This has the advantage of being an adjuvant recombinant subunit (ARS) vaccine with no live component. But is it as effective?

There are no head-to-head trials, but this new BMJ paper ([BMJ2018;363:k4029](#)) performs a network meta-analysis to compare the efficacies of both treatments against placebo, and a systematic review to assess for adverse events.

- 27 studies, including 22 RCTs, of >2,000,000 patients were included, aged 50y or over
- **Adjuvant recombinant subunit vaccine was superior to the live attenuated vaccine at preventing shingles** (risk ratio 0.15, 95% CI 0.02-0.69)
 - For lab or doctor-confirmed cases the ARS vaccine had a **vaccine efficacy of 94%**
 - The live vaccine had a vaccine efficacy of 57%
 - HZ ophthalmicus and post-herpetic neuralgia rates were inconclusive due to limited data

- There were almost twice as many injection sites (e.g. redness, swelling; RR 1.79) and systemic (e.g. headache, myalgia; RR 1.87) adverse reactions. However, there was no statistically significant differences in rates of death, serious adverse events or withdrawal from treatment.

The Joint Committee on Vaccination and Immunisation has recommended that Shingrix® is cost-effective and may be used as an alternative for patients who have a contra-indication to Zostavax. There had been issues with supply initially but as of [July 2022 the UK Government information](#) suggest this should be available again in the UK

But given this efficacy data should it be used for all individuals? In the US and Canada, it is now recommended in preference and from the age of 50.

A linked editorial ([BMJ2018;363:k4203](#)) sets about answering this question. There are several issues:

- There is a *trade-off between increased efficacy and “short-term cost”* - more minor adverse events, although serious adverse event and death rates are no different.
- The *ARS vaccine may be more effective in the target age group* (70-80yo) but this new study could not assess effectiveness by age group.
- The *ARS vaccine requires 2 injections 2-6 months apart*, and a single dose alone has a much lower immune response. This has *implications for compliance* and may reduce the overall effectiveness in the real world when people fail to return for their follow-up shot.
- *Follow-up data on the ARS vaccine is currently fairly limited* - 4-year data shows a drop in efficacy by that point, although 85%, however, it calls into question the longer durability of effect.

So overall this is a welcome addition in the prevention of shingles but as an alternative to Zostavax rather than a clear replacement for the time being.

NB Learning Points & Quality Improvement Ideas: Herpes Zoster

- The current vaccination programme has been effective at reducing shingles, PHN as well as reducing consultation rates and importantly hospitalisations as well
 - **We should be actively encouraging patients in the target age to have the shingles vaccination**
- Shingrix is a newly available option and may be used in immunosuppressed patients
- Early treatment of shingles with antivirals speeds recover and reduces PHN
- In **facial shingles look for Hutchinson’s sign** - a rash on side or tip of the nose:
 - This implies the nasociliary branch of the TG nerve is affected and is a strong predictor of ocular damage; 85% will develop ocular involvement
 - If present refer to ophthalmology for assessment

KISS: Management of Herpes Zoster

[BMJ2019;364:k5095](#), [BMJ2019;364:k5234](#)

The annual incidence in the UK in 3.9 cases/1000 population - increasing to 11/1000 in aged ≥ 80 y

Risk factors:

- **Strong:** age >50 , HIV (15x risk), immunosuppression; **Weak:** female, white ethnicity

Protective factors:

- It has been suggested that greater exposure to children may confer some protection due to a booster effect from their chickenpox, but this effect remains unclear. Equally rates of shingles appear to be increasing in countries regardless of vaccination of children against varicella.

Presentation - usually a clinical diagnosis.

- **Prodromal pain for 2-3 days** (rarely a week) **before developing a vesicular rash** in the dermatome of the infected ganglion.
 - T1 to L2 most commonly; usually sensory neurones, 5-15% have motor involvement
 - Pain intermittent or constant; burning/stabbing/throbbing is typical; may be severe
 - Rash initially erythematous, macular rash, then vesicles develop in 1-2/7, pustules within 1/52, then ulceration 3-5/7 later. A few lesions beyond the primary dermatome can occur. Healing occurs after 2-4/52, often with scarring.
 - The rash should not cross the mid-line. Dermatomal distribution is only seen with HZ.
- Consider differential diagnosis, e.g. rash ?dermatitis ?herpes simplex; pain ?abdo cause

Complications

- **Post-herpetic neuralgia is common**, risk increases with age. The duration can be into years.
- **Ocular complications** (see below)
- **Other complications are uncommon:** bacterial superinfection (1%), nerve palsy (1.8%), sensory loss (1.8%), encephalitis and disseminated herpes zoster (1.7%); rarely fatal but can be serious in immunocompromised patients

KISS: Management of Herpes Zoster (continued)

[BMJ2019;364:k5095](#), [BMJ2019;364:k5234](#) NICE PHE antimicrobial guidance CKS 2022

Treatment

- **Consider antivirals in all patients** - esp if moderate/severe pain, >50, immunocompromised, non-truncal or trigeminal nerve involvement.
- More effective if started within 72h of the rash appearing, but consider treating up to 1 week after development of rash if high risk of complications (eg older age, immunocompromised etc).
- **7 days of oral aciclovir, famciclovir or valaciclovir.**
 - 1st line = Aciclovir 800mg 5x daily.
 - 2nd line if poor compliance = Famciclovir 250-500mg TDS or 750mg BD, or valaciclovir 1g TDS.
- Treatment during **pregnancy** is usually the same but seek specialist advice first.
- Keep the rash clean and dry to prevent bacteria superinfection.
- **Acute herpetic pain:**
 - Simple analgesia, opioids if severe and necessary; topical lidocaine and nerve blocks are also effective
- **Post-herpetic pain:**
 - Mild > mod pain: simple analgesia (paracetamol, NSAID) +/- a weak opioid.
 - Mod > severe pain: consider a short-term stronger opioid and offer a tricyclic e.g. amitriptyline, or gabapentin/pregabalin or duloxetine. capsaicin cream (if available) is also an option. CKS also suggest to consider oral corticosteroids in the first 2 weeks following rash onset in immunocompetent adults with localized shingles if pain is severe, but only in combination with antiviral treatment.
- **Infectivity** is through direct contact with fluid from the rash, so covering reduces risk of transmission - however, immuno-compromised may also shed virus from the respiratory tract.

Referral

- **Immuno-compromised** - refer, treatment aims to reduce cutaneous and visceral dissemination.
- **Zoster ophthalmicus** - up to 20% of shingles cases
 - Potentially sight-threatening, treatment reduces the risk of ocular involvement.
 - May involve the eye and/or the surrounding skin, or occurs without eye involvement.
 - Typical shingles rash on the unilateral forehead may get bilateral periorbital swelling.
 - **Ask if pain in the eye** (suggests ocular involvement) **or periocular** (suggests not).
 - **Look for Hutchinson's sign** - rash on tip, side or root of the nose - implies the nasociliary branch of the TG nerve is affected and a strong predictor of ocular damage; 85% will develop ocular involvement ([BMJ 2014](#)).
 - **Examine the eye:** check acuity, assess for redness, consider fluorescein to assess for corneal changes - if present immediate referral is required.
 - **Refer all patients with ocular symptoms and if Hutchinson's sign +ve.**
 - Treatment may be for a longer duration at 7-10 days; oral aciclovir is still the first line.

CHICKENPOX

Chickenpox is the primary acute infection caused by varicella-zoster virus. It is highly infectious, particularly in the 48hrs before the characteristic rash develops, and until the lesions have crusted over, which usually takes 5 days. Transmission is via personal contact or droplet spread, with a 1-3 week incubation period. The majority of children will develop chickenpox; 90% are immune by age 15.

While an effective vaccination exists and is used in many developed countries, it has not been recommended by the JVCi for use in the UK due to an apparent lack of cost-effectiveness, driven by the theory that adults will lose a boost to their immunity with exposure to infected children, resulting in higher rates of shingles in older people, which is very costly to the health service.

The main change in the world of chickenpox is the updated [public health guidance 2022](#) recommending **antivirals (aciclovir) are the PEP treatment of choice (not VZIG) for pregnant women who are not immune and have significant chicken pox exposure.**

KISS: Chickenpox

[CKS 2022](#), [PHE Infections](#)

Diagnosis - clinical

- **Classic feature is characteristic pox lesions** - [click here for images of stages](#)
 - Initially small, erythematous macules on scalp, face, trunk, proximal limbs, which turn into papules, clear vesicles and pustules within 24 hours. Oral and genital lesions may also occur. Lesions are often intensely itchy.
 - Crusting usually occurs within 5 days (may take longer in immuno-compromised)
- A prodrome may occur before the rash appears: nausea, myalgia, anorexia and headache
- Fever, malaise, loss of appetite, feeding problems in younger children

Treatment - mostly supportive with symptomatic treatments

- If serious complications admit to hospital
- Symptomatic treatment includes topical calamine lotion, paracetamol, antihistamines
 - Avoid NSAIDs as they are associated with increased rates of severe skin infections
 - Aspirin should be avoided, especially in children <16yo
- **Consider oral aciclovir** 800mg 5x daily 7/7 for adults or adolescents (14y+) who present within 24h of rash developing, esp. if higher risk of complications e.g. severe, steroid use or smokers - given the increased rates of complications in older people. Not recommended in healthy children.
- Seek immediate specialist advice for neonates <4w old and those immunocompromised.
- Avoid contact with immuno-compromised, pregnant women, infants 4 weeks old or less
- Nursery/school exclusion until all the vesicles have crusted over

Complications - rare, but potentially serious

- **Children** - secondary bacterial skin infection may occur: suspect with erythema, tenderness and new fever after initial improvement - a wide range of other rare complications can occur including neurological (e.g. Reye's syndrome, encephalitis), myocarditis, glomerulonephritis, etc - neonates are at particularly high risk and may be given VZIG
- **Adults** - more serious infection - 80% of deaths from chickenpox occur in adults - pneumonia, hepatitis, encephalitis are more frequent than in children; smokers are at higher risk
- **Pregnancy** - infection more severe again, esp. in late gestation; 1 in 10 develop pneumonia



KISS: Chickenpox Infection or Exposure in Pregnancy

CKS 2022, [HPA Viral Rash in Pregnancy](#), [RCOG GTG13](#), [PHE Infections](#)

Pregnant women - primary infection

- Admit to hospital if suspected chickenpox and illness such as respiratory or neurological symptoms (excluding headache), haemorrhagic rash or bleeding, severe disease, immune suppression
- Consider/discuss need for admission if higher risk for complications e.g. approaching term, obstetric risk factors, smoking, chronic lung disease
- For all other women **seek specialist advice on further management:**
 - Oral aciclovir may be prescribed if presenting <24hr from development of rash and is 20 weeks gestation or greater. Women <20 weeks gestation aciclovir should be used with caution; [UKTIS](#) found inadequate data to allow evidence-based assessment of risk with aciclovir
 - There is a ~1% risk of developing fetal varicella syndrome with infection if <28/40
- Where primary care management is advised by a specialist, 'close monitoring' is recommended in the community (e.g. daily reviews) and careful safety netting is required
 - Women need to be admitted if fever persists, or new cropping of rash develops after 6 days
 - Consider bacterial superinfection if a new fever develops with pain/redness around lesions

Breastfeeding women - primary infection

- Consider aciclovir if presenting within 24 hours of rash development
- Seek specialist advice about whether to continue breastfeeding and if the baby requires treatment to minimise complication risk
 - VZIG is recommended for neonates if the mother develops chickenpox between 7 days before to 7 days after delivery
 - HPA advice is mothers should be allowed to breastfeed, but where lesions are close to the nipple expressing milk is advised, which can be used if the baby is covered by VZIG or aciclovir

Pregnant women - exposure and no known personal history of chickenpox

- Was it significant exposure?
 - 48hrs before rash developing until crusting (including disseminated zoster, exposure localised lesions or immuno-suppressed with shingles)
 - Continuous home contact, in same room for 15 minutes, or face-to-face contact
- If not immunocompromised and definite history of chickenpox (or 2 doses of varicella vaccine) then women can be reassured she is not at risk
- If no or uncertain history of chickenpox then check bloods for varicella-zoster virus immunoglobulin antibodies (VZV IgG) - [click here for HPA algorithm \(figure 1 P26\)](#)
 - If detected reassure, and advise review if rash develops
 - If VZV IgG equivocal retest (if time to do so)
 - **If negative PEP with antivirals is now recommended (NOT VZIG):**
 - Aciclovir 800mg 4x per day for 7 days, from day 7-14 after exposure
 - VZIG is now only recommended when antivirals are contraindicated (e.g. malabsorption, hyperemesis, renal toxicity)
 - The woman should be advised she is potentially infectious for 8-28 days after contact - up to 50% may develop a modified form of the disease
 - If further exposure a second course of antivirals can be given
 - Consider vaccination after the pregnancy



INFECTIVE DIARRHOEA

Infective diarrhoea is becoming more common courtesy of cheap foreign travel, however, most cases are actually acquired at home. The BJGP recently covered cryptosporidium infection and the BMJ highlighted that giardia is more common than we think and usually originates in the UK. But first, what about using 'poo in a pill' to treat C diff infections...

1 Faecal Transplant for Recurrent *Clostridium Difficile* [JAMA 2017;318:1985](#)

We are beginning to unravel the importance of the microbiome, our natural flora, from the digestion of food to its influence over allergic and inflammatory disease. Imbalances in this flora can lead to problems in the body, generally minor such as thrush or BV, but sometimes severe, as in the case of *Clostridium difficile*.

It has already been shown that faecal microbiota transplantation (FMT) is effective in preventing recurrent C diff. This requires donor stool to be infused into the patient's colon via colonoscopy or enema in most instances. Capsule versions are available but can 'poo in a pill' be as effective and will people use it?

This Canadian RCT of 116 adults with recurrent C diff infection were recruited:

- **Randomised to receive a single treatment of FMT by capsule or colonoscopy**
- **Primary outcome:** % without C diff at 12 weeks post-treatment
- **Secondary outcome:** adverse events, patient acceptability
- **Results**
 - Prevention of C diff was achieved in 96% of participants in both groups
 - 1 person in each group died due to underlying cardiopulmonary illness, not linked to the treatments
 - Minor adverse events were seen in 5.4% of capsule vs 12.5% of colonoscopy users
 - 66% receiving the capsule rated their experience as 'not at all unpleasant', compared to only 44% have a colonoscopy

The *authors conclude that among adults with recurrent C diff infection, FMT via capsules was not inferior to delivery by colonoscopy* for prevention of infection at 12 weeks.

Other new treatments for C diff have also become available: *bezlotoxumab* is a monoclonal antibody that neutralises C. diff toxin B, given with antibiotic therapy to treat the underlying infection. A DTB review ([2018;56:57](#)) comments that data shows a 10% improvement in C. diff rates vs placebo after 12 weeks, and at a cost of £2470, so "we cannot see a clear place" for the drug.

While new therapies are welcome, given how promising FMT appears, and this research showing that capsules can deliver the necessary 'good microbes' to the large intestine, perhaps very technologically advanced treatments are not required. FMT is also seeing promising results in other areas as well, such as the treatment of inflammatory bowel disease.

It may seem hard to swallow but we are going to see lots more 'poo in a pill'.

2 *Cryptosporidium* [BJGP 2017;67:425](#)

This recent and useful article in the BJGP ([2017;67:425](#)) reminds us about the importance of identifying cryptosporidium, a protozoal parasite and the 4th most common GI infection in the UK, and notifying Public Health to help prevent serious outbreaks. Most cases are due to the ingestion of contaminated water - swimming pools are a common source as chlorination is not strong enough to kill the infection - food, or direct spread from infections people or animals.

See our KISS below for diagnosis and management.

KISS: *Cryptosporidium*

[BJGP 2017;67:425](#)

- Highly infectious most commonly affects <5yo, incubation period usually 5-7 days
- Causes ***self-limiting acute illness lasting around 2-3 weeks***, potentially debilitating in very young or old or immunosuppressed
- ***Symptoms***: abrupt onset watery diarrhoea with mucous (rarely blood) +/- nausea, anorexia, fever
- If suspected ***collect a stool sample*** - genotyping can help determine the source of outbreak (e.g. type 1 is usually from humans, type 2 from animals)
- ***Supportive treatment***: fluids to prevent dehydration
- ***Notify public health*** of infection
- ***Advise on strict hand hygiene*** after contact with faeces, nappies and animals, separate towels and hot machine washes
- Affected patients should be ***excluded from school/work until 48hr after they are free from symptoms***. Food handlers do not require proven microbiological clearance. Swimming should be avoided until clear of diarrhoea for 2 weeks

3 **Giardiasis** [BMJ 2016;355:i5369](#)

As this excellent review in the BMJ highlights, contrary to established wisdom, Giardia in the UK is most commonly acquired in the UK and under-diagnosed due to a combination of clinical oversight (usually due to patients having no history of foreign travel) and the surprisingly poor sensitivity of microscopy.

See our KISS on the next page for full management of giardiasis.

4 **Oral Rehydration In Children** [JAMA 2016;315:1966](#)

NICE published guidance on the management on vomiting and diarrhoea in children back in 2009. One of its key recommendations is that children who are clinically dehydrated or at risk of dehydration should receive oral rehydration solution (ORS) therapy.

Recent research published in JAMA (2016;315:1966) theorised that dilute fruit juice may be as good as ORS therapy - as it turns out it is much better and should be the default treatment for children at risk of dehydration due to gastroenteritis.

See our KISS below for the management of children with gastroenteritis.

KISS: *Giardia*

[BMJ 2016;355:i5369](#)

- Flagellated protozoal infection transmitted via ingestion of cysts, either due to direct contact with faeces or in contaminated water, food or fomites (e.g. soil). Can survive 84D outside body
- Most commonly affects children <5yo and adults aged 25-44yo. Risk factors include:
 - Foreign travel, especially to low-income settings; toileting young children and changing nappies; attending childcare settings; drinking contaminated water (e.g. swimming pools or freshwater); eating fresh products raw; sexual transmission; dog ownership

Presentation

- ~50% of exposed individuals clear the infection without symptoms
- Of the persisting infections between 5 - 15% will be asymptomatic (with continued shedding lasting up to 6 months or more) and the rest exhibit typical symptoms within 7-14 days, which can manifest in acute and chronic stages:
- **Acute phase - 2-4 weeks**
 - Diarrhoea - often explosive, in the morning, and hard to flush; blood is uncommon - if present consider alternative pathology; abdominal pain, bloating, "eggy burps"
- **Chronic phase**
 - Diarrhoea becomes more intermittent; looser stools; weight loss due to malabsorption in >80% - 5kg loss over 4 weeks in adults may be seen; failure to thrive in children
 - Temporary lactose intolerance occurs in up to 40%, persisting for several weeks, so we should advise patients to avoid lactose-containing foods for 2-6 weeks after treatment

Investigation

- PCR is significantly better (sensitivity 90-100%; specificity 75-100%) cf. traditional MC+S (31%, 100%). If PCR not available, take 3 samples 2-3 days apart and rpt if -ve and Sx persist

Treatment

- **1st line: *tinidazole* or *metronidazole*** - former better tolerated and a single dose
 - **Tinidazole:** adults: 2g oral single dose; age 12-17y: 2g single dose, repeat if necessary; 1m - 11y: 50-75mg/kg (max. per dose 2g), repeat if necessary
 - **Metronidazole:** adult and children 10-17y: 2g OD 3/7, or 400mg TDS 5/7, or 500mg BD 7-10/7; age 7-9y: 1g OD 3/7; 3-6y: 600-800mg OD 3/7; 1-2y: 500mg OD 3/7
 - **Avoid both in 1st trimester of pregnancy, seek advice**
 - **If unsuccessful** repeat either course, exclude re-infection and consider lactose intolerance as an explanation for continuing symptoms (advise 2-6/52 of lactose avoidance)
- **2nd line options** include albendazole, nitazoxanide, paromomycin and mepacrine - all unlicensed in the UK for *Giardia* - specialist advice is recommended at this stage

Advice if giardia infection:

- Exclusion for 48hrs applies to children in nursery, food handlers, carers of vulnerable adults, healthcare (yes, including GPs!)
- Hand hygiene is very important, especially for parents of children in nappies
- Avoid sharing towels



KISS: Treatment in the Community of Gastroenteritis in Children

[JAMA 2016;315:1966](#), [NICE CG 84](#), [CKS 2015](#)

Gastroenteritis without dehydration

- Encourage normal fluids, continue mild feeds as normal, avoid undiluted fruit juice and carbonated drinks
- Offer ORS (or presumably diluted fruit juice) if at increased risk of dehydration
 - Children <1yo, low birth weight, 5+ stools or vomited 2+ in last 24hrs, not offered or refused supplemental fluids before review, infants who have stopped breastfeeding, signs of malnutrition

Gastroenteritis with clinical dehydration who can be managed at home

- NICE 2009 recommends offer ORS but diluted fruit juice is more effective in many children
 - Children <5y: give 50ml/kg over 4 hours + maintenance fluids (also ORS)
 - If breastfeeding this should continue and only use ORS in addition
 - Children >5y: give 200ml ORS after each loose stool + maintenance fluids
- Use fluids little and often; avoid solid food until dehydration is corrected
- If refusing ORS, offer their usual fluids (there more palatable and probably with the trial saw such good results)
- When rehydrated restart food, offer ORS 5ml/kg after further large watery stool, and restart milk products immediately – holding off doesn't seem to improve recovery

NB Learning Points: Infectious Diarrhoea

- Giardia is most commonly acquired in the UK and underdiagnosed due to very poor sensitivity of microscopy. Consider in patients with acute or persistent diarrhoea and weight loss
- Tinidazole is specialist's 1st line choice of treatment. Given as a single dose, it has less side effects than metronidazole
- Children with gastroenteritis and dehydration have better outcomes having diluted fruit juice and preferred fluids than with oral rehydration solution
- Consider traveller's diarrhoea in returning patients with acute diarrhoea - most will resolve within a few days but assess for troubling symptoms requiring stool culture and consideration of antibiotics

NB Quality Improvement Ideas: Infectious Diarrhoea

- Audit cases of suspected infectious diarrhoea
 - Has giardia been considered? If still using microscopy have the correct amount of samples been sent for satisfactory exclusion?

KISS: Traveller's Diarrhoea[BMJ 2016;353:i1937](#), [BMJ 2016;355:i536](#)**Risk**

- **High:** South/South East Asia, Central America, West and North Africa, South America, East Africa
- **Medium:** Russia, China, Caribbean, South Africa
- Increased in backpackers, all-inclusive holidays, children <6yo, immunosuppressed, acid-suppressant use

Cause

- **Bacterial** – Most common cause: E. coli, Campylobacter, Salmonella, Shigella
- **Viral:** Rotavirus, Norovirus
- **Parasitic:** Giardia, Cryptosporidium, Entamoeba

Presentation

- Usually, in 1st week of travel, repeat episodes are possible
- Dependent on causative organism, e.g. enterotoxigenic E coli causes watery, profuse diarrhoea preceded by abdo cramps, nausea and malaise
- Most episodes resolve in 1-7 days; 10% last >1 week, 5% >2 weeks, 1% >1 month

Prevention

- Lack of evidence but food/drink/hand hygiene should reduce risk
- **Antibiotic prophylaxis?**
 - **Usually not recommended.** Consider in people with severe immunosuppression (e.g. advanced HIV), underlying intestinal pathology (e.g. IBD) and conditions where reduced oral intake may be dangerous (e.g. sickle cell, diabetes)
 - PHE suggests consideration of standby Abx for remote areas or high-risk patients
 - Options include ciprofloxacin 500mg OD, norfloxacin 400mg OD, rifaximin 200mg OD/BD, bismuth subsalicylate 2tabs QDS (not in children <16 or with aspirin/warfarin use)

Treatment – symptomatic treatment is usually adequate and avoids antibiotic use

- **Self-treatment:** liberal intake of clear fluids, oral rehydration therapy for young children, elderly and those with co-morbidities
- **Mild symptoms** (1-2 unformed stools/24hrs): loperamide 4mg stat, then 2mg every BO, 16mg max. per day
- **Moderate symptoms** – antibiotics reduce duration of diarrhoea from 3 > 1.5 days
 - **South/Central America, Africa:** ciprofloxacin 500mg BD 3/7
 - **South/South East Asia:** azithromycin 1g stat or 500mg OD 3/7, rifaximin 200mg TDS 3/7
- **Severe symptoms** (high fever, severe abdo pain, bloody diarrhoea): seek local medical assistance and avoid anti-motility agents

Chronic Diarrhoea

- >14/7 unlikely bacterial infection – consider parasites, GI pathology, post-infective IBS (30%)
- **Investigations:** stool MC+S, FBC, E+C, LFT, CRP
- **Giardia:** metronidazole 400mg TDS 5/7 or tinidazole 2g stat



LYME DISEASE

Lyme disease, increasingly called Lyme borreliosis, is now endemic throughout the UK. [NICE has recently published guidance on Lyme disease](#) and is keen to raise awareness of the condition, promoting early investigation and treatment. Meanwhile, research is shedding some light on the epidemiology of Lyme disease ([BJGP2017;67:e329](#)) which has also revealed a likely causal link with Bell's palsy, and the likely sequelae from Lyme with neurological manifestations ([BMJ 2018;361:k1998](#)).

UK Epidemiology of Lyme disease and correlation with Bell's Palsy [BJGP2017;67:e329](#)

This interesting paper is covered in more detail regarding Bell's palsy in that chapter, but it makes a number of important points about the infection and how it is changing in the UK:

- ***Lyme disease is found throughout the whole UK***, especially following travel to forested areas but also affecting areas of pasture and heath where the ticks parasitise small mammals such as field mice and voles and larger ones such as sheep, horses and deer.
- The ticks are mostly present from April to October, but ***Lyme can present at any time of year*** as a low level of tick activity persists through winter in areas with a milder climate such as southern England. As climate changes, it is expected that this trend will increase.
- There are 2-3,000 new confirmed cases per year, but despite a 42% rise in hospital-recorded cases over 4 years to 2015, it is likely that ***Lyme disease is even more prevalent***, and that there remains significant under-diagnosis and under-recording.
- Different strains of Lyme cause different patterns of disease. For example, ***the European strain is particularly associated with neurological manifestations including facial palsy*** which is often misdiagnosed as a Bell's palsy thwarting the opportunity for it to be cured
- This study revealed an association of rising cases of Lyme in recent years with rising cases of acute facial palsy. Previous research has shown that up to 10% of cases of Lyme disease develop a facial palsy, and this association is thought to be causal. Bilateral facial palsy and mononeuritis multiplex are particularly predictive of Lyme disease
- ***If identified early, 95% of patients with Lyme disease can expect a cure*** with antibiotics
- A heightened awareness of risk is necessary to facilitate prophylaxis, diagnosis and treatment. They ***conclude that we should have a 'high index of suspicion for Lyme disease' following travel in these areas*** and that patients ***presenting with facial palsy should be tested for Lyme disease***.

Lyme Disease [NICE NG95 2018](#), [BMJ 2018;361:k1261](#)

This welcome guidance from NICE, summarised in the BMJ, has attempted to pull together the disparate data on Lyme and formulate a standardised approach to the disease.

Key points:

- Improving diagnosis by raising awareness amongst clinicians and public, and through the use of the best investigatory techniques: serology using a sensitive initial test (ELISA) then a more specific confirmatory test (immunoblot)
- Optimum treatment with extended courses of antibiotics (usually 21 days) tailored to the type of Lyme disease, e.g. erythema migrans only or non-specific symptoms or focal presentations
 - The BMJ have produced an [excellent infographic](#) with the recommended antibiotic therapy, duration and dosage for both adults and children for various presentations
- See our KISS at the end of the chapter for the full recommendations.

Long-term health after European Lyme Neuroborreliosis [BMJ 2018;361:k1998](#)

One of the big questions for primary care physicians is there credence to the public concern that Lyme disease could be the cause of long-term unexplained symptoms for patients?

This *recent Danish cohort study, published in the BMJ* ([2018;361:k1998](#)) doesn't conclusively answer that question but by looking at the longer-term outcomes of those patients with one of the most severe forms of Lyme disease - neuroborreliosis - it perhaps sheds a little light on what others diagnosed with the condition can expect.

- Using Danish national registers the authors identified all patients between 1986-2016 with Lyme neuroborreliosis (diagnosed with a +ve intrathecal antibody test)
- Data was collected on the patients for up to 10 years after diagnosis
- **Main outcomes:** mortality, co-morbidity incidence, differences in educational & social outcomes compared to a matched control population and also family members to reduce socioeconomic bias
- **Results**
 - Lyme group n=2067, control group n=20670
 - Mortality rates no different between groups
 - Haematological malignancies were 3x higher in Lyme group, non-melanoma skin cancers 1.5x (presumed due to greater sun exposure from outdoor activity)
 - At 5 years there was no difference in hospital contacts, employment rates, days of sick leave, income, rates of receipt of disability pension, or number of children
 - Development, health and education in children was the same in both groups.
 - There was a decrease in working capacity "in a small fraction of patients" and marginally less income, seen particularly when comparing affected patients to family members
- The authors conclude: "***Lyme neuroborreliosis had no substantial effect on long-term survival, health or educational/social functioning.***"

The *linked editorial* ([BMJ 2018;361:k2284](#)) commends the authors for publishing data which is high quality when the limited research there is on Lyme disease is so lacking and of low quality. It comments on the reassuring finding that if you have Lyme neuroborreliosis "***you will be able to make a full recovery and lead a normal working life***", albeit with a marginally lower-income and labour market activity.

So overall it appears generally good news for patients with Lyme neuroborreliosis, especially if you have it when you are young, however, this study will do nothing to convince the cohort of patients with ongoing chronic symptoms who believe they have Lyme but who have no diagnosis.

The reality for this group is that many, perhaps most, will not have and never have had Lyme disease but the uncertainties around the accuracy of current investigations is likely to provide a fertile ground for concern.

KISS: Lyme Disease

[NICE NG95 2018](#), [PHE](#), for Scotland: [NHS Highland Lyme](#)

Symptoms

- ***Erythema migrans rash is pathognomonic*** - [click for examples](#)
 - Usually occurs 1-4 weeks after a tick bite, but can appear from 3 days to 3 months
 - Usually at the site of bite, red, expanding, non-itchy rash; may have a central clearing
 - Bites can also cause a transient rash that usually recedes over 48hrs and is not EM
- ***1/3 patients do not get erythema migrans***. Absence of a rash, or presence of rash but no recollection of a tick bite does NOT exclude the diagnosis
- ***Non-specific symptoms***, e.g. flu-like symptoms, lymphadenopathy, fatigue, neck stiffness, point/muscle pain, cognitive impairment (e.g. brain fog), headache, paraesthesia
- ***Focal symptoms*** - occur due to early systemic borreliosis (usually <3 /12)
 - Neurological: cranial nerve palsy (consider in Bell's palsy or mononeuropathy), meningitis, neuropsychiatric changes
 - Joint: inflammatory arthritis affecting one or more joints (esp. knee or wrist)
 - Cardiac issues: heart block (rare, needs urgent cardiology referral), pericarditis
 - Eye symptoms (e.g. uveitis, keratitis) or other rashes (e.g. [lymphocytoma](#))

Diagnosis

- ***Do not diagnose Lyme disease in people without symptoms***
- ***Diagnose Lyme disease in people with erythema migrans - they do not need testing***
- ***Offer testing in people with non-specific or focal symptoms & clinical suspicion of LD***
 - Uncommon cause but explore history for possible tick exposure
 - Do not rule out Lyme disease if symptoms but no clear history of tick exposure
 - Send serum for antibodies to an NHS laboratory
 - If +ve treat as Lyme disease
 - If -ve and tested <4 /52 from symptom onset, repeat in 4-6/52
 - If -ve but symptoms persist, repeat the test after 12 weeks or more
 - If <18yo and any symptoms except simple erythema migrans d/w specialist
 - If focal symptoms and Lyme +ve, consider a discussion with/refer to an appropriate specialist
 - People with persistent but unexplained symptoms refer to a specialist (e.g. rheumatologist)
- If people have a +ve test from a non-NHS lab it may be unreliable - repeat via the NHS

Treatment

- If erythema migrans or Lyme +ve & non-focal symptoms treat for Lyme disease with antibiotics
 - In patients pending blood tests where clinical suspicion is high, treat while waiting
 - Patients with focal symptoms should be managed by a specialist, antibiotics may differ
 - See next page for antibiotic regimens
- Symptoms may persist for several weeks after treatment (or longer if organ damage)
 - In persisting symptoms, consider treatment failure. ***Treat with an alternative antibiotic.***
 - Do not offer further treatment after 2 different antibiotics. Consider specialist review
- If symptoms worsen rapidly after starting antibiotics consider [Jarish-Herxheimer reaction](#) - a sepsis-like reaction due to endotoxin release from spirochete death (also seen with syphilis)

KISS: Antibiotic Therapy for Lyme Disease

[NICE NG95](#)

Erythema migrans/non-focal symptoms

- **1st line:** doxycycline orally 21 days
- **2nd line:** amoxicillin orally 21 days
- **3rd line:** azithromycin orally 17 days

Lyme disease with cranial nerve or PNS involvement

- **1st line:** doxycycline orally 21 days
- **2nd line:** amoxicillin orally 21 days

Lyme disease affecting CNS: 1st line ceftriaxone IV, 2nd line doxycycline orally enhanced dose

Lyme carditis: 1st line doxycycline orally 21 days, 2nd line ceftriaxone

Lyme disease arthritis or acrodermatitis chronica atrophicans: 1st line doxycycline orally 28/7

Dosing:

- **Doxycycline**
 - ≥12yo & children ≥45kg = 200mg daily
 - Children 9-12y = day 1: 5mg/kg 2 divided dose, thereafter 2.5mg/kg
- **Amoxicillin**
 - ≥12yo & children ≥33kg = 1g TDS
 - Children <33kg=30mg /kg TDS
- **Azithromycin**
 - ≥12yo & children ≥50kg = 500mg daily
 - Children <50kg=10mg /kg daily
 - Avoid if cardiac abnormalities due to QT prolongation
- Note doxycycline and azithromycin are unlicensed for children <12yo, but use from aged 9 is “accepted specialist practice” - in this age we should be talking to paed about the diagnosis in any case.

NB Learning Points and Quality Improvement Ideas

- Lyme is increasingly common throughout the UK and can be found all year round
- Erythema migrans is pathognomonic; testing is not required and antibiotics should be given
 - NICE provides a [very helpful web page](#) with multiple different photos of EM
- Testing is merited where there is suspicion of Lyme even in the absence of tick bite or rash
- Oral doxycycline for 21 days is first-line treatment for most Lyme disease, amoxicillin 2nd line
 - For the great BMJ treatment infographic [click here](#)
- Audit patients diagnosed with erythema migrans or Lyme disease
 - If EM did they receive antibiotics without blood testing?
 - If no EM did they have appropriate testing? If negative did it require repeating? Was this done?

HIV

The UK government has announced the ambitious goal of [eliminating HIV transmission by 2030](#). The strategy is three-pronged: prevention of transmission, rapid diagnosis of those infected, and prompt treatment with antiretroviral therapy.

1 Pre-exposure prophylaxis (PrEP) [BJGP 2017;67:324](#), [BMJ 2016;353:i3515](#)

PrEP - Pre-Exposure Prophylaxis is a hot topic in HIV. PrEP is medication taken by those who are **HIV-negative before and after sex** to prevent the acquisition of HIV. When taken daily it **reduces the risk of contracting HIV through sex by 99%**, and the risk from drug injection by 74% ([CDC](#)) (note alternative harm reduction strategies such as needle exchange recommended for IDU ([BHIVA/BASHH guidelines 2018](#))).

PrEP has been around for a number of years (WHO guidance first issued in 2012). However, for many years it was not available through the NHS. As evidence and awareness of the benefits increased voluntary groups emerged to support those at high risk of HIV to purchase PrEP privately through online pharmacies. Sexual health clinics began to notice a decline in new HIV diagnoses. Some offered support and monitoring to those buying the medication privately and joined the push for NHS provision.

► *What is the evidence that PrEP is effective?*

Key studies supporting the effectiveness of PrEP at reducing the risk of HIV infection:

- **PROUD study** ([Lancet 2016;387:53](#))
 - UK RCT of 544 MSM
 - One arm of the study immediately took Truvada daily, the other commenced after 12m
 - Results showed an **86% reduction in the incidence of HIV infection** in the immediate PrEP group; ***NNT = 13***
- **IPERGAY study** ([NEJM 2015;373:2237](#))
 - French & Canadian RCT of 400 MSM and trans women
 - Participants used **event-based dosing** (see KISS)
 - Results also showed an 86% reduction in the incidence of HIV infection; ***NNT = 18***
- **Partners PrEP study** ([NEJM 2012;367:399](#))
 - African RCT of nearly 5000 heterosexual couples, with 1 partner HIV positive
 - Terminated 18 months early as it reported a **73% reduction in new HIV infections** in those taking PrEP correctly.

After years of campaigning and wrangling over funding, PrEP is finally available to high-risk patients on NHS throughout the UK. Access is via level 3 sexual health clinics in Scotland (since 2017), Wales and England (since October 2020). In Northern Ireland, local sexual health services provide initial assessment then refer centrally to Belfast, Derry/Londonderry for treatment. Patients can use these helpful resources to check their eligibility (iwantprepnnow.co.uk) and to locate their nearest provider (Prepster.info).

The preparation currently licensed in the UK contains Tenofovir disoproxil and emtricitabine (originally patented as Truvada, generics now available). It inhibits HIV reverse transcriptase thereby blocking viral replication. Tenofovir alone can be offered to heterosexual men/women where emtricitabine is contraindicated. An alternative preparation (Tenofovir alafenamide + emtricitabine, trade name Descovy) is approved in the US and can be purchased privately online in the UK.

► **Changing attitudes to PrEP: the [BJGP 2017](#) dispels some common myths relating to PrEP:**

- **PrEP does not increase risky behaviour**; PROUD study showed no difference in the total number of sexual partners/ incidence of bacterial STIs in those on PrEP vs controls
- It does not interact with contraceptives, alcohol, drugs such as methadone, feminising or masculinising hormones

KISS: PrEP [BHIVA/BASHH guidelines 2018](#)

Who is it for? (General principles, NHS access criteria varies geographically)

- HIV negative AND
- At high risk of acquiring HIV e.g:
 - Expect to have condomless sex with **HIV positive partner(s)** with a detectable viral load (note not needed if partner on treatment for >6m and viral load undetectable)
 - High risk group- cis/ transgender **MSM** or **transgender woman** having condomless sex
 - **Recent rectal bacterial STI**
 - A specialist assesses patient as very high risk (e.g. using recreation drugs for chemsex sex ('slamming' crystal meth, mephedrone, GHB) workers, patients from countries with high rates of HIV, recent need for PEP (post-exposure prophylaxis))
- **NOT SUITABLE IN THOSE ALREADY EXPOSED**- consider PEP if high-risk encounter within 72 hrs (see below), can transition to PrEP after PEP complete

How to access?

- NHS access via sexual health clinics
- Available privately through online pharmacies

Pre- PrEP tests:

- HIV antigen/antibody test: **PrEP should not be used if HIV-positive** ↑risk of developing resistance.
- Check for hx acute HIV infection- flu-like illness within previous 4 weeks.
- If hx high risk encounter within 4 weeks test viral load and repeat HIV screening in 4 weeks.
- Full sexual health screen including Hepatitis B (Hepatitis B vaccine offered if negative)
 - Taking PrEP can suppress Hepatitis B, with risk of flares of hepatitis if/when stopped
 - Specialist assessment of safety of starting/continuing/stopping PrEP needed for all with chronic hepatitis B
- Creatinine, eGFR, urinalysis: Tenofovir really excreted (dose reduction eGFR <50), caution in CKD
- Pregnancy test (note pregnancy not a contraindication)

How is it taken?

- **Daily PrEP**: suitable for both anal and vaginal (frontal) sex, effective after 7 days, must be continued for at least 7 days after last sexual risk.
- **On-demand/event-based: only advised for anal sex/ MSM** (current absence of data for frontal sex), not suitable for chronic hepatitis B due to increased risk liver inflammation when starting/stopping.
 - 2 tablets 2-24hrs before sex
 - 1 tablet 24hrs later
 - 1 tablet 24hr after that
- Other schedules include Holiday PrEP or 4 pills per week (for more info see [iwantprepnw](#))
- Note in the event of missed pills + high risk exposure PEP should be considered

KISS: PrEP continued [BHIVA/BASHH guidelines 2018](#)

Monitoring:

- **3 monthly:** 4th generation HIV test, STI screen and prevention advice (PrEP not effective against other STIs), Urinalysis (further ix if protein 1+)
- **Annual** creatinine/ eGFR (more frequent if >40, nephrotoxic drugs, proteinuria or abnormal baseline)

Side effects:

- Transient, affects 1/10, usually resolves within 1/12: headache bloating, diarrhoea, nausea, diarrhoea
- **Reduced kidney function**
- **Reduced bone density:** reduction BMD around 1.5-2% following 48weeks tx, no evidence increased fracture, absence of long-term data, consider FRAX assessment in those with additional risk factors

Resources for those considering PrEP

- <https://i-base.info/guides/prep>: comprehensive patient info on PrEP
- www.iwantprepnnow.co.uk: info on PrEP regimes, how to purchase online.
- **Prepster:** info on where to access free PrEP throughout the UK/ how to purchase online, videos from diverse communities promoting awareness to those who could benefit from PrEP. Interesting video charting the journey to availability of PrEP
- [Terrence Higgins Trust](#)



2 Post Exposure Prophylaxis (PEP)- new [BASHH/BHIVA guidelines 2021](#)

PEP is medication **given after a high-risk exposure** to reduce the risk of acquiring HIV. It should be started ASAP as delay reduces efficacy, maximum window = 72hrs. 24 hr access should be available (OOH often via A+E- check your local pathway).

The recommended first-line PEP regimen is **tenofovir disoproxil /emtricitabine plus raltegravir for 28 days**. Potential drug interactions can be checked using the [Liverpool University drug interaction website](#). Note commonly used medications such as antacids and iron supplements can reduce absorption of raltegravir.

► **What are the indications for PEP?** (See p34 [BASHH/BHIVA guidelines 2021](#) for useful summary table)

PEP should be routinely offered:

1. Following receptive anal intercourse with index partner of unknown HIV status/ known HIV +ve with detectable viral load
2. After receptive vaginal sex if index partner known to be HIV +ve with unknown/ detectable viral load
3. Following occupational exposure (sharp/mucosal splash) from an index case known to be HIV +ve with unknown/detectable viral load
4. People who inject drugs if index injecting partner known HIV +ve with unknown or detectable viral load

PEP considered if:

1. Insertive vaginal intercourse with index partner known HIV positive with unknown/detectable viral load
2. Insertive anal intercourse with partner of unknown HIV status

PEP generally not recommended: risk considered <1/10 000 unless extenuating factor (such as very high prevalence within index population/ breaches in mucosal barrier e.g. from sexual trauma/ genital ulcer disease/ concomitant STI/ multiple exposures within short period i.e. group sex).

Index case of unknown HIV status:

- Receptive or insertive vaginal intercourse
- Sharing of injecting equipment
- Occupational sharps injury (unless bolus of blood injected/ deep trauma)

Human bite (unless biter's saliva visibly contaminated with blood+ known high HIV viral load >3 log copies/ml+ severe/deep tissue injury)

Note PEP is not needed following any type of exposure if the index case has been on antiretroviral therapy for at least 6 months + latest viral load recorded undetectable within the last 6 months.

Baseline tests: HIV-1 Ag/Ab, creatinine and eGFR, ALT, pregnancy test, Hep B/C (+ sexual health screening if clinically indicated)

Follow-up testing for HIV should be a minimum for 45 days after completion of the PEP course (if 28-day course completed this is 73 days=10.5weeks). (Other tests dependant on type of exposure- see guidelines for details)

If there is ongoing high risk of exposure immediate transition to PrEP on completion of PEP should be discussed.

3 Test and treat: [BHIVA/BASHH/BIA 2020](#), [NICE NG60](#)

Excitingly **HIV prevalence in the UK is falling**. In 2018 there were an estimated 103,800 people living with HIV in the UK. 93% of these had a diagnosis. Of those accessing regular care an amazing 97% had an undetectable viral load; protecting both them and their sexual partners. However, **almost half of new diagnoses were late presentations (43% in 2018)**. People who are diagnosed late with HIV have a 10-fold risk of mortality within the first year of diagnosis compared to those diagnosed promptly. Therefore improving testing is a key opportunity for progress.

The importance of early treatment of HIV has gained extra momentum following the INSIGHT START study ([NEJM 2015;373:9](#)). The immune compromise caused by HIV is directly related to the loss of CD4 T cells and rates of opportunistic infections and death rise as the number of CD4 cells declines. In the past, RCTs that explored earlier ART largely enrolled patients with CD4 counts of <500, **INSIGHT START** was the first trial to show benefit of treatment with a normal CD4 count of >500.

- 5000 HIV +ve individuals with CD4 counts >500, no symptoms of HIV infection, and who had never taken ART; mean age = 36y
- Randomised to receive immediate ART, or start treatment with CD4 count fell <350 or an AIDS-defining illness was diagnosed; follow up 3 years
- Primary endpoint: serious AIDS-related event, any non-AIDS related event, death
- Results
 - 56% RRR in risk of AIDS, other serious illness or death in the immediate ART group
 - NNT = 43.5 at 1 year
 - The study was stopped early and all participants offered ART due to the initial results

► What do UK Guidelines say? [British HIV Association guidelines 2016](#)

INSIGHT START has driven a major change in HIV treatment guidelines: ***all those diagnosed with HIV infection should be reviewed by an HIV specialist within 2 weeks, and offered ART irrespective of CD4 count.*** Those with primary HIV infection (HIV acquired within 6 months of diagnosis) should be offered **immediate** treatment as it can help preserve immune function and reduce the particularly high risk of onward transmission seen in this phase of the infection.

► *Who should we be testing?* [BHIVA/BASHH/BIA 2020](#), [NICE NG60](#)

Many barriers to HIV testing have been identified. Patient factors include worrying about stigma and insurance implications. Clinician factors include time/resource constraints, lack of awareness of indications, concern around causing patient offence and the false assumption that lengthy pre-test discussion is always needed. Primary care has a key opportunity to break down stigma by bringing HIV testing into routine care, thereby reaching members of the community not accessing sexual health services. [NICE NG60](#) explicitly tells us we need to be offering more individuals HIV testing in primary care and gives guidance on who to target:

HIV testing should be routinely offered & recommended to the following individuals:

- People with symptoms or signs consistent with an HIV indicator condition (any medical condition associated with an undiagnosed HIV seroprevalence $\geq 1/1000$) or an AIDS-defining illness (See [NICE QS157](#) for list of conditions)
- People belonging to groups at increased risk of exposure to HIV:
 - Sexual partners of someone known to be HIV positive
 - Trans women
 - MSM (men who have sex with men) + their female partners
 - Black Africans
 - All those who disclose high-risk sexual practices e.g. chemsex, sex workers
 - All those diagnosed with, or requesting testing for STIs
 - People who inject recreational drugs (PWID)
 - All men & women from or reporting sexual contact with individuals from countries of [high HIV prevalence](#)
- Based on [local prevalence](#):
 - **High prevalence (>2/1000 aged 15-59y):** routinely offer at registration/ if having blood test for other reason and no test within last yr
 - **Extremely high (>5/1000 aged 15-59y):** consider opportunistic testing at every clinical contact
- Other services that should be routinely offering testing include prisons, clinics for sexual health, TB, hepatitis, lymphoma, antenatal care/ TOP, and addiction/ substance misuse.

At least annual testing is recommended for PWID, MSM and sex workers, three monthly if high risk.

► *How do we test?*

The majority of HIV tests conducted in the UK today are 4th generation laboratory blood tests looking for HIV antibodies & HIV p24 antigen. The window period for these (the time between potential exposure to HIV infection & the point when the test will give an accurate result) has recently been updated to **45 days (99% infections detected at 6 weeks vs 95% at 4 weeks)**. Test results are usually available between 2 days and 2 weeks after testing

Other options include point-of-care testing, self-testing and self-sampling. These tests have longer window periods (3rd generation 60 days, point of care tests 90 days).

KISS: HIV Pre-Test Counselling Essentials [BMJ 2014;349:g4275](#)

- **Explain the benefits of testing to the individual.**
 - Lengthy counselling unnecessary: explain testing is now routine, performing test will **not** affect eligibility for life insurance or mortgage applications (only have to disclose a positive test)
 - Explain window period (see above) and advise retest if exposure within window period.
- Document offer of HIV test & any discussion; **signed consent is unnecessary**
- For the anxious patient or with additional needs, consider referring to GUM for HIV testing
- Check contact details, and how & when the patient would prefer to get their results & reinforce confidentiality
- **Discuss other tests if clinically appropriate: e.g. chlamydia, gonorrhoea, syphilis, HBV & HCV**
- If the patient declines testing explore health beliefs and address any factual inaccuracies. Document reasons. Discuss alternative options for testing.

► **People living with HIV need long-term holistic care** [BHIVA Guidelines 2016](#)

- They are at increased risk of cardiovascular disease, metabolic disorders and osteoporosis
 - **CVD accounts for a significant proportion of excess morbidity.** [BHIVA guidelines](#) recommend those with **established CVD** or those with a **10y CV risk >10%** are screened annually for hypertension, diabetes, dyslipidaemia, CKD, BMI & smoking status.
- They should have an **annual influenza vaccine & a one-time pneumococcal vaccine**
- Women living with HIV require **annual cervical screening**
- There are many important drug interactions with ARTs; [click here](#) for a link to the Liverpool University HIV drug interaction database
- Mental health problems are more prevalent e.g. PTSD and depression

NB Learning Points: HIV

- PrEP is taken by those who are HIV-negative before and after sex to prevent the transmission of HIV.
- PEP is taken after exposure to reduce the risk of acquiring HIV
- **Early diagnosis HIV and prompt treatment with ART= improved prognosis.** Those who test positive should be referred urgently to an HIV specialist & be seen within 2w
- Treatment also protects others: **Undetectable = Untransmissible**
- **Reducing stigma and normalising testing is key**
- **HIV testing should be considered in primary care for a much broader range of people**, including all-new general practice registrants in geographical areas with a diagnosed HIV prevalence of >2 per 1000 population
- **HIV testing should be undertaken in a much wider range of clinical scenarios;** [NICE NG60](#) recommends HIV testing in line with [Europe's HIV indicator conditions](#)
- **HIV pre-test counselling** does not need to be lengthy; essentials to cover include benefits of testing, consent, & how a result will be given

NB Quality Improvement Idea: HIV

- Audit how many HIV tests has your practice done in the last year.
 - How many times have patients in at-risk groups with possible clinical indicators been tested?
- Identify your local prevalence of HIV & consider an "opt-out" policy for HIV testing for all new practice registrants if >2 per 1000 population

ACUTE COVID-19 INFECTION

Good news for higher-risk individuals is that we now have covid-19 treatments not just for hospitalised patients, but for those in the community as well. Prevention is generally considered better than a cure, so many patients who can't have or fail to respond to immunisation have been eagerly anticipating monoclonal antibody prophylaxis. Unfortunately, the news here is less good.

1 COVID-19 Prophylaxis

Monoclonal antibodies have become big business with covid but have been exclusively for acute management. However, 2022 saw the development of drugs with an extended half-life, and with it the possibility of prophylaxis, not just acute treatment. **Evusheld is a combination of two long-acting neutralising antibodies, tixagevimab and cilgavimab**, derived from antibodies isolated in human B cells in covid-infected individuals, which attach to distinct regions of the viral spike protein receptor-binding domain. It is administered as consecutive IM injections making a single dose. **The half-life is approximately 90 days, suggesting it could be used as a prophylactic treatment.**

Evusheld is [approved in the UK for adults](#) who are "unlikely to mount an immune response from COVID-19 vaccination or for whom vaccination is not recommended".

Initial data from the ongoing phase 3 RCT for Evusheld was published in the NEJM ([2022;386:2188](#)). This was conducted in **individuals >18yo with an increased risk of an inadequate response to covid vaccination or at increased risk for exposure to covid including healthcare workers, factory workers, students in dormitories, military personnel, etc.**

- 5197 participants were recruited, randomised 2:1 Evusheld 300mg vs placebo
 - 43% were ≥60yo, mean BMI 29, 74% white race/17% black race/3% Asian race
 - 73% were considered at increased risk due to covid, e.g. ≥60y, BMI >30, immunosuppression, significant co-morbidity e.g. CHF, COPD, CKD, chronic liver disease
 - Of note only ~4% of patients were immunosuppressed or on immunosuppressive therapies
- **Symptomatic covid occurred in 0.2% (8 of 3441) on active treatment vs 1.0% (17 of 1731) placebo by day 183 of follow up**
 - 6-month f/up showed a RRR 83%
 - 5 critical and 2 fatal cases of covid occurred, all in the placebo group
- Adverse events occurred in 35% of treatment vs 34% of placebo groups.
 - The majority were mild or moderate in severity, most commonly injection site reactions for both groups

The authors concluded that **"a single dose of AZD7442 had efficacy for the prevention of Covid-19, without evident safety concerns"** but noted that low numbers of serious event in smaller, important subgroups, such as the immunocompromised, meant efficacy for these individuals could not be estimated.

► *So why are UK patients not getting Evusheld?*

The [UK announced in 2022 that it would not be procuring stock of Evusheld for the NHS](#), although it has subsequently become available on a private basis. The rationale was that there were a number of methodological and reporting limitations in the data (beyond the NEJM paper) which meant government was unable to recommend it. It passed the buck to NICE for further assess the data and make a decision.

Meanwhile, the SARS-CoV-2 virus has continued to evolve and the dominant subvariants in the UK (and much of the world) are different from the original omicron strains the study was conducted on. Lab data suggests that Evusheld may have limited efficacy against current subvariants, and unfortunately, this issue is likely to be exacerbated further over time. AstraZeneca, the maker of Evusheld, presumably considers this a problem as it very recently [announced a new trial of a "next generation" combination monoclonal antibody prophylaxis treatment](#) which should be effective against the latest variants. Fingers crossed this is successful and makes it to market before the virus evolves once again.

2 COVID Acute Treatment in the Community

► Monoclonal antibodies and antivirals for COVID-19 in the community

Given the current restrictions on Evusheld, there is at least some comfort for high risk patients that a number of effective acute therapies are now available for patients with covid in the community. The [NHS Interim guidance can be found here](#).

What are the treatments?

There are currently 4 acute covid treatments available via the NHS for treating non-hospitalised patients (severely unwell hospitalised patients have a much broader palette available to them):

- **First line**
 - Nirmatrelvir + ritonavir (Paxlovid) - an oral antiviral given BD for 5 days, or
- **Second line**
 - Remdesivir - an IV antiviral, give OD for 3 days, double dose D1
- **Third line**
 - Molnupiravir (Lagevrio) - another oral antiviral drug given BD for 5 days

Sotrovimab (Xevudy) - an IV nMAB (neutralising monoclonal antibody) given as a single dose - is also an option but the most recent UK-wide NHS guidance has 'down-graded' it's use from joint first line to exceptional use where other options are unsuitable or contra-indicated. There is [some concern that sotrovimab may be less effective against current subvariants of covid](#), and they are no longer recommended in the USA.

How effective are these new therapies?

Initial data suggests they are mostly very effective. Paxlovid has recently grabbed the limelight as the preferred treatment for covid in the community due to a combination of high efficacy and ease of use as an oral preparation, but all of the options with the exception of molnupiravir have demonstrated good results, albeit, as the DTB ([2022;60:5:73](#)) points out, in single placebo-controlled trials by the manufacturers.

The NEJM has published all the major studies for these 4 treatments. The studies invariably are phase 3 double-blind RCTs conducted on unvaccinated symptomatic non-hospitalised adult covid patients at higher risk of disease progression due to age $\geq 55y$, obesity, DM, CHF, CKD, COPD or mod-severe asthma. All were treated within 5 days of the onset of symptoms.

An important exclusion from these trials is **SARS-CoV2 vaccinated people**. While these drugs are promising in the unvaccinated, we lack data on the degree of benefit in vaccinated individuals or populations.

To summarise their efficacy results and eligibility criteria:

	RRR of hospitalisation or death by D29	Treatment initiation within:	Age eligibility	Exclusion criteria
Nirmatrelvir/ritonavir	89% NNT = 18	5 days	18y+	Advanced decompensating liver disease, CKD stage 4 or 5, pregnancy, beware drug interactions
Sotrovimab	85% interim analysis, 79% final, NNT = ~20	5 days	12y+	Weight <40kg
Remdesivir	87%, NNT = 21	7 days	12y+	Weight <40kg, eGFR <30, ALT >5x ULN
Molnupiravir	30%, NNT = 33	5 days	18y+	Pregnancy

For the first three options, the absolute risk reduction for each is around 5-6% and importantly the active treatment groups showed no covid deaths, unlike the placebo groups.

Molnupiravir was initially touted as the first line therapy in the UK due to its oral preparation, but when the published data demonstrated a more modest efficacy and alternative treatments became available, it was quickly relegated to the bottom of the pack. It failed to prevent all covid deaths (there 1 one in the treatment group vs 9 on placebo), and subgroup analysis revealed that certain groups appear to have no benefit of molnupiravir compared to placebo: those with previous covid infection, low baseline viral load and people with diabetes.

Is there real-world data?

This is starting to emerge for some of the most commonly used options.

The [CDC recently published a retrospective analysis of Paxlovid](#) use in the US in people with mild-moderate covid. It examined a large cohort of 700k people with acute covid who were eligible for Paxlovid, of which 28% of patients were prescribed the drug within 5 days of diagnosis.

- **Relative risk reduction was 51% in the treatment group.**
 - Importantly, unlike the initial studies, this **included vaccinated patients** (69% had 2+ doses).
 - It was equally effective in this group.
- Mortality was 0.01% in the Paxlovid group compared to 0.04% in those without.

The CDC conclude: "*Paxlovid should be offered to eligible adults irrespective of vaccination status, especially in groups with the highest risk for severe COVID-19 outcomes, such as older adults and those with multiple underlying health conditions*".

Another paper ([BMJ 2022;379:e071932](#)) compared sotrovimab against molnupiravir in high risk community patients with covid using data from NHS England. Of 6000 patients, average age 52 and 88% having 3+ vaccinations, **the data showed a hazard ratio of 0.54 favouring sotrovimab, and 32 deaths in the sotrovimab group compared to 55 in the molnupiravir group.** This data was predominantly from Dec 2021 to Feb 2022 however they also conducted an exploratory review of the next 3 months after this, which was when omicron B2 variant was emerging in the UK and found similar protective responses, supporting its ongoing use.

The authors conclude: "*the results are consistent with current UK guidelines favouring the use of sotrovimab over molnupiravir*" although unfortunately the most current guidelines actually suggest the latter before the former.

What about adverse events?

The medicines appear from the current data to be generally safe and well tolerated, although the size of the trials (between 500-2200 participants) means rare events may be missed. Adverse events were generally slightly fewer with the active treatment than placebo owing to the improvement in covid-related issues. Each medicine has a number of exclusion criteria - in particular, Paxlovid has a list of almost 100 medications with which it interacts, many of which means it should not be used. [Click here for a list from Pfizer](#).

Also note, Paxlovid and molnupiravir are not recommended in pregnancy.

So we can prescribe them?

Unfortunately, it isn't quite that straightforward - we cannot prescribe them from primary care, although this is probably a good thing from a workload perspective, especially given the various caveats for each drug.

COVID Medicines Delivery Units (CMDUs) are to be set up by local health services (e.g. ICSs/CCGs) which will prescribe and where necessary administer the treatments where people meet the eligibility criteria. GPs will be able to refer eligible patients to CMDUs if they have not been identified by NHS Test and Trace (what could go wrong) after a positive PCR.

Patient information from the NHS can be found [here](#).

► *Inhaled corticosteroids in acute covid?*

The interest here evolved out of the observation that patients with asthma and COPD appeared under-represented in acute covid cases, with speculation there could be a protective effect from inhaled steroids.

While initial data from the STOIC trial ([Lancet Resp 2021](#)) was very positive, and then data from the Principle trial was somewhat positive ([Lancet 2021](#)), subsequent trials have failed to demonstrate benefit ([BMJ2021](#)).

Currently NICE recommends against the use of inhaled corticosteroids to treat covid in the community outside of the trial setting.

3 Pulse Oximeter Accuracy

These devices are now firmly embedded as part of an examination of unwell patients in primary care and have a particularly crucial role in the assessment of those with covid. However, in [2021 the MHRA issued guidance on the use of pulse oximeters](#), their accuracy and factors which may alter the accuracy.

Two editorials have recently published ([BMJ 2022;378:e071474](#), [BJGP 2022;72\(716\):132](#)) further discussing pulse oximeters. These editorials, both from the same trio of authors, highlight that these devices are not perfect and there are important issues regarding their use that clinicians must be aware of but most will never have considered. So what affects the accuracy of pulse oximeters?

► *Accuracy of device*

A common issue with medical devices is *the fallacy of accuracy* due to results being displayed as a single number, in this case a percentage. The MHRA specifies pulse oximeters must have an accuracy of <4% in laboratory conditions. Devices tend to have greatest accuracy with higher oxygen saturations, around 90-100% and is generally very poor <70%. Devices may claim higher accuracy, but to be allowed to market they simply need to meet the 4% threshold, which would mean a patient with sats of 90% may actually have sats anywhere between 86 and 94%. Devices should only be used if they are approved for clinical use, meaning they should display a valid CE, CE UKNI or UKCA mark and meet the technical standard BS EN ISO 80601-2-61:2019.

► *Factors affecting accuracy*

Pulse oximeters measure O2 sats by sampling the absorption of light by oxygenated and deoxygenated haemoglobin (Hb) that is travelling through the finger in a pulsatile manner. As range of factors may influence this:

- Haemoglobinopathies, high glycosylated glucose, carbon monoxide (e.g. smoking)
 - Heavy smoking may overestimate SpO2 by up to 8%
- Low perfusion, movement, nail polish, henna dye, tattoos, probe mispositioning, ambient lighting hitting the sensor

Darker skin colour has been raised as a potential cause of inaccuracy on pulse oximeters. Data supporting this idea was published in the NEJM ([2020;383:2477](#)), which found that in patients with sats of 92-96% on pulse oximeter almost 12% of black patients actually had sats <88% compared to 3.6% of white patients. A very recent study in the BMJ ([2022;378:e069775](#)) also found a racial bias with hospitalised black patients again demonstrating higher odds of occult hypoxaemia than white patients.

The caveat to this data is that **these are studies conducted on hospitalised patients ill enough to require arterial blood gas measurement**, an essential requirement to enable comparison with peripheral pulse oximetry. It is difficult to know how to apply this data in general practice. As a BMJ Practice Pointer paper ([2021;372:n677](#)) points out the NEJM data shows **inaccuracy "relates mainly to saturations below 90%—substantially below the level that would trigger a hospital referral"**, which may be less relevant to the majority of patients we see in general practice.

The MHRA states "it is not aware of any incidents where skin colour has had an adverse effect on the use of pulse oximeters when providing effective clinical care".

► *Interpreting pulse oximetry*

The editorial points out that "pulse oximeters identify hypoxia not its cause" and recommend that we consider any measurement in the context of the patient as a whole. What are their symptoms, signs, comorbidities, other physiological markers? The authors use the example that a fall in sats from 98% to 95% may indicate a covid infection which is expected, acceptable and may require no further action, but it could also be due to a PE or secondary bacterial infection, both of which need urgent intervention. Context is key.

As the MHRA points out: "the relative change in a patient's reading may be of greater significance to clinical management than the absolute value". This sounds particularly true for patients with existing respiratory disease and may have reduced bias in those with darker skin.

The BJGP concludes that clinicians should ensure they use a device which is medically certified and well maintained, use it in a manner that reduces the risk of sampling error, know the limitations of these devices and interpret the measurements with this in mind and in the context of the patient.

NB Learning Points: COVID-19 Acute Assessment

- Inhaled corticosteroids may not be effective for acute covid but we now have options of monoclonal antibody and antiviral drugs in those at highest risk, available via Covid Medicines Delivery Units
 - For a directory of CMDU services click [here](#).
- Evusheld is a long-acting monoclonal antibody that may be used as a prophylactic agent in very high risk patients, although it may not be as effective against current UK subvariants
- Oxygen saturation measurements form an essential part of the assessment. Exertional desaturation is a sensitive marker of early deterioration. Clinicians need to be aware about the limitations in accuracy of pulse oximeters and factors which can effect their accuracy.
- The Panoramic Trial continues to recruit community-based covid patients - patients can self-enrol via the website - click [here](#).
- NHS Patient information for acute covid management can be found [here](#).

KISS: Assessment, Monitoring and Management of COVID-19 in the Community

[NHS England 2022](#), [SIGN 2022](#), [BMJ 2020;368:m1182](#), [RCGP Guide to triaging patients](#), [Top Ten Tips for Telephone Consultations](#), [RCGP paper on Virtual Wards](#),

Before the call

- Have current national pandemic guidance to hand
- Scan the patient's notes to understand their existing medical history

Start the call

- Introduce yourself, establish you have the correct patient, ensure good connection and patient privacy
- Establish why you are calling early on, e.g. "I understand you might have coronavirus..."
- Rapid assessment of whether patient is very sick, e.g. too breathless to talk easily
- Establish why the patient is calling, e.g. for further assessment, for advice, for reassurance

History & Examination

- Check contacts - any with known COVID or currently unwell? high risk occupation?
- History of current illness, including date of 1st symptoms
- Assessment severity of symptoms
 - **SIGN notes the only consistently identified Sx that may distinguish severe disease is SOB but that some patients with hypoxia do not appear SOB. The best sign is low oxygen saturations.**
- Check respiratory function, e.g. talking full sentences, colour of lips
- "How is your breathing", "is it worse than usual?", "What does it stop you doing?"
 - If possible get basic observations from patient, e.g. temp, pulse, RR, ?BP, ?sats, ?peak flow
- Consider if video consult would facilitate assessment or if face-to-face (F2F) is required
- Consider co-morbidities, home situation and support, ability to manage ADLs, oral intake, etc
- Consider non-COVID pathology
- Lack of validated tools for COVID-19, consider use of familiar tool, e.g. NICE sepsis guide, NEWS2
- Safety net for all patients, advise on isolation, maintain fluids, advise when to seek help

Management

- Assuming initial remote triage NHSE recommends separating into non-severe and severe symptoms
- **Non-severe Sx**
 - If mild Sx & low risk advise on isolation, testing, self-management % safety netting
 - For others use clinical judgement - consider further assessment + treatment, e.g. F2F, O2 sats
 - **MILD**
 - O2 $\geq 95\%$ (also if no less than 1-2% below usual or any of RR < 20 , HR < 90 , no confusion)
 - Equivalent to NEWS2 0-2
 - *Unlikely to need hospital admission*
 - **MODERATE**
 - O2 93-94% (also consider if $> 3-4\%$ below usual RR 21-24, HR 91-130, new confusion)
 - Equivalent to NEWS2 3-4
 - *Consider community management or hospital admission*
 - **SEVERE**
 - O2 $\leq 92\%$ (also consider if $> 4\%$ less than usual, RR > 25 , HR > 131 , new confusion)
 - Equivalent to NEWS2 > 5
 - *Consider urgent hospital admission*
 - *For mild & moderate cases consider if eligible for a) acute covid treatment & b) oximetry@home*
- **Severe Sx**
 - Hospital referral

LONG COVID

Long COVID may be the defining condition for a generation of patients and primary care clinicians. But almost 3 years since the start of the SARS-CoV-2 pandemic, how much do we really know about it and how can we help our patients manage the problem?

1 **Prevalence of ongoing symptoms** [Office of National Statistics](#)

As of October 2022, the rates of long covid in the UK are significant: 2.1 million people reported ongoing covid symptoms for more than 4 weeks. Of course, the real question is how many people have truly prolonged symptoms? 83% affected had symptoms >12 weeks, 50% > 1 year and 24% at least 2 years.

Fatigue was the commonest reported symptom (70%), then difficulty concentrating (45%), shortness of breath and muscle aches (both 42%). 73% reported the symptoms affected day-to-day activities, and 16% said they were 'limited a lot'.

The most commonly affected groups were people aged 35-69y, female, living in areas of deprivation, working in social care, unemployed, and those with another activity-limiting health condition or disability.

2 **Long COVID - an update for primary care** [BMJ 2022;378:e072117](#)

The original BMJ paper from Greenhalgh et al set out a plan for assessing and managing people with possibly long covid. The team published an [updated version in the BMJ in 2022](#). A KISS of the recommendations as well as those from NICE, SIGN and the RCGP is at the end of the chapter.

As we already know, there are no significant advances in our management of long covid currently. The focus remains on:

- **Supportive, holistic care**
 - Having a HCP acknowledging the diagnosis and being empathetic has been shown to be beneficial for patients
- **Symptom control**
 - Including self-care as well as HCP-led
 - Referral to specialist covid clinics may be appropriate if ongoing, troubling symptoms
- **Detection of treatable complications**
 - Physical checks (e.g. for exertional desaturation, hypotension, POTs etc.), blood tests and CXR should be considered

► *Causes of long covid*

While this remains an area of great uncertainty several mechanisms have emerged which, individually or by interacting may play a role. These include a chronic, low-grade inflammatory response, multi-organ microvascular disease characterised by immunothrombosis and endothelial dysfunction, autoimmune reactions, and neurological damage, especially of the autonomic NS.

Long covid is more likely with increasing severity of disease, however, it may also occur after mild covid. A recent BMJ paper ([BMJ 2023;380:e072529](#)) using retrospective data from Israel's national healthcare database reviewed patients with mild covid and found a range of symptoms such as anosmia, cognitive impairment, dyspnoea, weakness and palpitations were all more frequent in adults in the first year after mild acute covid, while hair loss, chest pain, cough, myalgia, and respiratory disorders were only more frequent in the first 6 months. The data showed that the majority of symptoms resolved by one year.

► Symptom Clusters

Phenotypes are being characterised, with several patterns emerging including:

Fatigue, low exercise tolerance, deconditioning	Neurocognitive dysfunction
Post-exertional symptom exacerbation (PESE)	Dizziness and vertigo
Exertional breathlessness	Loss of smell/taste
Altered breathing/breathing pattern disorder	Allergic-type symptoms
Chest pain	Poor sleep
Throat and voice symptoms	Mental health
Autonomic dysfunction	Joint and muscle pain

► Recovery

The majority of people with long covid will improve. Those with troubling symptoms may benefit from additional input beyond general practice including rehabilitation, occupational therapy, and psychological support. However, a small group of patients clearly have ongoing symptoms despite this. For them as the BMJ paper explains: *"recovery" currently means developing the ability to manage limited energy, continuing pain, cognitive limitations, and ongoing flare-ups in what has become a long-term condition*".

Clearly, for this group general practice skills are crucial - the paper argues this needs to be managed like other chronic conditions - as is ongoing research into better treatments.

3 COVID Vaccine and Risk of Long Covid

The [UK Health Security Agency published a very interesting report in Feb 2022](#) reviewing published data on how vaccine may influence the risk of long covid.

Regarding preventing the development of long covid it found:

- **"Full" covid vaccination (2 doses of Pfizer BioNTech, Oxford Astra-Zeneca or Moderna, or 1 dose of Janssen) lowered the risk of developing long covid** in the short, medium and long term after covid infection in 6 out of 8 studies. The effect is likely to be an underestimate of the benefit as patients who had covid infection prevented completely were not included.
 - One medium-quality study of >6k UK patients found **the risk was reduced by around 50%**

A subsequent community-based cohort study conducted by the Office of National Statistics ([BMJ 2022;377:e069676](#)) of 28k UK adults from 18-69yo supported this conclusion, finding 25% reported ongoing symptoms at 12 weeks, with a 13% lower risk with one vaccine and a further 8% reduced risk after a second vaccine.

But what about patients who already had long covid prior to vaccination?

- 3 studies were reviewed showing that **patients with long covid who were subsequently vaccinated were overall less likely to have ongoing long covid symptoms** than those who were unvaccinated. A few cases reported a worsening of symptoms.
 - 1-month post-vaccination 23% reported improvement of long covid Sx vs 15%
 - Few people with long covid receiving the vaccine reported an increase in symptoms
 - Vaccination soon after reporting long covid was associated with the greatest chance of improvement
 - 70-71% of people in both vaccinated and unvaccinated groups reported no change in symptoms

These data suggest that covid vaccination may protect against long covid, is safe for the majority with long covid and may reduce existing long covid symptoms.

4 Duration of VTE and bleeding risk increase due to acute COVID infection

A BMJ editorial ([2022;377:o817](#)) explains that **incidence and mortality from VTE has doubled since the start of the pandemic compared to 2018 / 2019**. Early data established an increased risk of VTE during acute COVID infection but what was unclear was how long that risk remained elevated.

A new paper in the BMJ ([2022;377:o817](#)) examined national health records from Sweden to establish the duration that risk of PE, DVT and bleeding (due to the association with coagulopathy and DIC, independent of anticoagulation) is continued after covid infection. It compared >1 million patients with covid against 4 million controls. The data shows:

- Compared to controls, the incidence risk ratios were significantly increased for:
 - 70 days for PE
 - 110 days for DVT
 - 60 days for bleeding

But to what degree are the risks increased?

- In the 1-30 / 30-60 / 60-90 days from covid infection risk ratios were increased by:
 - 32x / 4x / 2.5x for PE, reducing to
 - 6x / 2.6x / 1.4x for DVT
 - 2.5x / 1.4x / 1.1x for bleeding
- The highest risks were seen in those suffering critical illness - the risk is much lower although still increased for patients with milder infection, e.g. RR ~6x for PE

These figures initially look rather alarming, however, the absolute risk to an individual remains relatively low:

- 0.17% PE
- 0.004% DVT
- 0.1% bleeding

On a population level, with high circulating levels of covid, this does become significant and highlights the continuing importance of vaccination to both prevent infection and serious illness should infection still occur.

As primary care clinicians, **we need to be aware of these increased risks in our patients with "long covid" symptoms - potentially these may be new acute serious illness even months after the initial infection.**

An interesting editorial in the BJGP ([2021;71\(709\):342](#)) also highlights the increased risk of cardiovascular events. 31 thousand CVD deaths were linked to covid in 2020 alone. But the authors highlight that **covid is not the only driver of increased CVD events - all infections may cause the same** - acute illness may destabilise plaques, cause arrhythmias and accelerate atheroma leading to MI or stroke. Pre-covid around 10k CVD deaths each year were due to infections.

The question is what can be done about this? The editorial makes the argument for more nuanced risk assessments and extending vaccination cover (such as influenza and pneumococcal) to people with higher risk but not established CVD (e.g. QRISK >10%).

Is there a role for short-course aspirin? The Recovery trial showed no improvement in mortality even in hospitalised patients due to an increase in bleeding cancelling out the CVD benefits, but the authors don't exclude aspirin as a possibility in high risk patients. Research in this area would be welcome.

5 Postural Orthostatic Tachycardia Syndrome [BJGP 2022;72:8](#), [POTSUK.org](#)

Autonomic dysfunction may be triggered by covid infection. It can result in insufficient vasoconstriction on standing. Blood then pools within the splanchnic vessels and legs reducing the venous return to the heart.

In postural orthostatic tachycardia syndrome (POTS) this mismatch of circulating blood results in an excessive compensatory response - tachycardia and noradrenaline release driving a wide range of symptoms, predominantly fatigue, palpitations, light-headedness, headache and nausea.

POTS is not a new condition but is highly under-recognised. It is often associated with a preceding viral infection (41%) and is a recognised complication post-COVID although the incidence is unknown.

The pathophysiology also remains unclear, and as the BJGP explains we need to consider POTS as a syndrome rather than a discreet entity; there is evidence of several different drivers "including hypovolaemia, autonomic denervation, hyperadrenergic stimulation, and autoimmune pathology" of which people may have multiple. Those seen in long covid-related POTS may differ from POTS pre-pandemic.

NICE highlights autonomic dysfunction as a problem for some post-covid. It recommends:

- If orthostatic features are present check lying & standing blood pressure and heart rate
- Patients will need a 3-minute active stand test for suspected orthostatic hypotension, and 10 minutes for POTS or other forms of orthostatic tolerance.

Diagnosis and treatment are set out in the KISS below.

KISS: Postural Orthostatic Tachycardia Syndrome [BJGP 2022;72:8](#), [POTSUK.org](#)

POTS can be a disabling condition caused by autonomic dysfunction in the upright position.

Clinical Features:

- **Orthostatic intolerance symptoms:** light headedness, palpitations (in >90%); breathlessness (>80%); atypical chest pain, tremulousness, blurred vision (>70%), syncope (>35%); sweating
- **Non-posture related symptoms:** brain fog, nausea, headache, fatigue (>90%); abdo pain, muscle pain/weakness, cold extremities (>80%); bloating/constipation, hand tingling (>70%); diarrhoea (69%); poor sleep, exercise intolerance
- **Signs:** tachycardia on standing; acrocyanosis - puffy, red/purple hands/feet (~50%)

Diagnostic criteria

- Sustained heart rate increase of 30bpm (teenagers: 40bpm) from lying to standing
- Associated with PoTS symptoms for at least 3 months
- In the absence of hypotension
 - **Perform Stand test** - rest supine for a minimum of 5 minutes, check HR (and ideally BP).
 - Recheck on standing at 2, 5 and 10 minutes. (Have support nearby in case of syncope)
 - Explore DDx: exclude cardiac causes, anaemia, hyperthyroidism, pheochromocytoma
 - BJGP recommends E+C, FBC, ferritin, TSH, morning cortisol and ECG
 - Note despite the absence of hypotension forming part of the definition patients with POTS have a tendency towards hypotension, and vasovagal syncope can co-exist

Continued next page...

KISS: Postural Orthostatic Tachycardia Syndrome [BJGP 2022;72:8](#), [POTSUK.org](#)

Treatment

- **Merits referral** to PoTS specialist due to impact on QoL
- **Conservative measures:**
 - Avoid triggers, encouraged fluid intake (2L+ daily) & salt intake (6g+ daily unless C/I), graduated exercise from supine to standing activity over weeks>months, compression hosiery, counter-pressure manoeuvres - cross legs/squeeze thigh muscles & buttocks/tightly fold arms, sleep in a head-up tilt positive >10degrees (expands blood volume), pacing, CBT/mindfulness
- **Medication can be helpful** - unlicensed indication
 - BB, CCB, ivabradine, midodrine, fludrocortisone, pyridostigmine, clonidine, methyldopa
 - GP could initiate if familiar after a discussion re: unlicensed meds, or refer
 - Withdraw drugs if possible that may exacerbate e.g. meds lowering BP or raising heart rate

6 Returning to exercise post-COVID

Many people will have had to stop exercise and sports during covid infection and many will be keen to restart as soon as possible. As covid can have a wide range of effects on the body including cardiac and respiratory injury official guidance has been published to help guide clinicians and patients how to do this safely.

KISS: Return to Exercise Post-COVID [BJSM 2020](#), [BMJ 2020;370;m3026](#)

- **After recovery from mild illness:** 1 week of low-level stretching and strengthening before targeted cardiovascular sessions
- **Very mild symptoms:** limit activity to slow walking or equivalent. Increase rest periods if symptoms worsen. Avoid high-intensity training
- **Persistent symptoms** (such as fatigue, cough, breathlessness, fever): limit activity to 60% maximum heart rate until 2-3 weeks after symptoms resolve
- **Patients who had lymphopenia or required oxygen:** needs respiratory assessment before resuming exercise
- **Patients who had cardiac involvement:** needs cardiac assessment before resuming

Learning Points & Quality Improvement Ideas: Long COVID

- There are a range of useful resources for patients:
 - The UK government has a good website for all aspects of recovery after covid: www.yourcovidrecovery.nhs.uk.
 - Phase 2 of Your Covid Recovery is now available - clinicians can refer patients for access to an interactive web programme which gives the option to set goals, track progress, plus has chat forums and live chat with a health professional
 - The Royal College of OTs has helpful [guidance on managed post-COVID fatigue](#).
 - [Disordered breathing](#) info on yourcovidrecovery.nhs.uk and [excellent PILS](#)
 - [Cardiac patient information](#) from British Heart Foundation
 - Anosmia info from [Fifth Sense](#) and [AbScent](#)
 - Support for children with long covid can be found at [Long Covid Kids](#)
- **Quality Improvement**
 - Keep a register of patients with COVID-19 pneumonia - they had repeat CXR at 12 weeks post-discharge?
 - If they have enduring symptoms have they been referred to the post-covid clinic/respiratory team?

KISS: Initial Assessment of Prolonged COVID-19 Symptoms in Primary Care Continued

[BMJ 2022;378:e072117](#), [RCGP Guide](#), [NICE NG188](#)

In the UK covid is described as:

- | | |
|--------------------------------|------------|
| ■ Acute covid | <4 weeks |
| ■ Ongoing symptomatic COVID-19 | 4-12 weeks |
| ■ Post COVID-19 syndrome | 12+ weeks |

Post COVID-19 Syndrome is defined as:

1. Signs and symptoms that develop during or after an infection consistent with COVID-19, continue for more than 12 weeks and are not explained by an alternative diagnosis.
2. It usually presents with clusters of symptoms, often overlapping, which can fluctuate and change over time and can affect any system in the body.
3. Post-COVID-19 syndrome may be considered before 12 weeks while the possibility of an alternative underlying disease is also being assessed.
4. A positive PCR or lateral flow test is not required to fulfil the definition of PCS (although it is notable that in some international guidelines, this is considered mandatory)

The Role of General Practice

- One of the most important acts we can make is to listen, believe, validate and empathise with our patients. For some, this will be all we need to do, for some it will be all we can do.
- *NICE: after holistic assessment use shared decision-making to plan future care*
 - **Self-management and supported self-management**
 - Offer advice on self-management and goal setting, signpost to appropriate services, e.g. [Your Covid Recovery](#), give safety net advice, consider the role of social prescribing/support groups/forums
 - Organise Fitness To Work certificates where unable to work or requires workplace adjustments
 - **Follow-up and monitoring**
 - This should be individualised and agreed with the patient
 - Consider supported self-monitoring at home, such as HR, BP, sats monitoring, if helpful.

Common Features - this list is far from exhaustive

- Profound fatigue, breathlessness, brain fog, recurrent fever, chest pain
 - Relapsing, remitting pattern is a common characteristic, often mimicking initial infection
 - For fatigue, the 3 P's are recommended
- Respiratory symptoms
 - Possibly increased risk of fibrosis
 - Disordered breathing (usually upper chest breathing using the shoulder and neck accessory muscles rather than the diaphragm; also hyperventilation) during the course of their illness and breathing exercises, physio and/or pulmonary rehab
- Cardiac symptoms
 - Increased risk of heart failure, AF, consider acute events during and after infection
 - Ongoing chest pain in the absence of demonstrable pathology is common
- Neurocognitive
 - Increased rates of dementia, also mental health problems

KISS: Initial Assessment of Prolonged COVID-19 Symptoms in Primary Care**BMJ 2022;378:e072117, NICE/SIGN/RCGP Guideline 2020***"Believe your patient, listen, show empathy and acknowledge the diagnosis"***Clinical Assessment**

- Take a comprehensive history and appropriate examination. Include:
 - The date of first symptoms and the nature and severity of current symptoms
 - Temp, HR & rhythm, BP, resp exam, functional status, O2 sats
 - NICE suggests an exercise tolerance test, if appropriate, e.g. 1 minute sit to stand test and monitoring breathlessness, HR & O2 sats - ongoing desaturation merits further assessment
 - If postural symptoms NICE suggests checking lying/standing BP and HR to assess for postural hypotension but also conditions including autonomic dysfunction and postural tachycardia syndrome
- Assess mental health, avoid unnecessary medicalisation
- Assess existing co-morbidities and optimise management
- Assess functional ability and changes from pre-COVID state
- Assess need for social, financial and/or cultural support
- Consider using a questionnaire to assess the effects of prolonged covid on the person:
 - [Click here](#) for the COVID-19 Yorkshire Rehabilitation Screening Tool - very comprehensive, too long for a GP consultation but could be printed/emailed/texted to patients to complete

Investigations - if indicated by clinical assessment

Order tests selectively & for specific clinical indications. Keep an open mind, exclude alternative causes.

- Offer tests & investigations tailored to the individual
- Offer blood tests: which may include FBC, E&C, LFTs, TFTs, HbA1c, CRP, ferritin (as inflammatory marker), creatinine kinase
- CXR is recommended in ALL patients with ongoing respiratory symptoms by 12 weeks post-COVID
 - Persistent changes merit further investigation but note a normal CXR cannot exclude all disease
- Check lying and standing BP & pulse if orthostatic symptoms (10mins for ?PoTS, and ECG)
- Consider additional tests:
 - Troponin (may identify ACS, myocarditis)
 - NT-pro BNP (may identify underlying or new COVID-induced heart failure)
 - D-dimer (a negative result helps exclude VTE, a positive result is very non-specific)
 - CXR if not already done
 - ECG
 - Urine analysis
- Further investigations are likely to require referral

Referral

- Refer immediately if suspicion of acute or life-threatening illness
 - NICE gives the examples of severe hypoxaemia or desaturation on exercise
- If troubling ongoing symptoms at >4 weeks from start of COVID consider referral
- If likely COVID-19 related then refer to a post-COVID clinic (if available) for multi-disciplinary input
- If specific system issue referral to relevant speciality (including psychology/psychiatry) may be appropriate (will need to necessitate a time-appropriate appointment)

MENTAL HEALTH AND COVID-19

'The mental health consequences are likely to be present for longer and peak later than the actual pandemic' Lancet 2020

Before the COVID-19 pandemic back in those heady, innocent days in 2019 of hugs, nights out with friends and going in person to Hot Topics courses we already had a mental health crisis. We had discussed it many times on the course with increasing rates of mental illness, especially in young people and the problems with limited access to specialist mental health services. And then came COVID-19...

A recent editorial in the BMJ [BMJ2020;370:m4352](#) has highlighted the evidence showing a deterioration in population mental health, increased rates of self-harm, problems accessing help and fears that suicide rates will rise. Research consistently shows that the mental health of children and young people has been disproportionately badly affected relative to older adults.

Anyone working in primary care over the last year, however, does not need a reference to tell them this. The increase in problems related to mental health is clear. The toxic combination of social isolation, loneliness, economic hardship, future uncertainty and loss of the usual supports, networks and nourishing activities that sustain us has led to both an increase in the severity of pre-existing mental health conditions and the emergence of new ones. On top of the 'lockdown' impacts of isolation and economic hardship, come the more direct effects of COVID-19 on mental health from fear of illness and health anxiety to recovery and post-traumatic stress and the impact of long covid to the consequences of bereavement. For people who have had and are recovering from COVID-19, there are also direct neurobiological effects of the virus on the brain. Recent research from the post-mortem series [Lancet 2020](#) shows that SARS-CoV-2 can directly cause brain inflammation, and there are case reports of new-onset psychosis following COVID-19 and emerging evidence of the neuropsychiatric impact in patients with long covid. And of course, we have the risk of burnout, both occupational burnout for healthcare professionals and also parental burnout for those struggling trying to home school children.

What is clear to us in primary care is that COVID-19 affects everything. If you have a pre-existing mental health condition (e.g. health anxiety, eating disorder, depression, generalised anxiety, panic disorder etc) it turns up the volume and makes it much worse. But COVID-19 also triggers new mental health problems, and as well as the common problems of depression, anxiety and alcohol/substance misuse we see new cases emerging of OCD, health anxiety and even psychosis.

To explore the impact of COVID-19 on mental health in more detail, including practical tips on helping people through this crisis, we have developed a **FREE one-hour webinar** on [Hot Topics COVID-19 and Mental Health](#) produced together with a clinical psychologist Dr Catherine Sykes and the mental health charity MIND.

So, what can we do to help?

Never underestimate the power of listening, of empathy and of being you. The WHO recommends an approach based on the principles of **psychological first aid** (PFA) as the most appropriate way to help people. PFA is a globally recommended approach for supporting people during emergencies and crises and offers guidance on delivering psychological care and support. It was discussed in a recent BMJ paper [BMJ202;369:m1815](#), and has been endorsed as an approach by the [UK Government and NHS England](#).

The WHO may call it psychological first aid, you may prefer to call it 'good general practice'. The international Red Cross have produced excellent guidance, see box below, as to how this can be delivered remotely during the pandemic. PFA is not professional counselling nor psychological debriefing, and it can be effectively delivered by anyone in a support role. If you are interested in learning more there is a free Future Learn course on [COVID-19: Psychological First Aid](#). See the KISS box below for the basic principles.

KISS: Psychological First Aid for COVID-19

[International Red Cross 2020](#) and [NHSE Psychological First Aid & COVID-19](#)

'Psychological first aid (PFA) is a method of helping people in distress so that they feel calm and supported to cope better with their situation. It is a way of assisting someone to make informed decisions. The basis of PFA is caring about the person in distress and showing empathy'

- **Listen**

- Active listening and showing empathy
 - *Listen, and probe gently with open questions*
 - *Be patient and calm; your calm will project and transfer to the patient whilst you listen*
 - *Acknowledge feelings and understandings of events*

- **Reassure**

- Reassurance that these feelings are normal for many people; normalise and show empathy
 - *'I fully understand why you feel like this. In this situation, your reaction is quite natural'*

- **Risk assessment**

- For self-harm, suicide, substance and alcohol misuse, domestic abuse, safeguarding etc
- For other significant mental illness

- **Manage**

- Suggest talking about how to acknowledge, accept and manage these feelings so that they are less overwhelming
- Have calming techniques and resources on hand to refer people to (see resources below) including breathing, relaxation and grounding exercises
- Practical guidance e.g.
 - Stay socially close even when physically distant
 - A daily routine
 - Look for and inject humour into situations; positive emotions boost immunity as well-being
 - Maintain hope & belief
 - Accept & acknowledge feelings of anger, frustration, anxiety, regrets, self-blame etc.
 - Focus on what's in your control
 - Use stress management techniques to help with feelings of emotional turmoil
 - Set realistic goals, keep active, create a **well-being plan**
 - [The Five Ways to Wellbeing](#)
 - **CLANG**: Connect, Learn, be Active, take Notice, Give
- *Help people to make informed choices appropriate to their unique circumstances*

- **Do not...**

- Pressure people to talk
- Be judgemental, or ask why this or that
- Try to analyse why people feel a certain way
- Talk about yourself or share others' personal stories
- Give false promises or false assurances

- **Signpost further resources**

- See full resources below
- Strongly recommended is [FACE COVID](#) by Russ Harris

KISS: Mental Health Resources

Adults

- **MIND** - mental health charity with fantastic resources, including a specific area for COVID-related mental health issues - <https://www.mind.org.uk>
- **Beating the isolation blues** - well-being pack developed in the North West to help deal with well-being during COVID - <http://documents.manchester.ac.uk/display.aspx?DocID=49000>
- **Coping with coronavirus** - Self-help guides and information written by psychologists from UCL - <https://www.copingwithcoronavirus.co.uk/self-help-guides.html>
- **Headspace** - mindfulness app - <https://www.headspace.com/covid-19>
- **Sleep council** - for all things sleep-related - <https://sleepcouncil.org.uk/advice-support/>
- **Sleepstation** - sleep improvement programme, now free in some parts of the UK via NHS support - <https://www.sleepstation.org.uk>
- **CalmHarm App** (good for young people too) - to help manage self-harm urges - <https://www.nhs.uk/apps-library/calm-harm/>
- **SHOUT** - 24/7 crisis helpline - just text to 85258 <https://www.giveusashout.org>
- **Bereavement support** - NHS search engine to find services in your area - <https://www.nhs.uk/service-search/other-services/Bereavement%20support/LocationSearch/314>
- **Bereavement support** through CRUSE - <https://www.cruse.org.uk>
- **Relationship difficulties** - free online sessions through <https://www.careforthefamily.org.uk/courses/marriage-courses-the-marriage-sessions>
- **Exercise** - free exercise plans/videos including pilates, yoga etc. through NHS - <https://www.nhs.uk/conditions/nhs-fitness-studio/pilates-for-beginners/>
- **Breathing with your belly** - abdominal breathing to reduce stress and anxiety - <https://www.sthk.nhs.uk/media/leaflets/611a834ae70b32.57539813.pdf>

Children and young people

- **Think Ninja** - mental health support app for 10-18-year-olds - <https://www.nhs.uk/apps-library/thinkninja/>
- **Princes Trust** - excellent signposting information for mental health issues for young people - <https://www.princes-trust.org.uk/help-for-young-people/who-else/housing-health-wellbeing/wellbeing/mental-health>
- **Happy Maps** - excellent website with information, resources and signposting for children of all ages and parents - <https://www.happymaps.co.uk>
- **Young Minds** - mental health support for children and young people, up to age 19 - <https://youngminds.org.uk>
- **CHUMS** - mental health support for children and young people - <http://chums.uk.com>
- **Book recommendation for children with anxiety** (very good for parents too!) - 'What to do when you worry too much. A kid's guide to overcoming anxiety.' by Dawn Huebner, published by the American Psychological Association

KISS: Mental Health Resources continued

Support for you and your colleagues

- **Headspace** - mindfulness training - free to all clinical and non-clinical NHS staff - <https://www.headspace.com/nhs>
- **Permitted to pause** - website to support wellbeing for health care workers - <https://www.permittedtopause.co.uk/#>
- **Free wellbeing apps for NHS staff** - <https://www.nhsemployers.org/news/2020/03/free-access-to-wellbeing-apps-for-all-nhs-staff>
- **Our frontline** - round the clock one-to-one support for key workers - <https://www.mentalhealthatwork.org.uk/ourfrontline/>
- **Looking after you too** - coaching support for primary care staff - <https://people.nhs.uk/lookingafteryoutoo/>
- **RCGP support for GP wellbeing** (especially have a look at the Note Cards from WorkWell doctors - really good brief simple reminders on how to deal with emotions - just what we tell our patients but sometimes aren't so good at doing ourselves!) - <https://elearning.rcgp.org.uk/mod/page/view.php?id=10501>
- **Book from Dr Catherine Sykes** (mentioned on the covid and mental health webinar) - The Exhausted Trajectory - deals with burnout and exhaustion - <https://www.amazon.co.uk/Exhausted-Trajectory-Tired-Energy-Purpose-ebook/dp/B087D49ZSN>
- **PTSD webinar** recently produced to help primary care clinicians - <https://vimeo.com/427005151/3a40d3edfb>

VITT: VACCINE-INDUCED THROMBOCYTOPENIA & THROMBOSIS

VITT, yet another new acronym that less than a year ago we knew nothing about...When cases first started emerging earlier in 2021 of rare cases of cerebral venous sinus thrombosis and other rare thrombotic events after administration of the Covid-19 AZ vaccine, we were of course inundated with patients worried about headaches and possible blood clots. We updated you with a revision of cerebral venous thrombosis and vaccine-induced thrombotic events associated with thrombocytopenia in an [NB Blog April 7th 2021](#). Since then Public Health England have produced useful [information for health professionals](#) June 2021 and NICE have published a new [COVID-19 rapid guideline on VITT](#) in August 2021. Evidence is sparse, and NICE are clear that their recommendations are based on expert consensus opinion.

KISS: Vaccine-Induced Thrombocytopenia & Thrombosis

Based on [PHE June 2021](#) and [NICE, COVID-19 Rapid Guideline VITT 29/07/2021](#)

■ Background

- VITT is rare with an incidence of 14 per million doses of COVID-19 vaccine; in the UK, the MHRA have reported 405 cases as of July 2021
- It is an immune-mediated complication associated with progressive thrombosis and thrombocytopenia
- The most common cases are of cerebral venous sinus thrombosis (CVST), the remaining cases a range of sites including DVT, PE and splanchnic or portal veins
- Arterial thrombosis has also been reported
- Typical laboratory features include low platelets and very high D-Dimer levels
- Most cases have been after the first dose of the AZ vaccine, but there have also been reports after the Johnson & Johnson/Janssen Vaccine; there is no current evidence to suggest they occur after the Pfizer or Moderna vaccines
- It is important to remember that acute COVID-19 infection also increases the risk of blood clots, 20% of hospitalised patients with COVID-19 have evidence of thrombosis; CVST is a known complication of acute COVID-19 with a higher incidence compared to matched cohorts who have received the vaccine
- This is a very rare immune-mediated complication, and patients with a previous history of thrombosis and also with antiphospholipid syndrome CAN receive the AZ vaccine; for these patients the risk of recurrent thrombosis with COVID-19 is far greater than the risk of VITT

■ Identification of possible cases in primary care

- Consider the possibility of the diagnosis in patients presenting with new or ongoing symptoms within 5 to 30 days of having had a COVID-19 vaccination with the following symptoms or signs
 - New onset of severe headache, which is getting worse and does not respond to simple analgesics
 - Headache suggestive of raised ICP or with neurological symptoms or signs e.g. an unusual headache which is worse when supine or bending, or accompanied by blurred vision, nausea, vomiting, difficulties with speech, weakness, drowsiness or seizures
 - New and unexplained petechiae or bleeding
 - Symptoms or signs suggestive of PE, DVT or persistent abdominal pain suggesting possible splanchnic vein thrombosis

■ Primary care management

- NICE make the rather obvious recommendation *'refer people who are acutely unwell to the emergency department immediately'*...
- Otherwise, if the patient is not acutely unwell AND same-day test results can be obtained, perform a FBC and refer immediately if it shows thrombocytopenia
- NICE do NOT recommend a D-Dimer in primary care but suggest that this is done in secondary care in association with other tests e.g. fibrinogen
- If the FBC is normal and the patient is not acutely unwell, reassure that VITT is unlikely, think of alternative diagnoses and manage symptomatically
- If high clinical suspicion of VITT remains, repeat the FBC in 2 to 3 days

ASTHMA

Asthma remains a huge hot topic with very recent [reports in the media](#) showing that asthma deaths have risen in England and Wales to their highest level for 10 years, with an upward trajectory which appears to be accelerating. The UK already has one of the highest rates of asthma in the world. A recent BJGP editorial (2019;69:405) explains why and what we can do about it, recommendations that are also mirrored in the latest [BTS/SIGN guidance](#).

Meanwhile on the global stage thinking about the most fundamental treatment for asthma, the short-acting reliever, is changing and might also provide a solution to the risk of severe acute asthma.

1 Digital Interventions for Asthma Medication Adherence [Cochrane CD013030](#)

With the growing number of health apps and programmes available, it is perhaps overdue that research can show whether these are truly useful tools compared with traditional care.

This new Cochrane review reviewed 40 RCTs with 15000 adults and children with asthma. These studies included a wide range of tools designed to aid asthma management including online or computer-based (off-line) platforms, mobile apps, SMS-based interventions, and electronic monitoring devices which may record the date and time of doses taken.

The result showed:

- **Digital interventions may result in a large increase in adherence**
 - **Adherence improved by 15% overall**, which the authors felt was likely to be clinically significant, especially in people with poor baseline medication adherence
- **The most effective interventions appeared to be using electronic monitoring devices (↑ 23% adherence) and SMS (↑ 12%)**
- Interventions were likely to lead to a small increase in asthma control and a reduction in exacerbations
- There was no data on cost-effectiveness
 - SMS interventions are generally cheap, while electronic monitoring devices may be considerably more expensive

The confidence in the data is somewhat reduced due to the heterogeneity of the interventions reviewed. As the number of tools continues to increase (there are dozens on the Apple App store alone) there is a need for a mechanism to ensure they are safe and effective, a challenge which will only increase over time.

However, utilising modern technology to improve medication adherence, patient education and empowerment, and self-management feels like an obvious direction of travel. [NHS Wales has produced the Astmahub and Astmahub for Parents apps](#) which includes personalised asthma action plan, educational & inhaler technique videos, control trackers and more, and looks to be an excellent tool, though sadly only available for patients in Wales. More like this would be welcome.

2 Combination Inhalers for asthma [BMJ 2022;376:o663](#), [Cochrane CD013518](#)

GINA, the Global Initiative for Asthma, in 2019 recommended a fundamental change in practice with **no longer treating adults or adolescents with asthma with short-acting bronchodilators alone**. In the [2022 guidance](#) it recommends ICS + formoterol as the preferred reliever option at all severities of asthma in patients 12 years and over (in the UK LABA/ICS is not licensed for this use in younger children).

Why the change? Excessive use of short-term reliever therapy is one of two major indicators of future risk of serious or fatal asthma (the other being previous history of asthma attack). Salbutamol is widely overused in the UK, with [possibly the highest rates of overuse in Europe](#), resulting in **failure to treat the underlying inflammation and exposing suffers to risk of severe asthma**. Frequent use of salbutamol can lead to rebound hyper-responsiveness, decreased broncho-protection, decreased bronchodilator response, and increased allergic response and eosinophilic inflammation.

► *PRN LABA / ICS for mild asthma*

A [new Cochrane paper](#), reviewed in the [BMJ as part of their National Institute for Health Research Alerts series](#), reviewed data from studies of people with mild asthma aged 12y and over using as required combination inhalers (containing formoterol for rapid-but-long acting reliever and budesonide for steroid) compared to either reliever-only therapy or standard therapy with regular ICS + PRN SABA.

The authors found:

- Compared to reliever therapy, PRN combination inhalers:
 - ↓ asthma attacks requiring emergency steroids by >50% in 1 year, effectively controlled symptoms and had fewer adverse effects
- Compared to routine therapy with ICS + PRN SABA, PRN combo inhalers:
 - Controlled symptoms as effectively, had a similar rate of adverse events, resulted in less overall use of inhaled steroids
 - Reduced the chance of emergency hospital or urgent care visit for asthma

So this approach can simplify patients' regimes while ensuring adequate treatment of the underlying inflammation driving asthma but still reducing overall steroid use, therefore, minimising the potential for steroid-induced adverse effects.

► *LABA / ICS as reliever therapy for all asthma severity*

The recent paper in the NEJM ([2022;386:2071](#)) showed that in adult patients already receiving regular inhaled steroids who developed moderate-severe asthma the **use of a combination inhaler with salbutamol and budesonide for rescue medication lead to a 26% fall in severe exacerbation episodes**. The linked editorial ([NEJM 2022;386:2139](#)) explains this is similar to the 32% reduction seen across 5 pooled studies of MART. Data clearly demonstrates combination inhalers are superior to salbutamol alone and may prevent the need for oral steroid rescue therapy.

► *So, is it time to stop the salbutamol?*

Clearly, the current manner in which salbutamol is used may help contribute to serious asthma attacks and some of the poor asthma outcomes we see in the UK. However, there still remains an important role in treating acute severe asthma with salbutamol via a metered-dose inhaler and spacer, still considered the most appropriate treatment. This does not mean we should abandon the idea of PRN or MART treatment with combination inhalers, rather we need to reframe the role for SABA use around acute severe attacks.

The Lancet Respiratory ([2019;7;5:380](#)) makes a strong argument for rescue packs in the home, with a salbutamol inhaler, spacer and plan describing appropriate use and when to seek help, possibly also inhaled or oral steroids.

There could be cost implications from using MART inhalers compared to salbutamol, but if we weigh this against the risk of severe asthma attacks and protection from permanent lung function decline any cost-related argument becomes hard to swallow and with lower overall use may be closer than we imagine.

③ **Asthma Phenotypes**

For years specialists have discussed the different characteristics in their asthma patients - some respond well to treat, some do not; some cough with exercise, some do not; some have severe asthma, some only ever mild. Phenotypes have emerged from these observations - GINA states the most common are allergic, non-allergic, adult-onset, asthma with persistent airflow limitations, asthma with obesity, but many others exist such as aspirin-induced asthma. This remains confusing because there is a lack of consensus on how to categorise phenotypes and even disagreement between individual types. This is compounded further by continuing scientific discoveries in immunology and airways pathophysiology.

More recently specialists have started talking about type 2 and non-type 2 asthma. This has increasing importance because many of the new biological therapies now available for patients with severe, uncontrolled asthma despite usual therapy is only effective in type 2 asthma.

Dupilumab, for example, endorsed by NICE in December 2021 ([NICE TA751](#)) is only for treating severe asthma with type 2 inflammation.

► *What is Type 2 Inflammation?*

Type 2 inflammation is cell-mediated with T helper type 2 cells of the adaptive immune system being a key component. In response to allergens (also sometimes viruses/bacteria/irritants), there is a release of specific cytokines including IL-4, IL-5 and IL-13 triggering a cascade of immune responses resulting in inflammation.

Typically Type 2 inflammation is associated with eosinophilia, increased FeNO and IgE. In contrast type 1 inflammation is linked to neutrophils.

A variety of conditions are linked to type 2 inflammation: 50-70% of asthma, 50% of nasal polyps, 80% atopic dermatitis, aspirin-related airways disease, sinusitis, rhinitis and also potentially GORD and OSA.

Patients with type 2 asthma will usually have a good response to inhaled corticosteroids, however, in severe asthma, it can be refractory even to high doses. Severe asthma with type 2 inflammation is associated with a higher risk of exacerbations, hospitalisation and death, and dependency on oral corticosteroids than if type 2 inflammation is not present. It is this group that is being targeted by the new biological therapies.

► *What are the new treatments available?*

There are now 5 biological treatments for severe asthma which are approved by NICE; all monoclonal antibodies which bind to various parts of the inflammatory cascade. Omalizumab targets IgE and is licensed for severe persistent allergic IgE-mediated asthma. Benralizumab, mepolizumab and reslizumab target IL-5 and are indicated in severe eosinophilic asthma.

Dupilumab is the new kid on the block - with [NICE approval in December 2021](#) after an initial rejection (it is already used for severe eczema at a slightly higher dose). Another monoclonal antibody this targets IL-13 and provides an alternative pathway for treatment than existing therapies.

RCT data in patients >12y with uncontrolled asthma ([NEJM 2018;378:2468](#)) showed a 48% reduction in exacerbations, which increased to 66% with an eosinophil count of >0.3 (300per cubic millimeter).

The manufacturer quotes the most common side effects to be injection site pain, pain in the throat, eosinophilia and parasitic infections. Serious allergic reactions have occurred (at what rate is not stated), as has vasculitis and joint issues.

Dupilumab is an add-on therapy for patients >12yo with severe asthma with type 2 inflammation, uncontrolled by high-dose ICS and other therapies, where the patient has raised eosinophils and FeNO, and at least 4 exacerbations in the previous 12 months. It is only for use in patients who are ineligible or have failed to respond to the existing biological therapies. It will only be continued where exacerbations are reduced by at least 50% per annum.

It is given by a 200mg subcutaneous injection every 2 weeks (with a 400mg initial dose). At around ~£600 per dose, it is expensive. NICE acknowledges dupilumab sits at the upper end of its cost-effectiveness threshold. As is common these days the manufacturer is giving an unspecified discount on the medication for the NHS to make it worthwhile. Nevertheless, for the small number of people with severe asthma which fails to respond to standard therapy, the current data suggests this may be a very welcome addition.

4 Inhalers and the environment

Inhalers account for around 20% of general practices' prescribing carbon footprint as a result of the propellants in metered-dose inhalers. These propellants are between 1000 and 3500x more potent greenhouse gases than carbon dioxide so even though they are small they have a hugely disproportionate effect on the environment. Worse still inhalers continue to leach propellant long after the medication has been all used up and yet most end up in landfills. The Government's Environmental Audit Committee has recommended a target of 50% of all inhalers to be 'low environmental impact' by 2022. What can we do?

[Greener Practice](#), a network of primary care professionals encouraging sustainable medicine, recently published an [Asthma Toolkit to help clinicians provide high-quality and low-carbon asthma care](#).

[Key elements](#) include optimising asthma care, considering dry powder inhalers in preference to metered dose inhaled where appropriate, using MDIs with a low carbon footprint, low carbon regimes (e.g. using high doses but fewer puffs) and correct disposal via recycling or incineration. The toolkit includes [educational videos](#) including a gold-standard asthma review and step-by-step [quality improvement projects](#).

The Primary Care Respiratory Society recently published [The Greener Respiratory Pathway](#), with detailed information about inhaler prescribing and environmental considerations. They signal a note of caution around prescribing for environmental reasons and rightly argue that we need to prescribe based on what is best for the patient. However all the points above remain important, but we need to remember to assess patients for their suitability for dry powder inhalers before prescription:

- If the patient can breathe in quickly and deeply over 2-3 seconds we can consider a DPI
- If the patient can breathe slowly & steadily for 3-5s we can consider a pMDI + spacer, SMI or BAI

Where there is doubt about inspiratory strength an inspiratory flow check device ([such as this](#)) can be useful.

For more information see our [NB Blogs on inhalers](#) and a fantastic website, [Greeninhaler.org](#)

NB Learning Points: Asthma

- Asthma is over-diagnosed in around 1/3 of adults and up to 1/2 of children
- There is an over-reliance on short-acting reliever therapy and GINA recommends avoiding SABA use without concomitant inhaled steroid use, with an emphasis on MART regimes.
- Guidelines all suggest an increased role for objective tests in most people with possible asthma
- [Personal Asthma Action Plans](#) are a really useful tool to aid asthma management and provide a clear guide for patients to follow, particularly if their control deteriorates
- [Rightbreathe.com](#) is a fantastic resource which details all asthma (and COPD) devices available in the UK, doses and indications, plus it has very helpful educational videos on how to use each inhaler.
- [Click here](#) for the PCRS-UK online slide rule tool, very useful for demonstrating excess SABA use
- Think Green - the [Greener Practice Asthma Toolkit](#) has excellent resources and QI project templates

NB Quality Improvement Ideas: Asthma

- Audit prescribing of salbutamol in asthma patients
 - Have 6 or more salbutamol inhalers been prescribed in the past year?
 - Invite these patients for an asthma review and explore why the high use.
- Audit asthma diagnosis from the past 12 months
 - How many people have had objective tests such as spirometry?
 - Consider the initial risk of asthma pre-diagnosis based on BTS criteria - did they merit spirometry?

KISS: Treatment of Acute Asthma

[BTS/SIGN Asthma Guideline 2019](#)

Adults

1. Assess severity

- **Moderate exacerbation** PEF>50-75% best or predicted and increasing symptoms
- **Acute severe** PEF 33-50% best or predicted, inability to complete sentences, RR>25, HR>110
- **'Life threatening'** - emergency admission if any one of the following features - PEF<33% best or predicted, SpO₂ <92%, silent chest, cyanosis, poor respiratory effort, arrhythmia, exhaustion or altered conscious level

2. Treatment

- Oxygen to all hypoxaemic patients - aim for SpO₂ 94-98%.
- **Use high-dose inhaled B2 agonists (salbutamol) as first-line** – inhaler via a spacer is an equivalent benefit but safer than nebuliser - only use a nebuliser for severe > life-threatening attack and always drive with oxygen rather than air ([BMJ 2017;359:j5763](#))
- **Give steroids in all cases in adequate doses, and continue prednisolone 40-50mg for at least 5 days or until recovery**
 - Overwhelming evidence for benefit in exacerbations but underused ([BMJ 2009;338:b1234](#))
 - Reduces admissions, relapse rates and symptom duration in asthma exacerbations
 - Increasing inhaler dosage is not effective in an exacerbation ([Cochrane 2016](#))
- ***Admit any patient with a severe attack persisting after initial treatment***

Children

1. Assess severity

- **Moderate attack**
 - Talking in sentences; SpO₂ >92%; PEFR >50% best or predicted; children >5yo: HR <125bpm, RR <30bpm; children aged 2-5yo: HR <140, RR <40; mild/mod recession and wheeze
- **Acute severe attack**
 - Inability to complete sentences in one breath, or too breathless to talk or feed; PEFR 33-50% best or predicted; SpO₂ <92%; >5yo: HR>125, RR>30; 2-5yo: HR>140, RR>40
- **'Life threatening'** (any one of the following in a child with severe asthma – admit at once)
 - PEFR <33-50% best or predicted, SpO₂ <92%, silent chest, cyanosis, poor respiratory effort, exhaustion confusion, hypotension

2. Treatment

- **Inhaled beta-agonists – give up to 10 puffs salbutamol via spacer**
 - If poor response add ipratropium bromide and repeat salbutamol every 20mins
- **If no improvement arrange transfer to hospital**, with further doses while waiting
- **Oxygen + nebulised salbutamol** should be available during transfer. High flow O₂ if sats <94%
- **Early high dose oral prednisolone for all children with a moderate attack or above:**
 - 20mg in children aged 2-5years, 30-40mg aged >5 years
 - If already on maintenance steroids, increase to 2mg/kg up to 60mg maximum dose.
 - 3 days is usually sufficient, but tailor to the individual; no need to wean if course <14/7

KISS: Treatment of Acute Asthma, continued

[BTS/SIGN Asthma Guideline 2019](#)

Mild-mod acute asthma in children >2 years (exacerbation but none of the above symptoms or signs)

- Assess as above, record vital signs and respiratory effort
- **Inhaled beta-agonists** via MDI + spacer are the preferred option, with a face mask if <3 years
 - Single puffs should be given one at a time and inhaled separately with 5 tidal breaths
 - If hourly doses are needed for more than 4-6 hours, switch to nebulised bronchodilators
 - Once stable and improving, switch to a decreasing regime
- **Prednisolone** (dosage as above) for up to 3 days is usually sufficient, but again, tailor to child

Always consider FUTURE RISK

- **A history of acute asthma attack is a major risk factor for future acute asthma**
 - Salbutamol over-use is the other
 - The 3 RCP questions used in QOF are ineffective for assessing future risk
- **Perform a post-asthma attack review, ideally at 48hrs post-attack**
 - Opportunity to check the acute episode has resolved, and if not modify treatment appropriately
 - Allows assessment of modifiable risk factors associated with poor outcome and optimise treatment
- **Refer high risk patients**
 - A history of multiple asthma attacks = poor control = very high risk. These patients should be referred to a respiratory specialist.

An exacerbation should be seen as an opportunity to discuss self-management and a PAAP - Asthma UK as a [good example of a PAAP](#)

KISS: NICE: Asthma Diagnosis [NG80, 2017](#)

- ***Undertake a structured clinical history and examination***
 - Check for wheeze, cough, SOB, daily or seasonal variation, triggers, Hx/FHx atopy
 - Examine for resp signs + expiratory polyphonic wheeze; may be normal
- If acute symptoms at presentation treat immediately and perform objective tests when able
- Do not offer skin prick, serum IgE, eosinophil count or exercise challenge tests for diagnosis
 - Skin prick and specific IgE tests may be useful to identify triggers after formal diagnosis

Investigations - Adults (aged ≥17 years old) [click here for adult flow chart](#)

- Check spirometry and fractional exhaled nitric oxide (FeNO) initially
- If diagnostic uncertainty remains monitor peak flow variability for 2-4w
- ***Spirometry shows reversible airways obstruction***
 - ***FeNO ≥40ppb = diagnose asthma***
 - ***FeNO <40 but peak flow variability present=diagnose asthma***
 - ***FeNO 25-39 and peak flow variability absent = suspect asthma and review diagnosis after treatment***
 - FeNO <25 and peak flow variability absent=consider alternative diagnosis
- ***Spirometry shows an obstructive picture but no reversibility***
 - ***FeNO ≥40ppb = diagnose asthma***
 - ***FeNO 25-39 and peak flow variability = suspect asthma, r/v Dx after treatment***

KISS: NICE: Asthma Diagnosis [NG80, 2017](#)

- • FeNO 25-39 and peak flow variability absent refer for airway hyper-reactivity test
 - If present = diagnose asthma; if absent = consider alternative diagnosis
- If FeNO <25=consider alternative diagnosis
- ***Spirometry shows no airways obstruction***
 - ***FeNO ≥40 and peak flow variability present = diagnose asthma***
 - FeNO ≥40 and peak flow variability absent refer for airway hyper-reactivity test
 - If present = diagnose asthma; if absent = consider alternative diagnosis
 - FeNO <40 and peak flow variability present refer for airway hyper-reactivity test
 - If present = diagnose asthma; if absent = consider alternative diagnosis
 - FeNO and peak flow variability absent = consider an alternative diagnosis

Investigations - Children (aged 5 - 16 years) [click here for child flow chart](#)

- Perform spirometry +/- reversibility; consider FeNO if diagnostic uncertainty
- ***Spirometry shows reversible airways obstruction = diagnose asthma***
- ***Spirometry shows an obstructive picture but no reversibility***
 - ***FeNO ≥35 but peak flow variability absent = suspect asthma***
 - FeNO <35=refer for specialist assessment
- ***Spirometry shows no airways obstruction***
 - ***FeNO ≥35 and peak flow variability = diagnose asthma***
 - ***FeNO ≥35 but peak flow variability absent = suspect asthma***
 - ***FeNO <35 and peak flow variability=suspect asthma***
 - FeNO <35 but peak flow variability absent=consider alternative diagnosis/refer
- If a child cannot perform the tests then treat based on observation and clinical judgement
 - Try performing tests again at a later date, e.g. after 6-12 months

Diagnosis of children <5 years

- Use observation and clinical judgement to suspect asthma
- If maintenance treatment is needed, use an 8-week trial of moderate-dose paediatric ICS
 - Stop at 8 weeks and monitor symptoms
 - If symptoms resolved then recurred in <4/52 restart ICS
 - If symptoms resolved but reoccurred >4 weeks repeat 8/52 trial of ICS
 - If symptoms never resolved to consider if an alternative diagnosis is likely

What is a positive test?

- **Spirometry:** +ve if FEV1/FVC <70% (or below the lower limit of normal if available - this is more accurate in children and elderly people and recommended by BTS/SIGN)
- **Bronchodilator reversibility:** +ve if FEV1 ↑ by ≥12% (+ ≥200ml increase in volume in adults)
- **Peak flow variability:** +ve if >20% variability over 2 - 4 weeks of measurement



GISS: NICE Chronic Asthma Treatment [NG80, 2017](#)

General Principles

- Review response to treatment 4-8 weeks after any changes; aim for the lowest effective dose
- If asthma is uncontrolled before changes consider alternative diagnoses, poor adherence, poor inhaler technique, smoking, occupational exposure, season/environment factors, psychosocial issues

Treatment - adults and children 5-16yo

- For children ensure the use of paediatric dose equivalent ICS
- Offer short-acting beta-agonist (SABA) as reliever therapy
 - If infrequent, short-lived wheeze and normal lung function consider SABA monotherapy
- **1st line maintenance therapy:** offer low dose ICS
- **2nd line:** offer an LTRA in addition to low dose ICS, review in 4-8/52
- **3rd line:** offer a LABA in combination with ICS
 - In adults consider stopping LTRA based on patient preference and response to drug
 - In children stop LTRA
- **4th line:** change inhaler(s) to a MART regime (maintenance and reliever therapy), e.g. Symbicort, Fostair, DuoPres, Spiromax - inhalers must contain a LABA with a fast-acting component
- **5th line:** increase the ICS to moderate dose (as either MART or fixed-dose)
- **6th line:** in children seek specialist advice, and for adults consider either 1) increase ICS to high dose fixed-dose regime, or 2) a trial of an additional drug (e.g. long-acting muscarinic receptor antagonist or theophylline) or 3) get specialist advice
- Monitor asthma control at every review, check inhaler technique, consider using a validated questionnaire such as the [Asthma Control Test](#)
- Consider decreasing therapy if asthma controlled for at least 3 months
 - Only stop ICS in those already using low-dose ICS monotherapy and symptom-free

Self-management strategies

- Offer an asthma self-management programme ([click here for Asthma UK PAAP](#))
- If asthma control deteriorates offer an increased dose of ICS for 7 days to children and adults using a single inhaler
 - Quadruple the regular dose but do not exceed the maximum licensed dose
 - e.g. if they are on 800mcg of Clenil ('200' 2p/day) then they can only double the dose
 - BTS/SIGN have a helpful table of steroid dose equivalence in their [Quick Ref Guide](#)
 - This recommendation is not intended as a treatment for people with a full exacerbation of asthma - NICE do not provide guidance on acute management, but oral steroids remains the most effective treatment option as per BTS/SIGN
 - Note that more recent data suggests this strategy may not be appropriate for children

KISS: SIGN/BTS Asthma Diagnosis and Treatment [July 2019](#)

Diagnosis

- Identify patients based on symptoms of cough, wheeze, breathlessness or chest tightness
- **Undertake a “structured clinical assessment” (SCA)**
 - Look for recurrent episodes, the variability of symptoms, recorded observation of wheeze, a personal history of atopy, consider a possible alternative diagnosis
 - Is there a historical record of variability on spirometry or peak flows?
- **High Probability = +ve for most criteria on SCA, DDx unlikely**
 - Code as suspected asthma and start a trial of treatment
 - Aim for objective measurement of improvement (e.g. improved lung function)
 - If present diagnose asthma and continue treatment as appropriate
 - If there is a poor response switch the patient to the intermediate probability pathway
- **Intermediate Probability = most patients are likely to be this category**
 - Test for airway obstruction using spirometry and bronchodilator reversibility
 - Other tests to consider include reversibility, PEF charts and challenge testing to assess for reversibility and FeNO, blood eosinophils and IgE skin prick testing to assess for eosinophilic inflammation
 - If investigations indicate asthma is likely then trial treatment and assess response if currently symptomatic, or watchful waiting if currently asymptomatic
 - If there is a poor response switch to the low-probability pathway
- **Low Probability**
 - Investigate and/or treat for a more likely diagnosis
 - If no other diagnosis proves likely then switch to intermediate probability pathway

Treatment

- **The aim of treatment is “complete control”** – no day or night time symptoms, no need for rescue meds, no asthma attacks or exacerbations, no limitation during exercise, normal lung function and minimal medication side effects
- All asthmatics should be **educated in self-management** and given a **personalised asthma action plan**
- Initial treatment should be at the level most appropriate for their severity
- **Use the lowest effective maintenance dose - step up as required, step down when control is good**
 - Reductions should be made at 3-month intervals, reducing the dose by 25-50% each time
 - Regular review when stepping down treatment is important
- **Combination inhalers are recommended** to aid adherence and **prevent LABA monotherapy**
- **Exercise-induced asthma is usually a sign of inadequate control** and treatment review is recommended
- Before changing treatment check adherence, inhaler technique and eliminate trigger factors
- In difficult to control asthma check the above and also psychological status, then refer to an asthma specialist team for further review
- In pregnancy, asthma may improve or worsen. SABA, LABA, LTRA and ICS should be used as for non-pregnant patients to maintain good asthma control. There is no known danger from their use during pregnancy.

KISS: Treatment of Chronic Asthma

[BTS/SIGN Asthma Guideline 2019](#)

Adults

Move up and down to control symptoms	Regular preventer	Start low-dose ICS
	Initial add-on therapy	Add LABA to low-dose ICS (usually combination inhaler)
	Additional add-on therapy	No response to LABA - stop and increase ICS dose If the response to LABA but inadequate control continue LABA and increase ICS to moderate dose +/- add consider trial of third agent: LTRA, LAMA, SR theophylline
	High dose therapies	Consider trials of increasing high dose ICS or add fourth agent: LTRA, LAMA, SR theophylline, beta-agonist tablet Refer patient to specialist services
	Continuous or frequent use of steroids	Use daily steroid tablet at the lowest dose to provide adequate control, maintain high dose ICS, consider alternatives to reduce steroid requirement

Children

Move up and down as required to control asthma - consider moving up if short-acting beta-agonist required >3x or more per week	Regular preventer	Very low dose ICS (50mcg 2p BD) or LTRA if <5y
	Initial add-on preventer	Very low dose ICS (50mcg 2p BD), plus 5y+ = add inhaled LABA, <5y add LTRA
	Additional add-on therapies	No response to LABA - stop and increase the dose of ICS to low-dose (100mcg 2p BD) If the benefit from LABA but control inadequate - continue LABA and increase ICS to low dose OR continue very low dose ICS and add LTRA
	High dose therapies	Consider trials of either: Increasing ICS up to medium dose (200mcg 2p BD), or Addition of 4th drug: SR theophylline *Refer to specialist care*
	Continuous or frequent use of oral steroids	Maintain medium-dose ICS, use daily steroid tablet in the lowest dose, consider other treatments to reduce steroid requirement *Refer to specialist care*

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Over the past few years, we have seen an explosion of inhaler options for people with COPD. New treatments are welcome, and improving convenience and therefore compliance for patients is important. However, the gains from many of these new treatments is limited compared to standard therapy - to improve outcomes perhaps we need a different approach. Could harnessing modern technology using computer programmes and apps be the answer? And what about repurposing drugs used for other conditions...? Like diabetes???

1 Novel Antihyperglycaemic Drugs & COPD Exacerbations [BMJ 2022;379:e071380](#)

While there's been a huge amount of data published on the benefits of the new anti-diabetes medications for a wide range of conditions, no one outside of the research setting would have anticipated they may have a role in managing chronic lung disease.

How? As the authors of a new BMJ paper ([BMJ 2022;379:e071380](#)) explain GLP1 agonists have an anti-inflammatory effect and reduce airways hyper-responsiveness. There may be additional benefit through a reduction in abdominal obesity and an easing of diaphragmatic pressure. DDP4i may cause a small increase in endogenous GLP1. But what about the SGLT2 inhibitors? Any mechanism here is currently unclear, but preliminary data also points towards benefit.

In this cohort study the authors examined patients stepping up diabetes treatment with one of three medications - GLP1 agonists, DDP4 inhibitors and SGLT2 inhibitors - or the active comparator - sulphonylureas. They used data from UK primary and secondary care databases to look at rates of severe or moderate COPD exacerbations in each group.

- N = ~1250 GLP1a, 8750 DDP4i, 3000 SGLT2i vs 42k sulphonylurea initiations
- **Results**
 - Compared to sulphonylureas:
 - **GLP1a ↓ severe exacerbations by ~30%** (3.5 v 5.0 events per 100 person-years) and moderate exacerbations by 37%
 - **SGLT2i ↓ severe exacerbations by ~38%** (2.4 v 3.9 events per 100 person-years) but not moderate exacerbations
 - **DDP4i showed a modest reduction in severe exacerbations but failed to reach statistical significance,** and no difference in moderate exacerbations

It concludes that *GLP1 agonists and SGLT2i were associated with a reduced risk of severe exacerbations compared with sulphonylureas in patients with chronic obstructive pulmonary disease and type 2 diabetes*, while DDP4i were not. These findings corroborate existing observational data but will, of course, need to be *replicated in RCTs before we start using them* in our patients.

An interesting finding was that the **GLP1 agonists only appeared to reduce exacerbations if the patient also had a history of asthma**, which fits with the anti-inflammatory mechanism. However, the more modest benefit seen with SGLT2i was independent of asthma history.

Either way, we may be seeing more of these medications for respiratory disease. The next question is whether they have a role in respiratory patients without diabetes...

2 Medical Apps for Managing COPD

And now for something completely different - it's not all about medicines. NICE has recently conducted its first review ([NICE MTG 68, March 2022](#)) of a digital application to aid COPD management: [myCOPD](#).

This digital tool is designed to support people in managing their COPD. It includes education on correct inhaler use, self-management plans, the ability to check prescriptions are correct, symptom trackers and a 6-week online pulmonary rehabilitation course. It requires an internet connection and can then be used on



smartphones, tablets, computers and televisions. Data is then fed back to a clinician portal which allows the clinician to monitor the patient's condition and provide any recommendations on management changes back to the patient via the app.

NICE reviewed available data. A number of small trials conducted within the NHS were included. Data suggests non-inferiority of the app-based pulmonary rehab course compared to traditional class-based courses. It also suggests benefits in clinical outcomes including improved COPD assessment test score, 6-minute walking test distances and inhaler technique. Acceptability was high, with >80% of patients (average age 74) reporting the app was easy to use and helped them understand and manage their COPD. 75% reported they felt confident in managing their COPD and 66% reported a reduction in exacerbations.

Despite the positive outcomes in the data, NICE felt unable to currently recommend myCOPD for widespread use in the NHS instead advising that further research was required to clarify uncertainties, driven by the size of the studies resulting in them being underpowered to answer questions on admissions and exacerbations, and the heterogeneity of the COPD populations included. NICE notes that the two main RCTs were designed as feasibility studies and show promise but larger trials are now needed.

Nevertheless, this is a window into the future in which patients and clinicians may work and manage disease very differently from traditional practice.

NB Learning Points: COPD

- New treatments GLP1a and SGLT2i may have a role in COPD management in patients with diabetes.
- There is a growing emphasis on identifying patients with COPD and asthma overlap. This confers a worse prognosis but also means treatments such as inhaled corticosteroids (and possibly GLP1a) may be more effective.
- Just like with asthma, patients may benefit from a personal action plan to manage exacerbations - CCGs in Nottingham have produced a useful, simple [COPD action plan for patients](#)
- The [British Lung Foundation](#) has a great website with lots of useful patient information

NB Quality Improvement Ideas: COPD

- Undertake a case review of patients using LABA + ICS
 - Has this been prescribed in patients with documented asthma/steroid responsiveness?
 - If not, could they be switched to LABA + LAMA combination?
- Audit patients with COPD exacerbations.
 - What duration of prednisolone has been prescribed?
 - Provide practice education on recommendations about duration of steroids

KISS: Diagnosis of COPD [NICE NG115 2019](#)

There is no single diagnostic test for COPD. The diagnosis is clinical, based on history, examination and the presence of airflow obstruction using spirometry.

- Airflow obstruction is defined as FEV1/FVC ratio of <0.7
- It is no longer necessary to have an FEV1 $<80\%$ predicted to define COPD: the diagnosis can be made if the FEV1/FVC ratio is <0.7 , but FEV1 is $\geq 80\%$ of predicted in the presence of symptoms such as breathlessness or cough.

Consider the diagnosis if:

- Age over 35, smokers or ex-smokers and any of the following symptoms
 - Exertional dyspnoea, chronic cough, wheeze; frequent sputum production; recurrent 'winter bronchitis'
- In patients without typical symptoms of asthma, such as
 - Chronic non-productive cough; significant diurnal or day-to-day variability; night-time waking with wheeze

Spirometric reversibility testing is not necessary to make the diagnosis or to plan therapy. However, asthma may be present if:

- There is $>400\text{ml}$ response to bronchodilators, or to 30mg prednisolone daily for 2 weeks
- Serial peak flows show a $>20\%$ diurnal or day-to-day variability

At initial evaluation, patients should also have:

- CXR (to exclude other pathology), FBC (to identify anaemia or polycythaemia), BMI calculated

Assessment of severity

There is a change in the 'severity staging' from the previous guideline back in 2004. It is important to assess the severity to guide further treatment. Current recommendations:

Post-bronchodilator FEV1/FVC	FEV1 % Predicted	Stage
< 0.7	$\geq 80\%$	Stage 1 (mild)
< 0.7	50 – 79%	Stage 2 (moderate)
< 0.7	30 – 49%	Stage 3 (severe)
< 0.7	$< 30\%$	Stage 4 (very severe)

The guideline recognizes however that disability can be poorly reflected by FEV1 and encourages the use of clinical rating scales, such as the MRC dyspnoea score (as is included in QOF).

Grade	Degree of breathlessness
1	Not troubled by dyspnoea except on strenuous exercise
2	SOB when hurrying or walking up a slight hill
3	Walks slower than contemporaries on the level ground due to SOB, or has to stop for breath when walking at their own pace
4	Stops for breath after approx. 100 metres or a few minutes on level ground
5	Too breathless to leave the house, or SOB when dressing/undressing

KISS: Treatment of Stable COPD[NICE NG115 2019](#)**Stable COPD**■ **Fundamentals of COPD management:**

- **Stop smoking** - offer treatment and support
- **Vaccinations** - pneumococcal and annual influenza
- **Pulmonary rehab** if indicated (MRC breathlessness scale 3+)
- **Self-management plan** - written with and personalised to the patient
- **Optimise treatment for co-morbidities**

■ **Inhaled Therapy**

- Short-acting bronchodilators (SABA or SAMA) as required
- If remains breathless or has exacerbations despite short-acting bronchodilators:
- **No asthma or steroid responsiveness features** - the majority of patients with COPD
 - **Offer LAMA + LABA**
 - **Still limited by symptoms?**
 - Consider 3/12 trial of LAMA + LABA + ICS - revert if no improvement
 - **Continued frequent or severe exacerbations?**
 - Consider LAMA + LABA + ICS
- **Asthma or steroid responsiveness features**
 - **Offer LABA + ICS**
 - **Still limited by symptoms or has exacerbations?**
 - Offer LAMA + LABA + ICS
- People on alternative regimes (e.g. LAMA monotherapy) who are well-controlled can continue with their current treatment
- Base choice of drugs and inhalers on how much they improve symptoms, patient preference and ability to use an inhaler, potential to reduce exacerbations, side effects, cost
- Minimise the number of inhalers and different types of inhaler used as far as possible
- Ensure patient has appropriate training for their inhalers

■ **Oral therapy**

- **Oral corticosteroids**: not normally recommended, use under specialist advice
- **Oral theophylline**: reduce dose if using macrolides/quinolones to treat exacerbations
- **Mucolytics**: consider if a chronic productive cough, continue if symptoms improve
- **Oral antibiotics**: azithromycin may be considered, needs a referral for workup, e.g. CT
- **Roflumilast**: if severe, FEV1 <50% and 2+ exacerbations in past 12/12 despite triple therapy

Asthma or steroid-responsive features:

- Previous “secure” diagnosis of asthma (e.g. good certainty of the diagnosis, ideally with supporting investigations) or atopy
- A higher blood eosinophil count,
- Substantial variation in FEV1 over time (of at least 400ml) or substantial diurnal variation in peak expiratory flow (at least 20%)

KISS: Treatment of Stable COPD (continued)

[NICE NG115 2019](#)

■ Oxygen therapy

- Refer to assess the need for long-term oxygen therapy if
 - O2 sats <92% on air, cyanosis, polycythaemia, peripheral oedema, raised JVP, or if severe airflow obstruction, i.e. COPD <50% predicted
- May be offered if PaO2 <7.3 or if <8 + polycythaemia, pulmonary oedema or pulmonary hypertension
- Patients should breathe supplemental oxygen for a minimum of 15 hours per day
- Do not use ambulatory or short burst oxygen therapy in those with mild or no hypoxaemia at rest

■ Self-care

- Patients should have an individualised self-management plan
- Offer people a short course of steroids and antibiotics to keep at home if they have had an exacerbation in the past 12 months, understand their use and inform us when they have used them (so we can check they are using them appropriately and are on optimised regular treatment)
- Ask people if they find breathlessness frightening - consider the role of CBT to manage anxiety and help cope

KISS: Treatment of Acute COPD Exacerbations

[NICE 2019, NG115](#)

Exacerbations

- Defined as an acute and sustained worsening of their usual condition, including increased breathlessness, cough and increased sputum production
- Start appropriate treatment with oral steroids (new evidence suggests 40mg pred 5/7)
- Give antibiotics if purulent sputum or signs of pneumonia
- Use pulse oximetry to aid assessment
- Admit if severely unwell, sepsis, SaO2 <90%, severe dyspnoea, significant co-morbidity, poor general condition, unable to cope at home, etc.

Treatment

- **OFFER oral prednisolone 30mg daily for 5 days (GOLD suggests 40mg)**
- **CONSIDER an antibiotic, but only after taking into account prescribing considerations**
- **Reassess if symptoms worsen rapidly or significantly**
 - Consider alternative causes and the possibility of antibiotic resistance
 - Send a sputum culture if symptoms fail to resolve and this has not been done already
- **1st line antibiotics:** amoxicillin 500mg TDS, doxycycline 200mg stat then 100mg OD, clarithromycin 500mg BD for 5/7
- **2nd line antibiotics:** if no improvement on 1st choice Abx after 3/7 switch to alternative 1st line, guided by sputum culture results if available
- **Alternative choice if at higher risk of treatment failure (e.g. repeated abx, resistance on past MC&S):** co-amoxiclav, levofloxacin, co-trimoxazole 5 days

KISS: [GOLD Guidelines 2023](#)► *Assessment of severity*

“The management strategy for stable COPD should be predominantly based on the individualised assessment of symptoms and future risk of exacerbations.”

Just as NICE has done, GOLD moves away from the use of spirometry for decisions on treatment. Spirometric assessment remains important for diagnosis and prognosis, but data shows symptoms are a better guide to which treatments are likely to be useful. As a result, GOLD uses a 2 part scoring system to determine severity but it is the “group”, based on symptoms and exacerbations, which determines treatment:

- **“Grade” - based on spirometry**

- This follows the same rating as NICE, e.g. FEV1 50-79% = grade 2 or moderate COPD
- Grade 2 indicates a 20% annual risk of exacerbation; this is substantially higher in grades 3 and 4

- **“Group” - based on symptoms and exacerbations in the past 1 year**

- The preferred option for assessment of symptoms is the [CAT assessment tool](#) but the mMRC dyspnoea scale can also be used
- GOLD uses its “ABE” assessment tool to combine symptom and exacerbation history
 - **Group A** = 0 or 1 exacerbation; mMRC 0-1 or CAT <10
 - **Group B** = 0 or 1 exacerbation; mMRC 2+ or CAT 10+
 - **Group E** = 2+ exacerbations or 1+ needing admission

► *Initial Treatment of COPD*

Choose initial inhaled therapy based on the patient’s ABCD group

GOLD is relaxed about the specific agents used preferring the choice to be individually tailored and will depend on “access, cost, prescriber, and most importantly, patient’s ability and preference”

- **Group A** - mild symptoms and rare exacerbations

- **Bronchodilator** (either beta-agonist or antimuscarinic)
- Long-acting versions are preferred except if only occasional dyspnoea
- If poor effect try an alternative class of bronchodilator

- **Group B** - more severe symptoms but rare exacerbations

- LABA + LAMA

- **Group C** - exacerbations common (i.e. 2+ per year) or severe

- LABA + LAMA
- Consider LABA + LAMA + ICS if blood eosinophils >300

- **Ongoing dyspnoea?** Consider switching inhaer device or molecule, 1x & treat other causes of dyspnoea, implement or escalate non-pharmacological interventions (e.g. pulmonary rehab, vaccinations, nutrition, home O2)

- **Ongoing exacerbations?** Consider roflumilast if FEV1 <50% & chronic bronchitis or azithromycin if former smoker, or

OBSTRUCTIVE SLEEP APNOEA / HYPOPNOEA SYNDROME

OSA is not a new disease so why is it a hot topic and what is new? Well, the name for a start - although specialists have actually been using the phrase OSAHS for many years. We also now have a broader appreciation of the complications caused by OSA and understand that it can affect people who do not have the classic symptoms or signs. Plus there's the small matter of related conditions Obesity Hypoventilation Syndrome and COPD-OSA overlap. NICE has recently published it's full guidance, complete with useful visual summaries ([NG 202 Aug 2021](#)) to advise on the management of these conditions.

First let's remind ourselves of OSA, thanks to a recent review in NEJM ([2019;380:1442](#)), BMJ Best Practice [2021](#), and Clinical Knowledge Summaries [2021](#).

NB Revision: Obstructive Sleep Apnoea/Hypopnoea Syndrome

[NEJM 2019;380:1442](#), [BMJ Best Practice 2021](#), [CKS 2021](#)

- **Apnoea** = complete airflow obstruction with temporary absence or cessation of breathing (>90% cessation in airflow for >10 seconds)
- **Hypopnoea** = decreased airflow (>30%) with reduced O2 sats ($\geq 3\%$) or arousals from sleep
- Due to **episodic sleep state-dependent collapse of the upper airway**, as the pharyngeal muscles are overwhelmed by collapsing forces due to decreased tone during sleep
- Repetitive reduction in ventilation results in hypoxia, hypercapnia and/or arousals from sleep
- **Prevalence** reported as 3% women / 10% men aged 30-49yo; 9% / 17% aged 50-70yo, although these figures may be very conservative
- **Risk factors include conditions which ↓ resting pharynx size or ↑ airway collapsibility:**
 - Obesity, neck circumference >40.6cm, ↑ age, male, FHx, smoking, alcohol, sleeping supine
 - Adipose deposition in tongue & pharynx; tonsillar/adenoid hypertrophy; craniofacial abnormalities
- **Complications**
 - Motor vehicle accidents; impaired glucose metabolism; CVD: hypertension, dysrhythmia, MI, heart failure, cognitive impairment. All are of high likelihood over time. Mortality is increased due to CVD.
- **Common symptoms & signs** are:
 - Loud or irregular snoring, daytime sleepiness, unrefreshing sleep, nocturia, witnessed apnoeas, choking/gasping in sleep, dry mouth on awakening, morning headaches, BMI >30, increased neck circumference, crowded oropharynx
- **Diagnosis**
 - Traditionally this involves a sleep clinic-based polysomnography sleep study. This monitors sleep and includes information such as airflow, heart & resp. rate, O2 sats, brain wave activity (EEG) as well as body position, muscle electrical activity and eye movement
 - More recent home polysomnography, using portable devices with fewer channels, have been shown effective at OSAHS assessment in high probability patients, with similar results to full sleep studies
- **Treatment**
 - Recommend lifestyle measures such as smoking cessation, weight loss and alcohol reduction
 - CPAP is the most common treatment except in the most mild cases. It has been shown to ↑ survival, ↓ sleepiness, ↓ blood pressure (by 2-3mmHg), ↓ risk of recurrence of AF, and may ↑ mood and cognitive function. It may also ↓ hyperglycaemia and insulin resistance although data here is more mixed. CVD does not seem to improve in non-sleepy patients.
 - Oral mandibular devices are less efficacious but still help around half of users, so are a good option for mild OSAHS where CPAP is not tolerated. They may not fit in patients with macroglossia, where tongue-retaining devices may be used, albeit with lower effectiveness again.
 - Surgery may be effective. Most commonly used for treating discrete anatomical lesions. Otherwise, it can improve but rarely resolves, and has the potential for complications.

OSAHS and Obesity Hypoventilation Syndrome [NICE NG 202 Aug 2021](#)

These are the first national UK guidelines on managing OSAHS but also make the surprise inclusion of obesity hypoventilation syndrome and COPD-OSAHS overlap syndrome. Key recommendations:

► When to Suspect OSAHS

- NICE encourages to take a sleep history and assess for OSAHS if 2+ of the following are present:
 - Snoring, witnessed apnoeas, unrefreshing sleep, waking headaches, unexplained excessive sleepiness, tiredness or fatigue, nocturia, choking during sleep, sleep fragmentation or insomnia, cognitive dysfunction or memory impairment
- We should be aware OSAHS is more likely in:
 - Obesity or overweight (including in pregnancy), treatment-resistant hypertension, T2DM, cardiac arrhythmia (esp. AF), CVA/TIA, CHF, mod > severe asthma, PCOS, Down's syndrome, non-arteritic anterior ischaemic optic neuropathy, acromegaly

► How to Assess for Possible OSAHS

- Use the [Epworth Sleepiness Scale](#) to assess the impact of sleepiness
 - We need to remember the ESS only assesses sleepiness and not risk of OSAHS. NICE repeatedly reiterates that *some people may have OSAHS and not report excessive tiredness (indeed it might be MANY - [click here for explanation](#))* so it should NOT be the sole determinant of referral. We should suspect OSA with a global clinical picture.
- Consider the [STOP-Bang Questionnaire in addition to the ESS](#)
 - 'consider' because the data is more limited for STOP-Bang although it suggests high sensitivity with low specificity for OSAHS. It may be less accurate with less common presentations and in women.
 - BMI and neck size (e.g. "what shirt collar do you take?") are required.

► When to Refer

- NICE does not make formal indications for referral, the focus being on clinical judgement incorporating symptoms, signs, risk factors and the questionnaire information. **Where OSAHS is suspected referral is appropriate to confirm the diagnosis.**
- NICE does however recommend **certain groups merit 'rapid assessment'**: (remember the "*Four-Week Wait Campaign*" a few years ago for professional drivers)
 - Those with a **vocational driving job or job where vigilance is critical for safety; unstable cardiovascular disease** (e.g. resistant hypertension, poorly controlled arrhythmia); **pregnancy; undergoing pre-op assessment for major surgery; non-arteritic anterior ischaemic optic neuropathy.**
 - It encourages us to include questionnaire scores, how sleepiness affects the individual, co-morbidities and occupation risk in the referral later to facilitate rapid assessment.

► Diagnosis

- This is confirmed by a sleep clinic. **Home respiratory polygraphy** is the 1st line investigation and only using lab-based when this is impractical or additional monitoring is required.
- This will class patients with mild, moderate or severe OSAHS.

Having made a diagnosis of OSAHS, what treatments are recommended?

NICE recommends lifestyle advice for all severities, including weight loss, smoking cessation, alcohol reduction and sleep hygiene as appropriate for the individual.

- Mild OSAHS without symptoms or where usual daytime activities are unaffected: treatment is not usually necessary but lifestyle measures may prevent OSAHS worsening.
- Mild OSAHS with symptoms affecting quality of life and activity: CPAP (continuous positive airway pressure) is recommended in addition to lifestyle and sleep hygiene measures.

- 'Consider' a customised mandibular advancement splint as an alternative to CPAP in patients ≥ 18 yo AND suitable dentition AND unable to tolerate or decline CPAP.
- 'Positional modifiers' may be considered for mild to moderate positional OSAHS (symptoms occur when supine but not with the patient on their side) where other treatments are unsuitable. Think tennis ball tapped to the back, although there are more dedicated tools such as special pillows and positional alarms. These methods are unlikely to be effective in severe OSAHS.

For moderate or severe OSAHS CPAP should be offered and is the most effective option.

Surgery is not often used but may be considered in those with large obstructive tonsils and BMI < 35 , or where CPAP and mandibular splints are not tolerated (despite "medically supervised attempts").

Does rhinitis contribute to OSAHS?

We have often tried empirical steroid nasal sprays in snorers and people with OSAHS. NICE found limited evidence to demonstrate benefits from this approach but does recommend people with nasal congestion are assessed for underlying allergic or vasomotor rhinitis, and treated with intranasal steroids for either type of rhinitis, or anti-histamines if allergy-induced.

Whilst this is unlikely to solve OSAHS it may improve sleep and make CPAP more comfortable. CPAP itself is a potential cause of rhinitis and an orofacial mask rather than nasal, plus humidification, may help.

What about lifestyle measures?

There is a long-held belief that overweight and obesity drives OSAHS. Given this hypothesis it seems then surprising that lifestyle measures such as weight loss don't feature prominently in the guidance.

In fact, things are more complex than this simple concept would suggest. Data suggests that in fact, most people with OSAHS have a normal or only mildly raised BMI. It has been argued that for some, OSAHS will be the driver for weight gain, not the other way around, and having OSAHS is a significant barrier to losing weight due to alterations in our brain-gut axis. To compound the problem, [weight loss has been shown to have limited impact on OSAHS](#), with even substantial falls in BMI leaving patients with ongoing respiratory symptoms, leaving weight loss very much an adjunct, not a cure. ([click here for a video explanation](#))

The guidance recommends that for all severities of OSAHS clinicians should discuss lifestyle changes: smoking cessation, preventing excess weight gain, weight loss and reducing alcohol intake. Lifestyle changes should not delay referral as data shows that patients especially find it very challenging to reduce body weight sufficiently to lead to significant improvements.

Remind me what obesity hypoventilation syndrome is...

The surprise inclusion in the OSAHS guidance is two additional related conditions which many of us will have never heard of before.

Obesity hypoventilation syndrome (OHS, also known as Pickwickian Syndrome) was summarised recently in the *European Respiratory Review journal* ([ERR 2019;28:151:180097](#)). It's defined as a combination of:

- **Obesity** (BMI > 30 , but often morbid obesity),
- **Daytime hypercapnia** (arterial CO₂ tension ≥ 45 mmHg), and
- **Sleep-disordered breathing** (usually severe OSAHS is present with 'hypersomnolence').
- **Other disorders causing alveolar hypoventilation must be excluded** (so this is very much a specialist diagnosis).

OHS is a form of chronic ventilatory failure and results from three main mechanisms linked to obesity. Firstly excess adipose tissue in the abdomen and chest wall reduce lung volume and impairs normal diaphragm motion, resulting in poor gas exchange and a need for increased work of breathing, compensated elevated respiratory drive. Secondly, for some, there is a failure to maintain the demand of this elevated respiratory

drive and hypoventilation ensures. Initially, this affects REM sleep but repeated hypoventilation drives a secondary depression of respiratory centres which continues during the daytime and hypercapnia develops, and OHS. Thirdly, sleep apnoea further compounds the previous two mechanisms.

Prevalence has been reported as 0.4% of adults. It may be found during a sleep study for presumed OSAHS or due to acute-on-chronic hypercapnic respiratory failure.

Diagnosis is mainly in the 5/6th decade of life and often delayed. COPD is a common misdiagnosis but alarm bells should ring because OHS does not show obstructive pathology on spirometry.

It is commonly associated with metabolic and cardiovascular co-morbidities, predominantly hypertension (55-88%), heart failure, ischaemic heart disease and pulmonary hypertension (~50%). Patients with OHS have greater morbidity and mortality than those with only OSAHS - untreated all-cause mortality may be as high as 24% over 1.5-2 years.

Formal diagnosis requires arterial PaCO₂ levels, but a serum venous bicarbonate level can be helpful. Bicarb increases with long-term raised CO₂ and is suggestive of OHS but is not sufficient to exclude, particularly where clinical suspicion is high. NICE recommend 'consider' measuring serum bicarb as a preliminary test if the pre-test probability is low - but the implication is still this is done by a sleep specialist who can decide on that probability. Assessment for co-existing OSAHS with polygraphy is also required.

Treatment is with some type of positive airway pressure, either via CPAP or NIV (non-invasive ventilation), and weight management strategies, and prompt specialist assessment during acute decompensation.

NICE: Obesity Hypoventilation Syndrome

Many of the recommendations for assessment and treatment are identical to that of OSAHS given that separating the two clinically is not possible.

- Suspect OHS if BMI ≥ 30 plus features of OSAHS or nocturnal hypoventilation (e.g. waking headaches, peripheral oedema, hypoxaemia sats $< 94\%$, unexplained polycythaemia)
- Use Epworth sleepiness scale to assess sleepiness but do not use this alone to determine referral as not all people with OHS have excessive sleepiness
- Prioritise for rapid assessment if severe hypercapnia (PaCO₂ > 7.0), hypoxaemia, acute ventilatory failure, or any of the criteria in possible OSAHS
- Offer lifestyle advice and manage rhinitis as per OSAHS
- If OHS and no acute ventilatory failure CPAP is 1st line, NIV alternative option
- If OHS and acute ventilatory failure NIV is 1st line
- Oxygen may be considered if they remain hypoxic despite CPAP/NIV

And what is COPD-OSAHS overlap syndrome???

This condition, also with guidance from NICE, is exactly what it says on the tin. Not a discreet disease this is a diagnosis for patients with both COPD and OSAHS. Given both are common NICE estimates a 1% prevalence and states it is under-recognised.

Hypoxia secondary to COPD is further exacerbated overnight by the OSAHS. Unsurprisingly this is linked to increased symptom burden and poorer prognosis.

A recent cross-sectional study published in Plos One ([9.7.2020](#)) reviewed almost 47k French people with moderate to severe OSAHS. In those with available spirometry, 13% had overlap syndrome. There were greater rates of CHD, CHF and PAD in the overlap group, which was linked to the severity of the COPD.

NICE recommends:

- Take a sleep history in patients with COPD and features of OSAHS or nocturnal hypoventilation.
- We can use the Epworth SS to quantify sleepiness, consider the STOP-BANG questionnaire, and offer spirometry to assess severity of COPD (the practice may be doing this routinely for COPD f/up already)
- Prioritise rapid assessment as per guidance on OHS
- Sleep clinics should assess for ventilatory failure and OSAHS
- Treatment is usually with CPAP but NIV should be considered if severe hypercapnia.

Anything else we need to tell our patients.....DRIVING!!!

NICE reminds us we need to counsel patients on the effects of excessive sleepiness on safe driving and occupational risk, and on DVLA rules. DVLA rules are clear: **a person with excessive sleepiness having, or likely to have, an adverse effect on driving SHOULD NOT drive**. This includes any severity of OSA, but the [British Thoracic Society also notes in a recent position statement](#) that patients may have symptom-less OSA and the guidance would not apply to this group. It explains a GP's responsibility is to:

- Recommend that a patient does not drive if excessive sleepiness is having or likely to have an adverse effect on driving, whatever the cause.
- Arrange onwards referral for specialist review if indicated and ensure prioritisation of cases as urgent where patients report an impact on vigilant critical activities such as driving, flying or operating dangerous machinery.

A patient also has responsibilities:

- Report symptoms honestly
- Notify the DVLA regarding a diagnosed medical condition when required to do so (the clinician is not responsible for informing the DVLA).
- Comply with DVLA recommendations.

KISS: DVLA Rules of OSA and Driving

[DVLA 2021, pg.108](#)

Excessive sleepiness due to a medical condition including mild OSA (including suspected OSA):

- Group 1 and 2:
 - **Must not drive**
 - Driving may resume only after satisfactory symptom control.
 - If symptom control cannot be achieved within 3 months the DVLA must be notified.

Excessive sleepiness due to moderate or severe OSA:

- Group 1 and 2:
 - **Must not drive and must notify DVLA**
 - Subsequent licensing will require:
 - control of condition
 - sleepiness improved
 - treatment adherence
 - DVLA will need medical confirmation of the above, and the driver must confirm review to be undertaken every three years at the minimum.

The DVLA have produced a leaflet for drivers with excessive sleepiness: [Tiredness can Kill](#)

NB Learning Points: OSAHS

- New NICE guidance encourages the referral of all patients with suspected OSA to a sleep specialist for formal assessment and diagnosis.
- Intranasal corticosteroids are unlikely to resolve OSAHS but may reduce the side effects of CPAP.
- We should be aware of Obesity Hypoventilation Syndrome and OSAHS/COPD overlap, both of which confer a worse prognosis than OSA alone.
- Any patient with excessive sleepiness should not drive, and if it continues for >3/12 notify the DVLA.
- For a good patient information leaflet [click here](#)
- For some interesting educational videos suitable for clinicians and patients, see the [Youtube channel of Vic Veer](#), head surgeon at the Sleep Surgery department of the Royal National ENT Hospital

MACROLIDES IN RESPIRATORY DISEASE

In an age of fear around antibiotic resistance and successful drives to reduce inappropriate prescribing, it may seem an unlikely time to publish national guidance on the role of long-term antibiotics for chronic respiratory disease, but the British Thoracic Society has recently published guidance on [the role of long-term macrolides in adults with respiratory disease](#), also published in-depth in [Thorax 2020;75;370](#). This provides clear guidance on when macrolides may be of benefit, when a benefit is unlikely, and, as important complications with this class of medication have emerged in recent years, how they can be prescribed to minimise risk.

Further weight is added in a very recent BJGP Clinical Practice paper ([BJGP 2022;72:42](#)) examining the latest data on macrolides in COPD, while the [DTB 2021;59:67](#) also discusses the subject with a firm view on the clinical application of this guidance.

Long-term macrolides will only be indicated for those with more severe chronic respiratory disease and initiation is very much the remit of specialists, but with a new, broadened national guideline it will be important for us to have an understanding of the rationale and regimen as we are likely to see increasing numbers receiving the treatment.

► *Why use long-term antimicrobials?*

While macrolides are effective antibiotics with strong tissue penetration and antimicrobial activity against Gram-positive and some Gram-negative bacteria as well as a range of atypical pathogens, the interest in this area is due to their possible immunomodulatory action.

The mechanism remains unclear but was initially proposed after observational data suggested a possible steroid-sparing effect in patients with asthma using macrolides for an extended duration. Further research shows a variety of effects including changes to airways secretions, reduction in inflammation and sublethal effects on bacteria including disruption of biofilms and reduced adherence, toxin production and mobility.

► *In which conditions are macrolides indicated?*

The guideline focuses on asthma, bronchiectasis and COPD, which as the DTB points out, is very welcome due to significant clinical overlap. It notably does not recommend in unexplained chronic cough and cystic fibrosis is outside of the remit of the paper.

In each disease, **the indication is frequent exacerbations**. Exacerbations are the only outcome for which we can be confident long-term macrolides do have a beneficial effect. Symptoms, quality of life, lung function either have mixed results or debatable clinical significance. Our understanding is further hampered by a surprising paucity of data given such a large population and heterogeneity between studies.

Definition of 'Frequent Exacerbations' [BTS 2020](#), [DTB](#)

- **Asthma**
 - Partial loss of control despite adherence to high-dose inhaled steroids; Asthma Control Questionnaire ≥ 0.75
- **Bronchiectasis**
 - ≥ 3 exacerbations per year
- **COPD**
 - >3 exacerbations requiring steroid therapy AND ≥ 1 hospital admission in the past year

Bronchiectasis has stronger evidence. Meta-analysis of three RCTs, all with exacerbations as the primary endpoint, patients with a baseline average of 3-5 exacerbations/annum (although the exact definition of an exacerbation varied) given macrolide use for 6-12 months found reductions in the number of exacerbations. One study showed **an incidence rate ratio of 0.57, another an ARR 33.5%** (despite a big drop in exacerbations in the placebo group as well).

For **asthma**, the principal data comes from the AMAZES trial (placebo-controlled RCT, N=420, adults aged 50-70 with asthma requiring high-dose inhaled steroids ($\geq 800\text{mcg}$ beclomethasone daily), mean 1 exacerbation in the past year, given azithromycin 500mg 3x weekly for 48 weeks). **During the trial, 61% on placebo and 44% on azithromycin had an exacerbation** ($p = <0.0001$). A second study, AZISAST (N=109) failed to show statistically significant improvements. There were differences in the study designs. It was also noted there may be differences between asthma phenotypes, e.g. eosinophilic and non-eosinophilic. Overall it seems that larger studies for longer durations would be helpful here for clarification of effect.

In **COPD**, **8 RCTs (N=1582) showed long-term macrolides had a significant reduction in the rate of exacerbations ($RR = 0.58$, $p < 0.01$) and time to first exacerbation**. [Cochrane](#) found the number needed to treat was 8 patients to prevent 1 exacerbation in 1 person.

However, the [BJGP](#) notes that in **patients requiring hospitalisation due to exacerbations there was no benefit** and proposes that in this group, where the airways disease is so far progressed there is little treatment can achieve. Following this logic, it also notes that there may be a benefit in mild COPD, with data suggesting an odds ratio of 0.31 for macrolides v placebo. Current guidance does not consider this group.

In fact, the BJGP notes that **the greatest gains may come in patients with COPD-asthma overlap** where data has shown not just a reduced requirement for oral corticosteroids, but improvements in quality of life, pulmonary function tests and reductions in the use of bronchodilators.

As the DTB notes, effectiveness in improving QoL and symptoms is less convincing and even when studies have shown statistically significant changes, "the size of the change is often very small".

When To Offer Long-term Macrolides [BTS 2020](#)

For management of exacerbations, BTS recommends long-term macrolides could be:

- **Offered to people with bronchiectasis** and high exacerbation rates to reduce exacerbations
- **Considered in adults with asthma** aged 50-70y with ongoing symptoms despite $>80\%$ adherence to high-dose inhaled steroids and ≥ 1 exacerbations requiring oral steroids in the past year;
- **Considered in patients with COPD** with ≥ 3 acute exacerbations requiring steroids and ≥ 1 requiring hospital admission in the past year

► Does this approach drive antibiotic resistance?

Yes (probably), but it hasn't been associated with adverse effects and may be temporary while receiving the treatment.

Data varies with each condition and study so it is hard to draw firm conclusions but as an example, the AMAZES trial found resistance to macrolides in 49% in the active treatment group vs 29% on placebo. AZISAST found an increase from 48% to 87% for those on azithromycin. However, it also found a reduction in resistance from 74% to 46% after a 4-week washout period at the end of the study.

This has prompted the idea about **drug holidays** in this group as a way of combating resistance in the individual and maintaining the benefits of treatment. **For both asthma and bronchiectasis, BTS suggest considering a drug holiday every year**, e.g. stopping over the summer for bronchiectasis when we would expect the risk of exacerbations to be lower.

What adverse events are linked with long-term macrolide use?

BTS found little evidence of significant adverse events with long-term macrolide use, though some of this is because studies may not have been examining and reports some events.

The main side effect of macrolides is GI upset which may affect up to 70% of erythromycin courses although less with azithromycin. Interestingly this is not due to alteration in gut flora but due to interactions with motilin receptors in gut smooth muscle driving increased contractility. Where GI effects are a continuing problem a lower dose of azithromycin (250mg 3x weekly) can lead to improvement.

There have been growing concerns about the adverse cardiac effects of macrolides. The [MHRA](#) strengthened its warning about erythromycin and clarithromycin in Dec 2020, advising the former should not be given to patients with a history of QT interval prolongation or ventricular cardiac arrhythmia, or electrolyte disturbances (e.g. hypokalaemia or hypomagnesaemia), and exercise caution in those with cardiac disease, conduction disturbances or concomitant use of medications associated with QT interval prolongation ([Credible Meds](#) is a very useful website for checking this). The MHRA does not make recommendations about azithromycin.

Hepatotoxicity has been [reported with macrolide use](#), and while acute changes in serum aminotransferase levels in 1-5% of treated patients, these are usually minor, asymptomatic, self-limiting and transient. More severe hepatotoxicity and long-term liver damage appears rare, although if bad cholestasis occurs this may lead to more persistent damage.

Ototoxicity with either hearing (tinnitus has been or balance difficulties is a recognised adverse effect of macrolides, but is generally uncommon and almost always reversible with cessation of the drug.

Nevertheless, cardiac assessment prior to and shortly after initiation, and LFT monitoring is recommended in all patients receiving long-term macrolide therapy (see KISS for details).

► *Should GPs be able to start prescribing macrolides?*

The guidance is very much about this being the remit of respiratory specialists. However, given the pressure on secondary care and often limited ability for specialist review for our patients, the closing words in the BJGP seem pertinent: "ultimately, the GP will be able to weigh up the risks and benefits of this therapy because of their close and longstanding relationship with the patient".

Whether we will want to take on the initial checks and monitoring that are ultimately required to safely prescribe long-term macrolides will also be an issue for GPs to weigh up.

NB Learning Points & Quality Improvement Ideas: Macrolides In Respiratory Disease

- Azithromycin may reduce exacerbations in patients with severe bronchiectasis, asthma and COPD.
- Improvements in other areas are uncertain.
- BTS has produced specific guidelines for when specialists should consider macrolides.
- Cardiac assessment and ECG prior to treatment initiation and repeat ECG at 1/12 is required
- A drug holiday may be recommended to reduce antibiotic resistance and possibly maintain efficacy
- Audit patients with bronchiectasis and 3+ exacerbations in the past 1 year
 - Are they having specialist follow up?
 - Have they been considered for macrolide therapy?

KISS: Long-term Macrolide Treatment and Safety

Treatment should be initiated by a respiratory specialist. Existing treatments should be optimised, e.g. best inhaler combination, good inhaler technique, smoking cessation, airway clearance techniques.

Macrolides are off-license for this indication.

Treatment

- **Azithromycin 500mg three times weekly**
 - Lower dose of 250mg 3x weekly can be used to reduce adverse effects (usually GI)
- Effectiveness should be assessed at 6-12 months and discontinued if it has not been successful
 - The DTB comments that given exacerbations rates vary year on year this effectiveness may be difficult to gauge and extended durations may be required
- A drug holiday of a few weeks may be recommended each year to combat resistance - sometimes over the summer is advised as the risk of infection/exacerbation is usually lower

Assessments required:

- **Cardiac**
 - *Requires cardiac risk assessment*
 - *Baseline ECG to check QTc interval*
 - Avoid if >470ms in women, >450 in men
 - *Repeat ECG at 1 month to check for QTc interval increase*
 - Stop if new QTc prolongation occurs
 - *Avoid drugs that may prolong QT interval*
 - [Credible Meds](#)
 - If medication is started which may affect QT interval then repeat ECG is recommended
- **Liver**
 - *LFTs at baseline, 1 month and 6 monthly*
- **Hearing/balance**
 - At review ask about hearing or balance issues or worsening of existing symptoms
 - If there is deterioration it is usually reversible on cessation
- **Microbiological screening**
 - *Sputum sampling prior to initiation to exclude non-TB mycobacterium* is advised
 - Long-term monotherapy can drive resistance and make it very hard to treat
 - If sputum is checked while on treatment a 2/52 washout period is recommended

How to manage an exacerbation on long-term macrolides

- Use a non-macrolide antibiotic to treat acute exacerbations
- Do not stop the macrolide while taking additional treatment
- Consider sending sputum MC+S where pt is expectorating ([NICE](#) recommends in bronchiectasis)



CHRONIC COUGH

Cough is one of the most common presentations in general practice. While for most it will be short-lived, for some it can be a persistent and troubling problem. This very useful, highly practical report in the DTB ([2019;57:74](#)) explores the assessment and management of chronic cough.

Chronic Cough: A practical approach to assessment and management [DTB 2019;57:74](#)

Chronic cough is defined as cough persisting for >8 weeks, reportedly affects 10% of people and may last for many years (data from specialist clinics show a mean duration of >10y). This presents us with the classic GP dilemma - the causes are diverse, most will represent minor illness, very occasionally it might be serious, even life-threatening and we need to identify and manage the lot.

In the past, we have always considered asthma, reflux and post-nasal drip as the primary causes of chronic cough, but by only considering these three misses a range of other important causes including drug-induced cough.

► **Assessment**

The main difference between this new paper and previous reviews on this subject and that, while a good history and examination are key, it also advises on a minimal set of investigations to exclude serious disease and guide towards a likely diagnosis:

1. **Exclude red flags**
2. **Questions to identify simple, common diagnoses**
 - **Timing** - diurnal variation/exercise-induced (asthma?), better at weekends (occupation asthma?), cough after meals or with speech (GORD?)
 - **Respiratory symptoms** - wheeze (usually asthma), sputum production (COPD or bronchiectasis), distinguish throat clearing (upper respiratory cough syndrome) from genuine cough
 - **Non-respiratory symptoms** - dyspepsia/reflux/metallic taste (GORD), rhinitis/anosmia (URCS)
 - **Smoking** - encourage to stop, consider COPD and cancer
 - **Drug history** - ACE inhibitors: 20% of users develop a cough, may start rapidly or after years of trouble-free use, can take >1y to resolve after cessation
 - **Quality of life** - impact on QoL can be substantial, ask re: mental health issues which can
3. **Examine the cardiorespiratory system, oropharynx and abdomen**
4. **Investigations**
 - Chest XR
 - Spirometry
 - Additional tests if felt clinically indicated (e.g. FBC to assess eosinophils, platelets, etc.)

► **Management**

Treatment, of course, depends on whether our assessment indicates a likely specific cause.

Smoking cessation should lead to improvements in cough (although there may be an initial worsening which is worth warning patients about). Stopping an ACEi will resolve related cough, however, the effects may take more than a year to wear off.

If no obvious cause promotes itself, the DTB suggests considering a sequential trial of treatment for asthma, GORD, and then upper airway cough syndrome.

- **Asthma and steroid-responsive airways disease**
 - In this group a trial of treatment, a usually inhaled corticosteroid is advised
 - Interestingly, although it may be the most common, asthma is not the only cause we should consider. Others include:

- **COPD with asthmatic features/features suggesting steroid responsiveness** (the phrase NICE uses in its latest COPD guideline, which we might know better as ACO or asthma-COPD overlap). Importantly there is no role for a trial of steroids in patients with probable COPD but no features of steroid responsiveness.
- **Eosinophilic bronchitis**: indicated by chronic cough and sputum eosinophilia but a lack of peak flow variability or airway hyper-responsiveness. Incidence is unknown. Inhaled steroids may only lead to a partial response, oral prednisolone may be required. An alternative option is to add montelukast to ICS, which leads to greater improvements than ICS alone in this group.
- **Gastro-oesophageal reflux disease**
 - A well-established cause but why it generates cough, and how best to treat remains unclear.
 - Guidelines usually suggest 3/12 high dose PPI + lifestyle modification (e.g. diet & weight loss).
 - Further additions of ranitidine then baclofen have been shown to provide additional benefit.
 - A significant proportion may be refractory to acid suppression - this group may have non-acid gaseous reflux (leading to cough hypersensitivity). PPIs will be ineffective - suspect in those without dyspepsia or regurgitation.
 - In this group, prokinetics may be effective (e.g. metoclopramide, domperidone) but their use is limited due to recent safety concerns, and prolonged treatment seems unwise.
- **Upper airway cough syndrome**
 - Triad of throat-clearing, post-nasal drip and blocked/runny nose
 - A 6-8/52 trial of nasal corticosteroids +/- antihistamine is indicated if atopic.

► What next?

If the cough fails to improve despite appropriate treatment then the DTB suggests we consider secondary care referral. But what is driving cough in these patients once "disease" has been ruled out?

The BMJ ([2015;351:h5590](#)) details the growing understanding of the pathophysiology underlying "chronic refractory cough" and the neuromodulatory component central to it. Some of this remains conceptual and based on expert opinion but several common themes have emerged:

1. Cough hypersensitivity syndrome and laryngeal hypersensitivity

- Innocuous irritants lead to up-regulation of upper airways mucosal cough receptors, in turn causing hypersensitivity of sensory airway nerves
- This is also seen with identifiable causes of chronic cough (e.g. asthma, GORD, etc)
- Laryngeal features include paraesthesia, cough, dyspnoea, dysphonia or spasm; also globus pharyngeus, paradoxical vocal fold movement (which can lead to inspiratory symptoms, e.g. stridor) and muscle tension dysphonia

2. Central neuronal sensitisation

- There is central as well as peripheral neuronal sensitisation and similarities exist between CRC and conditions such as neuropathic pain
- Variants on paraesthesia ("throat tickle"), hyperalgesia and allodynia ("allotussia" is cough triggered by low-level stimuli such as talking or cold air) may be seen in patients with CRC

The DTB reviews the two possible treatment strategies for this group:

- **Behavioural strategies**
 - Such as cough suppression techniques and vocal hygiene training, delivered by trained physios or speech and language therapists, have RCT data showing reduced levels of cough and improved QoL
- **Centrally acting drugs**
 - Off-label use of neuromodulating drugs has been explored recently, with the theory they may reduce the neural sensitisation and therefore reduce cough.

- Data is currently limited or weak but does suggest an improvement in cough frequency and QoL scores. However they also have significant side effect rates which may limit their use, and there are broader concerns about abuse/addiction of gabapentinoids and morphine.
- **Amitriptyline** - up to 25mg nocte; weak data showing benefit
- **Gabapentin** - up to 1800mg/day; small improvement in cough score vs placebo, 1/3 had s/e
- **Pregabalin** - up to 300mg/day; one RCT showed it was better than placebo for cough severity and QoL, s/e were more common
- **Morphine** - not discussed in the DTB, but the BMJ ([2015;351:h5590](#)) detailed an RCT that showed oral morphine SR 5mg BD lead to small improvements in cough score vs placebo. S/E were as expected with morphine. Despite the data, it was not recommended in the most recent US guideline on chronic cough, presumably due to concerns about opioid abuse. Note, codeine has no evidence in cough management and is not recommended, despite being considered anti-tussive for many years.

KISS: Assessment of Chronic Cough

1. **Red flags?** If present refer via 2WW pathway.
2. **Smoker?** Encourage cessation. The cough may worsen initially.
3. **ACE inhibitor?** Stop, switch to alternative antihypertensive. May take 1 year to resolve.
 - **Cough persists?**
4. **Organise Chest XR and spirometry, assess for GORD, asthma & upper airway cough syndrome**
 - Treat the most likely cause, otherwise, try a sequential approach: asthma > GORD > UACS
 - Use adequate duration and dose of the drug
 - **Complete response?** No further action.
 - **Partial response?**
 - Check compliance.
 - Consider alternative or dual diagnosis, treatment as appropriate.
 - Consider a high dose or longer duration of treatment if available.
 - **No response?**
 - Consider alternative diagnosis and start treatment trial.
 - Refer to a specialist clinic for further investigation.
 - Consider "neuromodulatory" drug, e.g. amitriptyline, gabapentin, morphine ([BMJ 2015;351:h5590](#))

► What about psychological causes? [Chest 2015;148:24](#)

Not covered in the BMJ but an important potential cause of cough and raised in the "Rapid Responses". This was discussed recently in Chest journal. Previously termed 'psychogenic cough' experts have renamed it 'somatic cough disorder'. Hard to diagnose, this is again a diagnosis of exclusion. Treatments include 'suggestion therapy', reassurance, counselling, hypnosis or referral to a psychologist/psychiatrist.

'Tic cough' (now called 'habit cough') is recognised, particularly in children, and may also respond to 'suggestion therapy'.

NB Learning Points & Quality Improvement Ideas: Chronic Cough

- For chronic cause assess for causes, exclude red flags using clinical judgment, CXR and spirometry
- A sequential trial of treatment for asthma > GORD > upper airway cough syndrome is appropriate in the absence of a clear cause
- Referral is advisable if this fails, with behavioural strategies and centrally-acting neuromodulators as treatment options
- Audit patients with chronic cough – has a cause been identified? Has a trial of treatment been effective? If not has referral +/- alternative treatment options been discussed?



BRONCHIECTASIS

Bronchiectasis has had two sets of guidance published in 2018: firstly, an update from the British Thoracic Society ([Thorax 2018;74:1](#), usefully [summarised by CKS](#)), and secondly, on [antibiotic prescribing from NICE](#).

The condition affects around 1 in 200 people, is more common with increasing age but is also recognised to affect children, particularly related to untreated protracted bacterial bronchitis or after pneumonia. It may be under-diagnosed, especially in people with existing airways disease such as COPD (data suggests up to 1/4 have a co-morbid disease).

KISS: Bronchiectasis

[BTS 2018](#), [CKS 2022](#), [NICE NG117 2018](#)

Suspect bronchiectasis in adults

- **Persistent productive cough** especially if accompanied by:
 - Large volumes of sputum daily, breathlessness, haemoptysis, chest pain, Pseudomonas, young age, smoking history or its absence, COPD with frequent exacerbations.
 - Clinically there may be coarse crackles in the lower lungs, wheeze, rhonchi, and clubbing.

Suspect bronchiectasis in children

- A **chronic moist/productive cough**, especially between viral colds or +ve sputum cultures for Staph. aureus, H. influenzae, Pseudomonas, non-TB mycoplasma or Burkholderia cepacia.
- Also non-responding asthma, severe pneumonia without full resolution, recurrent pneumonia, pertussis-like illness not resolving in <6/12, persistent & unexplained changes on exam or CXR, localised chronic bronchial obstruction, unexplained haemoptysis, exertional breathlessness.
 - An examination may show persistent inspiratory crackles, wheeze, clubbing, chest hyperinflation, cyanosis.

Differential diagnosis - asthma, COPD, pneumonia, lung cancer, interstitial lung disease, TB.

Suspected bronchiectasis

- **Refer all to a respiratory consultant** to confirmed diagnosis and appropriate treatment.
- Meanwhile arrange sputum culture, CXR and spirometry, bloods, other Ix if you feel appropriate.
- **HRCT (high-resolution CT) chest** is the investigation of choice in secondary care.
- Other tests may include CF if <40yo, screening for gross antibody deficiency, immunoglobulins (for allergic aspergillosis), ciliary function, bronchoscopy, endoscopy if possible GORD.

Long Term Management

- Offer annual review in primary care (high risk patients and children in secondary care):
 - **Smoking cessation**; **exacerbation frequency** (if ≥3/yr ?consider referral for prophylactic antibiotics as per [BTS 2020](#)); **breathlessness** (consider pulmonary rehab); sputum (send for MC+S yearly); **baseline O2 sats**; ?**spirometry** if available. Annual **CXR** is *not* required.
- All patients should know **airway clearance techniques** (should have been referred to physio via respiratory service), have **immunisation** (Strep pneumoniae and annual influenza).
- **Regular specialist follow-up** is indicated by: declining lung function, ≥3 exacerbation/yr, long-term antibiotic requirement, co-morbid RA/IBD/immune deficiency/primary ciliary dyskinesia/allergic broncho-pulmonary aspergillosis, advanced disease.
- For patient info, the British Lung Foundation has [good information](#)



KISS: Acute Exacerbation of Bronchiectasis

[BTS 2018](#), [CKS 2022](#), [NICE NG117 2018](#)

Indicated by:

- Acute deterioration of normal symptoms (usually over several days):
 - Worsening cough.
 - Increased sputum volume, viscosity or purulence.
 - Increased breathlessness.
- Also may have new symptoms, e.g. fever, haemoptysis, pleurisy.

Treatment

- Obtain a sputum sample for MC+S prior to initiating antibiotics (but don't await results).
- Offer an antibiotic for people with an acute exacerbation
 - Consider severity, previous exacerbation/admission history, risk of complications and previous culture results.
 - Advise on possible side effects and safety net - review if deteriorates.
- When sputum results are available review the choice of antibiotic BUT only change if the bacteria are resistant AND the infection is not improving.
- Reassess people if symptoms worsen rapidly or significantly
 - Consider DDx, e.g. pneumonia, sepsis, antibiotic resistance.
- Refer to the hospital if features of more serious illness.
- Seek advice if failure to improve despite repeated courses of antibiotics, if the infection is resistant to oral antibiotics or if the patient cannot take oral antibiotics.
- Do not routinely offer ICS unless on that for another indication (e.g. asthma or COPD).

Antibiotic Choice

- 1st line (adults)
 - Amoxicillin 500mg TDS 7-14 days, or if penicillin allergic:
 - Doxycycline 200mg D1, then 100mg OD for 7-14 day course or
 - Clarithromycin 500mg BD 7-14 days
 - Course length depends on the severity, previous history, etc. - use clinical judgement.
- If high risk of treatment failure (e.g. repeated courses, resistant bacteria, risk of complications):
 - Co-amoxiclav 500/125 TDS 7-14 days.
 - Levofloxacin 500mg once or twice daily 7-14 days.
- If on long-term antibiotics, stop temporarily and prescribe antibiotic from a different class.
- 1st line (children): Amoxicillin, clarithromycin or doxycycline (if age >12) 7-14 days - [click for dosage](#)

Prophylactic Antibiotics

- Are not routinely recommended but new [BTS guidance 2020](#) recommends consideration of a trial of prophylactic macrolides (e.g. azithromycin 500mg 3x per week) for **people with ≥3 exacerbations per year under specialist supervision**.
- Prophylactic antibiotics can reduce exacerbation rates for those with high exacerbation rates ≥3x per year, however before commencing **careful assessment of potential benefits/risks is needed**.

NB Quality Improvement Ideas: Bronchiectasis

- Audit your practice population for patients with bronchiectasis
 - Have they had an annual review? If not could a system be put in place to invite them?
- Perform a case review - is a sputum culture always sent before starting antibiotics?

DYSPEPSIA AND GORD

How many times have you prescribed treatment to your patients with dyspepsia or reflux only for you and them to be disappointed by the lack of improvement? In fact, a lack of response to treatment is by far the most likely outcome, which suggests a change in approach might be required.

Several new papers have published recently in this area. Firstly, an important update from the [British Society of Gastroenterology which completely changes our approach to managing dyspepsia](#). Secondly, a [BJGP paper on the holistic management of reflux](#), highlighting the overuse of PPIs. Thirdly, from Canada and the only one internationally, an [evidence-based guide to deprescribing PPIs](#).

1 Management of Functional Dyspepsia, BSG Guidelines [Gut 2022](#)

Two decades of research has revolutionised our understanding of the causes of dyspepsia and necessitated an entirely new approach. **Underpinning this new guideline is the fact that 80% of patients with dyspepsia will have functional dyspepsia. In this group, investigations are likely to be fruitless, anti-secretory medications are less likely to be effective, and symptoms may continue long-term.** While mortality is not increased but morbidity certainly is, and quality of life can be impaired.

► *What is functional dyspepsia?*

Functional dyspepsia (FD) consists of the classical features of dyspepsia such as pain or burning in the epigastrium, early satiety, and postprandial fullness, which have been present for at least 8 weeks, and where no abnormality can be identified on investigation. It may also be associated with belching, nausea and bloating. Heartburn is a separate issue but may co-exist, while vomiting is considered atypical.

The lack of identified cause does not mean that there is no underlying cause and FD is considered a disorder of the brain-gut interaction. There are a number of associations linked to FD. Not all patients will have all (or many) of these associations and how they interact is complex and poorly understood. They include abnormalities in motility, including delayed gastric emptying and impaired fundic accommodation, visceral sensitivity to both physical and chemical stimuli, central nervous system processing, psychopathology, immune function, changes in the gastric and small bowel microbiome, epithelial permeability, and genetics.

The BSG have adopted the ROME IV criteria for functional dyspepsia, which has two clinical subsets (see KISS), but recommends diagnosis can be made clinically after 8 weeks rather than the original 6 months.

► *When and what investigations are recommended?*

Traditionally FD has been considered a diagnosis of exclusion implying that at least 80% of patients with dyspepsia will necessitate an endoscopy. In recent years there had been a move away from using long-term anti-secretory medications without a formal diagnosis by endoscopy. This guideline suggests that endoscopy may only be required in certain situations, e.g. alarm symptoms (as per NICE) or treatment-resistant dyspepsia in older people. See KISS for full guidance. It also reminds us of features suggesting pancreatic cancer and biliary disease, which may require CT or USS of the abdomen.

The guideline highlights data demonstrating the most cost-effective strategy for initial assessment of dyspepsia is to test and treat for *Helicobacter pylori* (either with stool or breath test), responsible for an estimated 5% of all community dyspepsia cases, and recommends this for all patients. Limited tests are recommended for certain specific groups, e.g. >55yos should have a full blood count (to exclude raised platelets which would be a red flag), and if there is overlap with IBS symptoms check coeliac serology. After this, it suggests we can consider the need for further limited investigation, based on our clinical judgement.

In this situation the guideline encourages clinicians to "explain the diagnosis of FD is still likely" and where investigations are negative or not indicated suggests we "make a positive diagnosis of FD", explaining the condition in terms of the brain-gut axis.

This is about managing expectations, and the guideline interestingly highlights that empathy and active listening are key skills, and that "patient education and engagement is key to successful management".

KISS: ROME IV Classification of Functional Dyspepsia

Functional Dyspepsia (FD)

- One or more of the following:
 - Bothersome postprandial fullness
 - Bothersome early satiation
 - Bothersome epigastric pain and/or burning
- Plus no evidence of structural disease (including at upper endoscopy) that is likely to explain the Sx

Postprandial Distress Syndrome (PDS) - subset 1

- Bothersome postprandial fullness (i.e., severe enough to impact on usual activities)
- Bothersome early satiation (i.e., severe enough to prevent finishing a regular-size meal)

Epigastric Pain Syndrome (EPS) - subset 2

- Bothersome epigastric pain (i.e., severe enough to impact on usual activities)
- Bothersome epigastric burning (i.e., severe enough to impact on usual activities)

For both subsets, there should be no evidence of organic, systemic, or metabolic disease that is likely to explain the symptoms on routine investigations (including at upper endoscopy).

Supporting features: postprandial epigastric pain or burning, epigastric bloating, excessive belching, and nausea can also be present; vomiting may suggest an alternative disorder; heartburn is not a dyspeptic symptom but may often co-exist; Sx that are relieved by evacuation of feces or gas generally should not be considered as part of dyspepsia; other digestive symptoms (such as from GERD and IBS) may co-exist with EPS and PDS.

► *Does this mean there's no role for medication?*

No, quite the opposite, but first BSG recommend lifestyle advice including encouraging regular aerobic exercise and avoidance of dietary triggers.

Empirical acid suppression is recommended initially if HP eradication is not required. For the 20% with underlying disease driving the dyspepsia, this is likely to resolve their symptoms. Otherwise, data suggests that even in functional dyspepsia both PPIs or H2RAs are "an efficacious treatment" (relative risk 0.86-0.89). There does not appear to be a dose-response so the lowest effective dose is appropriate.

Prokinetics may also be effective but data is largely around medications unavailable in the UK (acotiamide, itopride, tegaserod) and efficacy varies dependent on the drug class.

Second-line therapy is focused on medications which influence the gut-brain interactions, such as low-dose tricyclics, also antipsychotics including sulpride or levosulpride, pregabalin, tandospirone, and possibly mirtazapine, but all these latter options are the remit of secondary care. There is no evidence that SSRIs or SNRIs are efficacious for FD.

Psychological therapies may be beneficial and data supports the use of interpersonal psychodynamic-informed psychotherapy, CBT, stress management and even hypnotherapy.

► *Is there any role for specialists?*

Yes, and the guideline supports a broader range of healthcare professionals to be involved in the management of patients with severe or refractory FD through multidisciplinary support teams, which may include specialists, dieticians and psychologists.

Clearly, in many areas, these services currently do not exist to deliver this, but the guideline will act as a framework to encourage service development and help to support us in managing these patients.

KISS: Management of Functional Dyspepsia [BSG 2022](#)

Assessment

- **FD-focused history and exam, assess for alarm features**
 - Alarm features for upper gastro-oesophageal indicating the need for 2WW referral:
 - Dysphagia (any age), ≥55 with weight loss and dyspepsia, upper abdo pain, reflux, or
 - Dyspepsia & >40 from an area at ↑ risk of gastric cancer or FHx of gastro-oesophageal cancer
 - Non-urgent endoscopy should be considered if:
 - ≥55 with treatment-resistant dyspepsia, or dyspepsia & raised platelets or nausea/vomiting
 - Consider urgent CT if abdo pain & weight loss & ≥60
 - Consider USS if epigastric pain <1y & characteristics of biliary colic
- ? reflux Sx in isolation (heartburn or regurgitation) = not FD
- ? epigastric pain related to change in bowel habit or relieved by defaecation = likely IBS
- ? medications, e.g. NSAIDs / opioids
- **Basic investigations**
 - **H. pylori test** - either breath or stool, serology is not recommended
 - FBC in ≥55y, coeliac serology if overlap of IBS-type Sx

Management

- **Make a +ve diagnosis of FD**, explain the condition in the context of the gut-brain axis
- **Discuss treatment options** - drugs and psychological - consider previous treatments & preferences
- **Manage expectations** and agree on a follow-up plan
 - Explain there is no cure, treatments aim to improve QoL, likely necessary in the long term
- **Lifestyle advice** including exercise and avoidance of simple dietary triggers
- **1st line treatments**
 - Trial of acid suppression, PPI or H2RA, at lowest effective dose
 - Trial of prokinetics (but recommended options unavailable in the UK)
- **2nd line treatments**
 - Tricyclic antidepressant, e.g. amitriptyline 10mg, titrating to 30-50mg if required and effective
- **Psychological therapies**
 - Refer for CBT, hypnotherapy, stress management, psychotherapy if patient is amenable & available
- **Referral**
 - Consider referral if treatment failure to discuss other options
 - Consider referral for severe or refractory FD

2 Holistic management of symptomatic reflux [BJGP 2022;72\(724\):541](#)

The close relation to dyspepsia (at least anatomically) is, of course, gastro-oesophageal reflux. This interesting paper in the BJGP predominantly discusses the overuse of PPI for its management, reasons behind it and strategies to reduce PPI use/prescribing.

It traces the route of the issue back to 2006, when international consensus resulted in the **Montreal Definition of GORD** stating that it is 'a condition that develops when the reflux of stomach contents causes troublesome symptoms and/or complications'.

A definition without requirement for investigation opened the door to long-term prescribing of acid-suppression therapy. Medics also commonly attribute extra-oesophageal conditions (e.g. laryngeal symptoms, recurrent AOM) to reflux. The result is lots of PPI prescribing, often prolonged - data suggests of patients prescribed a PPI almost 40% continue with the treatment for over 1 year, and around 1 in 3 cases of reflux are refractory to treatment. In fact, a 2022 paper in [Gut journal](#) found that 70% of people with heartburn in the community have "non-erosive reflux disease", a condition that remains poorly understood, and around half of which is due to functional heartburn.

► *Is there functional heartburn?*

In uncomplicated GORD a 4-8 week course of PPI is usually sufficient to resolve symptoms. If symptoms fail to improve this needs to be explored further; ongoing treatment is unlikely to be the answer. While the BJGP doesn't discuss this cause, functional heartburn is a possibility. The ROME IV classification defines this as:

- *Burning retrosternal discomfort or pain*
- *No symptom relief despite optimal antisecretory therapy*
- *Absence of evidence that gastroesophageal reflux or eosinophilic esophagitis is the cause of the symptom*
- *Absence of major esophageal motor disorders*
- *Present for at least the last 3 months twice a week or more, with initial symptoms for >6 months*

► *Is prolonged PPI use harmful?*

In recent years most clinicians have become concerned about the long-term effects of PPIs on patients. PPIs have been **associated** with increased risk of renal disease, pneumonia, electrolyte imbalance (particularly hypomagnesaemia), nutritional deficiency (e.g. low B12), GI infections, gastric fundal polyps and more. The paper does point out **"evidence of any association is weak and rigorous evaluation is lacking"**. Nevertheless common principles of prescribing apply here as they do elsewhere - only prescribe a drug if it is treating an actual condition or actually treating a condition, and always use the lowest effective dose.

► *What factors contribute to prolonged PPI use?*

Patient expectation, HCP wish to help control symptoms, time pressure, not being the initial prescriber (especially common from secondary care where no indication of duration is included), and misunderstanding amongst patients and HCPs about how PPIs work and when they may be protective are all important drivers.

Rebound symptoms have been commonly reported after PPI withdrawal typically 4-8 weeks after cessation although the frequency is poorly characterised in the data. There may be an assumption that the symptoms are due to the original underlying condition and so patients or clinicians may restart treatment, entering in to a vicious cycle of PPI use.

► *Discontinuation*

Some patients may question why we recommend changing a treatment they have been taking either for an extended duration or without apparent side effects, but a surprising number are keen to reduce their pill burden. Research suggests **around half of patients have concerns about long-term PPI use and a third actively want to stop** suggesting potential for change where safe.

[Canada produced the first evidence-based guideline on deprescribing PPIs](#) and the BJGP expands this further. See KISS on the next page for details.

Two UK primary care-based studies ([BJGP Open](#), [Qual Prim Care](#)) have demonstrated that there are effective strategies to overcoming rebound symptoms and successfully deprescribing PPIs. Nurse-led programmes encouraging self-management of rebound symptoms using rescue OTC alginate therapy resulted in 75-83% of patients stepping down or off their long-term PPI, with resulting financial savings on prescribing budgets.

KISS: Deprescribing PPIs

[Canadian Fam Phy 2017;63\(5\):354](#), [BJGP 2022;72\(724\):541](#)

- Deprescribe PPIs in adults who have completed a minimum of 4 weeks of PPI treatment for heartburn or mild to moderate gastroesophageal reflux disease or esophagitis, and whose symptoms are resolved.
- Deprescribing options include:
 - **Reduced dose:** data suggests no increased risk of symptom relapse on lower doses
 - **Stop PPI, or using “on-demand” dosing:** continue daily until symptoms resolve then stop +/- on-demand doses - only 1 in 10 will have symptoms relapse
- **The recommendations do not apply to high-risk patients**, e.g. Barrett esophagus, severe esophagitis grade C or D, or documented history of bleeding gastrointestinal ulcers.

The BJGP also recommends:

- **Emphasising holistic treatment instead of focusing on anti-secretory medications**
 - Includes behaviour change, and setting boundaries on PPI duration and framing them as temporary treatment for symptomatic reflux allowing time to address underlying drivers.
- **Educational programmes for patients** (and clinicians)
 - Data shows these are highly effective for driving change in this situation
 - Nurse-led programmes in UK general practice on deprescribing incorporating step-down guidance and rescue meds for rebound symptoms show 75-83% of patients can step down or off PPI with significant cost savings.
- **Active role in encouraging self-care and self-directed OTC use of antacids + alginates** (or even PPIs) to manage rebound or intermittent symptoms.

NB Learning Points: Dyspepsia & Reflux

- Functional dyspepsia and heartburn are common, and explain why many patients fail to respond to anti-secretory therapy.
- New guidance from the British Society of Gastroenterology on managing functional dyspepsia recommends limited investigations such as endoscopy unless patients have alarm symptoms or are in high-risk groups for gastro-oesophageal cancer
- Key management includes an empathic approach and moderating expectations
- Anti-secretory medication may still be effective, as can brain-gut neuro-modulators and psychological therapies
- Long-term PPI prescribing is common and guidance recommends low-risk groups should undergo deprescribing by stepping down or off medication following education and support.

KISS: GORD and Dyspepsia

[NICE CG 184, 2019](#)**Diagnosis**

- Keep an open mind and consider DDx – could this be a cardiac or biliary disease?
- Consider medications – NSAIDs, steroids, CCBs, nitrates, theophyllines, bisphosphonates

Referral

- Patients with dyspepsia and significant acute GI bleedings should be immediately referred (...!)
- Refer if red flags are present (see KISS and latest NICE guidelines)
- **Consider referral for people:**
 - *Of any age with gastro-oesophageal symptoms that are non-responsive to treatment or unexplained*
 - *With suspected GORD who are thinking about surgery*
 - Laparoscopic fundoplication may be considered for people with confirmed acid reflux and either adequate symptom control but do not want long-term suppression therapy or where it has responded to a PPI but cannot tolerate long-term treatment
 - With HP that has not responded to second-line eradication therapy
 - With GORD to diagnose Barrett's (do not offer routinely) based on patient preference and risk factors such as long duration/increased frequency of symptoms, previous oesophagitis/ hiatus hernia/stricture/ulcers or male gender

Common elements of care

- **Simple lifestyle advice** such as healthy eating, weight loss, smoking cessation
- **Avoid triggers** the patient has identified, e.g. smoking, alcohol, coffee, fatty foods
- Encourage those on long-term treatment to reduce their use of medication stepwise where appropriate (long-term suppression or gastro-protection may be important for some patients):
 - Use the lowest effective dose, try PRN, return to self-treatment with antacid and/or alginate therapy as able

Uninvestigated dyspepsia and reflux symptoms – primary care management

- Offer empirical full-dose PPI for 4 weeks OR HP “test and treat” for people with dyspepsia/reflux
 - Test and treat were associated with reduced dyspeptic symptoms after 1yr compared to empirical PPI. NNT = 7
 - The HPA suggest T&T as the preferred initial strategy in higher-prevalence populations
 - Leave a 2wk washout period after PPI use before HP testing for stool antigen or breath test as it suppresses levels of HP and makes the tests less accurate
 - There was no strong evidence showing one PPI to be more effective than another
- If symptoms persist after initial strategies encourage patients to step down to the lowest effective dose of PPI either regularly or on an “as required” basis, and advise it may be appropriate to return to self-treatment with an anti-acid or alginate therapy
- If an inadequate response to PPI offer H2 receptor antagonist to control symptoms
 - Note [MHRA withdrawal of ranitidine products in UK](#) due to finding raised levels of NDMA, a possible human carcinogen. [Alternative H2RAs](#) include cimetidine, nizatidine & famotidine.

KISS: GORD and Dyspepsia continued

[NICE CG 184, 2019](#)

Testing and Eradication of *Helicobacter Pylori* ([NICE](#), [HPA](#))

- **Test using urea breath test or stool antigen test > serology**
 - Serology not recommended by HPA for the elderly, children or post-treatment
 - If re-testing use a urea breath test
 - Home breath test kit for patients prescribable on FP10, approx. £20
- **First-line eradication therapy: 7 days of BD treatment**
 - Full dose PPI + amoxicillin 1g + clarithromycin 500mg or metronidazole 400mg
 - Take into account previous exposure to clarithromycin and metronidazole
 - If penicillin allergic offer PPI + clarithromycin + metronidazole
 - If previous use of clarithromycin substitute with bismuth and tetracycline
 - *Emphasise the importance of adherence to successful eradication*
- **Second-line eradication therapy: 7 days of BD treatment**
 - As per 1st line eradication switching in whichever antibiotic not previously used
 - If previous exposure to clarithromycin and metronidazole use:
 - PPI + metronidazole + tetracycline (or quinolone if tetracycline cannot be used - warn re: risks, [MHRA](#))
 - If penicillin allergic and no previous quinolone exposure use:
 - PPI + metronidazole + levofloxacin
 - If penicillin allergic and previous quinolone exposure use:
 - PPI + bismuth + metronidazole + tetracycline
- If eradication not successful with second-line treatment discuss with a gastroenterologist

IRRITABLE BOWEL SYNDROME

One of the biggest advances in the management of IBS has been the introduction of faecal calprotectin testing to exclude inflammatory bowel disease (and potentially other serious GI disease) - NICE DG11 recommends:

- FC is recommended as an option to support clinicians with the DDx of IBD or IBS in adults with recent onset lower GI symptoms for whom specialist assessment is being considered if cancer is not suspected ... and locally agreed care pathways are in place...and in children who have been referred for specialist assessment.

The other major advance has been the discovery that **dietary modification through reduction in FODMAP intake** (fermentable oligosaccharides, disaccharides, monosaccharides and polyols - short-chain carbohydrates) **provides significant benefit**. But given this approach has generally been dietitian-led via tertiary care, it has meant a significant limitation given the extent of IBS in the general population. Is there a way it could be applied more widely in primary care and how does it compare to drug therapy?

► **Diet or medication in primary care patients with IBS: the DOMINO study** [Gut 2022](#)

A [recent study published in GUT journal](#) randomised 459 primary care IBS patients to either a low FODMAP diet, guided via a smartphone app, or oral antispasmodic medication for 8 weeks and compared their IBS Symptom Severity Score before and afterwards, including at 24 weeks. The custom-made app did not advise a strict low FODMAP diet rather a FODMAP-lowering diet in combination with dietary recommendations from the NICE/BDA guidelines for IBS, with recipe guides and other interactive tools. The results showed:

- 71% in the diet group vs 61% in the antispasmodic group were classed as responders
 - Even by 4 weeks improvements were seen: 62% diet vs 51% AS
- The use of a smartphone app appeared very effective in this group (mean age 41, 76% female, 70% ROME IV +ve) with 94% adherence compared with 73% of those taking medication.

The results lead the authors to conclude that a FODMAP-lowering diet delivered via smartphone app should be considered the 1st line treatment for IBS in primary care. We can also conclude that the antispasmodic otilonium bromide is surprisingly effective (sadly not available in the UK, but perhaps alternatives could also prove beneficial) where this approach fails or is unacceptable.

NB Learning Points: IBS

- Faecal calprotectin is a useful diagnostic tool to exclude IBD in patients with suspected IBS, leading to fewer secondary care referrals and unnecessary colonoscopies - also, consider using the new [Never Too Young tool](#) to assess the risk of IBD/cancer ([risk tables here](#))
- A 2 step approach to dietary management in people with IBS can be very effective
- NICE now recommend a low FODMAP diet when general lifestyle and dietary methods have failed – data shows that smartphone apps are an effective technique to improve symptoms over 24 weeks with a high level of user acceptability and adherence.

KISS: Irritable Bowel Syndrome

[NICE 2015 CG61](#), [BMJ 2012;345:e5836](#), [BDA guideline](#), [NB Blogspot](#)

DIAGNOSIS

Consider in anyone with a history of ABC:

- Abdominal pain eased by defecation or with altered stool form or frequency ***and at least 2 of:***
- Bloating, distension, tension or hardness
- Change in stool passage – straining, urgency, incomplete evacuation
- Symptoms worse with eating
- Passage of mucous

Exclude red flags

- Unexplained or unintentional weight loss
- Rectal bleeding
- Family history of bowel or ovarian cancer
- In patients over 60 years old; change in bowel habit to looser/more frequent stools for >6 weeks
- Anaemia, masses (including rectal ie. PR recommended), ↑ inflam. markers (?IBD)
- Check CA125 in women >50yo with new symptoms

Make a positive diagnosis of IBS based on symptoms and confirmatory tests:

- Check FBC, CRP, coeliac serology.
- Check faecal calprotectin to exclude IBD if any doubt in diagnosis and considering referral
- No need for scoping, imaging, hydrogen breath tests, stool MC+S, FOB or TFTs
- **Identify subtype**
- Constipation predominant (IBS-C), diarrhoea predominant (IBS-D) or mixed (IBS-M)

TREATMENT

Offer general dietary and lifestyle advice

- Use a 2 step dietary approach – [click here](#) for NB's Blogspot patient leaflet on 'IBS and Diet'
 - Milk/lactose intolerant?
 - Step 1: Give the NICE/BDA advice for patients - [NICE](#) / [BDA Patient Leaflet](#)
 - Step 2: Trial a low FODMAP diet
 - Add linseeds if IBS-C, consider 4/52 probiotics trial (NNT=4)
- Consider referral to a dietician for elimination or empirical diet for 2-4 weeks
- Exercise – ↑ physical activity beneficial; evidence supports use in those with IBS assoc. with fibromyalgia/CFS

Drug treatments – based on subtype and severity of symptoms, encourage pt to titrate to symptoms

- **First-line:** symptoms-based single or combination medication
 - Antispasmodics (e.g. mebeverine/Buscopan, NNT=~5)) or peppermint oil (NNT=2.5)
 - Laxatives, e.g. isphagula/Fybogel if IBS-C +/- osmotic laxative e.g. Movicol
 - Only consider linaclotide when other laxatives have failed and >12/12 constipation
 - Loperamide if IBS-D

KISS: Irritable Bowel Syndrome continued

[NICE 2015 CG61](#), [BMJ 2012;345:e5836](#), [BDA guideline](#), [NB Blogspot](#)

• Second line

- Consider low-dose TCA nocte (start 5-10mg, max. 30mg amitriptyline) if 1st line ineffective
- Consider SSRI only if TCA ineffective
- Follow up these patients at 4 weeks, then every 6-12 months if continuing treatments

• Psychological therapies (e.g. CBT) - consider if refractory symptoms after 12 months of treatment

Follow up – agree with patient based on response to symptoms, at least annual review, reassess red flags

KISS: Dietary Management of IBS

We have produced this simple 2 step dietary plan on NB Medical's Hot Topics Blogspot - [click here for a patient information leaflet on 'IBS and Diet'](#)

▪ Step One

- Consider if this could be food allergy or intolerance
 - Lactose intolerance is fairly common; true allergies are fairly rare
- Follow the dietary advice from the British Dietetic Association and NICE such as:
 - Reduce intake of rich and fatty food
 - Reduced intake of manufactured foods – these often contain resistant starches
 - Reduce intake of some high-fibre foods (e.g. legumes) and eat soluble sources of fibre (e.g. linseeds)
 - Try lactose-free dairy products
- Keep a food/symptom diary, look to identify triggers and if no significant improvement after 4 weeks then move to Step Two
- [Food fact sheet](#)

▪ Step Two

- Consider trial daily supplementation with linseeds for IBS-C
- Consider a trial of probiotics
 - Trial for 4 weeks, if not helpful consider a different brand; NNT = 4
- For all IBS types advise reducing FODMAPs (ideally dietician-led as per the study protocols)
 - Research into low FODMAP diets for treating IBS has demonstrated a **significant symptomatic response** which ceases on re-introduction of a 'normal' FODMAP diet - up to **86% of patients with IBS achieve relief of overall GI symptoms** (compared to around 50% in comparator groups receiving standard dietary advice) – particularly bloating, flatus and abdo pain
 - For an exhaustive list of FODMAPs [click here](#)

If this is still not successful then consider alternative elimination or empirical diets, but by this stage, dietician supervision is definitely recommended.

CHRONIC DIARRHOEA

Diarrhoea is a common problem for many of our patients, but for around 4-5% can be especially protracted with a very real impact on the quality of life. While irritable bowel syndrome is the cause in many of our patients there is a much broader differential to consider as two recent papers demonstrate.

First, a new paper in the BJGP ([2021;71:42](#)) discusses *microscopic colitis and a useful diagnostic pathway for primary care*. Second, updated *guidance from the British Society of Gastroenterology on the assessment of chronic diarrhoea*, also recently reviewed in the BJGP ([2019;69:262](#)) reminding us of other causes such as bile acid malabsorption.

1 Microscopic Colitis

In early 2021 the BJGP ([2021;71:42](#)) published this fascinating article on microscopic colitis, a surprisingly common condition in older people. This coincides with [European guidelines](#) on its management in 2020. Most of us will have never made this diagnosis but will have seen it frequently reported, sometimes to our surprise, on our patients' colonoscopy reports.

► What is MC?

Microscopic colitis is a type of inflammatory bowel disease. It is often considered 'milder' than traditional Crohn's and UC because its primary symptom is 'only' chronic watery non-bloody diarrhoea, however, it is often associated with frequency (average BO 6-7x daily), urgency (55%), nocturnal stools (35%) and faecal incontinence (26%) so it can have a big impact on patient's lives. As the BJGP paper notes, it can be "severe enough to make patients effectively housebound".

For 1 in 3 the onset is acute (and may be mistaken for gastroenteritis), and 4 in 10 have diarrhoea for 6 months before diagnosis. Abdominal pain, bloating and weight loss have also been reported.

Extra-GI manifestations rarely occur as the disease appears confined to the bowel mucosa, but [fatigue has been reported](#) to affect up to 50%, probably due to a combination of inflammation and poor sleep, loss of appetite, etc...

The good news for patients with MC is it is not associated with increased risk of GI malignancy and surveillance programmes are not required.

► Who gets MC and how common is it really?

MC is typically a condition of older middle age, with the median age at diagnosis 60 years old.

Almost three-quarters of cases are in females, and it is more common in current and ex-smokers. MC is also associated with auto-immune conditions, such as coeliac disease (co-exists in 3-4%), and bile acid malabsorption (in ~40%).

It has been associated with a number of medications: PPIs, NSAIDs, SSRIs has been implicated where use is prolonged, although this is an association and causation has not been confirmed.

Incidence is 0.12% and it accounts for 12% of patients with unexplained chronic, watery diarrhoea.

► How is it diagnosed?

It is not possible to clinically diagnose MC, a biopsy of the colon is necessary. On colonoscopy the macroscopic appearance is normal but histology reveals inflammation in one of three patterns:

- *Lymphocytic colitis* - infiltration of lymphocytes in the intestinal wall
- *Collagenous colitis* - increase in collagen intestinal wall, also with some white blood cells
- *Microscopic colitis incomplete* - one or both of the above, but fails to meet full histologic criteria

MC is found on biopsy in almost 5% of colonoscopies which otherwise appear normal.

► *So what's the role of primary care?*

Clearly, we are not performing the colonoscopy, but before patients get to that stage we need to think about the broader differential: assess for red flags, consider iatrogenic causes, undertake basic investigations and faecal calprotectin, and refer when required. See our KISS below for a full pathway.

Faecal calprotectin remains a somewhat contentious issue - many of us have limited access, and even when we do there is often a cut-off above 40 years rather than the 60s suggested in the BJGP paper.

[NICE DG11](#), which recommends the use of FC in patients where there is diagnostic uncertainty between IBD and IBS, does not specify an age cut-off, but in a linked paper recommending a practical pathway for the use of FC in primary care it notes that there is utility in the test up to the age of 60y, after which the sensitivity and specificity deteriorate.

Despite being a variant of IBD, MC is not usually associated with a rise in faecal calprotectin (and if it is it is usually mild). This is potentially where some patients get missed - normal bloods and FC would suggest IBS as a likely cause, but, as we discuss in the management of chronic diarrhoea in section 2, ongoing symptoms indicate the need for further investigation. This is especially true in older patients with new symptoms.

► *How do we treat MC?*

Oral budesonide is the mainstay of treatment for the majority:

- **Initial treatment: 6-8 weeks of oral budesonide 9mg daily**
 - Response rates of 81-84% compared to 36-43% on placebo and may induce long-term remission.
- **Persisting symptoms: usually oral budesonide 3-6mg daily +/- loperamide**
 - Oral budesonide has a local effect within the GI tract and shows minimal systemic absorption so lacks the most troubling adverse effects of pred/dex/Hc with longer use
 - **Prednisolone and other corticosteroids are not recommended** - data shows them to be ineffective
 - **Mesalazine is also ineffective.**
 - Loperamide is an option for patients who fail to respond, are intolerant of or simply do not want to use budesonide.
 - Cholestyramine may be helpful, presumably in those with co-existing bile acid malabsorption
 - Those with ongoing troubling symptoms despite these measures may be considered for DMARDs: azathioprine and a range of biological agents have been shown to be effective in refractory disease.

► *What are the indications for referral?*

Any red flags need to be managed in the usual way, as do patients in whom initial investigations show an alternative cause for diarrhoea.

The faecal calprotectin result will then guide on referral. Moderate raises (100-250) indicate routine gastroenterology referral, higher levels (>250) indicate more severe disease and urgent referral is recommended.

KISS: Investigating Chronic Diarrhoea[BJGP 2021;71:41](#)*Sx of IBD or IBS, age 18-60y, with diagnostic uncertainty*

- **Red flags?** Refer via urgent cancer pathway or urgent IBD referral.
- **No red flags?** Perform initial screening investigations:
 - FBC, CRP, coeliac screen, stool MC&S, E+C, bone profile, TFTs (+ consider CA125 in females)
- Negative?
 - **Check faecal calprotectin**
 - FC <100: IBD unlikely (IBS likely Dx in 98%) , manage symptoms locally
 - *Symptoms persist?*
 - If FC <50 & age <50y: consider 2nd line IBS treatment before referral
 - If FC ≥50 or age ≥50y routine referral to gastro
 - FC 100-250: repeat within 2 weeks
 - FC <100 follow pathway above
 - FC 100-250 routine referral to gastro
 - FC >250 at any point: indicates severe disease - clinical review & urgent referral to gastro

2 Investigation of chronic diarrhoea [BJGP 2019;69:262](#), [BSG 2018](#)

The British Society of Gastroenterology have updated their guidance on assessing chronic diarrhoea, reviewed in the BJGP. The first fundamental change to the guideline is the very definition on diarrhoea - previously this had focused rather impractically on stool weight and frequency (how many of us have ever measured how heavy our poo is?), now there is a broader description which is much more clinically useful:

- *Chronic diarrhoea = the persistent alteration from the norm with stool consistency between types 5 and 7 on the Bristol stool chart AND increased frequency greater than 4 weeks duration*

► What are the causes of chronic diarrhoea?

- **Common**
 - **Diarrhoea-predominant IBS**
 - **Bile acid diarrhoea**
 - 1/3 with IBS-D actually have BAD and are misdiagnosed
 - There is a significant overlap of symptoms, but stools may be pale, greasy, hard to flush, sometimes green or orange-coloured. Bowel movements may be erratic, flatus are very pungent and cramps severe.
 - The SeHCAT nuclear medicine test is the investigation of choice, and only one really available of any merit.
 - **Diet-related**
 - FODMAP malabsorption, lactase deficiency, sweeteners, caffeine, excess alcohol or liquorice
 - **Colonic neoplasia**
 - **Inflammatory bowel disease**
 - Microscopic colitis accounts for 10% of IBS-D (mis)diagnoses - unlike UC there is no blood (the mucosa does not ulcerate) so chronic, watery diarrhoea is the predominant feature; linked to the use of a number of drugs, including NSAIDs, PPIs and SSRIs. Diagnosis is made on biopsy at colonoscopy.
 - **Coeliac disease**
 - **Medication**
 - Antibiotics, NSAIDs, Mg-containing drugs, anti-diabetics (metformin, gliptins), furosemide, chemotherapy
 - **Overflow diarrhoea**

■ **Uncommon**

- Such as small bowel bacterial overgrowth, mesenteric ischaemia, lymphoma, abdo surgery, pancreas disorders, hyperthyroidism, diabetes, giardiasis (and other chronic infections), cystic fibrosis...

► ***How can we assess people with chronic diarrhoea?***

The aim here is to distinguish possible organic GI disease from functional disease. For some this can be done exclusively in primary care, however, the BSG guideline acknowledges that for many (perhaps the majority) some type of lower scope will be required to complete an assessment.

Our role, therefore, is predominantly two-fold: 1) rule out serious causes through clinical assessment for red flags, and 2) gather data to guide us on whether referral is necessary and, if so, in what manner and urgency. There is a clear emphasis on using new technologies to aid us in general practice, specifically faecal calprotectin (FCP) and faecal immunochemical testing (FIT).

There is a noticeable discrepancy in the guidance regarding how to manage patients after these tests:

- *"In patients with typical symptoms of functional bowel disease, normal physical examination and normal screening blood and faecal tests, a positive diagnosis of IBS can be made."*
- However, it also states:
- *"In patients with chronic diarrhoea, we recommend colonoscopy... as the preferred investigation... In younger patients (<40yo) with normal FCP and in whom functional bowel disease is suspected, we recommend flexible sigmoidoscopy."*

This was clarified with the statement that "younger patients [are] recommended secondary care evaluation for patients with atypical, persisting, or severe symptoms of functional bowel disease even if the FCP is normal." The implication is a direct referral to the clinic for these patients rather than getting lower GI investigations in primary care first.

Using this information the authors of the BJGP paper produced an algorithm to aid assessment, which is very similar to the one above in the Microscopic Colitis section, so we recommend using the KISS above for both.

NB Learning Points & Quality Improvement Ideas

- Chronic diarrhoea affects 4-5% of the UK population
- Assessment involves excluding red flags for serious lower GI disease, then targeted blood and stool tests
- There is a strong role for faecal calprotectin and faecal immunochemical testing in this group
- The majority of patients will be recommended to have some lower GI endoscopy
- In younger patients, with normal tests and classic sx, we can make a +ve diagnosis of IBS
- Microscopic colitis is a variant of IBD causing chronic, watery, non-bloody diarrhoea. It can be managed very successfully with oral budesonide. Long-term remission may be induced by just 6-8 weeks of treatment

BOWEL OBSTRUCTION

Diagnosis of bowel obstruction is easy: colicky abdominal pain, vomiting, abdominal distension and absolute constipation. But as this very helpful *BMJ Practice Easily Missed paper* ([BMJ 2021;374:n1765](#)) explains patients often don't present with complete bowel obstruction - it may take time to evolve with, be subacute or chronic and then the diagnosis is considerably more challenging with diagnostic overshadowing a common issue.

► *Why is it easily missed?*

Complete bowel obstruction with classic features is usually a straightforward diagnosis. However, presentation can be very different with subacute or chronic obstruction. This can often mimic more common conditions we see in primary care (e.g. vomiting due to gastroenteritis, pain due to faecal constipation) or cause symptoms we don't associate with obstruction (e.g. watery diarrhoea). The presentation is also often protracted, subtle and patients not markedly unwell, with a lack of derangement in physiology and bloods.

Unfortunately, we all know the problems with a missed diagnosis. This is true with chronic or subacute obstruction as well, as delay may prevent easy treatment of the underlying cause (e.g. reducing a threatening hernia) or result in emergency surgery, which carries a significantly worse prognosis than elective surgery (e.g. laparotomy mortality is 15-19% v <5%).

► *How can it present?*

High obstruction (e.g. small bowel) usually presents with early vomiting and colicky abdo pain, and distension may be absent.

In contrast, low obstruction (e.g. colonic) has distension as a prominent feature, abdo pain is usually present but it may be less severe or frequent than expected (esp. in elderly) and vomiting may be absent/occur late.

Constipation will eventually be a feature of both, but confusingly diarrhoea can occur in the early stages of high and low obstructive pathology.

KISS: Presentation of Bowel Obstruction

- **Acute**
 - Small or large bowel, often adhesions (40% of all bowel obstruction) or obstructed hernia (12%)
 - Sudden onset severe colicky abdo pain, early profuse vomiting (high obstruction) or early abdo distension (low obstruction)
- **Acute on chronic**
 - Often partial colonic obstruction that then completes
 - Cancer (15%), diverticular stricture (unknown incidence), faecal impaction (8%)
 - Presents as acute obstruction but often the history reveals a chronic obstructive picture
- **Subacute small bowel obstruction**
 - High or low in small bowel, incomplete
 - May present over several days, symptoms (pain/vomiting) may be intermittent, less severe, constipation not absolute and distension not that marked
- **Chronic (large bowel) obstruction**
 - Initially incomplete, increases slowly; often colorectal cancer or diverticular stricture
 - Symptoms can evolve over months (esp. in the elderly), key features may be mild & intermittent or unnoticed. Weight loss and malnutrition can occur.

► *How do we confirm the diagnosis?*

Back to basics - keep an open mind and consider the possibility to combat possible diagnostic overshadowing. The paper makes a few useful clinical pointers:

- Subacute bowel obstruction diagnosis is largely clinical, but there are many 'classic features' which do not help us much either by their presence or absence.
- Colicky pain, vomiting and a history of abdominal or pelvic surgery shouts obstruction. The first 3 years post-surgery are especially high risk.
- Colicky abdo pain and ongoing vomiting (especially >24h) is unusual in infective gastroenteritis. GE is more likely with short-term vomiting as the predominant feature.
- The absence of abdominal distension does not exclude obstruction - it is mostly absent with higher obstructions and early low small bowel obstruction. It is also often not obvious in obese patients.
- Abnormal bowels sounds ('tinkling') has poor sensitivity and specificity and even specialists have 'minimal agreement' with each other when assessing.
- Peritonitis occurs late, after perforation or infarction, so its absence has no diagnostic value.
- Stool in the rectum does not exclude obstruction, especially higher ones.
- Remember to examine the hernial orifices - small irreducible hernias are easily missed, especially femoral hernia.

So if we suspect it we can confirm it with an abdo XR....?

Unfortunately not. The BMJ reports both sensitivity and specificity are low.

The best diagnostic tool is CT abdomen which is highly accurate at confirming obstruction and identifying the underlying cause at the same time. If only we could order those from general practice...

Clearly where we are suspicious of subacute bowel obstruction on-the-day referral to the local surgical team is most appropriate.

NB Learning Points: Subacute Bowel Obstruction

- Subacute bowel obstruction may present with subtle, mild and/or infrequent symptoms
- There is overlap with many common illnesses seen in general practice but persisting vomiting, lack of diarrhoea and previous abdominal surgery should prompt consideration of obstruction.
- The absence of classic signs such as tinkling bowel sounds, peritonitis and an empty rectum does not exclude obstruction.
- CT abdo is the test of choice - referral to the on-call surgical team will be required if the diagnosis is suspected

INFLAMMATORY BOWEL DISEASE

The prevalence of inflammatory bowel disease (IBD) is rising globally, including in the UK, with around a 0.3% prevalence in Western countries. This increase is especially true of children, with a recent BMJ evidence review ([BMJ 2017;357:j2083](#)) reporting a 76% increase in diagnosis in <16s in the past 15 years. The cause for this is not well understood but data shows an inverse correlation with environmental hygiene and a growing feeling that it may be linked to the "hygiene hypothesis" for allergy.

All the more reason for good, clear, update to date guidance based on the best data available. To that end in 2019 NICE updated its guidance on both ulcerative colitis ([NICE 2019, NG130](#)) and Crohn's disease ([NICE 2019, NG129](#)). Both guidelines were reviewed and summarised in the BMJ ([2019;367:15897](#) and [2019;367:15940](#) respectively).

Beyond NICE, there are several hot topics in IBD.

- In recent years *biological therapies* have provided a new treatment branch for patients in whom convenient DMARD therapy failed. NICE has very recently updated both its [Crohn's Disease](#) and [Ulcerative Colitis](#) guidelines, which are very much secondary care documents. It provides significant updates on the role of biological therapy, which is very much now an early "step-up" treatment when conventional drugs have failed to control the conditions.
- *Early and tight control*, just as we see in inflammatory arthritis conditions, leads to improvement in clinical outcomes ([Lancet 2017;390:2779](#))
- *Faecal microbiota transplantation* is an area of growing interest, with promising non-drug results ([JAMA 2019;321\(2\):156](#), [Lancet 2017;389:1218](#)) albeit in short-term trials.
- *Faecal calprotectin* has, of course, become commonplace in general practice following the 2013 recommendations from NICE on use to distinguish IBS and IBD. Nevertheless, the debate continues into how best to implement these recommendations and what is the optimal cut-off (given it is a quantitative test) to maximum sensitivity, specificity and cost-effectiveness in the NHS.

There is no doubt that much of IBD management falls to secondary care, but given this is a lifelong condition, and that 50% of patients have a flare each year and we are the first avenue of contact in the health service, there are important messages for primary care.

First is about how we can optimise treatment for a flare, with a new BJGP ([2018;68\(675\):497](#)) article about how management is not just about corticosteroids.

► **Using Corticosteroids Appropriately in IBD** [BJGP2018;68:497](#)

This useful article sheds light on the management of flares of both UC and CD. There is no doubt that oral corticosteroids can be extremely useful but careful prescribing is important for preventing side effects in the short term (increased risk of infection) and the longer term. Nevertheless, it is not uncommon for prednisolone to be initiated as the de facto treatment for flares prior to a patient being followed up in the IBD clinic. Up to 40% of IBD patients have excess steroid exposure.

There are alternative options to prednisolone - in UC mesalazine is significantly more effective, in both oral and topical preparations, and for Crohn's oral budesonide may be used in some cases to minimise systemic absorption.

The paper set out some simple recommendations which can guide our management of flares in general practice. It also points out that the management of this group is very multidisciplinary, and communication with the IBD team via IBD nurses is helpful and important. Flares requiring oral steroids more than once per year are an indication that stepping up maintenance therapy is required.

NB Learning Points: Inflammatory Bowel Disease

- Rates of IBD are on the rise, the hygiene hypothesis is the primary consideration at present
- Faecal calprotectin is helpful in distinguishing between IBD and IBS
- Up to 40% of patients with IBD have excess corticosteroid exposure
- In ulcerative colitis flares increase oral mesalazine to the maximum dose of 4-4.8g OD
- Topical mesalazine is more effective than topical steroids in distal/left-sided UC flares
- Flare management may need to be continued for up to 3 months
- If prescribing oral steroids for flares, start at 40mg prednisolone day, reducing by 5mg weekly
- Avoid prescribing more steroids than required for flare management (an 8-week course is 252 tablets of pred) - patients often stop early and keep supplies for future flares without informing us or the IBD team about the frequency of flares
- Crohn's usually needs oral steroids to manage, or more involved treatments such as anti-TNF agents or surgery - have a low threshold for referring early for rapid clinic review
- Communication with the IBD team is essential so they can track the number and severity of flares a patient has - this will guide maintenance treatment
- The RCGP has produced an excellent [IBD Toolkit](#) which provides background information, handy tips, treatments and doses, monitoring guidance and patient resources
- For patients, [Crohn's & Colitis UK](#) provides resources and local support information

NB Quality Improvement Ideas: Inflammatory Bowel Disease

- Audit patients with IBD for anaemia
 - If their Hb is <12 have they had iron studies to confirm IDA?
 - Has an assessment of alternative cause been made?
 - Has the patient been advised on diet and iron supplements to increase their Hb?
- Perform a case review of patients with IBD
 - Are patients on hospital-prescribed treatments?
 - Has this been documented in the primary care notes? This may be important information for other clinicians in the event of acute illness.
- Review prescribing for patients with a flare in the past 12 months
 - Have they required oral steroids? Were they prescribed the 8-week course of 252 tablets?
 - Has the flare been communicated with the local IBD team?

KISS: Management of IBD Flares

[BJGP2018;68\(675\):497](#), [RCGP IBD Toolkit](#), [NICE CD 2019](#), [NICE UC 2019](#)

- Acutely ill patients should be referred immediately (high fever, tachycardia, hypotension, etc.)
- **Communicate with the local IBD team** that the patient is having a flare
- **Ideally obtain “objective evidence of disease activity”**, e.g. CRP, faecal calprotectin, but don't delay treatment unless it is mild
- Mild > moderate UC or CD flares can have treatment initiated in primary care
 - Check for a documented plan for the patient from the IBD team
- **The recommended treatment depends on the site of disease**
- **Ulcerative colitis** - is this distal/left-sided disease or more extensive UC?
 - **Increase (or initiate) oral mesalazine to 4-4.8g once daily for all**
 - Once-daily dosing is as effective as multiple-daily dosing and more convenient
 - **Distal/left-sided disease**
 - **Topical mesalazine is most effective**
 - Mesalazine works through a local action on the gut mucosa (what little is absorbed into the body is quickly broken down and of no benefit), therefore topical treatment works better than oral
 - **Suppositories for rectal disease, otherwise enemas are required**
 - **Topical corticosteroids are less effective**, so not advised first-line but may be considered if there is a poor response to mesalazine
 - Ask the patient if they have responded to topical steroids in the past - in non-responders, it would be more appropriate to step up to oral treatment
 - **Extensive disease**
 - **Topical treatments may speed recovery** but by themselves are inadequate and should be used in **conjunction with increased oral mesalazine** dosing
 - **Flares may take 2-3months to control**
 - **If no response in 2-4 weeks start oral corticosteroids**
 - 40mg prednisolone daily, reduce by 5mg per week, 8-week total course (252 tabs)
 - Shorter courses have higher relapse rates; an initial dose of $\leq 15\text{mg/day}$ is ineffective
 - Co-prescribe vitamin D and calcium supplements for longer courses; consider bisphosphonate if >65y or established osteopenia/osteoporosis
- **Crohn's Disease**
 - Mesalazine lacks proven efficacy although NICE recommends it as an alternative in patients who do not want/cannot tolerate oral steroids
 - **Mild > moderate ileo-caecal disease**
 - **Budesonide 9mg once daily for 8 weeks**, with a 2-week taper
 - Budesonide has extensive 1st pass metabolism so is associated with significantly fewer classic steroid side effects
 - **Mild > moderate extensive/colonic disease**
 - **Oral prednisolone**
 - **Severe disease (bowels open >6x/day)**
 - **Consider admission**, oral prednisolone, admit if a poor response by 1 week
 - In poor responders, those requiring >1 course of pred/year or if more extensive disease increasing early treatment with anti-TNF or immunotherapies are considered

GISS: Maintenance of Remission of IBD

[RCGP IBD Toolkit](#), [NICE CD 2019](#), [NICE UC 2019](#), European Guidelines 2016/17 [CD/UC](#)

Ulcerative Colitis

- Step-wise treatment if control is poor
 - Mesalazine: oral ($\geq 2\text{g/day}$ OD), topical (3g/wk in divided doses) or combination
 - No sig. the difference was seen in efficacy or side effects between oral 5ASA options
 - Azathioprine or mercaptopurine ('thiopurines')
 - Consider ciclosporin or tacrolimus
 - Anti-TNF (e.g. infliximab, adalimumab) or immunotherapy (e.g. vedolizumab)
 - These may be used in combination with thiopurines
- Colectomy remains an option for severe or resistant disease

Crohn's Disease

- Step-wise treatment if control is poor
 - Thiopurines or methotrexate
 - Anti-TNF or immunotherapy

Monitoring

- **Medications**
 - **Aminosalicylates**
 - Mesalazine U&E annually; sulfasalazine FBC + LFT 3 monthly
 - Interstitial nephritis is a rare complication
 - **Immunosuppressants** (e.g. azathioprine, methotrexate)
 - Once stable FBC + LFTs 3 monthly, U&E 6 monthly (MXT 3/12)
 - ↑ risk of infections, lymphoma, non-melanoma skin Ca & cervical dysplasia
 - **Biologic agents** (e.g. infliximab, etc.)
 - Usually done in secondary care
 - FBC, CRP, U&E, LFTs 3-6 months, also lipid profile & hep B status periodically
 - ↑ risk of infections, melanoma, a lymphoproliferative disease in combination with thiopurines (although along with immunosuppressant risks, rates are low)
- **Bone health**
 - Risk increased due to malabsorption of both vit D and calcium
 - Check BMD if major risk factor (e.g. fragility fracture) or oral steroids for >3/12
- **Colorectal cancer** - see [NICE Pathways](#) for risks
 - Risk is 2% at 10 years, 8% at 20y, 18* at 30y
 - Surveillance should begin at 10 years after diagnosis then every 3-5 years
 - The more extensive or active the disease, the higher the risk - some merit annual scope
 - Consider CRC in younger patients with unusual symptoms for them
- **Iron Deficiency Anaemia**
 - 1 in 3 patients with IBD with have a Hb <12 due to a triad of malabsorption, blood loss and inflammation
 - Ferritin of <100µg/l may indicate a low deficiency in the context of inflammation
 - Consider other causes (e.g. CRC, menstruation, coeliac, etc.)
 - Dietary intake can be encouraged, iron supps if necessary

COELIAC DISEASE

Coeliac disease affects 1% of the population, but it is estimated that only 1 in 5 are currently diagnosed. For a lifelong condition with important complications, patients with a diagnosis may have surprisingly little long-term input. A new NICE guideline looks at improving diagnosis and long-term management.

NICE NG20: Coeliac Disease [NICE NG20 September 2015](#)

There are several key messages in this updated guideline:

1. ***We should offer serological testing in a wide range of situations.*** Patients may be considered 'asymptomatic' but after positive tests, it may be evident that subtle features present.

- a. First degree relatives – all should be tested as 10% will also have coeliac disease
- b. We should offer to test to patients presenting with:
 - i. Unexplained and persistent GI symptoms, faltering growth, prolonged fatigue, unexpected weight loss, severe or persistent mouth ulcers, unexplained iron/B12/folate deficiency, type 1 diabetes and autoimmune thyroid disease (both at diagnosis), suspected irritable bowel syndrome
 - ii. Also, consider in patients with the metabolic bone disorder (e.g. osteoporosis), unexplained neurological symptoms, subfertility or recurrent miscarriage, persistently raised LFTs of unknown cause, dental enamel defects, Down's syndrome, Turner syndrome

2. ***Patients should have an annual review.*** This may be undertaken by a range of health care professionals, with GP or specialist review if there are issues raised. 5-30% fail to have symptomatic improvement despite a gluten-free diet, of whom 10% have true refractory disease. Do not use serology alone to assess for gluten exclusion as titres may remain high for some even in strict dietary adherence.

- a. Measurement of height and weight
- b. Review of symptoms
- c. Consider assessment of diet and adherence to the gluten-free diet
- d. Consider the need for specialist dietetic and nutritional advice

3. ***In people with persistent symptoms despite gluten exclusion consider:***

- a. Review the certainty of the original diagnosis
- b. Referral to a specialist dietician to investigate gluten exposure
- c. Investigate potential complications or coexisting conditions that may be the cause
 - i. e.g. IBS, lactose intolerance, bacterial overgrowth, microscopic or inflammatory colitis

4. ***A healthcare professional with specialist knowledge in coeliac disease should tell people with a confirmed diagnosis about the importance of a gluten-free diet.*** This important part of management should not be underestimated and good information and support at diagnosis leads to better longer-term adherence to diet and follow-up as well as the quality of life.

KISS: Coeliac Disease

[NICE NG20 2015, BMJ 2014;348:g1561](#)

Presentation

- **GI symptoms**
 - Non-specific pain, bloating, IBS-like symptoms, diarrhoea (rarely malabsorption Sx)
- **Non-GI symptoms (atypical)**
 - Fatigue, weight loss, arthralgia, myalgia, depression, itchy rash/dermatitis herpetiformis, aphthous ulcers

Diagnosis

- **Offer serological testing to patients with any of the following:**
 - Persistent and unexplained GI symptoms
 - Prolonged fatigue
 - Unexpected weight loss, or in children failure to thrive/grow
 - Unexplained iron, vitamin B12 or folate deficiency
 - Autoimmune disease e.g. thyroid, type 1 diabetes
 - 1st degree relative with coeliac
 - Irritable bowel syndrome
 - Consider also with osteomalacia or osteoporosis, unexplained neuro Sx (esp. peripheral neuropathy and ataxia), unexplained subfertility or recurrent miscarriage, abnormal LFTs, dental enamel defects, Down's/Turner syndrome, aphthous stomatitis, and alopecia, etc.
- Patients must have a normal diet with gluten in >1 meal/day for 6 weeks before testing
- **1st line test is IgA tissue transglutaminase (tTG) plus total IgA**
 - Sensitivity and specificity in patients with normal IgA is 98%+
 - 2% of patients with coeliac have selective IgA deficiency which causes a false negative
 - If the tTG is weakly positive an endomysial antibody (EMA) should be performed
- **Referral all adults and children with positive serology to gastroenterology**
 - All adults will have endoscopy and biopsy to confirm; children may be scoped
- **If serology is negative but coeliac remains clinically suspected refer - 9% are seronegative**

Annual review – performed by a health care professional (but not necessarily a GP)

- **Measure height and weight**
- **Review symptoms**
 - If ongoing check adherence to a gluten-free diet
 - **If poor, consider referral to a dietician for advice and support**
 - **If good, consider if the diagnosis is correct, refer back to gastroenterology**
 - Continuing symptoms or ongoing high serology titres require re-biopsy

KISS: Coeliac Disease continued

[NICE NG20 2015](#), [BMJ 2014;348:g1561](#)

- Where there are concerns in the annual review patients should see their GP or consultant
 - Consider the need for:
 - DEXA scan – 32-80% have an abnormal bone mineral density
 - Scan at diagnosis, the onset of menopause or 55y in men, or if fragility fracture
 - Specific blood tests
 - Anaemia and micronutrient deficiencies are common but may be prevented by adherence to a gluten-free diet
 - Consider testing for Hb, vit B12, folate and serology annually
 - Assessment of long-term complications and comorbidities
 - Risk of hyposplenism – vaccinate for Strep pneumonia, men C, HiB and flu
 - Lymphoma – 2-4x risk vs general population but absolute risk increase is very small (incidence 1 in 2000/yr); risk possibly reduced by avoidance of gluten

NB Learning Points: Coeliac Disease

- Coeliac disease remains underdiagnosed – offer testing to first-degree relatives and those with autoimmune disease
- Audit of patients in your practice with autoimmune conditions
 - Have they had coeliac serology?
- There are several very useful resources available on gluten-free diets:
 - Patient information - www.coeliac.org.uk
 - Gluten-free recipes - www.bbc.co.uk/food/diets/gluten_free
 - For a more personal view, recipes and restaurant reviews – www.thehappycoeliac.com

EOSINOPHILIC OESOPHAGITIS

When is reflux not reflux? Every now and again one of those conditions comes along which has been under the radar for a surprisingly long time. Eosinophilic oesophagitis (EoE - a US acronym...) is one such condition - affecting up to 1 in 100 in Western populations, it initially presents in a similar way to classic GORD but the cause is very different, as is treatment. An excellent BMJ paper ([2017;359:j4482](#)) reviews our understanding of the causes, diagnosis and treatment of this fascinating disease.

► *What is it?*

Only discovered as a clinical entity around 20 years ago, eosinophilic oesophagitis is a chronic allergen-driven immune-mediated disease.

It affects both adults and children and the **majority of patients have a comorbid allergic disease** (up to 80%). However, it is **not caused by a classic IgE-mediated response but is driven by T helper type 2 cells**. These release a range of cytokines in response to trigger allergens leading to eosinophilic recruitment in the oesophagus (normally devoid of such cells) and subsequent mast cell activation. Histamine and transforming growth factor β cause chronic inflammation, fibrosis and reduced contractility.

Prevalence varies with the population - in Western countries, it is as high as 0.4-1%, in Japan, it is estimated to affect only 0.01%. Caucasians appear more susceptible, as are males (3-4x as frequently as females), typically in their 20-30s. A recent meta-analysis suggests that prevalence is higher in adults compared to children (~4-3 ratio). This prevalence is on the rise, for reasons not understood and not simply due to better recognition - review of historical barium swallows fails to show the same rate of concentric oesophageal rings which are a hallmark of the condition. One US study showed a rise of almost 30x in 15 years.

Risk factors include genetic predisposition (high concordance in twin studies), use of antibiotics at an early age, and dysbiosis of the gut microbiome (certain bacteria may inhibit the induction of Th2 cells). Interestingly H pylori infection appears protective, for reasons poorly understood.

► *How does it present?*

This varies with the age of onset:

- **Adults:** heartburn, chest pain, dysphagia, food impactions
- **Children:** difficulty feeding, vomiting, abdominal pain, failure to thrive

The key to diagnosis is in ***considering the possibility of EoE, particularly in younger men/boys with a history of atopic disease and persistent heartburn and/or difficulty swallowing***. It further highlights the importance of endoscopy in those with persistent reflux.

For diagnosis, endoscopy + biopsy showing characteristic histopathology, followed by an 8-week trial of PPI and re-endoscopy to identify persisting eosinophils. Endoscopy will also often reveal rings or strictures of varying severity caused by the associated chronic inflammation, explaining the dysphagia.

If the eosinophilia resolves with PPI therapy it is either due to GORD or “PPI-responsive oesophageal eosinophilia” (PPI-REE). At least 1 in 3 respond to PPIs. The mechanisms remain unclear, but may be acid or immune-mediated, and appear to be distinct phenotypes.

If the eosinophilia fails to resolve with PPI it is classed as true “eosinophilic oesophagitis”.

► *How is it treated?*

There are 3 main treatment options for patients with EoE, all considered first-line options:

- **Proton pump inhibitors** - useful for PPI-responsive disease
 - Effective by definition - >80% are successfully maintained on 'low-dose' PPI
 - Adverse effects are mild, but long-term data is lacking
- **Diet modification** - may be the preferred long-term option
 - A surprisingly effective tool. 2 types of use: elemental or food elimination diets
 - An **elemental diet**, i.e. ready-digested liquid food of amino acids, resolves symptoms in >90%. Cost, palatability and the misery of never eating normal food limits use.
 - The **6 food elimination diet** (SFED) removes food groups commonly associated with allergy: cow's milk, wheat, egg, soy, peanut/tree nut and fish/seafood. Remission is seen in >70%. Given the restrictive diet, compliance is understandably difficult.
 - The **4 food elimination diet** (FFED) uses the same principle but patients are allowed nuts and fish/seafood. It is effective in 54%. Non-responders may achieve remission if stepping up to the SFED, resulting in similar overall efficacy to an SFED first approach.
 - The **Step-Up approach** starts with just milk and wheat exclusion, then the FFED followed by the SFED if unsuccessful. Remission is seen in 40% initially and allows more nuanced differentiation of food triggers. Once identified avoidance is recommended to maintain drug-free disease remission.
 - **Why not simply test for allergies first?** Unfortunately, this strategy is poorly effective as standard allergy tests fail to correctly identify food triggers in this type of allergy.
- **Topical Corticosteroids**
 - No, not swallowing Betnovate cream... But is swallowing 'inhaled' steroids (e.g. fluticasone) or aqueous nebuliser solutions (e.g. budesonide, mixed with a sugary syrup) as used in asthma, thus coating the oesophagus. This is preferred to oral 'tablet' steroids as the long-term side effect profile is presumed to be better, although in truth there is no data on long-term topical treatment. None of the treatments are licensed for EoE.
 - An effective treatment - 58-82% have histological remission, although symptoms scores are the same as placebo treatment, probably due to a high placebo response.
 - Oral candidiasis is a potential issue - seen in up to 30% on endoscopy, although

EoE is a chronic disease. Recurrence is common on cessation of treatment. To date, no cases are known to have progressed to malignancy. However, progressive fibrosis can be a problem.

Oesophageal dilatation may be required for more significant rings and strictures. It is the only way to provide immediate and durable relief for dysphagia. Severe complications, such as an oesophageal tear, are rare at <1%. Retrosternal pain was seen in ~3/4 post-procedure but should improve.

NB Learning Points and Quality Improvement Ideas: Eosinophilic Oesophagitis

- Eosinophilic oesophagitis affects up to 1% and should be considered in patients with persistent or PPI-resistant reflux or food impaction, especially in younger males with a Hx of allergy
- Endoscopy and biopsy are required for diagnosis.
- Treatments with PPI, diet modification or topical steroids are very effective.
- Perform a case review of patients with reflux in the practice. How many have a recurrent disease and associated atopy? Have they had an endoscopy to confirm the cause?

BARRETT'S OESOPHAGUS

Oesophageal cancer is on the rise: the incidence is now 8 times greater than 30 years ago, with >9000 new cases each year in the UK. Over the same period, a shift in the type of cancer has been seen with 70% now adenocarcinoma rather than the squamous cell. ([BJGP2014;64:120](#)).

Barrett's oesophagus is a known premalignant condition which progresses in a stepwise fashion to oesophageal adenocarcinoma (OAC), with acid exposure due to reflux the main driver. Around 1 million people in the UK have Barrett's, and the associated lifetime risk of OAC risk is around 7-12%.

Unfortunately, even with regular surveillance, there are only modest improvements in cancer outcomes. PPIs are routinely recommended for patients with known Barrett's but the most effective regimen for cancer prevention is unknown and this is largely for symptomatic control.

Observational data has shown that PPIs may reduce OAC risk with Barrett's. It also shows long-term aspirin use may reduce the risk of adenocarcinoma. But there have been mixed findings in data, and neither has been tested in an RCT specifically in Barrett's patients and certainly not in combination.

1 **Aspirin Esomeprazole Chemoprevention Trial (AspECT) for Barrett's** [Lancet 2018](#)

The AspECT study is the randomised control trial which may help answer the questions around PPIs and/or aspirin for people with Barrett's. Conducted in the UK and Canada, published in The [Lancet in 2018](#), it provides the best evidence to date on the prevention of OAC in this group.

- **Patients with Barrett's >1cm length** (new or existing diagnosis) **were randomised to:**
 - **High or low dose esomeprazole** (40mg BD vs 20mg OD); **without or without aspirin** (300mg UK or 325mg Canada) on a 1:1:1:1 basis
- **Endoscopy was performed every 2 years**, follow-up > 8 years
- **Exclusions:** NSAID use at baseline, OAC, high-grade dysplasia
- **Primary outcome:** composite of all-cause mortality, OAC, or high-grade dysplasia, on an intention-to-treat basis
 - Outcomes were reported using "Time Ratio" - which indicates the difference in time to the outcome between the interventions; a TR >1 implies a beneficial effect
- **Results:**
 - n=2557, follow up median 8.9years, 313 primary events occurred
 - **High-dose PPI was superior to low-dose** (Time Ratio 1.27), NNT = 34
 - **Aspirin was superior to no aspirin if pts did not also use NSAIDs** (TR 1.29), NNT=43
 - **High-dose PPI + aspirin had the strongest effect**
 - TR 1.59 vs low-dose PPI, no aspirin
 - TR 1.38 vs high-dose PPI, no aspirin
 - Significant benefits were not seen by 5 years use, but were by 8.9 years use
 - 1% reported severe adverse effects, 15 related to aspirin, 13 to PPI
 - Haemorrhage was ~2x more common in pts taking aspirin but <1% had a high-grade bleed
- **Conclusion:** **"high-dose PPI and aspirin chemoprevention therapy, especially in combination, significantly and safely improved outcomes in patients with Barrett's oesophagus".**

The [linked editorial](#) acknowledges that "the risk reduction and dose-response relationship for PPI in the AspECT trial are rather convincing" and is likely to outweigh concerns over potential side effects from long-term PPIs in this group.

However, questions remain over aspirin and the most appropriate dose. The 300/325mg dose used in the trial led to a ~1% risk of serious adverse effects, and PPIs will not prevent complications such as ulceration in the small and large intestines.

But the observational data underpinning this study used predominantly 75mg - could a lower dose still prove effective and limit the negative effects? Further trials will be needed to answer this question.

The editorial does not touch on the significant discontinuation rate in the trial: ~35-50% in each of the four groups. This was not simply due to adverse events but the paper does not address what has happened to this group. If they had continued with the treatment we may have seen even greater benefits in the study but understanding why patients needed to stop is clearly going to be extremely important, especially in general practice as we are the long-term prescribers.

2 So where are we with Barrett's? [BMJ 2016;353:i2373](#), [BJGP 2014;64:120](#)

While this new data is scrutinised as to whether it should influence current practice, recent papers in the BMJ and BJGP discuss the current management of Barrett's.

► Should people be screened?

Population screening is not recommended (and would not be feasible!). However, the BMJ suggests endoscopic screening in patients with chronic or severe GORD and at least 3 risk factors (50yo+, Caucasian, male, obese, smoking) or a family history of Barrett's or oesophageal cancer). Interestingly there is no evidence that alcohol is associated with the risk of Barrett's.

► What happens then?

Patients diagnosed with Barrett's may require ongoing surveillance. Those with <3cm of Barrett's and no intestinal metaplasia on two endoscopies may be discharged as the risks are likely to outweigh the benefits. Longer lesions or the presence of metaplasia indicates long-term surveillance every 2 - 5 years depending on the severity.

Any dysplasia will be treated with eradication therapy, e.g. mucosal resection, ablation, cryotherapy, etc.

Data shows reductions in dysplasia and cancer with PPI use, therefore all patients diagnosed with Barrett's are recommended daily PPI therapy.

The Lancet paper clearly demonstrates higher-dose PPI to be more effective, with aspirin providing further gains. Perhaps the next question is that given the frequency and severity of reflux is linked to the risk of developing OAC (once weekly heartburn odds ratio = 8; severe, long-standing heartburn OR=44), should those with severe reflux and no Barrett's have high dose PPI + aspirin...?

NB Learning Points: Barrett's Oesophagus

- Combination of high-dose PPI and aspirin reduces the risk of progression to oesophageal adenocarcinoma in patients with Barrett's
- Current guidelines do not specify the best dose of PPI and may need to be revised
- Ongoing surveillance remains important as the rates of progression to cancer over a lifetime are significant

DIVERTICULAR DISEASE

Diverticular disease is hugely common but easily forgotten. There has been a recent surge of interest in the medical literature as we begin to understand that in acute diverticulitis the underlying pathology is often chronic inflammation and not infection, questioning the role of antibiotics.

But this is a spectrum condition and the majority of our patients will not present with the most extreme complications, so how can we best manage this large group? What should patients eat? (*"Can I never have my seeded loaf again?"*) How should we investigate? (*"I think I need a scan..."*) How should we manage diverticular pain? (*"Just a few more naproxen, doctor..."*)

A new NICE guideline ([NG 147, Nov 2019](#)), recent review of treatment in Inflammatory Intestinal Diseases ([2018;3:75–79](#)) and updates in [BMJ Best Practice](#) provide us with useful strategies for management.

Background to Diverticular Disease

[NICE NG147 2019](#), [CKS 2022](#)

Definition

1. **Diverticulosis**
 - Diverticula are present but the patient is asymptomatic
2. **Diverticular disease**
 - Diverticula cause symptoms such as abdominal pain and/or tenderness without systemic symptoms
3. **Acute diverticulitis**
 - Sudden inflammation or infection of diverticula causing marked, constant abdominal pain, predominantly in the left iliac fossa
 - Commonly associated with systemic features such as fever, malaise; occasionally rectal bleeding
 - 'Complicated' acute diverticulitis refers to a diverticular-associated abscess, fistula, sepsis, stricture perforation, etc. - access for suggestive features

Epidemiology

- The cause of diverticula remains uncertain but may be linked to raised intra-colonic pressure, e.g. low-fibre diets, obesity. There may be a genetic component as well.
- There is some uncertainty about the true prevalence but it is thought around 85% of people will develop diverticula in their sigmoid colon during their life. 15% have right-sided diverticula, which is more common in the Asian population.
- 75% of people have asymptomatic diverticulosis; 25% will be symptomatic.
- Age is a major risk factor. Diverticula are uncommon in <40yo, present in about 5-10% of adults >45yo and 80% in >80yo.
- The presence of diverticula confers a 4-25% lifetime risk of acute diverticulitis and 5% risk of complications.

1 Diverticular Disease: NG 147 [NICE Nov 2019](#)

This new guideline is concise and instructive with plenty of primary care emphasis. It acknowledges there is a definite lack of quality data to base recommendations on, so much of this is expert consensus. While not pushing the boundaries of diverticular disease management too far there are some useful key points:

- **Diverticulosis**
 - Requires no specific treatment but we should advise patients to eat a balanced diet with lots of fibre
 - There is no need to avoid seeds, nuts, fruit skins or popcorn (phew!)

- Exercise, weight loss if overweight and smoking cessation can reduce the risk of acute diverticulitis and symptomatic disease.
- **Diverticular disease**
 - Do not offer antibiotics.
 - Avoid NSAIDs and opiates if possible - may increase the risk of bowel perforation.
 - Advise a high-fibre diet for life - benefits may take several weeks to achieve.
- **Acute diverticulitis**
 - If systemically well *consider* a no-antibiotic strategy
 - Consider admitting patients whose symptoms persist or worsen

2 Antibiotics in uncomplicated acute diverticulitis: to give or not to give? [Inf Int Dis 2018;3:75](#)

That is the question currently of high interest in the research community. The authors of this new review ([Inf Int Dis 2018;3:75](#)) mirror the recommendation from NICE that antibiotics may not always be required for acute diverticulitis and suggest that in healthy patients with a non-toxic presentation (e.g. no fever, normal obs.) antibiotic use could be selective (e.g. considered case-by-case) rather than the default treatment.

What is acute diverticulitis?

This is the crux of the issue. As with much about diverticular disease in general, the pathogenesis of acute diverticulitis is not known for certain. Traditionally it is considered to evolve from obstruction of the neck of a diverticulum, causing bacterial proliferation (usually anaerobic organisms), ischaemia and possible micro-perforation. However, more recently it is proposed that the process is more inflammatory, less infectious. This, of course, changes the rationale for antibiotics.

Does the data support avoiding antibiotics?

Yes. A Swedish RCT N=623 found no statistically significant difference between antibiotic and non-antibiotic groups in outcomes including time to recovery, recurrence and complication rates. A further RCT from the Netherlands N=528 found the same. In both trials patients had been admitted to hospital, had CT-proven left-sided uncomplicated diverticulitis but also may have had fever, raised WCC/CRP, etc.

When are antibiotics recommended then?

The review comments on a lack of evidence for the treatment of groups which might be considered higher risk. Nevertheless, it suggests it is prudent to use antibiotic treatment in immunocompromised patients (including pregnancy, CKD, autoimmune disease, chronic corticosteroid users, and sepsis).

However, for "non-toxic" presentation of acute, uncomplicated diverticulitis antibiotics do not need to be the default option and can be used on a selective basis.

NB Learning Points & Quality Improvement Ideas: Diverticular Disease

- Diverticulosis is asymptomatic and does not require treatment.
- Acute diverticulitis may be inflammatory, not infectious, in origin and data suggests that in acute uncomplicated diverticulitis, antibiotics do not alter outcomes. For higher-risk patients, e.g. immunocompromised, immediate antibiotic use is recommended.
- A high-fibre diet is protective. Avoidance of seeds, fruit with skins, etc. is not required.
- NSAIDs and opioids may increase the risk of perforation and are ideally avoided.
- Quality improvement
 - Review patients with diverticular disease. Have they been given advice about high-fibre diets, weight reduction and smoking cessation?
 - Review patients with acute diverticulitis in the past year. How many have been prescribed antibiotics? Does this seem justified in light of new data and guidance? Could practice be changed?

KISS: Diverticular Disease

[NICE NG 147 2019](#), [CKS 2022](#), [BMJ Best Practice 2018](#) [Inflammatory Intestinal Diseases 2018](#)

■ **Diverticulosis** - asymptomatic

- Requires no specific treatment but we should advise patients to eat a balanced diet with plenty of fibre (wholegrain, fruit, veg) and adequate fluid intake.
 - For a useful patient leaflet on dietary management see [NHS Patient Info](#)
- There is no need to avoid seeds, nuts, popcorn or fruit skins.
- Consider bulk-forming laxatives if constipation.
- Exercise, weight loss if overweight and smoking cessation can reduce the risk of acute diverticulitis and symptomatic disease.

■ **Diverticular disease** - pain, systemically well

• **Investigations**

- Patients do not routinely require referral to secondary care for suspected diverticular disease unless: investigations cannot be organised from primary care, or colitis (IBD) is suspected, or they meet urgent cancer referral criteria.
- While not a formal recommendation the guideline committee suggest that normal practice would be to attempt to confirm the diagnosis with colon/sigmoidoscopy or CT colonography and it may be possible to organise this from the community.

• **Treatment**

- Do not offer antibiotics.
- Avoid NSAIDs and opiates where possible as they may increase the risk of bowel perforation.
- Consider simple analgesia, e.g. paracetamol, and/or antispasmodic, e.g. hyoscine butylbromide.
- Advise a high-fibre diet for life, and that the benefits may take several weeks to achieve.
- Consider bulk-forming laxatives (e.g. Fybogel) if a high-fibre diet cannot be achieved or the patient has persistent constipation or diarrhoea.
- If treatment fails to improve symptoms consider differential diagnosis and further investigation.

• **Acute diverticulitis**

◦ **Diagnosis**

- Largely a clinical diagnosis - suspect if the patient presents with constant abdominal pain, usually severe and localising to LIF (beware of right-sided disease in Asian population with RIF symptoms and signs) plus:
 - Fever, or sudden change in bowel habit and rectal bleeding and/or mucous, or LIF tenderness/ palpable mass/distension with a known history of diverticulosis/itis
- If mild and uncomplicated can be managed in primary care with careful f/up
- Consider checking FBC + CRP (others, as required, depending on the differential diagnoses)
- If managed in primary care re-assess if symptoms persist or worsen and consider referral
- Admit to hospital if suspect complicated acute diverticulitis e.g. suspected abscess, perforation, intestinal obstruction, sepsis or fistula into bladder/vagina

◦ **Management**

- If systemically well:
 - Consider watchful waiting with a 'no antibiotic strategy', simple analgesia, safety net (represent if persisting/worse)
- If systemically unwell, immunocompromised or significant comorbidity offer antibiotics
 - Co-amoxiclav is 1st line oral choice - 625mg TDS 5 days
 - Alternatives cefalexin 500mg BD/TDS plus metronidazole 400mg TDS - all 5 days, or
 - Trimethoprim 200mg BD plus metronidazole 400mg TDS - all 5 days

LIVER DISEASE

As a [BJGP editorial](#) comments: liver disease is the 3rd commonest cause of premature death in the UK and incidence of liver disease has increased by 400% over the past 30 years. So why is it not higher on the agenda?

As is becomingly increasingly clear, the problem is twofold: firstly, a lack of clarity among specialists on how to best assess patients with potential liver disease, and secondly, a lack of access in primary care to the resources required to achieve the various recommendations.

1 Non-alcoholic fatty liver disease [BJGP 2022;72:499](#), [BMJ 2018;362:k2734](#), [BSG 2017](#), [NICE NG49](#)

NAFLD is on the rise. Importantly 0.5% of people with NAFLD will progress on to cirrhosis. While this may sound low, this [new BJGP paper](#) quotes up to 46% of adults have NAFLD. This could mean huge levels of future cirrhosis. However, the authors also point out that this group is at increased risk of all-cause mortality and the "risks of non-hepatic ill health greatly outweigh risks of liver-related morbidity".

This is a metabolic condition, obesity and metabolic syndrome being key drivers. The paper suggests primary care has a crucial role - to identify risk, initiate investigations and advise on modifiable risk factors.

► *Where to start?*

The commonest presentation of NAFLD is an incidental finding of deranged LFTs (although most people with NAFLD will have normal LFTs) or on USS. Screening has been considered but it is unclear how useful this would be given the limited numbers progressing to severe disease and is currently not recommended by NICE. European guidelines advise screening people with metabolic risk factors. The paper also highlights that up to 7% of lean people may have NAFLD.

There is a lack of consensus among guidelines as to the optimum approach to manage patients with deranged LFTs. The BJGP recommends:

- If no metabolic risk factors (or RFs + alcohol excess) follow BSG guidelines (see next section)
- If at risk of NAFLD + deranged LFTs or steatosis on USS conduct non-invasive liver screen +/- USS
 - At risk ≥3 metabolic RFs: impaired fasting glucose or T2DM, hypertriglyceridaemia, ↓ HDL cholesterolaemia, ↓ waist circumference/↑ BMI >30, or hypertension
 - Screen includes LFTs, hep B&C, autoantibodies, immunoglobulins, caeruloplasmin, alpha 1 anti-trypsin, ferritin and transferrin saturation
- If abnormal refer as required
- If normal conduct a liver fibrosis blood test - they recommend initially FIB-4 or NAFLD fibrosis score
 - Low scores merit primary care management with review in 3 years (2y if <18yo)
 - Intermediate scores merit more advanced liver fibrosis assessment with either ELF (enhanced liver fibrosis score blood test) or Fibroscan which will clarify if the person can be managed in GP or needs referral
 - High scores indicate likely significant fibrosis and refer to hepatology is appropriate

► *How can we help manage NAFLD?*

There is no direct treatment for NAFLD so proactive management of metabolic risk factors is recommended.

The authors suggest LFTs <4x the upper limit of normal and in the absence of other clinical concerns, a trial of lifestyle modification and repeat LFTs in 3-6/12 initially would be a pragmatic way to manage these low liver-risk patients. Addressing the metabolic RFs may include sustainable weight loss where possible and consideration of statin therapy (only stopping if LFTs double at 3 months).

2 Managing Abnormal LFTs [BSG 2017](#)

The *latest guideline on management of abnormal liver blood tests from the British Society of Gastroenterology*, published in Gut ([2017;67;1:6](#)), provides an up-to-date strategy for this common situation - see our KISS on the next page. Fibroscans and ELF tests are also part of this guideline, and no doubt represents the future of liver assessment, but we are clearly not there yet, so in the meantime consideration of referral in concerning cases would seem appropriate.

It also has some highly valuable learning points buried within it which are useful for us all to be aware of:

- **20% of all LFTs are abnormal** - of which <10% is explained by existing disease
- **Repeating abnormal LFTs after a month or so rarely helps** - 84% remains abnormal at 1 month, and 75% at 2 years - so BSG advice is that even a first abnormality should trigger investigation as to the aetiology
- **Context is vital** - review of historical results and comorbidities/medications is important, e.g. a raised ALT may be due to a statin and does not require further investigation
- **The extent of the abnormality is not an accurate guide to clinical significance**, indeed liver disease is often seen in the presence of normal LFTs, and one person's normal may be another's significant change
- **The best predictor of NAFLD is an abnormal ALT**, rather than obesity and other metabolic features. Be particularly suspicious if risk factors are present along with abnormal ALT.

NB Learning Points and Quality Improvement Ideas: Liver Disease

- Rates of NAFLD in the UK adult general population have reached almost 50% driven by obesity and T2DM. 0.5% will eventually develop cirrhosis.
- Guidelines consistently recommend a non-invasive liver screen with USS and bloods to assess for alternative liver pathology.
- For the useful and clear BMJ infographic [click here](#)
- The latest BSG guidelines recommends if ALT or AST are persistently abnormal despite other tests proving normal then referral for further investigation is merited
- Perform a case review of patients in the past 3 months with abnormal LFTs
 - Have the LFTs been followed up? If they are persistently raised has a cause been identified? If NAFLD is diagnosed is there a practice protocol for a long-term follow-up?

KISS: Management of Abnormal LFTs and Assessment of NAFLDCombined from [BSG 2018](#) & [BMJ 2018;362:k2734](#)**Consider NAFLD** if abnormal LFTs & alcohol consumption within recommended limits**History and examination**

- Red flags? (e.g. suspected malignancy, jaundice, sepsis)
- Drug-induced cause? (e.g. methotrexate, steroids, amiodarone, tamoxifen)
- Other causes, e.g. infection, nutritional deficiency

Non-invasive liver screen

- Recommended for all - includes USS and bloods
 - Bloods include Hep B&C, autoantibodies, immunoglobulins, caeruloplasmin, alpha 1 anti-trypsin, ferritin and transferrin saturation
 - If positive refer to hepatology
- Consider non-hepatic causes of ↑ ALT e.g. coeliac, thyroid disease, muscle disease/heavy exercise

Dominant alk. phos. abnormality

- GGT normal? Consider vit D deficiency, bone disease, 3rd-trimester pregnancy
- GGT raised? If no acute features (e.g. gallstones) check USS for hepatic steatosis

Dominant ALT/AST abnormality

- Cirrhosis? If persistently ↓ albumin, enlarged spleen, ↓ platelets, ↑ bilirubin, ↑ INR - refer
- Check USS for hepatic steatosis
- Persistently raised ALT/AST is an indication for hepatology referral even in the absence of cause

Dominant bilirubin abnormality - see Gilbert's Disease chapter

- Normal ALT, ALP, INR and albumin?
 - Consider Gilbert's if conjugated bili <10 µmol/L
 - Consider haemolysis if unconjugated bili high and/or anaemia - refer to hepatology
- ALT or ALP raised? Refer to hepatology

NAFLD confirmed - assess the severity

- ELF test if available, if not consider [NAFLD fibrosis score](#) or FIB4 score or Fibroscan
- ELF >10.5 - advanced fibrosis likely - refer to hepatology
- ELF <10.5, repeat every 2-3 years and encourage lifestyle modification & check CVD risk

Suspected alcoholic risk

- ***Take alcohol history and AUDIT-C questionnaire***
- <35u/wk women <50u/wk men and Audit-C <5=lower risk, counsel on alcohol use
- <35u/wk women <50u/wk men and Audit-C ≥5=full Audit questionnaire
 - Audit >19 - higher risk > ELF test or Fibroscan (therefore we will need to refer)
 - Audit 8-19 - brief alcohol intervention, e.g. check GGT, nurse f/up 3/12, alcohol services
 - Audit <8 - lower risk
- ≥35u/wk women, ≥50u/wk men = harmful drinker - manage as per Audit >19

KISS: Abnormal Liver Function Tests in Pregnancy[BMJ 2013;347:f6055](#)

Interpretation of LFTs during pregnancy can be difficult as the normal ranges are different from nonpregnant women and alter with each trimester. Consider testing if:

- **Pruritis - commonest symptom:**
 - 25% of pregnant women will have pruritis (without a rash); in most cases this is benign.
 - Intrahepatic cholestasis of pregnancy needs to be considered (affects 0.7% in the UK)
 - Check LFTs and serum bile acids (latter is predictive of adverse outcomes)
 - Associated with preterm labour, foetal hypoxia and intrauterine death.
 - ALT and bile acids raised, bilirubin usually normal.
 - If present, referral for urgent gynaecology review is required.
- **Other conditions** which may be associated with abnormal LFTs include *pre-eclampsia and HELLP syndrome, acute fatty liver of pregnancy* (rare, very different from normal fatty liver, a medical emergency with mortality of 12-18%), and *hyperemesis gravidarum*.
- **Medications**
 - Certain drugs taken during pregnancy can cause hepatocellular damage (e.g. paracetamol, methyldopa, amoxicillin, co-amoxiclav, some antiretrovirals) or cholestasis (e.g. amoxicillin, flucloxacillin, progestogens, oestrogens, some antipsychotics, PPIs).
 - Affected women are usually asymptomatic and it is an incidental finding.
 - The drug should be withdrawn and an alternative used if required.
- **Testing is also indicated with a history of exposure to infectious contacts or risk factors for bloodborne infections, and in those at high risk of liver disease** (e.g. previously affected pregnancies, an FHx of liver disease in pregnancy, pre-existing liver disease).
- **Refer to gynaecology**

Abnormal LFTs may be due to a non-pregnancy specific disease such as non-alcoholic steatohepatitis (the most common cause in the UK), gallbladder disease, viral hepatitis, primary biliary cirrhosis, primary sclerosing cholangitis, etc. Specialist liver advice should be consulted.

Liver Enzyme	Not pregnant	Pregnant	1st trimester	2nd trimester	3rd trimester
ALT (IU/L)	0–40	–	6–32	6–32	6–32
AST (IU/L)	7–40	–	10–28	11–29	11–30
Bilirubin (µmol/L)	0–17	–	4–16	3–13	3–14
GGT (IU/L)	11–50	–	5–37	5–43	3–41
Alk. phos. (IU/L)	30–130	–	32–100	43–135	133–418
Albumin (g/L)	35–46	28–37	–	–	–

GILBERT'S SYNDROME

How often do we see a raised bilirubin and think: "it must be Gilbert's"? How often do we then look to prove our hypothesis, confirm our suspicions? Rarely in practice. But, as the DTB points out in its very recent, very helpful review of the condition ([DTB 2019;57:27](#)) this may be an error for several reasons.

NB Revision: Gilbert's Syndrome

- A benign hereditary disorder of bilirubin conjugation affecting 2-10% of Western Caucasians
- Causes isolated elevated blood unconjugated bilirubin, in the absence of haemolytic or abnormal LFTs
- Predominantly autosomal recessive due to defective hepatic conjugating enzyme leading to 60-70% reduction in bilirubin clearance. Asian populations are more likely to have an autosomal dominant variant of GS.
- Bilirubin pathway
 - Haem > biliverdin > unconjugated bilirubin > conjugated bilirubin
 - UDP-glucuronosyltransferase conjugates bilirubin in the liver and is affected by GS

► *Why make a diagnosis of GS?*

Perhaps the most important question given that GS is considered harmless with no link to subsequent liver disease and no reduction in life expectancy. However, while episodes of jaundice should be harmless for patients it can clearly be a cause for worry, they may question why a diagnosis has been "missed" if made later in life, and knowing a diagnosis could prevent unnecessary investigation for those with an incidental finding of abnormal bilirubin.

There has also been speculation that linked to the same impaired pathway, certain medications may not be metabolised normally. This has been examined with paracetamol, for which there is a theoretical mechanism by which toxic metabolites may accrue, although there are no known case reports of liver injury and no recommendations to alter use. There IS an increased risk of myopathy with gemfibrozil and some lipophilic statins (such as simvastatin) and this combination (thankfully uncommon) should be avoided in GS. A small number of specialist drugs (certain chemotherapies) may be affected and more toxic in GS, while others (certain antiretrovirals and biological agents) also lead to increased levels of unconjugated bilirubin rises, but what to do about this (if anything) falls outside the scope of general practice.

On the positive side, there may be benefits to having GS, with protective effects for some diseases. Unconjugated bilirubin may provide an anti-oxidant effect, and data suggests a lower incidence of Hodgkin's lymphoma, endometrial cancer and cancer-related mortality vs the general population. Even all-cause mortality is lower, with a lower prevalence of hypertension, T2DM, metabolic syndrome, obesity and CVD, plus possibly prognostic benefits in diabetes with slower progression of proteinuria and neuropathy.

► *What else could cause a raised bilirubin?*

GS is not the only cause of a raised bilirubin. GS is a diagnosis of exclusion and the aim of investigations is to rule out other important conditions.

■ **Causes of unconjugated hyperbilirubinaemia**

- Haemolysis, autoimmune haemolytic anaemia, G6PD deficiency, sickle cell disease, thalassaemia, hereditary spherocytosis, excessive haemoglobin breakdown, prosthetic heart valve
- Genetic disorders (rare) - Crigler- Najjar syndromes type 1 and 2
- Drugs: rifampicin, paracetamol poisoning, alcoholic hepatitis, erythromycin, oestrogen, sulfasalazine/methyldopa can cause haemolytic
- Physiological neonatal jaundice, esp in prep babies, breast-fed babies, thyrotoxicosis

► *How can we diagnose GS?***KISS: Diagnosing Gilbert's Syndrome**

- Consider assessment either due to recurrent episodes of jaundice, usually lasting only a few days precipitated by common stressors or an incidental finding of isolated raised bilirubin
- Exclude clinical features of liver disease other than jaundice
 - The DTB suggests patients should be asymptomatic, features of liver disease (e.g. pain, itch, pale stools, dark urine) are absent (and require Ix if present); however other sources report some patients (see [UpToDate](#) for example) may have mild symptoms during episodes of raised bilirubin such as fatigue, anorexia, abdominal discomfort
 - Precipitants may include dehydration, illness, heavy exercise, periods of fasting, tiredness
 - Mild jaundice may be present, usually just the sclera;
- Undertake blood tests to exclude alternative causes of raised bilirubin
 - LFTs: bilirubin >17 (jaundice occurs >40, >85 rare), other LFTs normal including albumin
 - Request conjugated and total bilirubin levels - former <20% of the total; unconjugated level raised
 - Clotting screen: should be normal
 - Hb, reticulocytes and blood film: should be normal (to exclude haemolysis/spherocytosis)
 - Genetic testing is rarely required but can be undertaken if uncertainty or significant pt concern
 - Provocation testing is no longer recommended due to poor specificity and impracticality
- Consider concomitant disease - e.g. fatty liver, gallstones, spherocytosis can co-exist with GS
 - This is usually apparent on clinical assessment and the blood screen
- Otherwise, make a positive diagnosis of Gilbert's Syndrome, reassure the patient, explain about potential stressors and seeking medical review for prolonged episodes of jaundice or if associated with other symptoms

NB Learning Points & Quality Improvement Ideas

- Gilbert's syndrome is a common but benign condition causing intermittent short episodes of jaundice usually without associated symptoms
- Diagnosis can prevent undue pt concern and avoid unnecessary investigation in alternative medical settings
- Normal LFTs, clotting, Hb, reticulocytes and a blood film with a raised unconjugated bilirubin indicate GS
- For patient information see [patient.info](#) and the [British Liver Trust](#)
- Perform a case review of patients with a diagnosis of Gilbert's syndrome
 - Do any have LFTs except bilirubin which is abnormal? Has this been adequately addressed?

ANAL FISSURE

Defined as a linear tear or ulcer in the anal mucosa, with an 11% lifetime incidence anal fissure can be a literal pain in the arse for patients and doctors alike.

KISS: Anal Fissure [BJGP 2019;69:409](#), [DTB 2013;51\(9\):102](#)

Cause – surprisingly not usually constipation; related to reduced blood supply due to the position of anal vessels, high internal anal sphincter tone and deficient production of nitric oxide

Presentation - typically seen in young and middle-aged adults; pregnancy and post-partum

- a Sharp pain on defaecation (“like broken glass”) followed by hours of burning pain in the anus
- PR bleeding during and after passing stools
- The examination usually reveals a fissure in the posterior midline; anterior is less common
- Often primary cause, but consider underlying pathology, e.g. Crohn’s, HIV, TB, syphilis, neoplasm – often a non-traditional appearance, e.g. non-midline, thickened, large skin tags, fistulae
- Fissures lasting <6 weeks = acute, >6 weeks = chronic

Treatment

- **Conservative treatment:** 87% of acute, 50% of chronic fissures resolve with these alone
 - Encourage patients to adopt measures early to prevent chronicity
 - Increase fluid and fibre intake; stool softeners
 - 1 RCT showed that 15g/day of fibre resulted in a marked reduction in recurrences at 1 year compared to lactulose (16% vs 60%).
 - Simple analgesia for pain, also buscopan/mebeverine can help for post-poo proctalgia; warm baths
 - Haemorrhoid creams lack evidence to support their use
 - Topical anaesthetics (such as lidocaine) limited use, can cause skin reactions
- **Chronic fissure treatment**
 - 2 treatments are available – they seem expensive but so is referral...
 - **Topical Glyceryl Trinitrate 0.4% ointment** - £39 per 30g tube, intra-anally BD for up to 8w
 - 1st line after conservative measures; licensed for pain associated with chronic anal fissure
 - Causes headache in up to 30% of patients, often severe. Settles in a few days
 - A Cochrane review ([2012, CD003431](#)) showed GTN resulted in more healed chronic fissures than placebo – 49% vs 36%; recurrence around 50%
 - GTN patches (applied elsewhere) were equally effective
 - **Topical Diltiazem 2% cream/ointment** - £94/£139 per tube, BD, unlicensed use, ‘special’.
 - Fewer side effects than GTN; has been associated with migraine and pruritus ani
 - Cochrane found the efficacy equal to topical GT with lower recurrence rates – RR 0.68

Referral

- **Refer:**
 - All children with ongoing fissure for >2 weeks despite (conservative) treatment
 - Adults with ongoing pain at 8 weeks or visible fissure at 16 weeks regardless of pain
 - If secondary fissure suspected and underlying cause needs assessment
- Treatment options for chronic fissures include:
- **Botox**- unlicensed use – as effective as GTN
 - Complications: incontinence (though should wear off eventually), haematoma, infection
- **Surgery** – lateral internal anal sphincterotomy is the first line
 - More effective than medical treatments – healing rates are 70-100%, recurrence 1/3 only
 - Complications: incontinence of flatus or seepage in 14%; of stools in <1%

HAEMOCHROMATOSIS

Haemochromatosis is a very hot topic, with lots of recent [media interest](#).

Of course, we already know it is the most common genetic disease in people of European descent, characterised by iron overload, with deposition in parenchymal organs, particularly the liver, pancreas, joints and skin leading to significant morbidity. It is mostly attributable to 2 gene mutations: 95% have the HFE p.C282Y mutation, 4% the p.C282Y/p.Hist63Asp compound mutation.

The former variant is present in around 10-15% of northern European populations. Around 1 in 150 are homozygotes. Other populations have lower rates - a US study showed rates of 2.3% in non-Hispanic black people and 2.8% of Mexican-Americans.

Regardless of varying rates in different ethnic groups, a **key issue has been the penetrance of the genes** - that is, those people who actually develop clinical haemochromatosis as a result.

There has been significant uncertainty about the rates of penetrance, with studies in general populations (rather than family studies which tend to be higher) **suggesting it may be as low as 1%**. However, this data was largely looking at young populations and there is a concern that haemochromatosis clinically manifests more commonly in older age (particularly true in women after they lose the protective effects of menstruation), and that we are missing a substantial burden for the patient and health care systems.

► Common conditions associated with Hereditary Haemochromatosis genetic variants

[BMJ 2019;364:k5222](#)

This interesting cohort study used data from the [UK Biobank](#) to compare morbidity and mortality between people with the most common genetic variant of haemochromatosis to those without.

- >450,000 volunteers of European descent aged 40 - 70 years had blood tests for the HFE p.C282Y gene. Homozygous patients were compared to controls.
- Morbidity & mortality was data collected from hospital inpatient records and death certificates.
 - Data on haemochromatosis diagnosis, liver disease/cancer, MI/angina, atrial fibrillation, diabetes, arthrosis, arthritis, osteoporosis, tiredness and susceptibility to infection (via Dx of pneumonia) were collected
- Mean f/up 7 years.

■ **Results**

- **2890 participants were homozygous for p.C282Y** (0.6% or 1 in 156).
 - At the start of the study, 12% of men and 3.4% of women had a haemochromatosis diagnosis
 - **At 7 years, this rose to 22% of men and 10% of women**
- There were higher rates of core features vs the control group for:
 - 15.7% and 10% of homozygous men/women vs 5% and 3.4% of non-homozygous men/women developed ≥1 incident-associated condition
 - Liver disease - 2.4% vs 0.5%; OR 4.3
 - Osteoarthritis - 14% v 7.5% ; OR 2.0
 - Diabetes - 4.7% vs 3.15%; OR 1.5
- Overall penetrance (considered to be a diagnosis of haemochromatosis or ≥1 core feature) was surprisingly high:
 - 38.3% of homozygous men vs 15.7% controls - an excess of 22.6%
 - 27.2% of homozygous women vs 16.6% controls - an excess of 10.6%

■ **Conclusions**

- The authors concluded that HFE p.C282Y homozygosity was associated with substantial excess prevalence and clinically diagnosed morbidity in both men and women.
- 1 in 5 men and 1 in 10 women will develop excess morbidity due to the condition.

- They suggest that haemochromatosis meets the criteria for disease screening and that given this data shows a high prevalence of the condition than traditionally thought, the case for screening should be re-considered.

What was also interesting is that the rates of diagnosis continued to rise in older age. The implication is that many people with C282Y homozygosity could have haemochromatosis inappropriately “excluded” at a younger age because they have no evidence of penetrance, when in reality the clinical disease may only manifest in people’s 50s, 60s or 70s.

A linked [opinion piece in the BMJ](#) from the lead author commented that a US study estimated that by age 90, nearly 50% of males and 25% of females with C282Y homozygosity would have a diagnosis of haemochromatosis.

For us in primary care, the key test remains transferrin saturation (an elevated level >45% is indicative) followed by an assessment of ferritin, which if raised is an indicator for bloodletting, and genetic testing (HFE gene testing is widely available in primary care). We must also bear in mind that there are at least 5 types of haemochromatosis and not all are associated with a raised transferrin saturation, so if you have a patient with suspicious symptoms and a raised ferritin level, a referral is still appropriate.

KISS: Hereditary Haemochromatosis

[BMJ2016;353:i3128](#), [BJGP 2013;63:331](#)

► Cause

- An inherited genetic disorder, autosomal recessive, which is associated with a defect in hepcidin, an iron-regulating hormone, resulting in increased intestinal absorption of iron and subsequent deposition in the liver, pancreas, heart, skin, joints and gonads leading to organ damage
- Prevalence is 1 in 150-250 in Northern European populations, the most commonly affected group.
- Up to 85% is due to a mutation in the human haemochromatosis protein (HFE) gene being expressed through C282Y homozygosity, though other genes are also implicated. Penetrance of the genes, that is those who are subsequently affected, as a result, is not 100%

► Presentation

- Around 30% of patients may be asymptomatic throughout life, especially women where blood loss through menstruation and childbirth is somewhat protective.
- When present, *symptoms may be non-specific*:
- Lethargy, weakness, somnolence, arthralgia, vague abdominal complaints
- **Abnormal LFTs**
 - This is the most common avenue to diagnosis in primary care
- Arthropathy, chondrocalcinosis, heart failure, erectile dysfunction, porphyria cutanea tarda
- “Classical presentation” due to end-organ damage only seen in 10% of cases
 - Liver cirrhosis, diabetes, skin pigmentation (a bronze tan in the UK winter is a warning sign)

KISS: Hereditary Haemochromatosis continued

[BMJ2016;353:i3128](#), [BJGP 2013;63:331](#)

► **Diagnosis**

- Be suspicious for haemochromatosis
- Ferritin levels are raised and transferrin saturation is >45%
 - Check serum ferritin
 - If ferritin is raised check transferrin saturation
 - Exclude other causes of raised ferritin e.g. infection, inflammation, alcohol, etc.
- If ferritin raised and transferrin saturation >45% refer to gastroenterology for genetic testing, phlebotomy and assessment of end-organ damage

► **Treatment**

- The mainstay of treatment is bloodletting. This is secondary care-led.
 - Maintenance aims to keep ferritin between 50-100ug/L
 - If no cirrhosis is present, patients should have a normal life expectancy
 - There is an increased risk of osteoporosis so a DEXA scan should be performed.
- Dietary restriction is unnecessary if having bloodletting but avoid iron and Vit C supplementation
- A ferritin of >1000ug/L at presentation is associated with a 20-45% risk of developing cirrhosis. These patients will be considered for liver biopsy as cirrhosis confers a 3-4% annual risk of hepatocellular carcinoma. Patients with cirrhosis should be enrolled in an HCC and oesophageal varices surveillance program.

► **Should the family have genetic testing?**

- Siblings should be screened (they'll have a 25% risk of having the condition) with serum ferritin and transferrin saturation
 - Ideally, they would have HFE genotyping with genetic counselling
- Partner testing can guide whether children require testing.

NB Learning Points: Haemochromatosis

- This new data shows that the penetrance of haemochromatosis is much higher than traditionally considered, increasingly so in older age
- Joint symptoms were the most common complications, then liver disease and diabetes - 1 in 5 men and 1 in 10 women will develop excess morbidity as a result of the condition
- For people with suggestive symptoms or a family history check iron studies - a raised transferrin saturation of >45% is the key marker, if raised assess ferritin and for HFE genes, then d/w hepatology

NB Quality Improvement Ideas: Haemochromatosis

- Perform an audit of your practice population
 - How many people have a raised transferrin saturation? Has this ever been assessed in more detail? Do they have a raised ferritin as well? Do they have any associated conditions such as tiredness, joint pain, abnormal LFTs or diabetes? If so they need genetic testing and consider referral.

ANOSMIA

Loss of smell is not a new symptom but has been a hot topic in the pandemic which has sparked a plethora of papers discussing olfactory dysfunction, which is a surprisingly common problem, even without covid. Articles in the BMJ ([2020;370:m2808](#)), JAMA ([2020;323\(24\):2512](#)) and the BJGP ([2021;71:135](#)) review the management of anosmia while the BMJ ([BMJ2022;377:e069860](#)) discusses the grossly unpleasant parosmia.

First, the latest research into the level of effect covid can have on the olfactory system.

1 **Prognosis and persistence of smell & taste dysfunction in patient with covid-19: meta-analysis** [BMJ 2022;378:e069503](#)

Around 50% of people with covid experience alteration in their smell or taste. This systematic review and meta-analysis included 18 studies with 3600 adult patients with covid-19 induced taste or smell dysfunction.

It showed that while around three-quarters could expect a recovery in the 1st month, at 180 days post-acute covid 5.6% still had persisting smell dysfunction and 4.4% taste dysfunction. Those less likely to recover included females, those with more severe initial dysfunction and the presence of nasal congestion.

2 **Understanding and Managing Olfactory Dysfunction**

Olfactory Dysfunction (Dysomia) [BJGP 2021](#), [BMJ 2022](#)

- | | |
|---------------------|---|
| ■ <i>Anosmia</i> | No perceivable sense of smell |
| ■ <i>Hyposmia</i> | Reduced sense of smell |
| ■ <i>Parosmia</i> | Distorted or unpleasant sense of smell |
| ■ <i>Phantosmia</i> | Smell perception in absence of a stimulus |

► **What causes anosmia?**

The [BJGP](#) and [BMJ editorial](#) refresh our memories on how the sense of smell works:

- Smell occurs when organic molecules in the environment are detected at the olfactory cleft, stimulating the olfactory neuroepithelium, a small area in the supero-medial nasal cavity
- This can occur when the molecules are carried into the nose on air (e.g. during breathing) or with retrograde airflow (during swallowing)
- An impulse is emitted by the neuroepithelium and transmitted to the brain (thalamus, hypothalamus, and frontal cortex) via the olfactory nerve. Some somatosensory input is provided by a small portion of the trigeminal nerve (CN V), allowing the sense of 'irritants', which may be preserved when normal smell is impaired.
- This pathway, especially when swallowing food, combines with senses of taste and touch, to provide flavour to food. The majority of people who have impaired taste actually have an impaired sense of smell.
- Anosmia may represent damage to the neuronal pathway, but data suggests the virus targets the neuroepithelium. Parosmia is considered as disordered connections, possibly during recovery from injury.

Olfactory dysfunction is common, with around 20% of the population affected. It is a very common complication of many infections (especially URTI) and other conditions, for example, it occurs in 30% after a serious head injury, but covid-19 is particularly associated with anosmia (meta-analysis suggests up to 55% develop some impairment), more commonly than other infections and with a more profound effect.

The BJGP suggests considering the causes in a similar way to hearing loss: conductive or sensorineural.

- **Conductive** - physical obstruction, inflammation or congestion
 - *Chronic rhinosinusitis, nasal polyps, allergic rhinitis*, gross septal deviation, intranasal tumours, granulomatous disease, previous nasal surgery
- **Sensorineural** - failure of the neurological pathway
 - *Viral / post-viral*, head trauma, medication-related, neurological (e.g. temporal lobe epilepsy, CVA, MS) or neurodegenerative (e.g. Parkinson's, Alzheimer's), space-occupying lesion, congenital
 - Medicines implicated include ACEi, CCBs, diuretics, statins, tobacco, alcohol, illicit drug use

We should be aware that mental health conditions are associated with anosmia - the sense of smell (and taste) is so important to human life that the loss can have a huge impact on an individual that we should explore this in affected patients and signpost to appropriate services.

► *What about parosmia?*

Parosmia, the misperception of an odour, results in a normal scent being smelt as an unpleasant one. There is a significant crossover between the causes of anosmia, and a marked association with covid infection. In contrast to covid-related anosmia, **parosmia develops on average 3 months post-infection, often after recovery from their anosmia**. Data suggests **>40% of people with covid-related anosmia had parosmia at 6 months**. Pre-covid data indicates spontaneous recovery rates are high, but the timeframe may be months to years.

The impact of parosmia should not be underestimated - imagine a world where food tastes of raw sewage and flowers smell rancid - and **the condition can be extremely distressing**. This needs to be acknowledged with the patient and mood disorders, weight loss and malnutrition should be actively screened for as part of a holistic assessment.

► *How can we manage anosmia?*

While many of us will have prescribed intranasal steroids for anosmia in the past, olfactory retraining is considered the most effective treatment for most. Thankfully it is very straightforward.

The concept is that, just as with any nerve pathway, **regular use and stimulation of the olfactory system can promote recovery**.

- Achieved through regular sniffing of strong smells - commonly lemon, rose, cloves and eucalyptus
- Patients smell each a few minutes apart for 20sec BD for at least 3 months (the longer the better).

Data shows this technique to be effective - one study found in patients with dysosmia for <12 months 63% improved vs 19% in a control group sniffing low-concentration odours. It was less successful where the olfactory dysfunction was more longstanding.

Patients can **make their own olfactory retraining kit at home** - there are useful guides on the charity [Fifth Sense](#) website and kits can be purchased from the [AbScent website](#).

Prognosis varies with the cause - post-infection it is between 35-66%, but 90% improve within 4 weeks of covid-induced dysosmia. Only around 30% with anosmia due to trauma will recover, usually in <3 months.

► *How can we help patients with parosmia?*

Genuine treatment options for parosmia are limited. Smell re-training should be considered to promote neuronal regeneration although data is lacking. Likewise, intranasal steroids have been tried but lack research. Simply acknowledging how unpleasant the condition can be for a patient is important. Practical measures regarding food, triggers, managing food intake via protein powders and dietician may be of benefit. Addressing mental health concerns is important. Also offering some hope - average time to recovery in patients seeking support for parosmia is 14-16 months, and recovery has been reported even around the 2-year mark.

KISS: Management of Anosmia

[BJGP 2021;71:135](#), [JAMA 2020;323:2512](#), [BRS/ENT UK 2020](#)

Assessment

- **Red flags?**
 - Dysomia with focal seizures (?temporal lobe epilepsy) or features of raised intracranial pressure
 - Cacosmia (foul smell detectable within the nose or to others - kids think foreign body [ENT], adults dental disease [ENT + dentist])
 - New neck lump + any nasal Sx (change in sense of smell, blockage, bleeding, dc) ?sinonasal Ca
 - Phantosmia - differs from parosmia as not reproducible with stimulus - may indicate a space-occupying lesion
- **History** - consider conductive or sensorineural causes
- **Examination** - external signs, e.g. gross deformity; internal, e.g. discharge, deviated septum, polyps, overt rhinitis, foreign bodies; neurological guided by history

Referral

- Referral to ENT recommended if >6 weeks duration, red flags, atypical presentation or no treatment response
- If covid-19 induced referral after 3 months is appropriate

Treatment - Anosmia

- Treatment should target the underlying pathology - consider meds, smoking, alcohol, drugs
- **Olfactory retraining is recommended for anosmia persisting >2 weeks**
 - Sniffing 4 strong scents for 20secs, several minutes apart, twice daily for ≥3 months
 - Patients should be given [anosmia information](#) (ENT UK) and directed to [Fifth Sense](#) and [AbScent](#)
- **Nasal Sx present? Intranasal steroids are recommended for anosmia persisting >2 weeks**
 - In GP intra-nasal pathology (e.g. chronic rhinosinusitis, allergic or acute sinusitis) are the most common causes and INCS may benefit, but if nasal symptoms have lasted >4-6/52 then ENT referral is still recommended
 - Intranasal drops or rinses could also be tried but are more difficult to administer effectively
- Oral corticosteroids are an option (short course) if anosmia of >2 weeks as an isolated or following resolution of any other covid symptoms (ENT UK)
 - Data supports use in chronic rhinosinusitis and (less strongly) post-viral anosmia
 - It is hard to marry this against recommendations to avoid oral steroids where possible in pandemic
 - JAMA recommends against routine use and comments success is unlikely post-covid as is the case (most of the time) for a post-infectious cause
- Food supplements lack efficacy in data, but ENT UK states omega 3 is an option
- **Consider mental health**
 - Half have concurrent MH issues, address where possible.
- **Warn about risks of lack of sense of smell** - e.g. unable to smell gas or gone-off food

Treatment - Parosmia

- Practical measures include knowing triggers and 'safe' foods, eating cold or room temperature foods, 'pushing through' bad taste may drive improvement, where food is intolerable protein powder can be tried and dietician referral is advisable
 - More good information from [AbScent](#) and [Fifth Sense](#)
- Lack of data to inform medical measures, but there may be a role for olfactory re-training
- There is a Facebook [support group overseen by AbScent](#) for people suffering with parosmia

MENIERE'S DISEASE

Could this be the most over-diagnosed cause of vertigo?

Historically this condition was diagnosed in many people with vertigo where there wasn't a clear alternative (we know BPPV, vestibular migraine are significantly more common, [BMJ 2019;366:l5215](#)), but as a new and helpful review on the condition in the [BJGP 2021;71:571](#) explains it remains an important diagnosis to make given the potential life-changing sequelae and available treatment options.

► *What is Meniere's Disease?*

MD is an idiopathic disorder of the inner ear leading to recurrent vertigo episodes and associated auditory features of tinnitus, aural fullness and fluctuating sensorineural hearing loss which becomes progressively more permanent.

The pathophysiology remains unclear. It has been linked to endolymphatic hydrops - raised pressures from extracellular inner ear fluid - found in postmortem in all people with Meniere's; however, it is also found in people without MD so it is speculated that this feature is one of several potential co-factors (e.g. autoimmune, age, vascular) required for MD to emerge.

Prevalence is estimated at 34-190 cases per 100,000, affecting predominantly adults between 30-70 years old. The BJGP highlights the significant effect it can have on individuals with up to 70% needing to adjust their working life as a result.

► *What are the key clinical features?*

There is no pathognomonic test for MD. Diagnosis is based on characteristic clinical features and the exclusion of alternative causes.

[International diagnostic criteria for MD](#) were established in 2015. It can be classified as 'Definite' or 'Probable', with duration of vertigo and audiometry-confirmed sensorineural hearing loss distinguishing the two.

Diagnostic Certainty	Definite	Probable
	≥2 spontaneous episodes of vertigo lasting 20 minutes to 12 hours	≥2 spontaneous episodes of vertigo lasting 20 minutes to 24 hours
	Audiometrically confirmed low-to-medium frequency sensorineural hearing loss in one ear, before/during/after an episode of vertigo	
	Fluctuating aural symptoms (fullness, tinnitus, hearing loss) in affected ear	Fluctuating aural symptoms (fullness, tinnitus, hearing loss) in affected ear
	Not better accounted for by another diagnosis	Not better accounted for by another diagnosis

Our KISS summary below detail the clinical assessment for possible Meniere's Disease.

► *How can we manage Meniere's Disease?*

MD is a diagnosis of exclusion, so for many patients referral is most appropriate.

Audiometry and tympanometry will identify sensorineural hearing loss while excluding middle ear pathology plus allows monitoring of progression and MRI identifies cerebellopontine angle or inner-ear pathology.

GISS: Clinical Assessment of Meniere's Disease

[BJGP 2021;71:571](#), [BJGP 2017;67:569](#)

- **Exclude red flags for acute vertigo**
 - Consider causes indicating central pathology: new unilateral hearing loss, focal neurology, new-onset headache, normal head thrust test
- **History & exam**
 - **Vertigo**
 - Distinguish from non-vertiginous dizziness
 - Episodes typically last 20 minutes - 12 hours
 - **Tinnitus, aural fullness and sensorineural hearing loss** classically accompany vertigo episodes
 - ***Predominantly unilateral*** - while bilateral cases have been reported symptoms should not occur in both ears simultaneously
 - ***Hearing loss starts as episodic and then becomes more permanent*** over time
 - Low-frequency hearing is predominantly affected
 - Examination may include focused neurological exam, otoscopy and Dix-Hallpike
- **Differential Diagnosis** - think broadly
 - ***Vestibular migraine*** - a possible DD but also present in 30% of people with MD
 - ***BPPV*** - the most common cause of vertigo, episodic but linked with movement
 - ***Superior semicircular canal dehiscence syndrome*** - 'third window syndrome', thinning of temporal bone +/- holes resulting in dizziness, vertigo, autophony, hearing loss, aural fullness, nausea, headache and fatigue. Can be treated by surgery.

Initial management can begin in primary care:

- **Lifestyle measures**
 - Dietary modification has been used for years and reported as beneficial although the evidence base is limited. Dietician input may be helpful for some and improve the chance of benefit.
 - **Salt restriction**
 - Thought to reduce endolymphatic pressure, reducing frequency & severity of vertigo attacks
 - Aim for <2000mg/day sodium intake (equivalent to 5g of salt)
 - **Caffeine restriction/avoidance**
 - **Increasing water intake**
 - Reported to improve vertigo and hearing
 - Aim for 35ml/kg/24h (e.g. ~2.5L/day for a 70kg person)
 - For those with co-existing vestibular migraine, adequate sleep, alcohol avoidance, etc. may benefit
- **Pharmacotherapy**
 - **Betahistine** is commonly prescribed but ineffective
 - Thought to improve labyrinthine circulation
 - The BJGP suggests it as an option, citing a [2016 Cochrane review](#) on the use of betahistine for acute vertigo (not specifically MD) which found low-quality evidence of a positive effect and good safety profile. Cochrane notes the evidence is low quality and the results should be interpreted with caution.
 - A RCT published in 2016 ([BMJ 2016;352:h6816](#)) compared placebo to low- and high-dose betahistine for MD and found no difference in rates of vertigo or improvement in other symptoms. There wasn't a control group so it was unable to assess the level of placebo effect.
 - The [associated editorial](#) counsels its use may divert management away from effective treatments.
 - **Diuretics**
 - Used to achieve the same effects as lifestyle modification, but there is a lack of data to support and given the potential side effects not advised to be initiated in general practice.



■ **Surgery** [BMJ 2016;352:i46](#)

- Injected intratympanic topical agents of vestibulotoxic medications (e.g. gentamicin) are very effective - 90% of patients will have their attacks controlled. However, 20% have persistent hearing loss and 20% chronic vertigo. Intratympanic steroids may have utility here and lower rates of adverse events - it is currently being researched.
- In very severe cases the vestibular system can be surgically removed or disconnected from the brainstem but chronic imbalance will affect 20% (which seems surprisingly low considering...).

Driving needs to be considered. [DVLA rules state that patients are obliged to inform them if they suffer from dizziness that is sudden, disabling or recurrent](#). Where patients will not inform the DVLA [healthcare professionals have a responsibility to inform the agency](#) themselves.

The Patient's Perspective www.menieres.org.uk

"An unpredictable and distressing condition"

It is useful for a more patient-orientated experience of conditions. For Meniere's, the picture that emerges can be quite different from how specialists describe classic disease.

Meniere's disease can be divided into 3 stages:

- **Stage one (early): unpredictable attacks of vertigo**
 - Intermittent bouts of tinnitus lasting minutes to hours, with variable amounts of hearing loss and aural fullness. Tinnitus may occur as well. Ear symptoms may precede attacks, but may happen without warning. In between attacks ear symptoms return to normal. Remissions can be days, months or years, making it very unpredictable.
- **Stage two (intermediate): attacks of vertigo; tinnitus; hearing loss**
 - Vertigo attacks continue intermittently. The vertigo may become less severe, but hearing loss may be permanent and tinnitus more troubling than in the early stages. Both will still increase during attacks. Imbalance & movement-induced giddiness can develop after, and/or before, attacks.
- **Stage three (late): hearing loss; balance difficulties; tinnitus**
 - Hearing loss increases and may be severe and highly problematic. Vertigo may decrease or cease. However, there is permanent damage to the balance centre commonly leading to persistent balance issues.

NB Learning Points & Quality Improvement Ideas: Meniere's Disease

- Meniere's Disease is an uncommon cause of vertigo associated with aural symptoms but can have a significant impact on quality of life.
- Vertigo for 20min - 12h plus tinnitus, aural fullness and fluctuating hearing loss of key features.
- Betahistine appears ineffective but is still frequently used.
- Other treatments such as intra-tympanic injections are highly efficacious.
- Referral is appropriate for the majority of patients to exclude alternative diagnoses, provide monitoring and discussion of therapy.
- Lifestyle measures including low salt diets and high water intake can be beneficial.
- **Quality improvement**
 - Audit patients receiving betahistine. Discuss if this has been beneficial. Have they experienced side effects? Would they consider a trial of withdrawal? A symptom diary during treatment and after cessation can help demonstrate if there has been a change.

THROAT SYMPTOMS

How often do we see patients with 'something in their throat'? While you might not imagine this as an area for significant discourse, amongst specialists there has been rigorous scientific debate about the causes of persisting throat symptoms and not much agreement.

Globus is a case in point - two recent articles in the BJGP ([BJGP 2015;65:554](#)) and the BMJ ([BMJ 2014;348:f7195](#)) discuss the management of globus pharyngeus. The historical term of globus hystericus has been consigned to the past as our belief about the underlying cause of this common problem has changed. Reflux is considered a primary driver as is post-nasal drip. And yet, how often have these treatments worked for your patient?

A recent BMJ Practice paper ([BMJ 2020;371:m4091](#)) on laryngopharyngeal reflux (LPR) suggests that the most common resulting symptom is the sensation of a lump in the throat and that PPIs are an effective first line drug therapy. But this commonly held notion has been seriously challenged by a new paper published in 2021 in the BMJ ([2021;372:m4903](#)) and may lead to a change in how we manage this condition.

1 Use of PPI to Treat Persistent Throat Symptoms [BMJ 2021;372:m4903](#)

Recent [meta-analysis](#) of PPIs for throat symptoms for LPR have suggested mild superiority over placebo but acknowledge this is based on very small RCTs which have significant heterogeneity. This new RCT aims to have a larger sample size to clarify the role of PPIs in patients with persistent throat symptoms (including hoarseness, globus, repeated throat clearing, mucus in the throat, or "catarrh", cough and throat discomfort), as well as a pragmatic design to make it useful in day-to-day practice.

- 346 adults with persistent throat symptoms recruited from 8 UK ENT outpatients clinics
- Randomised to either lansoprazole 40mg BD or placebo for 16 weeks
- **Primary outcome:** symptomatic response on total 'reflux symptom index' (RSI)
- **Secondary outcomes:** response at 12 months, QoL and throat appearance
- **Results:**
 - *No significant difference in RSI score between active and control groups*
 - Both groups saw minor improvements in symptoms, with a trend towards greater benefit with placebo
 - There was no difference in secondary outcome measures.
- **Conclusion:**
 - *No evidence was found of benefit from PPI treatment in patients with persistent throat symptoms. RSI scores were similar between the lansoprazole and placebo groups after 16 weeks of treatment and at the 12-month follow-up.*

In their discussion, the authors acknowledge that this trial provides data on what doesn't work for throat symptoms but leaves unanswered questions, principally what alternative strategies may be effective? Currently, we don't have these answers although there are some areas discussed in the KISS below which may provide other avenues for patients to explore.

2 Globus Pharyngeus [BMJ 2020;371:m4091](#), [BJGP 2015;65:554](#), [BMJ 2014;348:f7195](#)

45% of the population experience this misunderstood condition at some point.

► What causes globus?

Globus pharyngeus (or commonly just 'globus') remains poorly understood. While psychological factors do play a role – 96% of patients with globus have worse symptoms in times of stress – the condition has an unclear aetiology, with several physical theories postulated.

Chronic inflammation from reflux of acid into the laryngopharynx is a contentious issue (see above). Studies show 23-68% of patients report reflux. However, the same studies often report similar rates of reflux in asymptomatic controls. It is still thought that reflux accounts for an important subgroup of globus patients.

Oesophageal motility may account for some cases – some patients may have elevated upper oesophageal sphincter pressure resulting in abnormal sensation.

Persistent cough and/or post-nasal drip have the potential to irritate the laryngopharynx.

KISS: Assessment of Globus/Throat Symptoms

Clinical assessment - globus is usually a clinical diagnosis

■ History

- The **painless** sensation of a lump in the throat is the key feature, often described as a lump, ball, itch or swelling; typically felt centrally and suprasternally
- Symptoms are usually felt when swallowing and often come and go
- May be associated with throat clearing, chronic cough, hoarseness and catarrh
- Ask about reflux symptoms, anxiety and other psychological symptoms

■ Examination - thorough examination of head and neck

- Palpate the neck including the thyroid and cervical lymph nodes
- Assess oropharynx and oral cavity – look for asymmetry and ulceration, check under the tongue
- Examine the nose for inflamed mucosa, polyps or pus – all might suggest a post-nasal drip

Referral

■ Presence of red flags

- Pain on swallowing – not typical of globus
- Persistent hoarseness (or change in voice), progressive dysphagia, dysphagia for solids, haemoptysis, weight loss
- Worsening globus in patients with high consumption of alcohol or tobacco
- Abnormal head/neck examination
- Unilateral symptoms or asymmetrical examination

■ Recurrent, persistent and troubling symptoms merit consideration for ENT referral

Treatment

■ **Vocal hygiene** – avoid smoking, alcohol, caffeine

■ **Resist the urge to dry swallow or clear the throat** – try sipping chilled carbonated water when they feel the urge to swallow

■ If **symptoms and/or signs of reflux consider aggressive treatment**

- BD PPI and alginate for 4 months+
- NB if there are no symptoms of reflux PPIs are no better than placebo

■ **Speech and language therapy** has shown to be of benefit, but not proven that this is due to anything more than the added reassurance this provides patients

■ **Consider anxiety. CBT has been shown to help symptoms;** there is little data regarding medications

NB Learning Points: Throat Symptoms

- PPIs are no more effective than placebo for persistent throat symptoms and are not recommended for treating globus in the absence of upper GI/reflux symptoms
- Globus is a common problem that remains poorly understood
- Non-drug options such as vocal hygiene, behavioural changes and SALT can be effective
- For patient information explaining management [click here](#)



TINNITUS

Tinnitus can be a miserable condition but despite affecting 1 in 10 of the population, only a small minority are severely troubled by the condition. Nevertheless for this group morbidity can be severe and our help is required. We all will have an approach to how to manage tinnitus for patients, but new data and guidance provides strategies and suggestions for ways we can make a meaningful difference to our patients who find tinnitus problematic.

KISS: Background Tinnitus

CKS 2020

■ Definition:

- Perception of sound in the absence of sound from the external environment which can manifest as ringing, hissing, buzzing, whistling, etc. and may be constant or intermittent, unilateral or bilateral
- **Subjective:** affects 99% - only heard by the individual, results from inner ear or CNS
- **Objective:** affects 1% - also audible to examiner, caused by physical cause in or around the ear

- Tinnitus is heard when there is a failure of the auditory filtering or environmental masking of the normal background cochlear activity or there is a change to this background activity and exacerbated by focusing attention on the sound.

■ Epidemiology:

- Affects 1 in 10 people, occurs at any age but is more frequent as we get older
- **Subjective tinnitus** is more common in people with hearing loss:
 - Including age-related, loud noise exposure, impacted wax, Meniere's and otosclerosis
 - Also ototoxic drugs, including loop diuretics, aspirin and NSAIDs (all reversible), tetracyclines, macrolides, aminoglycosides, valproate, quinine & chloroquine, cytotoxic drugs (e.g. cisplatin)
- Also associated with DM, HT, obesity & dyslipidaemia; ear infections; acoustic neuroma, MS; thyroid disorders, zinc deficiency; anxiety & depression; TMJ disorders & head/neck trauma
- **Objective tinnitus:** usually due to vascular cause:
 - AV malformation or benign venous hums, vascular tumour, carotid/vertebral artery stenosis/dissection/aneurysm, aortic stenosis or mitral regurgitation
 - Also high cardiac output, e.g. anaemia, eustachian tube/palatal/middle ear muscle dysfunction

Tinnitus: assessment and management [NICE NG 155 March 2020](#)

NICE has just released its first guidance on the management of tinnitus. But does it add anything to our existing management strategies?

In fact, it does! The guidance is summarised in the KISS at the end of the chapter, but the key novel recommendations include:

- Patients should receive 'tinnitus support' from the very first consultation and this can be achieved in general practice
- Acknowledging the significant impact on mental health associated with tinnitus in some patients sets out specific timelines for assessment and the benefit of psychological therapies in this group
- Offering audiological assessment to every patient with tinnitus

► *What does NICE mean by tinnitus support?*

This recommendation is central to the guideline and the understanding that the vast majority of people with tinnitus (it is so common we might refrain from labelling them as 'patients') do not want or need significant input. The way we handle the first consultation sets expectations for the long term so there is a benefit in being positive about tinnitus.

'Tinnitus support' is "a 2-way process of information giving and discussion to help the healthcare professional understand the difficulties and goals of the person with tinnitus". Part of this is, therefore, an assessment of the severity of the impact of tinnitus on the person, then the crucial information NICE wants to impart is that a) tinnitus is common, b) it may resolve itself, c) while it is commonly associated with hearing loss, other underlying disease is uncommon, and d) there are management strategies that can help manage the condition if required.

This is our opportunity to de-pathologise the condition for patients before they get trapped in a vicious circle of focusing on the problem and exacerbating it further.

The [British Tinnitus Association](#) has a good website to direct people to for further information.

► *What are the mental health impacts of tinnitus?*

While it is possible to have a chicken and egg argument about tinnitus and mental health issues, the guidance importantly acknowledges the significant impact the condition can have on mental health and makes specific recommendations which may speed up specialist assessment in those truly troubled by it. This includes:

- Immediate referral to a mental health crisis team of people with tinnitus and a high risk of suicide. If necessary provide a safe place while waiting.
- Refer for review within 2 weeks people with tinnitus associated with distress affecting mental well-being (e.g. preventing normal daily activities) where tinnitus support at 1st contact is not adequate.

► *Why should every person with tinnitus have a hearing test?*

While NICE acknowledges there is no evidence regarding audiology there was significant consensus on the committee that patients with tinnitus should be offered a hearing test. Particularly age-related changes may be insidious and reduced hearing drives the central auditory system to work harder producing more noise. Hearing aids can be useful in reversing this. Assessment will also identify conductive causes such as middle ear effusion or eustachian tube dysfunction and allow appropriate management.

► *What other treatments are recommended?*

Indeed hearing aids are one of the only therapies NICE has actively recommended due to a lack of evidence for most other interventions. However, there are a number of useful options available to patients:

- **Psychological therapies**
 - NICE recommends 'consider' a stepped approach to treat tinnitus-related distress
 - Step 1 - digital tinnitus-related CBT
 - Step 2 - group based tinnitus-related psychological interventions, e.g. Mindfulness, acceptance and commitment therapy, CBT
 - Step 3 - individual tinnitus-related CBT, tinnitus retraining therapy
- **Tinnitus Retraining Therapy**
 - Comprises of educational counselling and sound therapy, delivered in specific protocols
 - Very limited data, but results are positive; widely used in specialist settings but standard protocols lack the ability to personalise for patients
 - NICE was unable to make a recommendation regarding TRT but added it as a 'research question'
- **Sound therapy**
 - Essentially having background noise to 'distract' from tinnitus sounds, e.g. leaving open the window to allow outside noise, having the radio on, wearable 'masking' devices which play white noise
 - Easy to do and advise from primary care

What about medications and supplements? NICE is clear it does not recommend betahistamine ([Cochrane 013093](#)) and zinc supplementation (considered due to deficiency being linked with tinnitus) has been found not helpful ([Cochrane 009832](#)). There are a wide range of herbal therapies purported to be beneficial but lack any supporting data.



NB Learning Points & Quality Improvement Ideas: Tinnitus

- New guidance encourages positive messages regarding tinnitus from the initial contact
- Hearing loss is a common co-pathology - in these patients hearing aids can help so NICE recommends all patients with tinnitus have an audiological assessment
- There is a lack of supporting data for many treatments other than hearing aids, but psychological therapies for tinnitus-related distress is effective, and sound therapy is widely considered to be sensible initial management
- Audit patients coded with hearing loss and tinnitus - have they been offered audiology referral?
- Perform a case review of patients with troubling tinnitus - have they been referred for psychological support?

KISS: Tinnitus

[NICE NG155 2020](#), [CKS 2020](#)

Assessment

- **Character** - uni/bilateral, episodic/continuous, pulsatile (with heartbeat?); objective/subjective
- **Associated symptoms** - hearing loss? dizziness/vertigo, balance issues, TMJ dysfunction, neurology
- **Psychological** - suicidal ideation, depression, anxiety, catastrophising, stress
- **Impact**: consider [Tinnitus Functional Index](#) or a visual analogue scale (or "out of 10"...)
- **Triggers**: think medication (e.g. loop diuretics, aspirin, NSAIDs, macrolides, tetracyclines), environment
- **Examination**: otoscopy, cranial nerves, BP +/- routine bloods (metabolic disease may present with tinnitus – consider FBC, LFTs, lipids, vit B12, zinc). If objective listen for bruits in neck, pre-auricular and mastoid

Initial Management

- **Provide 'tinnitus support' at all**
 - Positive and reassuring approach to discussing tinnitus, identify needs, assess affect on life and mental state
 - Information to share: common, may resolve, hearing loss common cause but other disease rare, various strategies can be helpful
- **Refer**:
 - Immediately to mental health crisis team if high risk of suicide
 - Immediately to appropriate speciality if tinnitus associated with:
 - Sudden onset significant neurological features, acute uncontrolled vestibular Sx, or suspected CVA
 - Within 24h if tinnitus & rapid hearing loss (in <3/7) within the last 30 days
 - Within 2 weeks if tinnitus & distress affecting mental well-being or rapid hearing loss within 30-90 days or rapid worsening over the last 4-90 days
 - Routine if persisting bothersome tinnitus, persistent objective tinnitus, unilateral/asymmetric hearing loss
 - Consider routine referral for persistent pulsatile or unilateral tinnitus
- Offer audiological assessment; consider tympanometry if suspect middle ear effusion/ETD



KISS: Tinnitus (continued)**Treatment**

- Offer amplification devices to people with tinnitus and hearing loss which makes communicating difficult
 - Consider if hearing loss but no communication issues
- Consider a stepped approach to tinnitus-related distress using psychological therapies
 - Digital CBT > group CBT/other techniques > individual CBT
- Options NICE were unable to comment on due to lack of data include sound therapy (e.g. distraction) and tinnitus retraining therapy
- Do not offer betahistine. Evidence of lack of effect from zinc supplements or herbal remedies.

Further resources

- The “Eight-Minute Primary Care Consultation” is a fantastic 2-page proforma from GP and ENT services in Liverpool – [Click here to read](#)
- British Tinnitus Association has an excellent website, a patient helpline and self-help material. www.tinnitus.org.uk

HEARING LOSS

Hearing loss leads to marked isolation, a significant reduction in quality of life and has a huge economic impact. With an ageing population, it is a growing issue so up-to-date guidance published by NICE ([NG98](#)) in mid-2018 is very welcome.

While most hearing loss is gradual and age-related we shouldn't forget acute hearing loss - often an unknown cause, possibly permanent life-changing effects, also potentially reversible with early treatment. But, as a new BJGP ([2020;70:144](#)) paper points out, referral to ENT can be a significant barrier to patients getting this treatment. The authors suggest how we can solve this.

1 **Sudden Sensorineural Hearing Loss** [BJGP 2020;70:144](#)

This intriguing paper in the BJGP ([2020;70:144](#)) reminds us that sudden-onset sensorineural hearing loss (SNHL) is an otological emergency. [CKS](#) reports that between 3-6 out of 10 people will have spontaneous return of their hearing but that leaves a substantial proportion with long-term hearing loss.

The mainstay of treatment is early oral steroids although evidence of benefit remains inconclusive ([Cochrane CD003998](#)). Regardless there is consensus that while the data is uncertain, complications are unlikely so treating remains the preferred option. There is also consensus that benefit is most likely within the first 48h of hearing loss and very unlikely after 2 weeks.

Therein lies the problem - speedy referrals to ENT can be difficult both for primary care and the specialist and patients are the inevitable losers. So is there another option so our patients receive timely treatment?

► *Starting steroids in practice*

This paper makes it clear that investigation is not the time-sensitive part of management. 70% of cases are idiopathic, and most other causes will resolve themselves, will be clear from the history and state of the patient, or can be undertaken on a non-urgent basis. The urgency of assessment is driven by the desire to initiate steroids as soon as possible.

Why not start steroids in primary care? The paper argues there is very little reason not to given the rarity of significant side effects from short courses of oral prednisolone.

It proposes a simple algorithm for the identification of patients who merit treatment, which involves a simple history, examination and Weber's test to separate conductive and sensorineural hearing loss. See next page for KISS.

For patients with normal otoscopy, no current URTI and Weber's suggesting SNHL steroids can be initiated in primary care immediately, which removes the time pressures of ENT assessment and a routine referral can then be made to ENT to try and identify the cause.

► *Which steroids are best?*

There is no consensus on which steroids are 'best'. Studies suggest some marginal benefits in different causes for both dexamethasone and prednisolone (if indeed either truly benefits) but overall there is no preference as long as something is started ideally within 48hrs.

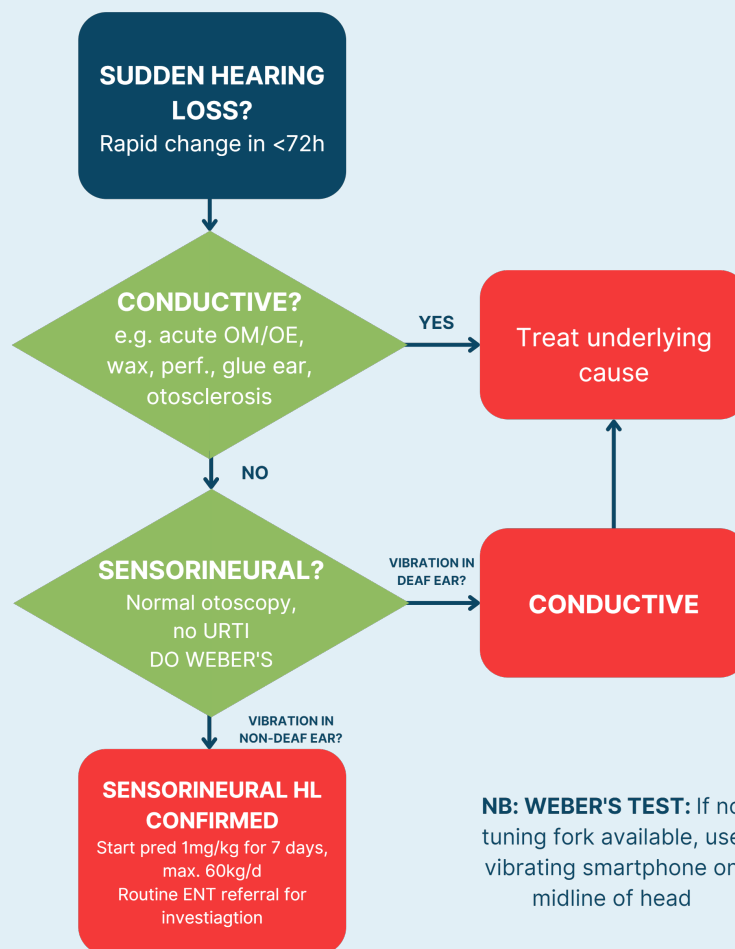
Pragmatically prednisolone would be most straightforward in primary care. The recommended dose is 1mg/kg OD for 7 days, with a max. dose of 60mg/day. There is no need to taper, which had been standard practice several years ago.

► *I can't find my tuning fork*

Don't worry, you are not alone. The authors give a useful tip - use your smartphone. Traditionally the Weber's test uses a tuning fork pitched at 256 Hz placed in the midline of the head (it can anywhere from forehead to occiput). However, a vibrating smartphone (which will be a different frequency) works as well. A prospective study showed >97% agreement between phone and tuning fork tests. They used the Real Razor app (free to download) for their vibrations.

The authors remind us that with Weber's if the patient hears the vibration in the deaf ear it suggests a conductive hearing loss. If it is predominantly heard in the non-deaf ear it implies a sensorineural cause.

KISS: Managing Sudden Hearing Loss Within Last 72 hours
[BJGP 2020;70:144](#)



Causes of Sudden Sensorineural Hearing Loss

- **Idiopathic** (70%)
- **Infective** - viral (e.g. measles, mumps, rubella), bacterial (e.g. syphilis, Lyme, meningitis, acute labyrinthitis)
- **Trauma** - head injury, blast trauma, barotrauma
- **Pharmacological** - ototoxic medication (e.g. IV aminoglycosides)
- **Autoimmune** - Wegener's granulomatosis
- **Central** - ischaemic (e.g. CVA), multiple sclerosis, cerebellopontine angle tumour
- **Other** - Meniere's

2 Hearing Loss In Adults [NICE NG98, June 2018](#)

While this recent NICE guideline is for the most part based on expert opinion, there are some useful messages for clinicians, particularly:

- *When does hearing loss require a referral?*
- *Which specific presentations necessitate urgent specialist review?*
- *Screening for hearing loss in certain groups such as patients with learning disability or memory loss*
- *Managing ear wax - more interesting than you might think!*

► *How should we manage ear wax?*

NICE recommends that ear wax can and should be sorted in out in primary care - obvious but also controversial given the current state of funding for such procedures. Hopefully, these recommendations will help drive funding back to where it's most appropriate:

- *Offer to remove ear wax in adults in primary care if it is contributing to hearing loss or other symptoms, or needs to be removed to examine the ear (or mould hearing aids)*
- *Use pretreatment wax softeners either immediately or for up to 5 days before ear irrigation*
 - *If unsuccessful repeat use of ear wax softeners, or instil water into the ear canal 15 minutes before repeating irrigation*
 - Lack of data means any difference between oil, sodium bicarbonate and even water is essentially unknown. As such NICE suggests all are appropriate.
 - While commonly recommended, there is a lack of data favouring longer courses of wax softener prior to ear irrigation, and authors felt that even on-the-day softening may be useful and prevent further delay on helping a patient's hearing loss
- *Consider ear irrigation using an electronic irrigator, microsuction or another method (e.g. manual removal with a probe). Do not off manual syringing.*
 - Manual syringing is now contraindicated due to the risk of trauma; electronic irrigation is much safer.
 - 1 low-quality study suggested clinically important benefit of ears drops + irrigation vs ear drops + home syringing (n=206, 63% vs 48% with no impacted wax at 1-2 weeks, NNT=~6) - although this data obviously shows that almost half have a good result with just home syringing - for a patient 'how to' guide [click here](#)
- *Advise adults not to insert small objects (e.g. cotton buds, Lego...) into the ear...*

NB Learning Points: Hearing Loss

- Sudden loss of hearing is a medical emergency - assess for sensorineural deficit and consider initiation of steroids in the community followed by routine ENT referral
- Patients with gradual hearing loss should be referred to audiology after wax and infection are excluded
- We should be proactively referring adults with LD and dementia for hearing assessment every 2 years
- Ear wax can be softened with any commonly used drops (or water) and irrigation undertaken on the same day without the need to wait for extended periods

NB Quality Improvement Ideas: Hearing Loss

- Audit your practice for patients coded with hearing loss
 - Was an acute cause excluded? Have they been offered a hearing assessment?
- If earwax removal is not done at your practice, could patients be offered advice on self-care - [click here](#) for a link to some useful guidance

KISS: Red Flags in Hearing Loss[NICE NG98](#), [BMJ 2018;361:k2219](#)■ **Immediate Referral** (<24hrs)

- Sudden hearing loss ($\leq 3/7$) within the past 30 days - but consider BJGP recommendations
- Acquired unilateral hearing loss and altered sensation/facial droop on the same side
- Otolgia + otorrhoea not responding to treatment in 72h in immuno-compromised

■ **Urgent referral**

- Sudden hearing more than 30 days ago or hearing rapidly worsens (over 4-90/7)
- Adults of Chinese or South-Asia family origin with hearing loss and middle ear effusion not associated with URTI
- Naso-pharyngeal tumours are common in these populations, the risk is >3% with these presenting features

KISS: Indications for Non-Immediate Referral

Consider referral to ENT, audiovestibular medicine or specialist audiology service for diagnostic investigation:

- Unilateral or asymmetric hearing loss as a primary concern
- Hearing loss that fluctuates and is not associated with an upper respiratory tract infection
- Hyperacusis causing significant distress and affects a person's day-to-day activities
- Persistent tinnitus that is unilateral, pulsatile, has significantly changed in nature or is causing distress
- Vertigo that has not fully resolved or is recurrent
- Hearing loss that is not age-related

Consider referral to ENT if after initial treatment for ear wax or acute infection:

- Tympanic membrane cannot be fully visualised due to obstruction
- Pain affecting the ear lasting >1/52 and not responding to 1st line treatment
- History of discharge for >1/52 and not responding to 1st-line treatment
- Abnormal appearance of the outer ear or the eardrum including:
 - Inflammation, polyp formation, perforated eardrum, abnormal bony or skin growths, swelling of the outer ear, blood in the ear canal
- Middle ear effusion in the absence of, or persisting after, acute upper respiratory tract infection

Screening for hearing loss

- Consider referring adults with diagnosed or suspected dementia or mild cognitive impairment to an audiology service every 2 years if not previously diagnosed with hearing loss.
- Consider referring people with a diagnosed learning disability to an audiology service when they transfer from child to adult services, then every 2 years.

VERTIGO

Vertigo has been a huge hot topic in the past year. It forms possibly the most relevant, important and surprising recommendations in NICE's recently published guidance on Suspected Neurological Conditions: Recognition and Referral ([NICE NG 127, July 2019](#)). However, demonstrating that not everyone always agrees, a very recent BMJ Practice paper ([BMJ 2019;366:l5215](#)) suggests a very different approach to the assessment of the vertiginous patient. Meanwhile, a further recent paper ([BMJ 2019;366:j4213](#)) counsels clinicians on a condition that is often forgotten and often missed...

1 Assessment of Vertigo [NICE NG 127](#)

Essentially a key message for us is: ***always think stroke in sudden onset vertigo, then seek to disprove it***. This is, of course, increasing likely with increasing age, and new-onset "labyrinthitis" in the elderly should be treated with suspicion. NICE has very helpfully provided a new framework for the assessment of sudden onset vertigo (also see Neurology Section).

KISS: Assessment of Sudden Onset Vertigo

- **Sudden onset dizziness + focal neurological deficit**
 - Think hypoglycaemia if diabetes, test and treat as appropriate
 - If PPV or postural hypotension is unlikely cause then assume posterior circulation stroke and refer immediately for assessment
- **Sudden onset vestibular syndrome** (vertigo, N&V, gait unsteadiness)
 - **Perform the HiNTs test** (if trained to do so)
 - **HiNTs test** - consists of 3 components, and all must be performed for it to be valid
 - **Head Impulse test** - an abnormal test is where there is a visible corrective saccade (drift of line of sight from a fixed point) when turning the head rapidly from the midline to one side - ***abnormal test suggests peripheral cause and is therefore reassuring***
 - Be very careful in patients with neck pathology, we want to avoid injury esp. arterial dissection and performing the test may be inappropriate for some
 - **Nystagmus** - an abnormal test is bi-directional, vertical or isolated rotational nystagmus, whereas ***uni-directional nystagmus, i.e. fast-beating to one side only regardless of looking left or right, suggests vestibular neuronitis*** or a rotational component with the head down is usually BPPV
 - To assess nystagmus properly ask the patient to look to the side themselves rather than fixing on your finger, as the latter can mask the sign
 - **Test of skew** - an abnormal test is deviation of the eye while being covered, visible when alternating cover to the other eye. Any vertical deviation is particularly suspicious.
 - So a ***positive head impulse test, uni-directional horizontal nystagmus and negative skew test is a negative HiNTs test, indicating the risk of a central cause is very low***
 - If HiNTs shows signs of stroke refer the patient immediately for further assessment
 - Refer immediately if unable to perform HiNTs and no other identifiable cause
 - For a useful detailed review of HiNTs [click here](#) and a video of the examinations [click here](#)
- **Sudden onset vertigo without imbalance or focal deficit**
 - NICE comments this is unlikely to be due to a serious neurological cause
- **Vertigo on head movement** (transient rotational vertigo)
 - **Perform the Hallpike manoeuvre to check for BPPV, if positive offer a "canolith re-positioning manoeuvre"** (e.g. Epley's), or refer for it if untrained or the patient has "unstable cervical spine disease"
 - NICE comments that BPPV is common after head injuries and labyrinthitis
 - For a video demonstrating the Dix-Hallpike manoeuvre [click here](#)
- **Vestibular migraine**
 - May cause recurrent episodes of dizziness lasting 5m > 72h in patients with recurrent headaches.
 - **See below for an in-depth review of vestibular migraine from the [BMJ 2019;366:j4213](#)**

KISS: Assessment of Sudden Onset Vertigo continued

- **Dizziness and altered consciousness**
 - Recurrent episodes of fixed pattern dizziness and change in consciousness, think epilepsy and refer for further assessment.
- **Recurrent dizziness and functional neurological disorders**
 - For some patients no cause can be identified for their neurological issues, dizziness may be a feature for some, which can get worse at times of stress, and in the absence of new signs doesn't require re-referral.

A key recommendation in this guideline is ***the use of the HINTS test to differentiate between peripheral and central causes in sudden onset vestibular syndrome*** (vertigo, nausea/vomiting, gait unsteadiness,). This consists of 3 tests: 1) head impulse test (sometimes called the 'head thrust'), 2) check for nystagmus and 3) test of skew. See the KISS for details of each component. ***All 3 must be conducted and considered together for a valid HINTS test.***

The HINTS test is an impressive tool. Data shows higher sensitivity and specificity (97% and 99% respectively) for the detection of central causes of vertigo than MRI and the ABCD2 score. (There is also a HINTS-plus which adds an assessment of unilateral hearing loss which increases the sensitivity to 99% with a sensitivity of 97%.) For an in-depth review [click here](#).

However, ***there are some caveats***. These ***impressive figures come from secondary care where patients have had an assessment by a neuro-ophthalmologist***, highly trained in observing subtle eye signs. There is a lack of data for its use in the primary care setting. It is also only validated for use within 72h of onset of vertigo and in adults.

NICE recommend that patients with acute onset sudden-onset acute vestibular syndrome in whom BPPV and postural hypotension do not account for the presentation, ***refer immediately for the exclusion of stroke if a healthcare professional without training and experience in the use of the HINTS test is not available.***

2 Acute Vertigo [BMJ 2019;366:I5215](#)

This paper authored by 2 specialists and a GP has a slightly different take on the assessment of acute vertigo in general practice.

Central to assessment remains a thorough history with specific consideration of red flags (e.g. focal neurology, diplopia) the presence of which merits immediate referral. It encourages us to consider vascular risk factors (e.g. age >60, DM, HT, dyslipidaemia, smoking Hx) and refer patients in whom they are present with prolonged continuous vertigo.

The paper then ***recommends a focused examination, including assessment of gait, neurology and hearing. It recommends performing the Dix-Hallpike manoeuvre in ALL patients presenting with vertigo, dizziness, falls or imbalance***, except where vertebral dissection is suspected or the patient has severe cervical spine instability.

The rationale is that the test can provide a definitive diagnosis of BPPV, which it considers the most common cause of vertigo (perhaps with the exception of vestibular migraine) and effectively excludes a central cause. The sensitivity of the test is 48-88%, so a negative test cannot exclude the condition. The figures likely reflect that this is a paroxysmal condition and patients may present in an asymptomatic phase, during which you may not elicit nystagmus.

There are a number of very useful points regarding BPPV in this paper:

- BPPV does not always present with typical positional vertigo, particularly in older adults, and so the Dix-Hallpike test may be the only way to identify the condition.
- BPPV causes rotational nystagmus with the head down during the Dix-Hallpike manoeuvre - this is a normal, confirmatory finding compared to rotational nystagmus during the HINTS test.
- A variant of the Dix-Hallpike can be used for where access around a bed (e.g. stuck in the corner of your room) is limited which avoids the need for having the head below the bed end. Easy to perform, this is best demonstrated in the [excellent video on the BMJ webpage](#), find it at 7min30.
- Linked with the above alternative manoeuvre is the alternative treatment - the Semont manoeuvre. This is demonstrated in the above video at 9min30.

But, isn't labyrinthitis the most common cause of vertigo...?

Well, this is somewhat complex. The data quoted in the [helpful visual summary](#) gives an incidence of BPPV of 600 per 100000 but of vestibular neuritis only 4 per 100000. And yet every GP will feel they see more cases of 'labyrinthitis' than BPPV in practice. There is clearly a discrepancy between epidemiological data and general practice. Perhaps this is no surprise quoted data is often from ENT clinic attendees, while we know that in reality, we refer very few patients with vertigo to specialist care.

There is further confusion over the definitions for vertigo disorders. In fact, there are many more causes than we might usually have on our differential ([click here for the very useful www.balancedisorderspectrum.info website detailing the very wide range](#)). While we might describe labyrinthitis in a patient with acute vertigo and the absence of other causes, strictly speaking, this is likely to be vestibular neuronitis resulting from (probable) viral inflammation of the vestibular nerve. Labyrinthitis has hearing loss in addition to the vertigo and is driven by inflammation again but of any causes, not just infection. Then there's vestibular hypofunction, auto-immune inner ear disease, superior canal dehiscence syndrome... And that's before we start on the extra-aural...

3 Vestibular migraine: easily missed [BMJ 2019;366:j4213](#)

Give NICE's recent guidance on neurological disease, very helpful but lacking in detail on vestibular migraine, this was a very timely published in the BMJ ([2019;366:j4213](#)).

The main messages are:

- ***Vestibular migraine is common*** - data suggests it's twice as common as BPPV - but infrequently diagnosed
- ***It can be hard to diagnose*** (although mostly appropriate diagnosis fails due to lack of awareness) but should be considered in people with recurrent vestibular symptoms and a history of migraines
- ***Vestibular suppressants are largely ineffective but migraine treatments are highly effective***

► Why does it occur?

Only recognised as a distinct clinical entity in 2012, most of us won't consider vestibular migraine very often. However data suggests the prevalence is high - a US survey of the general population showed it as high as 2.7%, almost double rates of BPPV and 5-10x more common than Meniere's. Further data shows only 10-21% of patients with vestibular migraine receive the correct diagnosis, with misdiagnoses ranging from anaemia to somatisation. This is clearly very unfortunate as these patients experience reduced quality of life, more missed days off work or school, and higher rates of depression/anxiety due to the condition.

There is a strong association between migraine and vestibular symptoms - both affect similar areas in the brain. As with stand-alone migraine, the pathophysiology is poorly understood. One interesting feature in patients with migraine, and even more so with vestibular migraine, is a poor tolerance for motion - sickness and sensitivity are reported in 9 out 10, and while not in the ICHD-3 criteria are a useful diagnostic marker.

KISS: ICHD-3 Diagnostic Criteria for Vestibular Migraine

A. ≥ 5 fulfilling criteria C and D

B. A current or past history of migraine with or without aura

C. Vestibular symptoms of moderate or severe intensity lasting 5 minutes to 72 hours

- Symptoms may be spontaneous, postural after a head position change, visually induced by moving stimulus, or induced by head motion

D. At least half of episodes are associated with ≥ 1 of 3 migrainous features

- Headache with 2+ of unilateral location, pulsating, moderate/severe intensity, aggravated by routine physical activity
- Photophobia and phonophobia
- Visual aura

E. Not better accounted for by another ICHD-3 diagnosis or other vestibular disorder

► How does it present?

Consider vestibular migraine in people with 1) recurrent vestibular symptoms, 2) a history of migraine, 3) a temporal association between the two for some attacks and 4) absence of other likely causes.

As ever, we must exclude red flags for vertigo (see above) and consider alternative benign causes of vertigo (and seek to exclude them if the history is suggestive). Having done this the key is the temporal association between migraine symptoms and vestibular symptoms. Of note:

- 74% of patients report migraine onset preceding vestibular symptoms; therefore this tells us that 1 in 4 do not get this association
- Just as with migraine, presenting symptoms can be highly variable - vestibular symptoms do not have to be frank vertigo but may be more subtle sensations of movement (that 'just got off the boat' feeling) or intolerance to head movements
- Examination is usually normal between episodes, although positional nystagmus may still be seen in 41% even when asymptomatic. In acute episodes, 70% will have nystagmus and 70% a +ve Romberg test.

The condition can co-exist with other vestibular disorders, particularly Meniere's disease. 45% of those with MD have migraine features during an attack, but most patients with VM do not demonstrate the prominent aural features seen with MD. BPPV should be readily distinguishable on history, but consider the possibility of VM where patients fail to respond to canolith re-positioning manoeuvres, especially if young.

► Do patients require neuroimaging?

Short answer: no. In patients with classical features of vestibular migraine and an absence of red flags or likely alternative diagnosis, imaging is not required. Any signs of central disease, especially with CVD risk factors, merits urgent imaging. Other vestibular neurophysiology tests are largely unhelpful and not usually recommended.

► How can we manage vestibular migraine?

Data is limited specifically for vestibular migraine, but management is largely the same as for simple migraines, e.g. avoidance of triggers, acute management (zolmitriptan and rizatriptan show some reduction in vestibular symptoms in small RCTs), anti-emetics if required for nausea (but used sparingly to avoid longer-term vestibular suppression), and then prophylaxis.

Prophylaxis options include tricyclics, venlafaxine, antihypertensives (beta-blockers and candesartan), flunarizine (helpful in high-frequency vertigo attacks), and anticonvulsants (e.g. topiramate and Na valproate).

Vestibular rehabilitation is helpful as an adjunct to medication in those with chronic symptoms. This aims to habituate the brain to visual motion stimuli and improve postural stability.

4 **Treatment of chronic vertigo** [BMJ 2019;367:l5922](#), [BMJ 2017;358:j3727](#)

While there are clear and simple treatments for BPPV and vestibular migraine, there is has uncertainty over the years on how to manage chronic vertigo.

Chronic vertigo is generally defined as symptoms persisting for > 1 month and is common - 30-40% of patients with vestibular neuronitis have symptoms at 6 months.

Betahistine is still widely prescribed for chronic vertigo yet Cochrane reviews now suggest [only weak evidence for the use of betahistine](#). So, what are the alternatives? There is growing evidence to support vestibular rehabilitation. A 2017 BMJ paper concludes that **all patients with chronic vertigo should be offered vestibular rehabilitation**. But, of course, there are always the same old issues of cost and accessibility.

A new Dutch primary care RCT published in the BMJ ([BMJ 2019;367:l5922](#)) showing that in adults >50yo with chronic vestibular syndrome a 6-week internet-based vestibular rehab intervention had clinically significant improvements in vertigo at 3 and 6 months compared to usual care from the GP and was at least as effective as a 'blended programme' which also contained face to face physiotherapy. All methods were found to be safe.

KISS: Vestibular Rehabilitation

- Vestibular rehabilitation is an exercise-based treatment consisting of varied eye, head and body movements designed to stimulate the vestibular system and optimise vestibular compensation
- It involves daily low-intensity exercises for 6-12 weeks
- Good news is that it is now widely available for free via an internet-based programme - [click here for link](#); patients can also access booklet versions of vestibular rehab - [click here](#)

NB Learning points And Quality Improvement Ideas: Vertigo

- NICE have provided clear guidance on assessing patients with sudden onset vertigo - always think stroke and then seek to exclude it using the HiNTs tests - head impulse test, nystagmus, test of skew.
- Consider a Dix-Hallpike test for all vertiginous patients to rule in BPPV. If +ve continue and perform the Epley or Sermont manoeuvre to treat the condition.
- Vestibular migraine is common and underdiagnosed. Consider in patients with recurrent vertigo and a history of migraine. Successful treatment targets migraine, not vestibular symptoms.
- Acute treatment of vestibular neuronitis - prochlorperazine or antihistamines can be used but only for 3 days; ongoing use suppresses vestibular compensation and slows recovery
- Vestibular rehabilitation should be offered to all patients with chronic vertigo - [click here for free internet-based programme](#)
- **Quality Improvement Ideas:** perform a case review of patients with vestibular codes. How many also have migraine? Has a link been considered? Are patients receiving vestibular suppressants for their symptoms?



TEMPOROMANDIBULAR DISORDERS

According to a new BJGP review ([BJGP 2019;69:523](#)) temporomandibular disorders (TMD) affect 60-70% of the population at some point, although only a minority (5%) will seek help. [Best Practice](#) was also updated on TMD at the end of 2019 providing further insights into the management of this often complex, heterogeneous condition.

► *What causes TMD?*

TMD can be divided into three types (see KISS). The **most common is myofascial pain and dysfunction**. Here the muscles of mastication become painful and tender, often due to bruxism (teeth grinding) or clenching. This can prevent maximal opening of the mouth, cause temporal headaches, and bruxism may lead to tooth pain and wearing and is associated with cyclical pain due to occurring in the nighttime. Patients will often report more diffuse pain as well, usually intermittent and in several areas around the head/neck.

Internal derangement occurs when the articular disc becomes dislocated from its ideal position in the fossa, then relocates during movement of the mandible. This results in clicking and pain and, where relocation is prevented, locking of the jaw.

In the older population osteoarthritis is common. Crepitus is described in Best Practice as pathognomonic for OA, and this coupled with pain is highly characteristic.

Depression and anxiety can often drive pain and prevent improvements, so worth exploring early in the condition.

KISS: Causes of TMD		
BJGP 2019;69:523		
Myofascial pain and dysfunction	TMJ Functional Derangement	TMJ Degenerative / Inflammatory Joint Disease
Fibromyalgia	Internal derangement - disc displacement	Osteoarthritis
Neuropathic pain	Hypermobility disorders, e.g. dislocations	Rheumatoid Arthritis
Chronic Pain Syndrome	Hypomobility disorders, e.g. Ankylosis	Psoriatic and juvenile arthritis
Myositis		

► *How can we assess patients with TMD?*

- **Examine the head and neck thoroughly**
 - Exclude neurology of the neck and cranial nerves
 - Observe the opening and closing of the mouth - monitor for limited movement, asymmetry, deviation
 - Palpate the condyle of the TMJ - place index and middle fingers anterior to the tragus and ask the patient to slowly open and close the jaw - feel for crepitus, clicks or abnormal movement
 - Palpate the muscles of mastication - ask the patient to gently clench their jaw and feel the temporalis and masseter muscles
- **Exclude red flags**
 - Potential causes are diffuse - consider possible neoplastic (primary and mets, including nasopharyngeal Ca), inflammatory (e.g. giant cell arteritis esp. in >50yo, RA, etc.), infective (esp. with lymphadenopathy), intra-cranial, trauma causes, etc.

► How can we manage patients with TMD?

The good news for patients is that 68-95% of patients with TMD will improve with conservative measures and so early self-management is key. This includes:

- **Limiting mandibular function** - advise a soft diet for 7-14 days
- **Avoid parafunctional activities** - e.g. chewing gum, biting nails
- **Massage tender muscles and regular hot compresses**
- **Try to reduce stress** - leads to less clenching. Data shows CBT can be helpful for TMJ pain. Also, some limited data supports acupuncture in this group.
- **Simple analgesia**, e.g. paracetamol, ibuprofen
 - Best Practice highlights that NSAIDs have a limited role outside of pain relief as the chronic facial pain is not due to inflammation. It suggests a role for *up to 10 days of diazepam* (some of us may feel a little uncomfortable about that), while [Clinical Knowledge Summaries](#) suggests in *chronic pain trialling neuropathic analgesics*, e.g. amitriptyline off-label (an approach favoured by NB's local specialists)
- **Consider signposting patients to their dentist** - if bruxism is suspected they can provide an 'occlusal splint' (mouthguard)

Conspicuous by its absence is a recommendation for jaw exercises in both the BJGP and Best Practice reviews. Perhaps this is explained by the lack of supporting evidence (although not evidence of lack), and, as the US [TMJ Association](#) puts it: "*Remember less is best!*" It seems that the rest of the joint in the early stages is favoured over more active treatments these days.

At the end of the paper, it acknowledges that 'home physiotherapeutic exercises' would be part of treatment, and the British Association of Oral Surgeons has a [patient information leaflet](#) which recommends jaw joint and muscle exercise to relax the jaw. More specific advice on how to perform the exercises is given in the [PILs from Oxford University Hospitals](#).

Where symptoms persist (Best Practice suggests for >6/12) referral to the oral/maxillo-facial team is recommended. Prompt referral is merited where patients have a history of inflammatory joint disease, progressive difficulty opening the mouth, inability to eat a normal diet or recurrent dislocation of the TMJ.

Specialist treatments include arthrocentesis (joint washout), condylotomy (to re-position the joint) and, apparently, total TMJ replacement (who knew). While there is a paucity of data supporting these, anecdotal evidence suggests each may be beneficial.

NB Learning Points & Quality Improvement Ideas: Temporomandibular Disorders

- TMD affects a significant proportion of the population at some point in their lives. Early self-management leads to resolution in two-thirds or more.
- Examine the patient, exclude abnormal neurology and other red flags.
- Where teeth grinding is suspected, signposting to their dentist is advisable for a mouthguard.
- For patient information:
 - General information on TMD: [British Association of Oral Surgeons](#) and [TMJ Association](#) (USA)
 - For conservative treatment including psychological aspects [click here](#)
 - For TMJ exercises [click here](#)
- **Quality Improvement Idea**
 - Perform a simple audit of patients at your practice with TMD. What percentage have chronic pain lasting >6/12? How many of this group have been offered referral to a specialist?

KISS: Management of Temporomandibular Disorders (TMDs)

[CKS December 2021](#) & [British Journal of General Practice 2019; 69 \(687\)](#)

■ Background:

• Symptoms include:

- Focal pain and tenderness in the muscles of mastication, but can present with diffuse pain in head/neck; the commonest site of pain is anterior to the tragus of the ear and extends down the border of the mandible and up into the temporal region.
- Limited mouth opening and 'locking' of the jaw, with clicking/crepitus in TMJ.
- Painful teeth.

• Signs suggestive of TMD include:

- Instability/crepitus while palpating over the TMJ and opening the jaw.
- Tenderness over the muscles of mastication with teeth clenched.
- Assess also for facial/neck asymmetry/swelling, cranial nerve signs and lymphadenopathy which may indicate an alternative cause.

• Serious causes of orofacial pain to consider include head and neck malignancies and infection

- The **most important differential is giant cell arteritis**; jaw claudication can mimic TMD pain - consider GCA in the >50s if associated with headache/scalp tenderness and systemic symptoms.

• Causes of TMDs:

- Myofascial pain/dysfunction (most common cause).
- TMJ derangement - e.g. disc displacement, hypermobility disorders/dislocation.
- Degenerative/inflammatory joint disease - OA, RA, psoriatic arthritis.

■ General management:

- Encourage self management (see PILs below) - 68-95% respond to conservative therapy.
- Reassurance about benign nature; TMDs rarely progressive, symptoms can fluctuate but usually improve.
- Eat a soft diet for 7-14 days and rest jaw if acute pain.
- Avoid **activities that may exacerbate symptoms**, e.g. wide yawning, teeth grinding or jaw clenching, chewing gum or pencils, and nail-biting.
- Local measures e.g. ice pack, warm flannel, heat pad, or massaging affected muscles.
- Identify and address stress, anxiety or depression.
- Advise dental review if poor dental health, suspected malocclusion or dental pathology, or for **consideration of an occlusal splint** (usually worn at night; particularly **useful for people who grind or clench their teeth**).
- Introducing exercises after the initial pain has started to improve may be of benefit - [click here for PILs on home jaw exercises](#)

■ Medication:

- Usually, self management with paracetamol or NSAID is sufficient.
- If symptoms are acute and severe consider diazepam e.g. 2mg TDS max 2 weeks.
- If symptoms are chronic consider a neuropathic agent e.g. amitriptyline, gabapentin (but note gabapentin recommendation based on expert consensus with little evidence, and known risks).

■ Patient information:

- Excellent leaflet from British Association of Oral Surgeons - [click here](#)
- Good website with PILs on orofacial pain inc. TMDs - [click here](#)

■ Refer (most cases to maxillofacial surgeons) if:

- Symptoms persistent or worsening despite primary care management for 3 months.
- Marked psychological distress and/or occlusal preoccupation (persistent hyperawareness or hypervigilance of their bite).

ACNE VULGARIS

Recent research on this common and frequently distressing condition has focussed on the increasing rates of **antibiotic resistance** both with oral and topical antibiotics. This was the subject of a BJGP editorial back in 2018 [BJGP2018;68:64](#) and hot off the press we have **new UK guidelines in June 2021** from the [Primary Care Dermatology Society \(PCDS\) acne guidelines 2021](#) and [NICE NG198](#) which propose what for many GPs will be a **significant change in practice to rationalise and reduce antibiotic use**. The NICE guidelines were helpfully summarised in the [BMJ 2021;374:n1800](#) and [BJGP 2021; 71 \(713\): 568-570](#). Whilst there are inevitable subtle differences between the NICE and PCDS guidelines they are broadly very similar.

► **Initial management and antibiotic use for acne.** Good infographic from [BMJ here](#)

By way of background (which I'm sure will chime with many in General Practice), interesting recent qualitative research from [BJGP 2021; 71 \(702\)](#) investigated GPs attitudes towards oral antibiotics in acne, given that prolonged courses of oral antibiotics are still commonly prescribed, despite [NICE/PHE guidance dating back to 2017](#) that courses of oral antibiotics should generally be limited to 3 months. The research found a number of ongoing challenges to addressing oral antibiotic use in primary care for acne:

- GPs generally perceived topical treatments to be less effective than oral antibiotics or felt pressure to prescribe oral antibiotics from patients who perceived topical treatments to be less effective, as well as concern with side effects of the topical treatments.
- Whereas GPs have a good handle on antibiotic stewardship with acute infections, many do not view the use of oral antibiotics in acne as a particular concern.
- Some GPs were unfamiliar with the guidance that oral antibiotics should only be used for 3 months.

Research [Lancet ID 2016](#) reports **resistance rates of >50%** (and 45% to 91% for topical antibiotics [BJGP2018;68:64](#)) and the Lancet makes the recommendation that antibiotics should only be for short-term use and always in combination with benzoyl peroxide (BPO) or a topical retinoid to limit resistance.

As a result of accumulating evidence, the [BJGP2018;68:64](#) together with the [NICE NG198](#) and [PCDS 2021 Guidelines](#) advise that **oral and topical antibiotics should never be used as monotherapy and should generally only be used for relatively short periods of 3 months** before reverting to non-antibiotic maintenance therapy. The evidence suggests that there is little benefit in using antibiotics for > 3 months and prolonged use increases resistance. The [NICE 2021](#) and [PCDS 2021 Guidelines](#) advise:

- Antibiotics still have a role in inflammatory papular and pustular acne, particularly if widespread, but always in conjunction with topical benzoyl peroxide (BPO) or retinoid (ideally in combination).
- In mild to moderate papular/pustular acne **topical adapalene + BPO (Epiduo®) is first-line**, with second-line/other options being topical clindamycin combined with BPO (Duac®) or tretinoin (Tretin-A®).
- In more extensive papular/pustular acne or if not responding to topical treatment, use **systemic oral antibiotics in combination with a topical agent**:
 - Preferably use oral antibiotics with combined topical adapalene/BPO - If not tolerated use single-agent BPO or topical adapalene (NICE also recommend topical azelaic acid as an option if adapalene/BPO contra-indicated or not tolerated but evidence base not so good).
 - **Lymecycline 408mg OD or doxycycline 100mg OD are first-line choices** (both C/I in pregnancy and those <12) - PCDS suggest both can be increased to BD if partial response.
 - **Second line** (may depend on local guidance/resistance rates) macrolide e.g. clarithromycin 250-500mg BD or trimethoprim; PCDS recommend clarithromycin over erythromycin due to lower resistance rates and also note that trimethoprim can rarely cause severe skin reactions and agranulocytosis and that FBC monitoring is recommended by BNF if using trimethoprim long term.
- **Antibiotics should be reviewed and generally stopped at 3 months** and the **patient should remain on their topical agent**. If patients relapse quickly or there is only a partial response a longer 6m course can be considered. NICE recommend courses >6 m '*should only be used in exceptional circumstances*'.
- NICE are clear that **we should NOT** use oral or topical antibiotics as monotherapy (they should ALWAYS be used with a non-antibiotic topical agent) and **we should NOT** use oral and topical antibiotics together.

► *What about general skincare advice?*

This was usefully summarised in both the [BMJ 2021;374:n1800](#) and [BJGP 2021; 71 \(713\): 568-570](#), as we are often asked for skincare advice, and this is an important topic for people with acne:

- Advise people with acne to use a non-alkaline (skin pH neutral or slightly acidic) **synthetic detergent (syndet) cleansing product twice daily on acne-prone skin**. Syndet is a blend of synthetic surfactants, formulated to have a neutral to slightly acidic pH and high fatty acid concentration, which is less irritant than traditional soap, helps hydrate the skin, and rinses off easily. Syndets are widely available in solid and liquid form as skin-cleansing products.
- Advise people with acne who use skincare products (for example, moisturisers) and sunscreens to avoid oil-based and comedogenic preparations.
- Advise people with acne who use make-up to avoid oil-based and comedogenic products, and to remove make-up at the end of the day.
- Advise people that persistent **picking or scratching of acne lesions can increase the risk of scarring**.

► *What about hormonal treatment and acne?*

For women, we have the additional option of hormonal treatment. The [PCDS 2021 Guideline](#) advises that:

- Hormonal treatments can be used on their own, or in combination with treatments above - may be particularly useful for women with PCOS; consider tricycling for 9 weeks then 4-7 day break.
- Dianette has historically been used for acne, but the PCDS now suggest that alternative COCPs with lower oestrogen content (20mcg) have a better safety pile and may benefit some women.
- Eloine or Mercilon can be considered 1st line, or alternatively Dianette if not C/I.
- NICE also advise a COCP as an option for women, and to avoid the progesterone-only pill; NICE recommend co-cyprindiol for those with PCOS.

► *What about pregnancy?*

One of the snags with a number of the treatments above is that they are contra-indicated in pregnancy, which means we need to be **careful when counselling women of childbearing age**. This issue is specifically addressed by NICE who recommend we **advise women of childbearing age that topical retinoids and oral tetracyclines are contraindicated in pregnancy** and that **they will need to use effective contraception if considering these options**. If considering pregnancy or the woman is pregnant options include:

- Topical BPO
- Topical erythromycin
- Oral erythromycin (with topical BPO) if benefits outweigh risks (e.g. scarring acne).

► *What about acne in pigmented skin?*

The [PCDS 2021 Guideline](#) reminds us that while acne is no less or more common in people with pigmented skin, there are additional factors we need to be aware of:

- Post-inflammatory hyperpigmentation can be significant and it can persist for months or years, so earlier more aggressive treatment is advocated including early referral for consideration of isotretinoin.
- Topical retinoids, BPO and azelaic acid play an important role but they can cause dryness and irritation which in pigmented skin can lead to further pigmentation. Therefore these agents should be used for shorter periods of time and less frequently (e.g. every second or third day). Used in this way, as well as treating comedones they may help to treat hyperpigmentation.
- Azelaic acid may be the best topical choice to reduce hyperpigmentation.
- Be aware that some cultural practices may cause secondary acne e.g. hair oil causing forehead acne, or skin-lightening products causing steroid-induced acne.

► ***Don't forget the mental health impact***

The NICE guideline together with the BMJ and BJGP reviews above **all highlight the known effects that acne can have on mental health**, with [NICE specifically recommending](#) we consider mental health support for people presenting with acne. It is likely to be at least part of the reason we see so many people with acne (currently thought to be about 3.5 million GP consultations per year). How we do this will depend on the severity of the mental health impact and local resources.

► ***When should I refer?***

Although we can manage the majority of patients with acne in primary care, there are a number of circumstances when we should consider referral:

- All patients with Nodulo-cystic acne or Acne conglobata (a severe form of nodulocystic acne with interconnecting sinuses and abscesses).
- Mild/moderate acne not responding to 2 completed courses of treatment.
- Moderate/severe acne not responding to treatment including an oral antibiotic.
- Severe and scarring acne - refer early whilst initiating treatment in primary care.
- Acne with persisting pigmentary changes (as above more common for people with coloured skin so consider early referral for them).
- Acne of any severity that is causing/contributing to severe psychological distress (refer both to a dermatologist and mental health services).

► ***What anabolic steroid misuse?***

The PCDS review a number of potential triggers for acne, but one particular one worth bearing in mind is the increasing use of anabolic steroids, particularly in young men to increase muscle bulk. The [latest statistics from the Home Office](#) suggests a significant increase in anabolic steroid use in males aged 16-24 and is now estimated to be used by 0.4% of that group. This is due to social trends increasing the prevalence of [muscle dysmorphia](#) ('bigorexia') which may present for the first time to the GP with secondary acne so is well worth asking about. There is a useful page on [anabolic steroid misuse on NHS Choices](#).

► **Other questions/practical points:**

- ***Why is combination adapalene/BPO being recommended first line?*** There are a number of strands to this question:
 - There is now strong evidence that prolonged use of oral (and topical) antibiotics in acne is contributing to antibiotic resistance so all guidance now recommends that for mild/moderate acne non-antibiotic topical treatments are used first line.
 - Despite our anecdotal misgivings on topical treatments (as highlighted in the qualitative evidence above), there is good evidence that topical treatments are effective (if used properly - see below).
 - NICE cite evidence that combined topical treatments are more effective than single agents (hence the move to use combination adapalene/BPO rather than stand-alone agents).
- ***How long should we trial treatments?*** For all treatments (including non-antibiotic and topical treatments) NICE advise a 12-week trial - we need to warn patients whatever treatment is trialled it will take at least 6-8 weeks before effects may be seen.
- ***Why no oxytetracycline?*** The NICE/PHE guidance on antibiotic prescribing acne section dates back to 2017, where oxytetracycline is still recommended as an option, but both the new NICE acne guidance and PCDS do not recommend oxytetracycline now due to concerns over concordance as a) it cannot be taken with food and b) it is BD dosing.
- ***But my patients say they can't tolerate the non-antibiotic topical treatments...*** this is a common issue and one highlighted by the qualitative evidence above and is a reason why these treatments are under-used. NICE recommends that to reduce skin irritation patients start with alternate day applications, or short contact applications (e.g. washing off after 1 hour) then progress to standard application.
- ***For acne that is mainly comedonal*** (i.e. no papules/pustules) the PCDS recommend topical retinoid, either alone or in combination with BPO.

NB Learning Points & Quality Improvement Ideas: Acne Vulgaris

- Don't forget to give **general skincare advice**:
 - Avoid oil-based and comedogenic skincare, sun-cream and make-up products.
 - Remove makeup at the end of the day.
 - Use syndet cleansing products twice daily.
- For mild/moderate papular/pustular acne **first line treatment should be topical combination adapalene/BPO** (Epiduo ®).
- If using topical antibiotics clindamycin is recommended first line and should always be used as fixed-dose combined treatment with either BPO or tretinoin.
- If using **oral antibiotics always use a non-antibiotic topical agent as well** to reduce antibiotic resistance (preferably Epiduo ®, or if not tolerated single-agent BPO or adapalene).
- **Oral antibiotics should only be used for 3 months in most cases** - there is little evidence to support use beyond this; when stopping oral antibiotics continue with a non-antibiotic topical agent.
- The PCDS now recommend Eloine or Mercilon as 1st line hormonal treatments due to the better safety profile, although Dinaette can still be considered; consider tricycling for 3 months with 4-7 day break.
- Be aware of the increased risk of post-inflammatory hyperpigmentation in **people with pigmented skin** - treat more aggressively and lower referral threshold.
- Ask muscular young men and teenage boys about anabolic steroid misuse.

Quality improvement ideas:

- Remember to **direct all your patients with acne to the excellent information from BAD** ([click here](#)) - this gives a very good overview of acne and treatment options, as well as a very good section on self-care.
- Practice **audit of antibiotic use (both topical and oral) in acne**:
 - Review any patients who have been on antibiotics >3 months and aim to stop.
 - Are all patients having concurrent topical treatment with BPO, adapalene, or Epiduo ® combination?

KISS: Acne Vulgaris

Primary Care Dermatology Society 2021 & BJGP2018;68:64 & NICE NG198 2021

	Mild	Moderate	Severe
Description Click here for images	Comedonal acne - open comedones (blackheads) and closed comedones (whiteheads)	Inflammatory papules and pustules and may be more widespread distribution	More severe and widespread papules/pustules; may have nodules or cysts; scarring or hyperpigmentation may be present
First line Give info, including on skin-care Click here for patient information	► Topical retinoid +/- benzoyl peroxide (BPO); options include: - Adapalene - Adapalene + BPO (Epiduo®) - Isotretinoin Avoid topical retinoids in pregnancy	► 1st line: Combination topical agent: Topical adapalene + BPO (Epiduo®) ► 2nd line: Fixed-dose topical clindamycin combined with BPO or tretinoin. Stop antibiotic after 3 months & continue BPO +/- topical retinoid ► Consider a COCP for women - Eloine or Mercilon 1st line; alternatively, Dianette if not C/I; consider tricycling with a 4-7 day break after 9 weeks	REFER all for specialist assessment and consideration of oral isotretinoin In meantime, commence treatment with oral antibiotic and BPO + topical retinoid
Second line (or more widely distributed)		CONTINUE topical BPO/retinoid (ideally combined) and add oral antibiotic for 3 months then step down to topical treatment alone (Do NOT use topical/oral abx together): ► 1st line: lymecycline 408mg OD or doxycycline 100mg OD but can double both to BD if partial response ► 2nd line: Clarithromycin 250-500mg BD (but check local guidance/resistance patterns) ► Children <12: Clarithromycin ► Pregnancy: Topical BPO or erythromycin; only consider oral erythromycin if benefits outweigh risks e.g. scarring acne	► Consider a COCP for women (see above) ► Acne in skin of colour: post-inflammatory hyperpigmentation can be more significant and persistent - more aggressive treatment and earlier referral for isotretinoin recommended

Refer:

- Severe acne (especially those with nodulocystic acne) - start treatment and refer early.
- Moderate acne not responding adequately to treatment that has included an oral antibiotic, or that is starting to scar or cause hyperpigmentation (esp. those with pigmented skin - consider early referral).
- Patients with associated severe psychological distress, irrespective of severity of acne.

HIDRADENITIS SUPPURATIVA

Hidradenitis suppurativa has been the subject of recent papers in the BJGP [BJGP2023;73:43](#) and the BMJ [BMJ 2022;379:e068383](#). It is a chronic relapsing and remitting skin condition which can have a profoundly negative effect on quality of life and yet is 'easily missed' as a diagnosis in primary care, as patients are frequently misdiagnosed and mismanaged as having 'recurrent boils'. Late diagnosis is common with an average time from onset to diagnosis of 7 years, and the longer it remains untreated the more challenging it is to manage making the early diagnosis a key priority.

It is a relatively common condition. It affects about 1% population and typically presents between the ages of 20-40 years. Smoking is a significant risk factor and genetics also play a role as 40% of patients have a family history. Women are three times more likely to be affected than men. *So, how do we recognise and treat this condition?*

► *What is Hidradenitis suppurativa?*

Hidradenitis suppurativa affects the apocrine glands in the axillae, groin, perianal, perineal and inframammary skin. Follicular plugging obstructs the glands leading to the formation of inflammatory nodules. In more severe disease deep abscess form and may join under the skin with tunnels or form drawing sinuses. Over time band-like scars can form. The pathophysiology is not completely understood. It is **not related to poor hygiene nor is it contagious**. This should be discussed with patients as there can be a significant stigma attached.

Diagnosis is based on the clinical triad of:

- **Skin changes**
 - deep-seated painful nodules, abscesses, tunnelling, comedones and rope-like or tombstone scarring
- **Location of the lesions**
 - axilla, groin, chest, abdominal folds, perineum
- **Duration**
 - lesions are chronic with regular relapses, often at previous sites and ≥ 2 lesions over the last 6 months

There is no diagnostic test. Patients are likely to have seen 3-5 clinicians before receiving their diagnosis. Common misdiagnoses include acne, folliculitis and ingrown hairs. *We should have a high index of suspicion in patients who have been treated for recurrent 'boils' or infections previously.*

The pain caused by HS can be very severe. Over time neuropathic pain can develop that is particularly challenging to treat. Patients are at increased risk of substance use disorder. It is hypothesised this may be due to attempts to self-medicate. Therefore pain management is an important aspect of management.

► *Psychosocial aspects*

A patient affected by HA was involved in the production of a [recent BMJ article](#) and writes about her experience. She highlights her difficulties with body image and shame due to open wounds and scarring. She attributes the breakdown of relationships to delays in diagnosis. She writes 'the emotional and psychological impact hurt more than the physical pain of the condition itself'. This underscores the importance of timely diagnosis and specifically asking about psychosocial factors a part of reviews. Anxiety and depression are common in HA and it is also associated with an increased risk of suicide.

Common emotional effects of HS include embarrassment over the visible signs of illness such as wounds, scars and odour from discharging sinuses. People may miss work or school opportunities. Additionally, some find HS a barrier to physical intimacy and sexual relationships. There can be difficulties in planning activities due to the unpredictability of flares and times when movement is restricted due to pain associated with abscesses.

In some cases, people may benefit from input from a clinical psychologist. Signposting to support groups can also be of benefit. [Changing Faces](#) is a charity for everyone with a scar, mark or condition on their face or body that makes them look different. The website contacts advice and guidance on mental health and well-being, coping with people's reactions and relationships and social life. The Hidradenitis Suppurativa Trust is a condition-specific support group accessed through Facebook.

► *Physical comorbidities*

Hidradenitis suppurativa is associated with a number of inflammatory and metabolic conditions. One of the key educational drives being the recent BJGP paper was for improved screening for comorbidities in people diagnosed with HS. They recommend annual review of the following:

- **Chronic inflammatory disease:**
 - Spondyloarthritis: patients have a 65% increased risk of ankylosing spondylitis, 44% increased risk psoriatic arthritis and 16% increased risk rheumatoid arthritis. Ask about joint symptoms.
 - IBD: in particular consider a Crohn's disease in patients presenting with perianal HS. Consider faecal calprotectin.
- **Metabolic syndrome:** dyslipidaemia, hypertension, obesity, insulin resistance, NAFLD. Also increased risk of MI/ CVA. Consider annual blood pressure, HBA1C and lipid profile. Ask about smoking and alcohol and offer appropriate support.
- **Hormonal:** consider PCOS and hyperthyroidism
- **Dermatological:** squamous cell carcinoma linked particularly to perianal scarring in HS

► *Management*

See KISS summary below. Patient education and lifestyle advice are crucial. Adequate analgesia is important and often overlooked. Steroids, antibacterial wash, topical and oral antibiotics including 12 week courses of tetracyclines all have a role in management. Patients should be referred to dermatology if they have mild to moderate disease that has not responded to this first line treatment in primary care, or if they have severe disease. It is important to screen for cardiovascular risk factors, mental illness and the effect on quality of life.

NB Learning Points: Hidradenitis Suppurativa

- Early diagnosis is key to reducing the risk of severe disease. Consider the possibility of hidradenitis suppurativa in patient with lesions in characteristic locations (axillae, groin, perianal, perineal, inframammary) Could that recurrent 'boil' actually be HS?
- Mild to moderate disease can be successfully managed in primary care, see KISS below
- The psychosocial morbidity can be very significant. Ask specifically about mental health and impact on social life/ relationships. Consider signposting to sources of support such as:
 - Hidradenitis Suppurativa Trust - Patient Support Group Facebook
 - [Guy's and St Thomas' patient resource](#) on managing psychosocial aspects of HS
 - [Changing faces](#): advice and guidance for people living with visible difference

Quality improvement idea:

- HS is associated with an increased risk of metabolic syndrome and annual screening is recommended. Could you audit patients to see if they are receiving an annual assessment of cardiovascular risk factors such as BP, HBA1C, lipid profile

KISS: Hidradenitis Suppurativa

[BMJ 2022;379:e068383](#), [BJGP 2023; 73:43-45](#), [Primary Care Dermatology Society](#)

► Diagnostic triad: diagnosis is clinical

1. **Typical lesions:** inflammatory nodules or abscesses, tunnelling, tombstone scar/ rope-like scar
2. **Characteristic locations:** axillae, submammary, neck, abdominal fold, groin, genital, perianal
3. **Recurrent pattern** of lesions: >2 within 6 months or >5 anytime scale

► Assess severity using the Hurley scale (see helpful pictures in [BMJ article](#) for examples):

- **Stage 1** - solitary or multiple, isolated abscess formation without scarring or sinus tracts
- **Stage 2** - recurrent abscesses, single or multiple widely separated lesions, with sinus tract formation
- **Stage 3** - diffuse or broad involvement, with multiple interconnected sinus tracts and abscesses

► Holistic primary care assessment:

- **Cardiovascular risk assessment**, screen for joint disease (due to increased risk spondyloarthritis). Bloods including LFT, fasting lipid profile, HBA1C
- **Screen for Crohn's disease:** consider faecal calprotectin if isolated perianal disease or associated bowel symptoms.
- **Mental health:** screen for depression/ anxiety

► Primary care treatment

- **Lifestyle:** support smoking cessation, weight loss, advise loose clothing
- **Analgesia:** pain can severe and chronic, consider pain clinic involvement
- **Dressings:** provide plentiful highly absorbent dressings, avoid adhesive dressings (risk irritation)
- **Acute treatment of large painful lesions:**
 - Lesions are often sterile - consider intra-lesional steroid injection or oral prednisolone (30-40mg for 3 days)
 - If very explosive flare consider infection, avoid steroid and treat with high dose flucloxacillin (or macrolide if penicillin allergic) and consider lancing
- **Long term management**
 - Regular **antibacterial wash** to reduce bacterial colonisation: chlorhexadine gluconate 4%
 - Mild disease (Hurley stage 1): **topical clindamycin 1% BD**.
 - If topical alone is ineffective or more severe disease add **oral tetracyclines:** doxycycline 100mg OD or lymecycline 408mg OD (double doses in obesity or lack of response at 12 weeks)
 - Reassess after 3 months, refer if inadequate response. Consider treatment break after 1 yr if disease controlled and no flares
- Consider **metformin** for associated metabolic syndrome (off-license)
- Patient information: [patient information leaflet](#) (produced by British Association of Dermatologists)

► When to refer?

- Severe disease (Hurley stage 3) at presentation OR
- When mild to moderate disease has not responded to maximum tx in primary care after 3 months
- **Secondary care management** (see [British Association Dermatologists flow chart](#)):
 - An MDT approach is increasingly offered with dermatologists, surgeons, clinical psychologists, pain teams and tissue viability services.

TOPICAL STEROID WITHDRAWAL

KISS: Topical Steroid Withdrawal

[MHRA Sept 2021 Drug and Therapeutics Bulletin 2021;59:184](#) [BAD/NES PIL](#)

► Background:

- Topical corticosteroids (TCS) are a crucial part of managing many skin conditions, most notably eczema, but there has long been concern about the long-term effects of topical steroids.
- TCS withdrawal is an increasingly recognised condition as flagged by the MHRA in Sept 2021.
- However, the MHRA highlight right at the beginning that we need to **put these reactions into context** - they are rare (although not well recognised) but as they state '*Topical corticosteroids are safe and highly effective treatments for skin conditions such as eczema, psoriasis, and atopic dermatitis when used correctly*'.
- **Reminder of TCS potencies** ([click here](#) for BNF link with all the brands delineated by potency):
 - Mild e.g. hydrocortisone
 - Moderate e.g. clobetasone
 - Potent e.g. beclometasone
 - Very potent e.g. clobetasol

► Who gets this?

- TCS withdrawal reactions are a group of conditions that generally occur after prolonged application of moderate to high potency TCS, usually after a year or more of regular daily application.
- However, this may happen in children after as little as 2 months of daily application.
- It is most common in people with eczema.
- It's important to re-iterate this is a rare condition and **does not appear to happen if TCS is applied appropriately** (depending on body site) and for appropriate time lengths (see our eczema chapter under the topical steroids section, which guides what potency steroid to use/where and for how long).

► How does it present?

It can be difficult to differentiate between a flare of the underlying skin condition or a TCS withdrawal reaction, however, the following points can help:

- Symptoms/signs occur within days/weeks of discontinuing long-term TCS, usually mod/high strength.
- **More common on sensitive areas** e.g. face, genitals.
- Skin redness may **extend beyond the initial area of treatment** - may be **confluent** rather than patchy.
- Often described as '**burning**' or '**stinging**' rather than itchy, and patients describe it as being worse and different to their original skin condition.
- It is variously known also as Red Skin Syndrome and Topical Steroid Addiction.

► Other skin conditions associated with long-term TCS use/part of TCS withdrawal phenomenon

- **Skin atrophy** - thinning of the skin exposes small vessels underneath giving a red appearance.
- **Rosacea** - can occur on the face if moderate/potent steroids are used, worsening when TCS stopped.
- **Peri-oral dermatitis** - Can be triggered by inappropriate use of TCS on the face.
- **Allergic reactions** - may not necessarily be the TCS but maybe a preservative in the cream.

► How do we manage this? NB excellent PIL from BAD/NES [click here](#)

- If suspected we should refer to dermatology for further assessment.
- Further investigation (e.g. patch testing) may be needed before they manage the TCS withdrawal.
- Topical calcineurin inhibitors (e.g. pimecrolimus), UV therapy and immunosuppressants may be considered as steroid-sparing agents.

BITES AND STINGS

For something seemingly so simple, how best to manage bites and stings is surprisingly uncertain. *'I've been to see the pharmacist and they think this bite is infected and I need some antibiotics'* is a consultation we have had many times, particularly over the summer months. Balancing the appropriate use of antibiotics for secondary infections, without over-prescribing for simple reactions is an age-old GP conundrum. So my heart was somewhat lifted when I saw both a BMJ review article ([BMJ 2020;370:m2856](#)) and NICE guideline ([NG182, Sept 2020](#)) on bites and stings. *Finally*, I was going to get some good evidence-based guidance as to when to prescribe antibiotics.....or not. As this BMJ article summarised at the top of the paper...

► **Insect bites and stings** [BMJ 2020;370:m2856](#) [NICE NG182 Sept 2020](#)

- *'There is little research evidence on best practice for insect bites'*
- *'Insect bite reactions vary widely and depend on the insect and the patient's response'*
- *'Secondary infection may be indicated by fever, systemic symptoms, and worsening reactions with spreading erythema. It can be difficult to know if mild secondary cellulitis has occurred'*

Are they common?

- Simple answer, yes. And much of it we don't see in primary care. It inevitably varies depending on the season, and geography. I was particularly taken with the questionnaire survey of a duathlon event in the west of Scotland where only 14.2% of participants had NOT experienced midge bites!

Is it actually an insect bite?

- For many people, they will have no idea what has bitten them, or indeed if it is a bite at all, but it is obviously important to check any recent foreign travel that might suggest a vector-borne disease.
- **Single prominent blisters or marks** may indicate horsefly, mosquito or a sting, often painful.
- **Multiple small red papules** suggest fleas, midges, mites, mosquitoes or bed bugs; **bed bugs** may occur anywhere on the body but are often multiple and occur overnight in exposed areas, e.g. face, neck, arms.
- **Flea bites** tend to occur below the knees.
- Remember **tick bites** which are often painless; NICE reminds us to consider the erythema migrans rash typical of Lyme disease.
- **Important differentials include:**
 - Don't forget **scabies**, which is very common. Typically tiny red multiple areas of itch (often intensely itchy), especially on hands but can be anywhere.
 - **Contact reactions** also important to consider - poison ivy or **giant hogweed** (giant hogweed is common in the UK in country areas in summer months, especially around river banks, and can look like giant cow parsley; it can cause nasty reactions - see some [useful information from the Woodland Trust](#)).

Is it infected and do I need antibiotics?

- This is the million-dollar question and the one for which we don't have definitive evidence to guide us.
- Some signs will obviously point to infection (systemic symptoms, pus etc.) but how to differentiate a mild or early secondary infection drew a complete blank in this BMJ literature review.
- The BMJ paper suggests we look for **spreading erythema** after the expected time to improvement, as a sign of secondary infection developing.
- There is no evidence to suggest what 'usual' resolution is for bites/stings, as it can be highly variable, but the general consensus is that **most non-infected bite/sting reactions will peak at 2-3 days** then start improving; NICE point out that **skin redness and itching are common for up to 10 days**.
- The implication is that we are still dishing out far too many antibiotics, with flucloxacillin prescriptions rising by a third in the summer months in the UK, much of which may be for 'infected' bites.
- **NICE advise that 'most insect bites or stings will not need antibiotics'.**
- The BMJ paper suggests that the **threshold for when to offer antibiotics** should be partly driven by risk factors for complications (e.g. diabetes, leg ischaemia) or high risk areas (e.g. peri-orbital).

KISS: Bites and Stings

BMJ 2020;370:m2856 NG182 Sept 2020 & [click here for visual summary](#)

Assessment:

- Ask about foreign travel and whether vector-borne illnesses need considering.
- Single prominent blisters or marks may indicate horsefly, mosquito or a sting, often painful.
- Multiple small red papules suggest fleas, midges, mites, mosquitoes or bed bugs.
- Flea bites tend to occur below the knees.
- Consider erythema migrans rash typical of **Lyme disease**; tick bites often painless.
- **Important differentials include:**
 - **Scabies** - typically tiny red multiple areas of itch (often intensely itchy), especially on hands but can be anywhere.
 - **Contact reactions** e.g. poison ivy or giant hogweed (giant hogweed is common in the UK in summer months, especially around river banks; it can look like giant cow parsley and can cause nasty reactions - [see here](#) for more information from the Woodland Trust).

Is it infected?

- Rapid onset skin reactions are usually inflammatory or allergic, not infected.
- **Most insect bites or stings will not need antibiotics** - be aware skin redness and itching are common and can last 10 days.
- Time to improvement for non-infected insect bites/sting reactions can be very variable, but general consensus suggests that most will peak and start to improve 2-3 days after exposure.
- Consider secondary infection if erythema continues to spread beyond usual improvement times.

Management:

- Most people with insect bites and stings can be directed to a community pharmacist.
- Warn patients that itching and redness are common and can last 10 days and that antibiotics are usually not needed.
- **Try to avoid scratching** - this reduces inflammation and reduces the risk of secondary infection.
- **Antihistamines** (orally) can be considered but the evidence is limited.
- Cold compress and topical steroids can be considered to help reduce itching.
- **Thresholds for antibiotics** should be lowered if risk factors for secondary infection present e.g. co-morbidities (e.g. diabetes), oral steroids, high risk reaction site (e.g. peri-orbital, areas of lymphedema, area of leg ischaemia).
- **If infected:**
 - Assess if systemically unwell or bites/stings in very high risk areas/high risk patients that may need consideration of referral/admission (e.g. in mouth or throat, severely immunocompromised, previous systemic allergic reaction, caused by unusual or exotic insect or occurred whilst outside UK).
 - **Flucloxacillin** 500mg-1g QDS for 5-7 days (or clarithromycin 500mg BD 5-7 days if penicillin-allergic; erythromycin if pregnant).
 - **If near eyes or nose use Co-amoxiclav** 625mg TDS 7 days (or clarithromycin 500mg BD with metronidazole 400mg TDS for 7 days if penicillin-allergic).
- **Human or animal bites?** See KISS below.

Safety netting:

- Draw around area of erythema to track spreading, and review if erythema continues to spread.
- Consider asking patient to take serial pictures to monitor (very useful for remote consultations).
- Review if any systemic symptoms develop.

GISS: Human and Animal Bites**NG 184 Nov 2020 & [click here for a visual summary](#)****General considerations:**

- Assess site and depth of bite and Irrigate (and debride) as necessary.
- Be aware and **consider if this may be a safeguarding issue**.
- Seek specialist help from microbiology for bites from wild or exotic animals (including birds and non-traditional pets) or farm animals or other unfamiliar domestic animals - the spectrum of bacteria may be different and there may be risks of non-bacterial infections.
- If the bite is infected and **there is discharge, take a swab** from the wound.
- If antibiotics are given tell patients to re-consult within 24-48 hours if not improving.

Human bites:

- Do not offer antibiotics if the bite has not penetrated the skin.
- **Offer antibiotics** if the skin is broken and it has drawn blood.
- **Consider antibiotics** if the skin is broken but it has not drawn blood if high risk of infection:
 - High risk area - hands, feet, face, genitals, skin over cartilage structures or area of poor circulation.
 - Person at high risk of complications e.g. diabetes, immunosuppression, asplenia, etc.

Cat bites:

- Do not offer antibiotics if the bite has not broken the skin.
- **Offer antibiotics** if the bite has broken the skin and drawn blood.
- **Consider antibiotics** if the bite has broken the skin but not drawn blood if the wound could be deep.

Bites from dogs and other traditional pets:

- Do not offer antibiotics if the bite has not broken the skin or broken the skin but not drawn blood.
- **Offer antibiotics** if it has broken the skin and drawn blood if it:
 - Has penetrated bone, joint, tendon or vascular structures.
 - Is deep, is a puncture or crush wound, or has caused significant tissue damage.
 - Is visibly contaminated (e.g. if there is dirt or a tooth in the wound).
- **Consider antibiotics** if it has broken the skin and drawn blood if high risk of infection:
 - High risk area - hands, feet, face, genitals, skin over cartilage structures or area of poor circulation.
 - Person at high risk of complications e.g. diabetes, immunosuppression, asplenia etc.

Consider referral/admission/specialist advice:

- If systemically unwell, have lymphangitis or have deteriorated despite taking antibiotics.
- If they are at risk of serious complications due to pre-existing medical conditions.
- If the bite is in an area of poor circulation.
- **Admit ALL people** with penetrating wounds involving arteries, joints, nerves, muscles, tendons or bones.

Antibiotics for adults ([click here](#) for detailed prescribing guidance for children):

- For prophylaxis of infection give 3 days, for treatment of infection give 5 days.
- **1st line** - co-amoxiclav 325mg or 625mg TDS.
- **If penicillin allergic** - doxycycline 200mg stat then 100-200mg daily WITH metronidazole 400mg TDS; if pregnant and penicillin-allergic seek specialist advice.
- **For children** 1st line co-amoxiclav; 2nd line co-trimoxazole if <12 years old, doxycycline plus metronidazole if ≥12 years old. See above for detailed prescribing for children.

ECZEMA

Eczema is a miserable condition and is becoming increasingly prevalent. 'Eczema' comes from the Greek word 'to boil' and helps describe the characteristic itchy red skin we see. For such an important and common condition we see precious little primary research on eczema. However, there have been some helpful recent studies. One should prompt a change in our prescribing of bath emollients in children ([BMJ 2018;361:k1332](#)) and recent primary care based trials published in 2022 help us answer some important questions: which emollients are most effective? ([The Lancet Child & Adolescent Health, Volume 6, Issue 8, 522 - 532](#)); and which are preferred by parents/ patients ([BJGP 2022; 72 \(719\): e390-e397](#)). Further, there is an exciting new online tool to help support children and parents to manage eczema effectively ([ECO](#)). On the back of this, we have new NICE guidance on antimicrobial prescribing in secondary bacterial infection of eczema ([NG190 March 2021](#)...spoiler, antibiotics often not needed...), as well as an updated version of NICE guidance on atopic eczema in the under 12s ([CG56 March 2021](#)), together with updated guidance from the brilliant [Primary Care Dermatology Society](#) (PCDS) and [CKS](#), both in 2021.

1 Emollients in Eczema

► *All types of emollients are equally effective, but people like different things!*

Which moisturiser is best for eczema? Given that emollients are the mainstay of eczema maintenance you would think that this would be well-established. However, up until now, there have been no trials comparing the emollients commonly used for childhood eczema. Enter the '[Best Emollients for Eczema](#)' or 'BEE' study. This was a pragmatic, individually randomised, parallel-group, phase 4, superiority trial of four types of emollient (lotions, creams, gels, and ointments).

- 550 patients were recruited from 77 GP surgeries in England
- Age 6 months- 12yrs
- They were randomly assigned to use lotions, creams, gels or ointments
- Their POEM (patient-orientated eczema measure) scores were assessed weekly over 16 weeks
- **CONCLUSION:** there was no difference in eczema severity between emollient types

So if there is no difference in the effectiveness of the emollient can we be guided by acceptability to the patient? Is there one type more popular than others? A qualitative interview study nested within the BEE trial looked into this ([BJGP 2022; 72 \(719\): e390-e397](#)). They interviewed 44 parents and essentially concluded that **people like different things**. Some prioritised thickness, some favoured easy absorbency. **There is no one emollient suitable for all.**

From a practical perspective, it would seem most helpful to have a resource which allows the patient/ parent to reliably compare the types of emollient so that they can choose the one most suited to their preferences from amongst a bewildering range of choices. This excellent resource [eczemacareonline.org.uk](#) has a really handy section which lists the common emollient brand names by thickness and gives advice about how to apply them.

► *Bath emollients do not improve eczema outcomes in children* [BATHE Study BMJ 2018;361:k1332](#) and associated editorial [BMJ2018;361:k1791](#)

Bath emollients are estimated to account for about 1/3 of the total cost of eczema treatment in the UK, but we now have evidence they are ineffective for most children:

- This RCT randomised 482 **children aged 1-11 years old with eczema** (mean age 5 years and most had moderate eczema) to usual care (including regular leave-on emollients, using regular emollients as soap substitutes and prn topical steroids) **with or without regular bath emollient additives**, for 12 months.
- There were **no significant differences in disease severity between the groups in any of the outcome measures**.
- **Conclusion:** This trial found no evidence of clinical benefit from including bath emollients in the standard management of eczema in children aged over 1.

The authors conclude that this research should prompt a change in prescribing to more cost-effective treatments and **reduce the amount of bath emollient prescribed**. As an associated commentary stresses, however, it is crucial these results are not extended to all emollients for leave on use and soap substitution, and uncertainty remains for children under 1.

This evidence tallies with the updated [PCDS guidance](#) regarding **bath/shower formulations**:

- In general, there is no good evidence to support the use of specific products to use in the bath/shower.
- The same emollients used to moisturise with can be used as a soap substitute.
- The one exception discussed by the PCDS is the '*occasional patient*' who gets frequent skin infections, in whom we can consider anti-septic emollients e.g. Dermol ® 600 bath emollient or Dermol ® 200 shower emollient, but they warn that frequent use of these products can irritate the skin.

2 **Secondary bacterial infection of eczema**

[NG190 March 2021](#) inc [summary page](#) [PCDS June 2021](#)

This NICE guideline is the latest in a long line aiming to rationalise antibiotic use and reduce risks of resistance and may be somewhat practice-changing for some of us. **Summary points:**

- Symptoms and signs of secondary infection include weeping, pustules, crusts, rapidly worsening eczema and no response to usual treatment (as well as the obvious systemic features e.g. fever, malaise).
- The key message for us is that **we should not routinely offer topical or systemic antibiotics for people with secondary bacterial infections of eczema** (assuming they are systemically well). **Why?**
 - Not all eczema flares are caused by bacterial infection even if weeping and crusting is present.
 - Eczema can be colonised with bacteria but may not be clinically infected.
 - Although the evidence is somewhat limited NICE cite studies where the addition of oral or systemic antibiotics to standard emollient and steroid therapy was no more effective than emollient and steroid therapy on its own.
- **Swabs?** Not routinely, consider for eczema that is worsening or not responding to standard therapy. For people with **recurrent bacterial infections** take both skin swabs and nasal swabs - if *S. aureus* present on nasal swabs use a decolonisation regime.
- **If antibiotics are considered:**
 - Use topical treatment for localised/not severe infections - fusidic acid 2% TDS for 5-7 days.
 - For more widespread or severe infections use oral antibiotics (adult doses below) - [click here](#) for a summary and dosing inc children:
 - 1st line flucloxacillin 500mg QDS 5-7 days
 - Alternative if unsuitable or penicillin-allergic clarithromycin 250-500mg BD 5-7 days
 - Pregnancy Erythromycin 250-500mg QDS 5-7 days

This NICE guidance very much tallies with the PCDS approach, which agrees that antibiotics generally have a limited role in eczema management, but if the eczema continues to flare take swabs and treat as appropriate. **If recurrent infected flares take nasal swabs:** if *S. aureus* present use de-colonisation regime e.g. nasal Bactroban ® cream BD for 1 week.

3 Patient information:

The elephant in the room with all of this (as is often the case in General Practice) is time. One of the biggest (if not the biggest) factor in poorly controlled eczema is concordance, and both NICE and the PCDS stress the importance of giving patients and parents time to discuss the condition and the ins and outs of how (and when) to apply treatments. This is often not feasible in a routine appointment but the good news is that there are lots of patient resources available:

- **ECO: Eczema care online** - [click here](#). This is a fabulous new evidence-based resource for managing childhood eczema. It was developed by a mixed panel of dermatologists, nurse specialists, GPs, psychologists and patients following systematic review of studies and interviews with parents/carers, as described in a recent [BJGP paper](#). They identified barriers to optimum management of eczema such as incomplete knowledge, concerns about topical agents and limited skills for co-managing eczema. This comprehensive resource is intended for both those new to eczema and experienced patients. It explains the rationale behind treatment and highlights new evidence. Further, it contains practical tips for encouraging compliance at specific ages and stages- from toddlers to teens.
- PIL from British Association of Dermatologists (BAD) - [click here](#)
- National Eczema Society - [click here](#)
- Advice on how to reduce the itch-scratch cycle - [click here](#)
- Finger Tip Units for steroid creams/ointments - [click here](#)
- How to apply emollients and topical steroids - see the top two videos on this [link here](#)

NB Learning Points & Quality Improvement Ideas: Eczema

- **New evidence shows all types of emollients are equally effective. The key is to find one the patient is happy to use regularly**
- For patients with **frequent flares** (e.g. every 1-2 months) consider the 'steroid weekend regime' - once eczema controlled use the usual topical steroid on two consecutive days each week (e.g. Saturday and Sunday) to areas that tend to flare. Apply steroid even if the skin is not inflamed.
- New NICE guidance recommends we should **not routinely use topical or oral antibiotics for eczema with secondary bacterial infection** - although somewhat limited there is evidence that the addition of oral or topical antibiotics to standard emollient/topical steroid therapy in patients with infected eczema does not improve outcomes.
- We should not routinely swab infected eczema; however, for people with **recurrent infected flares of eczema** we should take skin swabs and **nasal swabs** - if St Aureus present on nasal swab use a decolonisation regime e.g. nasal Bactroban ® cream BD for 1 week.
- **Quality Improvement Ideas:**
 - Current evidence does not support the use of bath emollients in children aged 1-11 years - **Audit children aged 1-11 being prescribed bath emollients** - can they switch to using their usual emollient as a soap substitute? Make yourself popular with your medicines management team!
 - **Too busy to discuss all the practicalities of eczema management?!** Direct parents/ children to [ECO](#) or use our KISS below which patients can [download/access from our website here](#)

KISS: Eczema - Background and initial assessmentPCDS June 2021 CKS 2021 NICE CG57 updated March 2021Background

- Prevalence in children is estimated to be 10-30% and it can lead to itchy children that don't sleep, as well as poor school performance, behavioural problems and poor self-esteem.
- The good news is that it does tend to improve over time with 65% of affected children clearing by age 7, and 75% by age 16.
- However, it still affects 2-10% of the adult population with significant morbidity including poor self-esteem, reduced quality of life and possibly increased cardiovascular risk.

Presentation

- It is an itchy skin condition with ill-defined patches of erythema, commonly with a personal or family history of atopy.
- Presentation can be variable but generally **distribution changes with age** with facial eczema common in infants then progressing to flexural areas, then more widespread distribution in older age groups. In people with darker skin extensor surfaces may be commonly affected, and discoid patches or follicular involvement are common.

Identify triggers and address where possible

- **Irritants:**
 - Use of soaps and detergents.
 - Clothing - synthetic and wool clothing tend to be more irritant than cotton, but some dyes in cotton clothing can cause sensitivity reactions.
- **Heat** - consider clothing and be aware of the long hot bath/shower!
- **Contact allergens** - perfumes, topical medications, metal, latex.
- **Inhalant allergens** - are symptoms linked to **seasonal flares**? Particularly consider in children >3 years and adults with **facial eczema especially around the eyes**, and in those with eczema associated with asthma or allergic rhinitis. If present, consider referral for patch testing.
- **Additional stresses?** Emotional upset, concurrent illness and lack of sleep may all contribute to symptoms of atopic eczema.
- **Food allergy/dietary factors?**
 - Most people do NOT need allergy testing, with PCDS stating that *'Most children with mild-moderate eczema that respond well to treatment do not have a food allergy. In such patients, and in the absence of any close temporal link between a particular food causing a flare of eczema, investigations and dietary restrictions are not recommended. Irrespective of the clinical context, if there appears to be a close temporal link between a food and the symptoms the diagnosis can be confirmed by dietary exclusion, often without the need for any investigations.'*
 - We should **suspect food allergy** (and refer) children with atopic eczema who have reacted to food with immediate symptoms, or in infants/young children with moderate/severe atopic eczema not controlled by optimum management, particularly if associated with gut dysmotility (colic, vomiting, altered bowel habit) or failure to thrive.
 - A 6-8 week trial of extensively hydrolysed or amino acid formula is recommended for bottle-fed infants <6 months with moderate/severe eczema not responding to optimal treatment, but in practice these children are likely to have been referred by this stage.
 - Milk, egg, wheat, soy, and peanut account for ~75% of food-induced atopic eczema.
- **Scratching!** The 'itch scratch' cycle is a common problem for people suffering with eczema and is difficult to address - [excellent booklet here](#) from the national eczema society.

KISS: Eczema - General Management**PCDS June 2021 CKS 2021 NICE CG57 updated March 2021****Emollients - use lots! - see also KISS below on practical tips for managing eczema**

- As discussed above there is no evidence that one type of emollient is better than another.
- The PCDS which recommends **'complete emollient therapy to the whole skin every day'**.
- Rub emollients gently into the skin until no longer visible, **downwards in the direction of the hairs** (reduces risk of folliculitis).
- Aqueous cream is not recommended due to higher rates of skin reaction.
- If possible **prescribe as a pump dispenser** to reduce the risk of cross-contamination and prescribe in generous quantities e.g. 500g pump dispensers.
- As above, **current evidence does not support the use of separate bath emollients for children > 1 year**; standard emollients (particularly ointments) make effective soap substitutes.

Topical steroids - see also KISS below on practical tips for managing eczema

- Use the lowest **appropriate** potency and apply only to inflamed skin OD for 1-2 weeks (can $\uparrow$ to BD if not settling), however, the PCDS note that for **flare-ups it is more effective and safer to 'hit hard'** using more potent steroids for a few days than to use milder steroids for longer periods of time.
- **Strength of steroid** to be determined by the age of patient, site and severity:
 - **Child face:** mild potency e.g. 1% hydrocortisone (only consider moderate potency steroids for severe flares and for max 3-5 days).
 - **Child trunk and limbs:** moderate potency e.g. Eumovate® (clobetasone butyrate 0.05%) or Betnovate-RD® (betamethasone valerate 0.025%)
 - **Adult face:** mild or moderate potency e.g. Eumovate®
 - **Adult trunk or limbs:** potent e.g. Betnovate® (betamethasone valerate 0.1%), Elocon® (mometasone)
 - **Palms and soles:** potent or very potent e.g. Dermovate® (clobetasol propionate 0.05%)
 - **Vulnerable sites (axillae/groins):** generally use mild potency steroid; if used, moderate/potent steroids should only be used for short periods (e.g. 7-14 days).
- **How much to use?**
 - The amount of steroid should be determined by the [Finger Tip Unit](#) method.
 - As a rough guide, adults are likely to need (for 1-week use):
 - Both hands 15-30g
 - Face and neck 15-30g
 - Both arms 30-60g
 - Both legs 100g
- **Skin atrophy** is rare if topical steroids are used appropriately but should be avoided around the eyes, and regular use should be avoided on the face.
- Creams may be preferred, especially on visible areas e.g. hands/face; ointments may be more effective but are greasy but may be a good option to use at night.
- See also below KISS for further practical tips on topical steroid use.

What about antihistamines?

- Antihistamines are not recommended for routine use in eczema management as the eczema itch is NOT histamine-mediated, however, if there is severe itch or urticaria consider a trial of non-sedating antihistamines.
- If there is severe itch affecting sleep consider sedating antihistamines (e.g. adults hydroxyzine 25-50mg, children chlorphenamine 5-15mg).



KISS: Eczema - Ongoing Management and Referral

[PCDS June 2021](#)
[CKS 2021](#)
[NICE CG57 updated March 2021](#)
[NG190 March 2021](#)

Secondary infection?

- Symptoms and signs of secondary infection include weeping, pustules, crusts, rapidly worsening eczema and no response to usual treatment (as well as the obvious systemic features).
- The key message for us is that **we should not *routinely* offer topical or systemic antibiotics for people with secondary bacterial infections of eczema** (assuming they are systemically well).
- If antibiotics are considered:
 - Use topical treatment for localised/not severe infections - fusidic acid 2% TDS for 5-7 days.
 - For more widespread or severe infections use oral antibiotics (adult doses below) - [click here](#) for summary and dosing inc children.

How can we manage frequent flare-ups? (i.e. flares more than every 1-2 months)

- Check concordance - are they using enough emollient? Are they using enough topical steroid?
- Are there triggers that need to be addressed?
- Consider the '**steroid weekend regime**' - Once the eczema is under control use the usual topical steroid on two consecutive days (e.g. Saturday and Sunday) of each week to the areas that tend to flare. **The treatment must be applied even if the skin is not inflamed** - the aim is to reduce the frequency of flare-ups; use strategy for 3-6 months then review.

Topical calcineurin inhibitors

- Whilst often not thought of as a 'GP treatment' the PCDS state that these treatments can be initiated by health care professionals experienced in treating eczema 'this of course includes General Practitioners', however NICE do take quite a cautious approach stating that (for children <12) these treatments should be '*initiated only by physicians (including general practitioners) with a special interest and experience in dermatology*'.
- NICE recommend these treatments as an option when eczema has not been controlled by topical corticosteroids where there is a serious risk of important adverse effects from further topical corticosteroid use, particularly irreversible skin atrophy.
- Because they are not steroid based they do not cause skin atrophy, and therefore **can be considered as 'steroid sparing' agents**, for people with eyelid and peri-orbital eczema, people using regular steroids to the face, or for people with signs of skin atrophy.
- **Local adverse effects** include stinging, burning, irritation and photosensitivity (sun protection recommended), and they should be avoided/discontinued with infected skin, but data has shown no serious adverse effects if used appropriately.
- **Formulations:**
 - Tacrolimus (Protopic ®) 0.03% ointment and pimecrolimus cream are licensed for ages ≥ 2 years.
 - Tacrolimus (Protopic ®) 0.1% ointment licensed for ages ≥ 16 years.

When should we consider referral?

- **Refer acutely/admit if eczema herpeticum is suspected**
- Diagnostic uncertainty or suspected food allergy or inhalant allergy triggering eczema (see above).
- Severe eczema or Eczema causing significant social or psychological problems (e.g. sleep disturbance).
- Eczema only partially responding to standard treatment (e.g. recurrent secondary infection, very frequent flares 1-2 times/month despite management above)
- Steroid atrophy or concerns regarding the amount of topical steroids being used. Possible cases of contact allergic dermatitis.

KISS: Practical Tips for Managing Eczema

[Primary Care Dermatology Society/ ECO](#): Eczema care online

How much emollient (moisturiser) should I use and which one should I use?

- **Use as much emollient as you can!** Ideally, an emollient should be applied 3-4 times per day (but at least twice per day).
- People with moderate/severe eczema should get through about 500g every week.
- Use whichever types work best for you, but you **can vary different types** e.g. use creams (which are thinner and easier to apply) in the day and ointments (which are thicker and greasier) at night.

How and where do I apply emollient?

- Emollient should be applied to the whole body (not just affected/red or inflamed areas).
- Apply **downward in the direction of the hairs** to reduce the risk of infection/inflammation of the hair follicles (folliculitis).
- The emollient may sting initially when you use it but this should improve in a couple of days.
- Here's a [link](#) to a video on how and how much to apply

How do I use steroid cream/ointment and how much should I use?

- Apply steroid cream or ointment only to affected (inflamed/red) areas of skin.
- One Finger Tip Unit (FTU) is the amount of cream squeezed from a standard tube along an adult's fingertip from the very end of the finger to the first crease. One FTU is enough to treat an area of skin twice the size of the flat of an adult's hand with the fingers together - [click here](#) for more explanation.
- Allow at least 20 minutes between applying emollient and steroid
- Here is a link to a [video](#) explaining how to apply steroid to treat a flare.

What about bathing and showering?

- **Do not use soaps or shower gels** as they generally make eczema worse.
- Specific bath/shower emollients/gels are rarely helpful - just **use your usual emollient as a soap substitute** (ointment emollients dissolved in hot water work well).
- Pat dry after bath/shower then put plenty of emollient on.

How do I prevent my skin from becoming infected?

- Ideally, use **pump dispensers** for emollients.
- If you have thicker emollients (e.g. ointments) these may come in tubs - **don't put fingers or hands into the tub** but use a clean spatula or spoon to scoop out the ointment.

My skin is really itchy and I can't stop scratching!

- Itchiness can be one of the most distressing symptoms of eczema and unfortunately scratching then makes the skin worse, so **trying to break the 'itch-scratch' cycle is really important**.
- It is not easy to do, but there are lots of useful tips from the National Eczema Society [click here](#).

For more general information and support see links to the [British Association of Dermatologists](#), [Eczema Care Online](#) and the [National Eczema Society](#)

ACNE ROSACEA

Rosacea is a common condition, and as with acne vulgaris, there has been a move away from oral antibiotics to use non-antibiotic topical treatments, to reduce antibiotic resistance. We have some background evidence from a few years ago but [CKS](#) and the Primary Care Dermatology Society [PCDS June 2021](#) have recently updated their guidance on rosacea. See KISS below for diagnostic features, differential diagnoses and triggers.

Just remind me what Rosacea is...

Rosacea is a chronic inflammatory skin condition affecting the centro-facial region (i.e. forehead, cheeks, nose, chin). It mainly affects those aged 45-60 and is more common in women and in those with fair skin/blue eyes. The aetiology is unclear, but genetic factors, vascular and neuronal dysfunction and colonisation with *Demodex folliculorum* mites may all be factors. **Relapse rates are high** - studies suggest ~20-30% of people can get long-term remission, with the remainder having a variable relapsing/remitting course of the disease.

Show me the evidence...

A Cochrane review in 2015 included 106 studies, n= 14,000 patients and evaluated topical and oral interventions. [Cochrane2015; CD003262](#)

- There was **high-quality evidence to support topical azelaic acid, ivermectin cream, brimonidine gel and oral doxycycline and isotretinoin**.
- There was moderate-quality evidence to support topical metronidazole and oral tetracycline.
- For topical metronidazole and azelaic acid, the benefits start to appear after 3 to 6 weeks. There was no clear evidence to say that one treatment was more effective than the other, although metronidazole appeared slightly better tolerated.

What are ivermectin cream and brimonidine gel?

These are actually not new treatments, but seem not to have made their way into routine General Practice for many people, despite a pretty reasonable evidence base. Their obvious advantage is that they are not antibiotics, and can therefore help to stem the tide of antibiotic resistance. One important point to note from the PCDS guidance is that if no papules/pustules/nodules are present and **flushing/erythema/telangiectasia are the main symptoms antibiotics do not work and should be avoided**.

Brimonidine gel ([DTB2014;52;138](#)) is a treatment for the **symptomatic treatment of facial erythema** in rosacea. It is applied once daily and can be used on an 'as needed' basis. It is an alpha-agonist that acts through vasoconstriction. The gel starts to reduce erythema within 30 minutes of application and lasts up to 12 hours. Two short-term trials have been performed comparing it to placebo. In some people it didn't work, in some, it made it worse but a noticeable benefit was seen in 70% of patients compared to around 30% to 40% in the control group. It is to be used for symptom control and does not influence the disease, and given the side effect profile its place in rosacea is questioned by some, but if tolerated can be very effective and has the advantage it can be used on a 'PRN' basis. [NICE](#) conclude that the drug **may be an option for adults with a diagnosis of rosacea and moderate to severe erythema**. We should warn patients that **brimonidine gel may exacerbate or produce rebound symptoms of rosacea** in up to 16% of patients [Drug Safety Update 2016](#); advise patients to start with a small amount of the gel and increase gradually based on tolerability and response to treatment. There have also been reports of **systemic effects** such as bradycardia, hypotension and dizziness [Drug Safety Update June 2017](#), and it should not be applied to broken skin to minimise the chance of systemic absorption.

Ivermectin 1% cream ([DTB2015;11;0364](#)) is a member of the avermectin class of drugs, used to treat parasitic worms, and in its topical form has been shown to be effective in papulo-pustular rosacea, and indeed is **recommended as a first-line treatment by the PCDS if papules/pustules/nodules are present**. Its name came to prominence during the COVID pandemic (in its oral form) as a possible COVID treatment, gaining notoriety for its use particularly in Latin America and South Africa despite no evidence base, leading the [EMA](#) to advise against use unless in clinical trials, although it is now one of the drugs being studied in the [PRINCIPLE trial](#). Ivermectin cream is a once-daily treatment option. [NICE](#) reviewed the evidence in 2016:

- In 2 RCTs ivermectin cream was more effective than placebo, with 40% of patients successfully treated compared to 15% for the placebo.
 - 2/3 people rated their improvement as 'excellent' or 'good' compared to 1/3 on placebo.
- In another RCT, it was marginally more effective than topical metronidazole (10% fewer skin lesions) and in another better tolerated than azelaic acid.
- The cost is comparable to other topical treatments and DTB conclude (NB very unusually for them with a new treatment!) that it is a 'useful addition to the range of treatments for papulopustular rosacea'.

Ocular rosacea

What may be a surprise to many in General Practice (or it was to me!) is that **ocular involvement is common with Rosacea** - the PCDS state that up to 50% of people with rosacea get ocular involvement. They usually present with gritty eyes, conjunctivitis, blepharitis, episcleritis and chalazion. **Keratitis** can occur and is obviously a much more serious complication. How do we manage these symptoms:

- **Lid hygiene** - clean the eyelids using cotton wool soaked in cooled, boiled water.
- **Artificial tears** - should be applied liberally throughout the day. If necessary a lubricating ointment, sometimes containing an antibiotic preparation may be used at night.
- **Systemic tetracyclines** are the most effective treatment for ocular rosacea. Erythromycin can be taken orally for patients intolerant of tetracyclines.
- **Retinoids should be avoided** in patients with significant ocular problems as they can worsen symptoms and lead to a severe keratitis.
- We should consider referral to an ophthalmologist if persist symptoms despite the above measures. **Patients with potentially more serious symptoms such as keratitis should be seen urgently.**

NB Learning Points & Quality Improvement Ideas: Rosacea

- Rosacea presents with facial flushing, erythema with papules and pustules in the centro-facial region (i.e. forehead, cheeks, nose, chin); it spares the naso-labial folds and lacks comedones.
- **Topical ivermectin 1% cream** has a good evidence base, is well tolerated and is the **1st line treatment for papulo-pustular rosacea**.
- **Brimonidine gel** is an option for facial flushing/erythema but we need to warn patients it can cause worsening symptoms in some people as well as rebound symptoms, it can be used on a prn basis.
- **Antibiotics do not work for facial flushing, erythema, or telangiectasia** in the absence of papules/pustules/nodules.
- Don't forget that **addressing triggers** (see KISS below) and using **high SPF suncream** to reduce UV exposure are key parts of managing rosacea.

Quality Improvement Idea:

- Review all your patients on **oral antibiotics for rosacea**:
 - Have they had >3 months of oral antibiotics? If so recall for review.
 - If they have no papules, pustules or nodules the antibiotics will be ineffective - stop.
 - If they do have papulo-pustular rosacea consider a trial of ivermectin cream 1% for 3 months whilst stopping the antibiotics.

KISS: Rosacea

[CKS 2021](#), [PCDS 2021](#), [Cochrane2015; CD003262](#), [NICE 2016](#), [Drug Safety Update 2016](#)

Diagnosis:

- Chronic inflammatory skin condition affecting the centro-facial region (i.e. forehead, cheeks, nose, chin); mainly affects those aged 45-60, more common in women.
- **Diagnostic features (one required for diagnosis):**
 - Phymatous change - facial skin thickening due to fibrosis/sebaceous gland hyperplasia (commonly the nose = 'rhinophyma').
 - Persistent erythema in centro-facial region which intensifies with trigger factors (below).
- **Major clinic features (two required for diagnosis):**
 - Flushing/transient erythema, often with sensation of prickling, tingling or burning/stinging of skin.
 - Inflammatory papules and pustules, usually centro-facial region.
 - Telangiectasiae.
- **Ocular involvement** is common, in as many as 50%, and maybe present without other skin features - lid margin telangiectasia, blepharitis, conjunctivitis, anterior uveitis, keratitis.
- **Key differential diagnoses:**
 - Acne vulgaris - generally younger people, comedones present (absent in rosacea), more widespread distribution.
 - Seborrhoeic dermatitis - yellow scaling, affects eyebrows and naso-labial folds (spared in rosacea).
 - Lupus 'butterfly' rash - malar distribution, rarely presents with pustules.

Management:

- **Avoid triggers** - Caffeine, alcohol, spicy foods, hot drinks, emotional stress, calcium channel blockers.
- **Avoid UV radiation** - use high SPF 50 with UVA/UVB cover (e.g. Anthelios®, Sunsense®, Uvistat® - NB expensive, can be prescribed under 'ACBS' rules).
- Useful information on self-help [NHS Choices](#) & [British Skin Foundation](#) & [BAD, Rosacea](#)
- **Facial flushing and moderate to severe erythema (NB will NOT respond to antibiotics):**
 - **Topical brimonidine gel** (alpha-adrenergic agonist) can be used on an 'as required' basis; works within 30 minutes, lasts for up to 12 hours; but may exacerbate rosacea/symptoms in some or lead to rebound symptoms - so use small amounts for 1 week and gradually increase as tolerated; other side effects include alpha-adrenergic effects - dizziness, dry mouth, headaches.
 - **Other options** - carvedilol 12.5mg OD has given significant improvement in small studies; propranolol 40mg BD; clonidine 50 mcg BD; laser therapy is an option but unlikely on NHS.
- **Mild to moderate papulopustular rosacea:**
 - **First line:** Topical ivermectin cream 1% once daily for 3 months.
 - **Second line:** Topical metronidazole 0.75% gel or cream BD, or azelaic acid 15% cream BD.
- **Moderate to severe papulopustular rosacea (use together with non-antibiotic topical treatments) - in all cases use for 8-12 weeks then step down to topical treatment only:**
 - Doxycycline (100mg OD or MR 40mg OD - both equally effective, possibly lower side effects with 40mg MR dose with reduced antibiotic resistance, but expensive).
 - Alternatively lymecycline 408mg OD, oxytetracycline 500mg BD, erythromycin 500mg BD.
 - If very severe, refer for consideration of oral isotretinoin.
- **Ocular disease** - lid hygiene and artificial tears/lubricants; consider oral tetracyclines; refer if troublesome symptoms to ophthalmology or urgently if anterior uveitis or keratitis suspected.

MASKNE

Masks, facial coverings and PPE have become a way of life since the outbreak of the COVID pandemic and while PPE and facial masks have been a crucial component of slowing the spread of the virus, they have not been without their problems for both patients and professionals. Aside from the communication problems, masks can be hot and uncomfortable and have resulted in their very own **new increasingly referenced condition - 'maskne'**. 'Maskne' is not a condition or diagnosis in its own right, but is increasingly being used to describe a wide range of facial dermatoses triggered by the use of facial PPE, including face shields, visors, safety spectacles as well as masks, facial coverings and respirators. They can be triggered de-novo or can be an exacerbation of a pre-existing dermatosis. It was the subject of an excellent BMJ article ([BMJ 2021;373:n1304](#)) which also gave a whistle-stop review of all common causes of facial dermatoses, and as with all things skin the [primary care dermatology society](#) has more in-depth information on each condition.

Background

'Maskne' has been well described but as a new phenomenon data is inevitably limited. **Mask-related acne and irritant contact dermatitis are thought to be the commonest causes.** Masks cause a warm, moist occlusive environment around the face which can worsen skin conditions, and added to this may be an allergy to components of the mask or friction from the mask and its strapping/support.

How should we assess it?

- As with any facial dermatosis we need to review past history of skin disease, prescribed and non-prescribed medications, and symptoms/signs - itch, soreness, appearance of papules or pustules, as well as the impact of the condition on mood, work and social life.
- **Specific to 'maskne' we should enquire about the temporal relationship between the dermatosis and the wearing of facial PPE** - do periods without the mask alleviate or improve symptoms? How often do they remove their mask? Are they allowed mask breaks at work?
- Examination then obviously involves looking at the morphology of the eruption, distribution and whether present at other sites other than the face.

What are the common causes of maskne? ([click here](#) for a good visual summary of causes)

Irritant contact dermatitis (ICD)

- Is the most common cause and is a form of exogenous eczema caused by direct physical or chemical injury is associated with prolonged mask-wearing >6 hours and is more common in people with pre-existing atopic dermatitis.
- Pressure ICD is commonly described over the cheeks and nasal bridge.
- Presentation may vary from discrete dry scaly patches to oedema, vesicles and erosions/ulceration.
- Treatment involves general measures (see KISS below) and taking regular breaks from mask-wearing.

Allergic contact dermatitis (ACD)

- Much less common than ICD, and caused by delayed type IV hypersensitivity reaction, most commonly due to preservatives, but has been reported from a rubber accelerator found in the elastic straps of surgical masks. Metal wires used to mould masks have also been reported causes.
- Can be clinically indistinguishable from ICD but well-demarcated areas of inflammation may be more suggestive of ACD.
- Can be treated with mild/moderate potency steroid e.g. hydrocortisone 1% ointment.
- If **suspected refer for patch testing** for identification and avoidance of causative agent.

Atopic eczema

- Inevitably wearing of facial PPE can cause flares of atopic eczema; treatment involves the usual treatment regimes for eczema and general measures below.

Seborrhoeic eczema/dermatitis

- As with atopic eczema, the warm occlusive environment of masks can predispose to seborrhoeic eczema.
- It usually starts in young adult life.
- It presents with a dermatitis with greasy yellow scale affecting the scalp, eyebrows and naso-labial folds.
- Treatment for the face ([see here](#) for further advice from primary care dermatology society):
 - Ketoconazole cream; if this causes irritation alternatives include clotrimazole or miconazole creams.
 - Topical steroids e.g. hydrocortisone 1% ointment or eumovate ® cream can be added for flares but short term only e.g. few days only.
 - Avoid combination products e.g. daktacort ® as high risk of overusing steroid component.

Periorifacial dermatitis

- Patchy erythema with tiny papulopustules affecting periocular and perioral areas sparing the vermilion border, typically in young women.
- Can be triggered by cosmetics or topical steroids and can be triggered or exacerbated by masks, for similar reasons to atopic/seborrhoeic eczema.
- Treatment involves avoiding causative topical treatments (including being careful with cross-contamination after applying topical treatments elsewhere on the body) and general measures in KISS below - warn patients that withdrawal of topical steroid will exacerbate symptoms in the short term
- Topical antibiotics e.g. erythromycin can be used
- For more serious cases lymecycline 408mg OD for 4 weeks.

Urticaria

- Pressure and contact urticaria are rare causes.
- Urticaria may develop either immediately or delayed (4-6 hours).
- For pressure urticaria checking mask fitting (or use alternative PPE) is key.
- Contact urticaria typically resolves within 24 hours and antihistamines can be used.

Acne

- PPE-related acne can occur in people with a past history of acne or de-novo.
- It has been associated with surgical masks and respirators.
- Again thought to be due to the warm moist microclimate, bacterial overgrowth and friction from the mask.
- Treat as usual for acne - see our separate chapter.

Rosacea

- Heat can be a trigger for rosacea so it's no surprise masks can cause rosacea.
- Rosacea is an inflammatory skin condition affecting the centro-facial region (forehead, cheeks, nose and chin) and presents with flushing/erythema, papules/pustules and telangiectasia; commonly age 45-60.
- For more detail including treatment - see our rosacea chapter.

Folliculitis

- More common in men, presents with small papules and pustules, although can be larger/deeper.
- May be infective or non-infective.
- Causes include bacterial infection from shaving (folliculitis barbae) or inflammation and ingrowing hairs (pseudofolliculitis), but occlusion (hence mask) is also a well-recognised factor.
- Treatment involves gentle daily cleansing with soap-free cleanser and gentle exfoliation.
- If persistent take swab of pus from pustule - st. aureus commonest bacteria - if present consider either antibacterial treatment e.g. Dermol ®, or for more persistent or deep-seated infection 4-6 weeks of antibiotics based on culture results.
- If shaving is a factor give relevant patient information leaflets on [folliculitis barbae](#) or [pseudofolliculitis](#).

KISS: Maskne

[BMJ 2021;373:n1304](#) and [Primary Care Dermatology Society](#)

Background

- 'Maskne' is not a diagnosis in its own right, but refers to a wide range of facial dermatoses triggered by the wearing of facial PPE, including masks, facial coverings, visors, spectacles and respirators.
- Masks cause a warm, moist occlusive environment around the face which can worsen or trigger skin conditions; allergy to components of the mask or friction from the mask and its strapping/support can also be factors.
- Mask-related acne and irritant contact dermatitis are thought to be the commonest causes.

Assessment

- History as usual, including affect on the quality of life and mental health.
- Ask about the **temporal relationship** between the dermatosis and the wearing of facial PPE - do periods without the mask alleviate or improve symptoms? How often do they remove their mask? Are they allowed mask breaks at work?
- **Examine the morphology** of the eruption, distribution and whether present at other sites other than the face - [click here](#) for excellent visual summary of causes/distribution.
- **Causes include:**
 - **Irritant contact dermatitis (ICD)** - thought to be commonest cause - wide variation in presentation with dry scaly patches to oedema and vesicles/erosions; signs from pressure on nose/cheeks may be present.
 - **Allergic contact dermatitis (ACD)** - rarer - similar to ICD but may be more discrete patches; refer for patch testing if suspected.
 - **Atopic eczema.**
 - **Seborrhoeic dermatitis** - dermatitis with greasy yellow scale affecting scalp, eyebrows and naso-labial folds.
 - **Periorifacial dermatitis** - patchy erythema with tiny papulopustules affecting periocular and perioral areas sparing the vermillion border, typically in young women.
 - **Urticaria** - may be immediate or delayed (e.g. 4-6 hours later).
 - **Acne** - comedones with papules and pustules, may be more widespread.
 - **Rosacea** - affects the centro-facial region (forehead, cheeks, nose and chin) and presents with flushing/erythema, papules/pustules and telangiectasia.
 - **Folliculitis.**

Management - General Measures

- Cleanse skin with gentle soap-free cleanser.
- Apply light emollient at least 30 minutes before applying PPE.
- Apply silicone-based barrier tape (e.g. siltape) to nasal bridge and cheeks.
- Wipe skin under PPE with silicone-based barrier wipe to provide a film and protect skin from microenvironment.
- Take time to fit mask and that not over tight.
- Take regular breaks from mask-wearing to relieve pressure and prevent moisture build-up (may need to be every hour if using respirators).
- Maintain oral hygiene (regular teeth brushing and interdental flossing/brushing).
- For **condition-specific management** see text in chapter above.

URTICARIA

CKS has recently updated its guidance [CKS 2022](#) but the basis for much of this comes from the British Society for Allergy and Clinical Immunology [UK Guideline 2015](#) which reviewed this common and difficult to treat the condition. Standard doses of antihistamines are often ineffective at licensed doses and guidelines commonly recommend using higher doses which may make us nervous...so, a recent DTB article summarising the evidence for higher-dose antihistamines is welcome [DTB2018.4.0641](#)

Background

- Urticaria is a mast-cell driven condition characterised by red, itchy raised wheals or hives that blanch with pressure. Typically the individual wheals come and go within 24 hours.
- When there is deeper swelling this is angioedema.
- It is extremely common. There is a female-to-male predominance of 2:1 and a point prevalence of 0.5% to 1%, and the lifetime prevalence for acute urticaria is 20%.
- An underlying cause is never found in most patients with chronic urticaria, but up to 30% of them may have an associated autoimmune disease.

How do we diagnose urticaria?

By definition, urticaria is characterized by the sudden appearance of wheals, angioedema or both. Wheals have a fleeting nature, with skin returning to a normal appearance in 1 to 24 hours whereas angioedema is characterised by deeper erythematous swellings (frequently involving below the mucous membranes) which are slower to resolve, lesions typically lasting up to 72 hours.

How is urticaria classified?

Acute urticaria	Spontaneous wheals, angioedema or both for < 6 weeks
Chronic spontaneous urticaria	Spontaneous wheals, angioedema or both for > 6 weeks
Chronic inducible urticaria	Urticaria or angioedema for >6 weeks, induced by known, specific triggers e.g. pressure, dermographism, cold, solar, heat, vibratory, water, exercise etc

Be aware of important differential diagnoses

We need to be aware that other rarer diseases and syndromes may present with urticaria so that if the clinical picture is not typical they should be referred. The key one to be aware of is **urticarial vasculitis**, which may be associated with autoimmune disease, but may also be triggered by infection (e.g. hepatitis B or C, glandular fever, streptococcal infection) or drugs (e.g. penicillins, thiazides, allopurinol, quinolones) or paraproteinaemia.

Typically the urticaria lesions last >24 hours, they 'burn' and are painful, rather than itch and leave a mark. Suspected cases should be referred for a biopsy.

What about urticaria in children?

- Acute spontaneous urticaria is common in infants and children, particularly in those with atopy
- It is commonly triggered by:
 - Viral infections
 - Food allergy especially milk, egg or peanut. Refer these children for allergy testing as food-induced urticaria may be a harbinger to anaphylaxis

GISS: UrticariaBased on [CKS 2022](#), [BSACI Guideline 2015](#) and [DTB2018.4.0641](#)**Diagnosis:**

- A clinical diagnosis is characterised by the sudden appearance of superficial wheals (red, raised, itchy) which generally resolve within 24 hours.
 - Angioedema is characterised by deeper, more painful swellings under skin & mucous membranes often of the face, genitalia, hands and feet, but these 2 conditions can co-exist in as many as 40%.
- **Acute urticaria** < 6 weeks.
- **Chronic urticaria** > 6 weeks. This can be one of 2 types:
 - Chronic spontaneous urticaria (previously idiopathic urticaria), includes autoimmune urticaria.
 - Chronic inducible urticaria (previously physical urticaria) e.g. cold, heat, pressure, solar etc

Assessment:

- Try to identify triggers with a [symptom diary](#) e.g. drugs (inc. topical and complementary), stress, exercise, physical triggers etc.
- Check for symptoms suggestive of or history of autoimmune disease.
- **Investigations - usually not required**, but consider if the cause is unclear to identify trigger factors or causes, particularly in those with chronic urticaria:
 - **Bloods:**
 - LFTs +/- viral hepatitis screen (hepatitis can trigger urticarial vasculitis).
 - TFTs with autoantibodies (presence of autoantibodies suggests autoimmune urticaria).
 - FBC - raised WCC may be seen in infection or inflammatory triggers.
 - ESR/CRP - may be raised in inflammatory/autoimmune causes inc vasculitis; ESR disproportionately raised vs CRP may suggest paraproteinaemia.
 - **Others** - Consider Helicobacter (weakly associated with chronic urticaria) & urine dip (?vasculitis).

Management:

- Acute urticaria with identified trigger - is self-limiting without treatment.
- Identify and manage the underlying trigger with avoidance if possible.
- **Drugs:**
 - **First-line:** non-sedating antihistamines (e.g. loratadine, cetirizine etc) at standard dose for up to 6 weeks. If no response, **progressively increase dose up to fourfold if needed**. Once control achieved continue for 3 to 6 months or prn if intermittent symptoms.
 - An additional sedating antihistamine at night may be used if interfering with sleep.
 - Oral montelukast is an option if resistant to antihistamines.
 - **Steroids:** If symptoms are severe, short course of oral prednisolone (40mg daily for up to 7 days) may be used. **They should not be repeated or used for long-term use.**

Referral?

- If urticaria is painful or persistent (this suggests urticarial vasculitis) +/- investigations suggestive.
- If not controlled on antihistamines (may be treated with montelukast, or immune suppressants).
- If thought to be due to food or latex allergy for allergy testing (more likely if intermittent attacks).
- If chronic urticaria that may be difficult to manage e.g. solar or cold urticaria.

Patient information: [NHS Patient information](#) and [British Assoc of Dermatol Information](#)

CHILBLAINS

Chilblains are a good example of a relatively common, annoying problem that patients ask us about and yet rarely appears in the literature. So, it has been welcome this year to see *new primary care-based randomised controlled trials* published [BJGP2017;67:e187](#) and [Ann Family Med 2016;14:453](#) and [Dermatol Ther 2016;29:263](#).

Often considered by patients as a vascular problem, chilblains and perniosis (chronic chilblains) are better considered as a dermatological condition as is a cold-induced dermatosis, although there is a link with Raynaud's and vascular disease.

Just remind me what chilblains are?

Chilblains are itchy or painful lesions on the extremities (fingers, toes and ears most commonly) but can also affect anywhere on the face and legs. It is a cold-induced dermatosis, also known as primary perniosis. The lesions are typically purple or purple-red nodules and the pain, itching or burning can be intense. They may ulcerate and scar.

Chilblains, also known as pernio, represent a *localised inflammatory dermatological response* resulting from an abnormal response to the cold. They occur most commonly in young women but can occur in children and older people. They can occur in non-freezing cold weather once the temperature drops below 15 degrees and can develop up to 24 hours after the cold exposure and may last for one to two weeks. *Chronic perniosis* develops on repeated exposure to cold resulting in persistent lesions which may scar.

Chilblains are generally idiopathic but may occur secondary to other diseases e.g. SLE. They are more common in people with Raynaud's phenomenon, 25% of people with chilblains also report Raynaud's. In older people, there is a link with peripheral arterial disease, noted by the presence of acrocyanosis (the dusky blue-red mottled skin we often see in older people with poor peripheral circulation).

Although the condition is benign, patients can report serious limitations in their ability to function and feel the need for treatment beyond cold avoidance.

What treatments are effective?

Obviously, cold avoidance is key, but in patients who develop chilblains what treatments work? GPs most commonly prescribe topical corticosteroids or vasodilators, based on previous observational evidence and expert opinion. For such a common problem there has been precious little evidence to guide us, but three recent randomised trials have been published:

- **Topical Steroids** [BJGP2017;67:e187](#)
 - These are widely used by GPs to treat chilblains
 - This small RCT (n=34) compared topical betamethasone valerate 0.1% once daily for 6 weeks compared to placebo cream in patients with chronic chilblains
 - There was no difference between the topical steroid and placebo, however, the response rate was high in line with other studies which typically show a placebo therapeutic response rate of 60% to 70%
 - Conclusion: topical steroids should not be used for chronic chilblains
- **Vasodilators**
 - **Nifedipine** is the most widely used oral drug prescribed for chilblains on the basis of older observational evidence
 - This was a new [RCT Ann Family Med 2016;14:453](#) comparing nifedipine 30mg controlled release bd to placebo in patients with chronic chilblains. It was well designed but again was a small trial (n=32)
 - After 6 weeks, there was no difference between nifedipine and placebo
 - Conclusion: These findings do not support the use of nifedipine in chronic chilblains

- **Pentoxifylline** is a peripheral vasodilator which has been around for years and is licensed (but not recommended by NICE) for the treatment of peripheral vascular disease
 - A recent study [Dermatol Ther 2016;29;263](#) comparing pentoxifylline 400mg tds found it to be superior to placebo and 'a safe and effective treatment'
- However, again this was a very small study (n=110) with limited follow-up (3 weeks!) so caution is advised!

Chilblains: Learning Points

- Common cold-induced dermatosis, producing painful, itchy, tender violaceous lesions on extremities, face and limbs
- They often appear 24 to 48 hours AFTER the cold exposure, and lesions may last for up to 2 weeks
- Although they are benign, chronic chilblains (perniosis) may develop with recurrent and persistent lesions which may ulcerate and scar
- There is a link with Raynaud's, SLE and other connective tissue diseases and proteinopathies however chilblains are generally idiopathic and most common in young women
- In older people, there is a link with peripheral vascular disease
- Previous observational studies have suggested that vasodilators such as nifedipine and topical corticosteroids are useful; recent randomised trials do not show efficacy for topical betamethasone or oral nifedipine, but one small trial has shown some efficacy for pentoxifylline
- All studies exhibit a high placebo response rate, of 60% to 70%
- Cold avoidance and lifestyle advice (e.g. avoid rapid re-warming with hot water bottles etc after cold exposure) is key
- Patient information sources
 - [patient.info chilblains](#)
 - [NHS Choices Chilblains](#)

SCABIES

KISS: Scabies

CKS 2022 & PHE/NICE antimicrobial prescribing

Background:

- Consider scabies as a differential in all patients presenting with itch.
- Infestation of the skin by the mite *Sarcoptes scabiei*.
- Incubation period is long (3-6 weeks).
- Most commonly spread through skin-to-skin contact - it is thought to be possible to be passed by sharing bedding, towels etc.
- In the UK it is most common in the winter and:
 - In Institutionalised communities (e.g. nurseries, schools and care homes).
 - Amongst the most vulnerable - the very young, the very old and those with immune deficiency.

Assessment:

- Typical sites of the infestation are between the fingers and wrists but also skin folds (e.g. buttocks, breast creases, genitals) and flexor surfaces where visible burrows may be seen.
 - It causes intense itching, especially at night which is much worse if immunocompromised.
 - The burrows typically appear as small, greyish lines in the web spaces and flexor surfaces of hands, wrists, ankles and feet.
 - The secondary eczema-like rash spreads over all the skin or secondary infection.
- In older people, it may present as a diffuse truncal eruption, in infants and children the face, scalp, palms and soles are most commonly affected.
- Severe, persistent itch and secondary infections can be debilitating.
- Occasionally, crusted ('Norwegian') scabies may develop which is then resistant to routine treatment
 - If crusted scabies, consider the possibility of immunodeficiency and investigate if appropriate.
- **Contacts/spread** - Ask about all people with whom skin-to-skin contact has been made for a prolonged period within the previous 2 months - they are at risk of scabies and should be informed.

Management:

- Treat all members of the house (ideally within 24 hours), sexual partners over the last month plus other close personal contacts:
 - **Topical permethrin is the most effective treatment** (1a) and highly effective at increasing cure rate at 28 days; use **two applications 1 week apart**.
 - Topical crotamiton is less effective than permethrin, but there is evidence of effectiveness.
 - Both of these preparations are considered superior to topical malathion and benzyl benzoate which have insufficient evidence to judge their efficacy.
 - Malathion is recommended if permethrin allergy.
- Clothing and bed linen and towels should also be washed at ≥ 60 degrees.
- Warn patients even with effective treatment itch may last up to 4 weeks.
- **Patient information** from [NHS Choices](#) and [British Dermatol Assoc](#)
- **OTC purchases:**
 - Insecticides can be purchased over the counter.
 - Malathion 0.5% aqueous liquid can be purchased for children/adults over 6 months.
 - Children under 2 years old require a prescription for permethrin 5% dermal cream.
- **Referral/seek advice:**
 - Resistant or severe cases (inc crusted scabies - may need consideration of Ivermectin).
 - Children <2 months.
 - Ongoing symptoms 2-4 weeks after the last treatment application, or concern over immune deficiency.

KISS: Guidance for Topical Treatment of Scabies

[CKS 2022](#) [BNF](#)

- Use permethrin 5% cream first line.
- Apply 5% preparation to the whole body and wash off after 8 to 12 hours:
 - Apply twice, one week apart.
 - For a single whole-body application, you need 30g of cream (average adult/child over 12 years) - larger adults may need double that amount.
- Do not apply after a hot bath, as this may increase absorption into the blood.
- All members of the household should be treated simultaneously.
- Treatment should be applied to the whole body, including the scalp, neck, face and ears (NB despite the manufacturer's recommendation to exclude head and neck).
 - Particular attention should be paid to the webs of the fingers and toes, and lotion brushed under the ends of the nails.
 - Avoid contact with the eyes and do not use it on broken skin.
- People with hyperkeratotic crusted scabies may need 2 or 3 applications of treatment on consecutive days.
- The itch and associated eczema may persist for some weeks after the infestation has been eliminated
 - Crotamiton lotion may help the itching.
 - A topical corticosteroid may be used after the scabies has been successfully treated.
 - Oral sedating anti-histamine at night may be helpful.
- Patient information from [NHS Choices](#) and [British Dermatol Assoc](#)

NB Learning Points: Scabies

- Consider this in the differential diagnosis of all patients presenting with an itch, and bear in mind that the incubation period is long (3-6 weeks).
- Permethrin 5% cream is the most effective treatment, applied as above.
- All household contacts should be treated at the same time.
- Continued itch may last several weeks, but topical crotamiton can be a useful adjunct and may even help cure rates.
- After the second application, topical corticosteroids can be used for the secondary itch.
- In treatment failure in severe cases, consider referral for oral ivermectin.

PSORIASIS

Psoriasis is a common condition that we will all see in General Practice, and one that can have a substantial impact on the quality of life and can result in significant functional, psychological, and social morbidity. Alongside the psycho-social impact, there is increasing evidence of an excess cardiovascular burden with psoriasis, especially in those with more severe disease, and we need to be ever-alert to the possibility of psoriatic arthritis, which if treated early can lead to significantly better outcomes. The national guidelines are looking somewhat out of date now - the SIGN guidance dates back to 2010, and the NICE guidance to 2012 (although it was updated in 2017), but excellent practical help has recently come along in the form of an updated guideline from the [Primary Care Dermatology Society \(PCDS\) May 2019](#).

Psoriasis: an overview and chronic plaque psoriasis [Primary Care Dermatology Society May 2019](#)

► Background

- Autoimmune inflammatory disease affecting ~3% of the population
- Genetic factors important - family history is present in 40-50% of cases, and in 75% of cases in those that develop the disease before age 20
- May develop at any age but most commonly in young adults and in the 6th/7th decades

► Triggers

- Stress - strongly associated with psoriasis
- Smoking and heavy drinking of alcohol are both associated
- Trauma - psoriasis can occur at sites of skin injury (Koebner phenomenon)
- Infection - streptococcal infection may trigger guttate psoriasis but can also be associated with chronic plaque psoriasis
- Drugs - many drugs affect psoriasis, the commonest are lithium, antimalarials inc. chloroquine, beta-blockers, NSAIDs, ACEI

► Clinical features

- Plaques tend to develop symmetrically, more commonly on the extensor surfaces, or can be widespread; common sites include scalp, elbows, umbilicus, gluteal cleft (in between buttocks), genitals, knees, nails
- Plaques tend to be ruby-red and well-defined with silvery scale, but lesions on lower legs may be less typical; typically described as either large plaque or small plaque
- [Click here](#) for excellent images from the PCDS website.

► Management - assess for associated co-morbidities

- **Psoriatic arthritis:**
 - May be present in up to 30% of people with psoriasis and can have a very varied presentation, but is **strongly associated with those with nail disease**
 - Refer anyone with psoriasis who presents with:
 - Signs of **inflammatory arthritis** (refer urgently if so)
 - **Dactylitis** (inflammation of an entire digit)
 - **Enthesitis without a mechanical cause**, especially if in multiple sites (that is inflammation of a tendon/ligament - typically characterised by pain and stiffness at big tendon insertions e.g. Achilles and patellar tendons)
 - The assessment of suspected psoriatic arthritis is covered in more detail by [NICE guideline NG 65 Feb 2017](#), which is summarised in **our chapter on SPONDYLOARTHRITIS** in the musculoskeletal section
 - Consider screening for psoriatic arthritis on an annual basis using the PEST screening tool ([click here](#)):

PEST Screening tool for psoriatic arthritis

- Have you ever had a swollen joint?
- Has a doctor ever told you that you have arthritis?
- Do your fingers or toenails have holes or pits?
- Have you had a pain in your heel?
- Have you had a finger or toe that was completely swollen and painful for no apparent reason?

Score 1 point for each affirmative answer.

A score of ≥ 3 may be indicative of psoriatic arthritis and should prompt referral.

- **Cardiovascular risk factors:**

- There is a strong association with CVD, particularly for those with more severe psoriatic disease
- Assess for, address and treat CVD risk factors - smoking, hypertension, hyperlipidaemia, diabetes

► **Management of psoriasis**

- **General:**

- Assess psycho-social impact - The **dermatology life quality index (DLQI)** score is one option to help with this and is recommended as an option by NICE - [click here](#)
- Provide patient information - [good information from BAD here](#)

- **Emollients:**

- Prescribe copious quantities - should be used both during flares and in-between times

- **Topical treatments:**

- The PCDS suggest starting with combination products of betamethasone and calcipotriol to encourage rapid improvement and better adherence (in contrast to NICE that recommends individual components initially) - options include Dovobet ® gel or ointment as well as the **new spray foam** (Enstilar ®):
 - Stop treatment when skin feels smooth even though it may be pink-red, and drop down to maintenance treatment with a vitamin D analogue or simple emollients
 - **Scalp** - Consider Dovobet ® gel as it has a more accurate applicator mechanism
- **Thin skin areas (face, flexures, genitals):**
 - Avoid betamethasone and consider calcitriol instead of calcipotriol as the vitamin D analogue, as it is less irritant to sensitive areas
- **Tar preparations** may be preferable for large thin plaques e.g. Exorex lotion ®
- If a **thick scale is present** consider treatments combined with salicylic acid e.g. Diprosalic ointment ® (a combination of betamethasone and salicylic acid)
- If not responded to first-line topical treatments above refer for consideration of second-line treatments - these include phototherapy, ciclosporin, methotrexate, acitretin, and the biologic agents

► **What about psoriatic nail disease? [CKS March 2018](#)**

- Psoriatic nail disease is a difficult condition to treat and is **strongly associated with psoriatic arthritis**.
- **Typical nail features** include ([click here](#) for images from the PCDS website):
 - Pitting
 - Onycholysis - yellow/white discolouration where the distal nail plate has separated from the nail bed.
 - Subungual hyperkeratosis - thickening of the nail bed
- **General management:**
 - Consider alternative diagnoses, especially fungal nail infection - consider sending clippings
 - Keep nails short - this reduces the risk of onycholysis
 - Avoid prosthetic nails
 - Consider nail varnish
 - If nail disease is having a significant functional or psychological impact refer to dermatologist for further input, but we need to warn patients that treatment options are limited, often with poor results
 - [Click here](#) for good patient information from Psoriasis Association on nail treatment

KISS: Psoriasis

[NICE CG 153 Sept 2017](#) [Primary Care Dermatology Society \(PCDS\) May 2019](#) [CKS 2018](#)

► **Assessment**

- Assess **physical, psychological, social well-being** - consider using DLQI ([click here](#))
- Assess **disease severity** - body surface area affected (consider Lund and Browder BSA chart to help estimate - [click here](#)), nail involvement and difficult-to-treat sites (face, scalp, palms, soles flexures)
- Assess for **psoriatic arthritis** (may be present in up to 30%):
 - Nail involvement is strongly associated with psoriatic arthritis
 - **Refer anyone with dactylitis or enthesitis without mechanical cause** especially if in multiple sites (inflammation of a tendon/ligament - typically characterised by pain and stiffness at big tendon insertions e.g. Achilles and patellar tendons)
 - Consider **yearly assessment with PEST screening tool** ([click here](#)) - refer if 3/5 +ve answers
- Assess for and **address/treat CVD risk factors** e.g. hypertension, smoking, diabetes, obesity - psoriasis is associated with excess CVD burden, especially those with more severe disease.

► **Treatment** NB TCS = topical corticosteroid VDA = vitamin D analogue

- **Emollients:** Prescribe copious quantities - should be used both during flares and in-between times
- **Trunk and limbs:**
 - 1st line: Potent TCS plus VDA daily (one applied morning, the other evening) for 4-8 weeks
 - 2nd line: VDA applied twice daily for 8-12 weeks
 - 3rd line: Potent TCS or coal tar preparation applied twice daily, or daily preparation of combination betamethasone/calcipotriol (options include ointment, gel or the new foam Enstilar®). NB the **PCDS suggests that we should consider the combination preparations first line** aiming to achieve quicker resolution of disease and aid concordance (which is in contrast to NICE advice)
 - If a **thick scale is present** consider treatments combined with salicylic acid e.g. Diprosalic ointment® (a combination of betamethasone and salicylic acid)
 - Stop TCS when skin feels smooth, but may still be pink-red and drop down to maintenance VDA or simple emollients
 - Aim to have a 4-week break between courses of potent TCS
- **Scalp:**
 - 1st line: Potent TCS for 4 weeks - consider different formulation if ineffective (shampoo/mousse)
 - 2nd line: Add scale removing agents before TCS application (e.g. with salicylic acid e.g. Sebco®)
 - 3rd line: Combined betamethasone/calcipotriol or VDA daily for 4-8 weeks (NB **PCDS recommend first-line Dovobet® Gel applicator** as it has a precise delivery system - massage into scalp, leave overnight and wash out with shampoo then water the next day)
 - Coal tar-based shampoos as well for long-term management (don't use alone for severe disease)
- **Face, flexures, genitals:**
 - Do NOT use potent TCS
 - Offer short-term (max 2 weeks/month) mild-moderate TCS and warn about skin atrophy
 - Consider VDA to reduce steroid requirement (calcitriol may be better than calcipotriol as less irritant)
- **Nails** ([Click here](#) for good PIL from Psoriasis Association):
 - Difficult to treat with limited treatment options - we need to warn patients even if referring
 - Consider differential diagnoses, esp. fungal nail infections - consider sending clippings
 - Keep nails short (reduces risk of onycholysis), avoid prosthetic nails, consider nail varnish

► **Refer to dermatology**

- Severe or extensive disease (e.g. >10% body surface area).
- If not controlled with topical therapy or having a significant impact on psycho-social wellbeing
- If a nail disease has a major functional or cosmetic impact
- Options include phototherapy, methotrexate, ciclosporin, biologic agents

FEVERISH ILLNESS IN CHILDREN

For one and a half decades primary care clinicians have relied upon a simple traffic light system to identify unwell children. When in 2007 NICE introduced its guideline "[Fever in under 5s: assessment and initial management](#)" despite the usual cries of turning the nuance of general practice in to arbitrary box ticking many of us favoured the easy-to-understand structure, defined physiological parameters and clear management pathways. Most of all, given it was produced by NICE, representing one of the most developed evidence-based approaches to medicine in the world, we believed in it.

It will therefore be a surprise to most of us that this traffic light system has never been validated in UK primary care. This is important because if we examine the data underpinning the guidance it is based on secondary care research, mostly in <6 month-olds. Step in a new paper published in the BJGP ([2022;72\(719\):e398](#)) which aims to determine the accuracy of this system.

► **Accuracy of the NICE traffic light system in children presenting to GP** [BJGP 2022;72\(719\)e398](#)

This is a retrospective cohort study analysing data from the DUTY (Diagnosis of Urinary Tract Infection in Young Children) study. The DUTY study recruited children <5yo in primary care who were acutely unwell from any cause and then evaluated for UTI.

This broad inclusion criteria plus the careful collection of physiological observations meant it was possible to apply these patients retrospectively to the NICE traffic light system (TLS) (with the exception of some focal features such as stiff neck and leg swelling). The researchers then accessed hospital data to check for admission over the next 7 days to indicate severe illness and the ultimate diagnoses.

- Numbers included = 6703
 - **Prevalence of serious illness** = 0.3%, just 17 children
 - **Admission within 7 days of initial presentation** = 2.1% (139 children)
 - **Of the unwell children presenting in GP the TLS categorised:**
 - Red = 31.6%
 - Amber = 62.7%
 - Green = 5.7%
 - **Identification of children admitted to hospital with serious illness, comparing red indicators to amber and green gave:**
 - Sensitivity = 59%
 - Specificity = 68.5%
 - Negative predictive value = 99.8%
 - However, **using only red indicators would miss 41% of children with serious illness.**
 - The NPV appears good but is flattered by the low prevalence of serious illness.
 - The positive predictive value for serious illness is a poor 0.5%.

The authors conclude "**this system is not suitable for use as a clinical tool in general practice**". They found the NICE traffic light system did not accurately detect children admitted with a serious illness, nor those not seriously ill who could have been managed at home and called for further research to update or replace it.

While this is only one study, existing published research supports the findings - a [smaller primary care study from the Netherlands](#) found similar results in febrile children and [emergency care data](#) also highlights the limitations in sensitivity and specificity.

Which red indicators were common triggers?

Deranged physiology including temperature, pulse rate, O2 sats and respiratory rate contributed to an amber classification in 28.8% of children in the amber category and contributed to a red classification in 1.7%. This surprisingly low level may be explained by the limited number of physiological markers in the "red category" - only severe tachypnoea and fever in <3 month olds.

The most common trigger was "abnormal social cues" - present in 97% of amber cases and 61% of red cases. This is a very clinician-subjective indicator and one wonders if this has not translated easily to retrospective analysis, resulting in skewed results towards greater numbers of higher-risk cases.

Does the lack of severely deranged vital signs as high risk indicators limit the tool's ability to discern higher-risk cases? Perhaps, but arguably it may not be that simply and it may be more appropriate tool to simply say those with deranged physiology are "intermediate risk" implying that further clinical assessment and judgement is required.

Can the accuracy be improved?

For general practice, the key to this type of tool is to have a high sensitivity so that we can be confident we don't miss serious illness. A high sensitivity is necessary for 'rule out' tools. The limited sensitivity seen here is perhaps unsurprising given NICE recommends children with 'amber features' may be admitted or managed in the community with safety netting advice - these children, or indeed indicators, could go either way.

The authors addressed this by also calculating the sensitivity and specificity comparing red and amber indicators to green. This improved the sensitivity to 100% but with a specificity of 5.7%, resulting in almost 20 children being referred to identify 1 with serious illness. Few of us would feel this would be appropriate or manageable for our paediatric colleagues.

What about other clinical decision tools?

Notable alternatives include [NICE sepsis guidance](#) and [PEWS](#) - the Paediatric Early Warning Score. There are aspects which may promote this tools as having greater utility. Both have stronger focus on levels of deranged physiology and indicators which clearly suggest a child is in seriously unwell.

This is a complex area because while these tools have a greater number of physiological markers we cannot assume this translates in to greater accuracy. While data supports NICE sepsis in the [emergency department](#) neither of these tools (or any other currently) have been validated in primary care.

What can we learn from this?

As the linked editorial ([BJGP 2022;72\(719\):251](#)) explains because truly serious illness in young children is rare "we don't see [the tool] failing" and "the system works when it comes to sending away 'green' lit children, reassuring parents and GPs."

On the positive side, the NICE guidance has engendered within PHCPs the act of carefully monitoring and documenting vital signs which may have benefits beyond a flawed assessment system but this important paper highlights the limitations of the algorithmic approach to medicine and necessity of validating such tools in the setting for which they are intended.

There are other important points we can pull from this paper:

- Serious illness is uncommon (0.3%) even in constitutionally unwell children
- The presence of only green indicators in unwell children makes serious illness extremely unlikely, but the presence of only green indicators is uncommon (5.7%) in unwell children presenting to primary care

The main message to come out of this paper is that clinical judgement remains paramount in any assessment of an unwell child and that face-to-face assessment to allow thorough assessment is important. An algorithm cannot successfully differentiate between a serious and not serious illness - a clinician is needed. However there is still a place for these tools - they can supplement our judgement, focus our minds through methodical, thorough assessment including physiological markers, and so help us find that needle in the haystack.

In the NICE guideline appendices, page 585, the original 2007 evidence summary notes that **"in isolation, none of these symptoms or signs are reliably associated with serious illness" but "a child identified as 'ill' when assessed by an experienced healthcare professional is likely to have a serious bacterial illness"**. Perhaps this is the most important message we were meant to learn from this guideline.

NB Learning Points: Feverish Illness in Children

- The latest data challenges the accuracy of the NICE traffic light system for assessing fever in children <5 years old, demonstrating that it fails to identify almost half of children with serious illness.
- Serious illness is uncommon in young children but can only be reliably excluded by the traffic light system when children only have green indicators.
- This paper highlights the importance of clinical judgement and not over-reliance on clinical decision tools for identifying acutely ill children.

TIC DISORDER

A recent BMJ Practice Pointer paper ([BMJ 2022;376:e069346](#)) discusses this interesting problem, which is surprisingly common in childhood and often goes unrecognised.

► What is a primary tic disorder?

Tics are an involuntary motor or vocal action, varying in type, frequency and severity in individuals and over time. The person often has a premonitory sensation prior to the tic such as a feeling of needing to stretch a muscle, which is then temporarily relieved (often for only a short period) by the tic occurring.

Tics are outside of a person's control but a person may be able to suppress it for some time (variable from seconds to hours, the paper gives the example that some children may be able to reduce them during school). This is typically followed by rebound in frequency. Frequency reduces when relaxed, performing tasks (e.g. school work, sports, music) and distracted, and increases with anxiety, stress and tiredness.

By their nature, tics can be bothersome physically (and may be painful due to repetitive movements) and psychologically (due to living with a long-term condition, embarrassment, impact on self-esteem & QoL, etc.).

Tourette's is the most well-known tic disorder - it is at the more severe end of the spectrum comprising of both multiple motor tics and at least 1 vocal tic lasting for at least 1 year.

KISS: Primary Tic Disorder

Definitions

- **Transient or provisional tic disorder:** tics persisting for less <1 year
- **Chronic or persistent tic disorder:** multiple motor or vocal tics lasting >1 year
- **Tourette's syndrome:** multiple motor and ≥1 vocal tics lasting >1 year

Associated co-morbidities

- **ADHD is most common** (in 90% of people with Tourette's attending specialist clinics)
- Also ASD (less commonly), OCD/OCB
- Anxiety, depression, sleep disorders, self-injurious behaviours, anger issues, poor impulse control & behavioural disinhibition, disruptive disorders, learning disorders

► How common is primary tic disorder?

The BMJ paper reports a prevalence of around 1% in school children, but up to 20% may be affected at some point, albeit only for short periods of a few weeks or months before resolving.

Tics often improve with age. By adulthood 30-50% will be in remission, 20% have ongoing moderate to severe tics, but 5% will experience debilitating or refractory tics.

► Is primary tic disorder associated with other conditions?

Yes. ADHD is a very common co-morbidity (in up to 90% of people with Tourette's). Tics are often mistaken as bad behaviour when in fact the child has a combination of pathologies.

It is also associated with autistic spectrum disorder, obsessive-compulsive disorder/behaviour, and a range of mental and behavioural disorders.

► How do we assess for primary tic disorder?

See KISS below.

KISS: Assessment of Tics in Children [BMJ 2022;376:e069346](#)

Goals of initial assessment:

- Distinguish from other involuntary movements, e.g. neurological (dystonia, chorea, etc.), stereotypies
- Assess for co-morbidities, e.g. ADHD, ASD, learning disability, OCD/OCB, depression, anxiety, etc.
- Assess for psychological and QoL impact on patient, e.g. school, family, friends, relationships
- Family history?

Characterise the tic

- Type, nature, frequency, intensity, impact of tics
- Motor and/or vocal, simple or complex
- There are some specific types of tics:
 - Coprolalia - involuntary swearing (20-30% with Tourette's)
 - Copropraxia - involuntary obscene gesturing or socially inappropriate touching, e.g. genitals of self/others
 - Echolalia: repeating word, phrase, sentence
 - Echopraxia & suggestibility: copying/repeating other people's actions/movements; needing to tic when seeing/talking about tics

Differential diagnosis - is it a stereotypies?

- *Defined as involuntary, patterned, coordinated, rhythmic, repetitive, non-reflexive movements that occur in the same fashion with each repetition*
 - Includes rocking, hand flapping, waving
- More commonly linked with ASD and learning disability
- Features that separate these from tics include:
 - Movements being more rhythmic, co-ordinated & purposeful, usually of limbs or entire body, lacking a sense of urge prior or relief after, non-voluntary, constant without much variation, onset usually in <3y with no specific change relating to age
- Also consider if not a stereotypies but neurological, e.g. dystonia, chorea

► How can we help manage tic disorder?

Acknowledging the problem and explaining tics are outside of the child's control is crucial, and providing parents and school with the same information very important - children should not be reprimanded for their tics. The BMJ has a number of very helpful strategies which can help alleviate or manage tics day-to-day for children - see KISS on next page.

Watchful waiting may be appropriate for simple tics causing no distress or dysfunction. This recommendation is supported by NICE guidance ([NG 127 - Suspected Neurological Conditions](#)). However, dysfunction, distress or pain is an indication for intervention and a range of options exists. Children will require referral via local pathways. Both the BMJ and NICE suggest:

- If associated with anxiety, OCD/B or other mood disorder refer to paediatric psychiatry
- If associated with autism or ADHD refer to neurodevelopment team
- If tic disorder is severe refer to neurology

Treatment options include non-pharmacological treatments (e.g. CBT, habit reversal, exposure & response prevention), pharmacological treatments (alpha-adrenergic drugs e.g. clonidine, 2nd gen anti-psychotics e.g. risperidone or aripiprazole, benzamides or 1st gen anti-psychotics, also botox, cannabinoids and deep brain stimulation for refractory disease) and treatment of co-morbidities such as ADHD or OCD.

KISS: Self-help Strategies for Children with Tic Disorder

[BMJ 2022;376:e069346](#)

- Psychoeducation of young person, family, teachers and peers
- Provide proactive (not reactive) support and predictable routines
- Provide breaks, with opportunities for physical movement
- Designate a safe place to go to when tics are severe, with a pre-agreed pass/signal for time out
- Preferential seating (e.g. front of the class, close to the door)
- A buddy for learning and social support (may help prevent bullying)
- Planning/managing communication about tics (to school, peers/friends, employer) and self-advocacy (how to respond if asked about tics)
- Allow scribe/computer if writing is affected; special provisions for examinations—seating in a separate room, allowing extra time
- Identify and avoid triggers; manage stress, anxiety, or boredom
- Encourage relaxing activities and foster strengths and hobbies to boost self-esteem
- Join support groups to share experiences and attend activities or events such as camps
 - [Tourette's Action](#) is a charity with lots of resources, links for local groups and professional resources
 - The [NHS has clear information](#) as well on clinical features, diagnoses and management

NB Learning Points

- Tic disorder is a common and under-recognised issue with up to 20% of children experiencing tics at some point and 1% being affected in the long term.
- Primary tic disorder should be differentiated from stereotypies
 - For more patient/parent information on stereotypies [click here](#)
- Tic disorder is commonly associated with ADHD - assess children for the condition if not yet diagnosed
- Watchful waiting is appropriate in mild forms of the condition - progress is good for most in this group with resolution of tics as they grow older
- Troubling tic disorder merits referral as a range of non-drug and drug treatments may be helpful.



VIRAL INDUCED WHEEZE

Despite being the most common of presentations during winter there has been considerable uncertainty as to the best management for these children. What is the best treatment during an acute episode? Can any treatment prevent future episodes? And, crucially, when is this simply viral-induced wheeze and when is it asthma?

Previous data has suggested some answers to these questions, but newer research challenges much of what went before. A [new update of the Global Initiative for Asthma in 2021](#) reviews the best knowledge to date and provides strategies for managing this group of children ≤ 5 years old.

► *What causes wheeze in younger children?*

Wheeze is common in younger children. It is typically linked to upper respiratory tract infections and some viruses are particularly associated with recurrent wheeze during childhood (e.g. RSV, rhinovirus). These can trigger wheeze in children without asthma, but of course, children with possible asthma may be more likely to be affected by wheeze during infection with the potential for more severe or longer-lasting symptoms. Asthma in its own right is the other common cause of wheeze in children.

In the past viral-induced wheeze in children has been characterised in a number of different ways such as the phenotypes "episodic wheeze" and "multiple-trigger wheeze". Subsequent prospective data examining this found uncertain clinical utility in the labels in the real world.

The GINA guidance suggests considering whether an affected child may have underlying asthma and adopting a "probability-based approach".

► *What features make asthma more probable?*

Using this approach clinicians can look at a range of clinical features to build a picture of the chance of asthma. This can then guide on whether to trial controller therapy is indicated or if PRN short-acting beta-agonists during acute episodes is all that is required.

Sounds straightforward, but we know this can be challenging and rarely clear-cut in reality. It is made more difficult due to the lack of merit of investigations in this younger population. Diagnosis is therefore based on clinical features such as:

- **Symptom patterns** (recurrent episodes of wheeze/cough, nocturnal Sx), **presence of risk factors** (e.g. FHx, atopy, food allergy), **response to controller therapy** and **exclusion of alternate diagnoses** (important to examine the child to elicit +ve and -ve findings)

GINA - Probability of Asthma: Symptom Patterns [GINA 2021](#)

Few have asthma	Some have asthma	Most have asthma
Sx for <10 days during URTI	Sx for >10 days during URTI	Sx for >10 days during URTI
2-3 episodes per year	>3 episodes per year, or severe episodes +/- night worsening	>3 episodes per year, or severe episodes +/- night worsening
No Sx between episodes	Between episodes child may have occasional cough, wheeze or heavy breathing	Between episodes child has cough, wheeze or heavy breathing during play or when laughing
		Allergy sensitisation, atopic dermatitis, food allergy, or FHx of asthma

► What is the differential diagnosis?

Keep the differential wide. Most commonly the alternative is recurrent viral URTI but there are other important conditions that could also cause similar features.

- **Recurrent viral infections:** mainly cough, coryza, no Sx between episodes
- Also consider GORD, foreign body, persistent bacterial bronchitis, tracheomalacia, TB, congenital heart disease, cystic fibrosis, primary ciliary dyskinesia, vascular ring, bronchopulmonary dysplasia, immune deficiency

► Is there an age when it can't be asthma?

GINA does not state any age restrictions for this diagnosis but notes that wheezing and coughing are common in children aged 0-2 years old without asthma.

► How should we manage this group?

Management can be considered as immediate and longer-term treatments.

Oral corticosteroids for acute wheeze have been an area of significant debate over the past few years. GINA recommends only using in mild to moderate exacerbations where symptoms are worsening. Data shows no benefit of oral corticosteroids when given to children in the outpatient setting, and some reduced risk if given from an emergency department. In severe exacerbations, the rationale is stronger, and they are essential for children who actually have a diagnosis of asthma rather than simply viral-induced wheeze.

KISS: GINA Management of Acute Wheeze / Asthma in Children ≤5 years

- **Severe/life-threatening illness**
 - **Indicators:** child unable to speak or drink, cyanosis, RR >40, sats <92% air, silent chest, lack of response to 6p salbutamol or persisting tachypnoea after 3 rounds of SABA
 - **Immediate hospital transfer via 999.** While waiting for ambulance:
 - Give salbutamol MDI via spacer 6 puffs every 20 minutes as needed (or 2.5mg via nebuliser)
 - Oxygen if required to maintain O2 sats 94-98%
 - Prednisolone 2mg/kg - max. per day 20mg for <2yo, 30mg for 2-5yo
 - Consider 1-2 doses of nebulised ipratropium bromide
- **Mild > moderate illness**
 - **Salbutamol MDI 2puffs via spacer or 2.5mg via neb every 20 mins for 1st hour**
 - O2 if needed, consider ipratropium if needed
 - Spacers will require a face mask for <3yos and a mouthpiece for most 3-5yos
 - **Monitor closely for 1-2h - refer if lack of response to treatment, any signs of severe exacerbation, increasing resp rate, decreasing O2 sats**
 - **If improving continue to monitor, if Sx recur within 3-4h give 2-3p salbutamol per hour**
 - **If worsening**
 - **Give 10p salbutamol over 3-4h, prednisolone for 3-5 days** (dose as per severe illness) **and treat as per severe illness**
 - **Discharge and f/up**
 - Ensure adequate resources at home, continue reliever, continue/adjust/add controller, check inhaler technique, provide an asthma action plan
 - **Follow up 1-2 working days and at 1-2 months**

BTS / SIGN management of acute asthma exacerbation is subtly different - with larger numbers of puffs being recommended if necessary (up to 10 over 3-4h) and lower doses of pred (10mg <2y, 20mg 2-5y)

Regular controller treatment is recommended where asthma is suspected or where acute treatment is required more than every 6-8 weeks it should be trialled to confirm if symptoms are due to asthma. Where there is no response to treatment alternative diagnoses should be considered.

The pathway is comparable to standard asthma treatment:

- Step 1 - PRN short-acting beta-agonist via MDI + spacer
- Step 2 - Daily low-dose inhaled corticosteroid + PRN SABA
 - Daily montelukast or intermittent ICS at onset of resp illness are other options (but less preferred)
 - ICS - LABA combination inhalers are not recommended due to lack of safety data in <4yo
- Step 3 - Double low-dose ICS
- Step 4 - Continue treatment and refer

So for children with only infrequent wheezy episodes during viral illnesses having a salbutamol MDI at home to use via a spacer is appropriate management.

► *Is there a role for montelukast?*

Historical papers have suggested montelukast may be helpful in children with viral-induced wheeze. However, a [Cochrane review in 2015](#) found that neither intermittent montelukast for acute episodes or maintenance treatment for frequent episodes reduced risk of future exacerbations requiring oral steroids compared to placebo. Concerns have also been [raised about the side effects of leukotriene receptor antagonists](#) - neuropsychiatric effects may occur with insomnia, depression and agitation occurring in up to 1 in 100.

Subsequently, GINA makes no mention of their use for acute treatment, and for recurrent viral wheeze (which GINA would consider more likely to be an indicator of underlying asthma) low dose ICS is the preferred option and LTRAs reserved as an add-on therapy where symptoms are inadequately controlled.

NB Learning Points & Quality Improvement Ideas

- Previous phenotypes describing viral-induced wheeze in children are unhelpful and clinicians should simply consider whether these children have patterns of illness which may indicate asthma.
- Acute management is with salbutamol MDI via a spacer with mask or mouthpiece - doses recommended are lower than many of us may have used in the past.
- Oral corticosteroids are only advised in hospitalised children with acute viral wheeze.
- For children with features suggestive of asthma use the step-wise approach to management. Frequent recurrent episodes or symptoms between attacks are an indicator for low-dose inhaled steroids.
- Follow-up is crucial after an acute episode - children should be reviewed in 1-2 days and then in 1-2 months. All children, parents and carers should have an asthma acute plan discussed.
- [Click here for an excellent patient information leaflet](#) including instructions on how to use a spacer, what doses of salbutamol to use during various severities of acute episodes and a short-term management plan for future episodes
- **Quality improvement**
 - Audit children <5yo with viral-induced wheeze - do any have frequent attacks? have they been asked about persisting symptoms? have they trialled ICS?

RESPIRATORY ILLNESS IN CHILDREN

Having dealt with a global pandemic and the highest-ever patient demand on general practice you would think we were due a break. Predictions however suggested it would be quite the opposite with exceptionally high levels of viral respiratory illness, particularly in children.

During the pandemic cases of viral respiratory illness other than covid dropped dramatically - [RSV levels were around 98% lower, influenza 99%](#). [Data from the southern hemisphere shows a marked bounce back](#) in their summer when levels are usually very low. Areas with stricter lockdowns have had a higher and more prolonged peak.

In the UK, winter 20/21 was unusually quiet for non-covid respiratory infections with a similar increase in the summer as seen in the southern hemisphere. It may be that that small unseasonal summer peak actually protected us from a much greater spike throughout the winter period as levels have thankfully been around the seasonal average, albeit with an additional huge spike in coronavirus cases.

► *What infections will we see?*

Nevertheless, the concerns drove the discussion on how children could be managed in the community to reduce pressures on hospitals. **The Royal College of Paediatrics and Child Health** produced [guidance, updated in July 2021, on assessment of bronchiolitis, cough and breathlessness in children](#), with useful pathways for community settings including on remote clinical assessment. This provides a more nuanced 'traffic light' risk stratification than the recently updated [NICE Bronchiolitis Guideline 2021](#) (see specific Bronchiolitis chapter) and more *general guidance on children with cough/breathlessness including remote assessment*. They also have great parent/carer resources.

Key areas for general practice to be aware of include:

- Just as we have found with covid, O2 sats are a crucial part of bronchiolitis assessment, but as we will all have discovered in the past, standard finger sats probes are hopeless in small children. *Prioritising access to paediatric oxygen saturation monitor probes in general practice is a key recommendation from the RCPCH.*
- *Bronchiolitis may present in older children than we typically see* - rather than the usual <1y, it may be in 1-2 year-olds as well ([mean age in Oz/NZ has been 18m, usually 7-12m](#)) due to a combination of RSV naive babies, older infants with no prior RSV exposure and a lack of herd immunity.
- In children who are very unwell, it is highly unlikely to be covid (almost always a mild illness in young children) and more likely to be one of the 'traditional' illnesses such as bronchiolitis, croup, pneumonia, etc. Remote assessment is possible but many will require face-to-face assessment, and waiting for a covid swab result is not safe for the child.

The RCPCH calls for support for community services which would be very welcome although hard to see beyond education and sats monitors what this would be.

NB Learning Points and Quality Improvement Ideas: Respiratory Illness in Children

- Updated guidance on the assessment of children with respiratory illness includes strong support for measuring oxygen saturations
- Infants <2yo require a dedicated paediatric pulse oximeter to avoid inaccuracy
- Quality improvement
 - Does your practice have a paediatric sats monitor? If not can you purchase one?
 - Staff will need training on how to successfully use a paediatric probe reliably.

KISS: Assessment of Cough/Breathlessness RCPCH 2021

Clinical support tools for remote clinical assessment children [under 1 year of age](#) and [over 1 year of age](#)

Clinical Findings	Green - low risk	Amber - intermediate risk	Red - high risk
Colour / activity	~Normal colour skin/lips/tongue ~Responds normally to social cues ~Content/smiles ~Stays awake or awakens easily ~Strong normal cry/not crying	~Pale/pallor ~Reduced response to social cues ~Wakes only after prolonged stimulation ~No smile ~Decreased activity/lethargy	~Blue/grey colour ~No response ~Unable to rouse or if rouse does not stay awake ~Clinical concerns about cry Confused
Respiratory	~None of the amber or red symptoms	~Resp rate: <1y = 50-70bpm 1-2y = >40bpm 2-5y = >35bpm 5-12y = >30bpm >12y = >25bpm ~Mild/mod resp distress ~Audible stridor with distress or exertion	~Resp rate: <1y=70bpm, or >1y = 10+bpm over amber levels ~Severe respiratory distress ~Apnoeas ~Audible stridor at rest ~Unable to complete sentences
Circulation /hydration	~None of the amber or red symptoms	~Cold hands/feet in absence of fever ~Reduced urine output ~In <1y: reduced fluid intake: 50-75% usual over 3-4 feeds ~In >1y: not tolerating fluids / repeated vomiting ~Unable to swallow saliva	~In <1y: markedly reduced fluid intake: <50% usual over 2-3 feeds
Other	~None of the amber or red symptoms	~In 3-6 month old: temp ≥39.0oC ~Fever ≥5 days ~Risk factors for severe illness ~Additional parent/carer support required ~Lower threshold for face-to-face review if significant chronic co-morbidities	~In 0-3 month old: temp ≥38.0oC ~Seizure ~Sudden onset and parental concern about inhaled foreign body

Action Required

- **Green** - provide safety net advice (PIL for coughs and colds [<1y](#), [>1y](#)), confirm happy with decision, consider safeguarding
- **Amber** - consider video consultation and/or bring in for F2F assessment
- **Red** - refer immediately to emergency care, consider if 999 or parent/taxi most appropriate

BRONCHIOLITIS

In August 2021 NICE updated their bronchiolitis guideline [NICE 2021, NG9](#) in time for the expected surge in cases over Autumn and Winter 2021/22. The Royal College of Paediatrics and Child Health has produced [RCPH Bronchiolitis Guidance 2021](#) with a useful pathway for community settings.

In 'normal' pre-COVID years, bronchiolitis was always common over the winter months, affecting 1 in 3 children by the age of 1 year and hospitalises 2-3% of all infants. During the first winter of the pandemic, there was a huge 98% fall in cases of RSV related illness, leaving large numbers of small children non-exposed. ***It is predicted that there will be a huge surge in cases (NB already underway, September 2021) over Autumn and Winter 2021/22*** and we are likely to also see more cases in children older than we usually see with bronchiolitis due to a combination of RSV naive babies last year, older infants with a lack of exposure and lower herd immunity. A key update in the NICE 2021 bronchiolitis guideline compared to the previous version in 2015, is that ***measurement of oxygen saturation using appropriate paediatric pulse oximeter*** is a key part of the assessment. They have updated recommendations on O2 saturation thresholds for admission e.g. and children with a persistent oxygen saturation of <92% when breathing air be considered for admission and also management in secondary care. A recent [NHS Safety Alert](#) on pulse oximeters has warned that if an adult probe is inappropriately used on a child, it can produce a reading 50% lower than the real value.

► How do we assess for bronchiolitis?

Bronchiolitis remains a clinical diagnosis. Investigations are not required. It affects children under 2 years old, usually in the first year of life and most commonly in the 3 – 6 month age group although it is predicted for Autumn/Winter 2021 to occur more often in children age 1 to 2 years.

KISS: Assessment for suspected Bronchiolitis [NICE 2021](#)

Diagnosis is made if children have prodromal coryzal symptoms for 1-3 days followed by:

1. Persistent cough, AND
2. Tachypnoea and/or chest recession, AND
3. Wheeze and/or crackles on chest auscultation

Associated symptoms include:

- Fever (30%, usually low grade – fever >39°C is unusual, consider alternative diagnosis e.g. pneumonia)
- Poor feeding (most common days 3-5)
- Apnoea may be the presenting feature in very young children (especially <6wk old)

Examination

- Measure temperature, heart rate, respiratory rate, capillary refill time
- Check oxygen saturations, using appropriate paediatric sat monitor
 - *if an adult probe is used for a child, it can produce a reading of 50% lower than the real value* [NHS Safety Alert 2018](#)
- Look for signs of respiratory distress (recession, nasal flaring, exhaustion, etc)
- Check for dehydration – ask re: oral intake and wet nappies

Alternative diagnosis?

Consider pneumonia as an alternative diagnosis if T>39 and/or persistently focal crackles
Consider viral-induced wheeze or early-onset asthma rather than bronchiolitis in older infants (over 1) and young children if they have persistent wheeze without crackles, recurrent episodic wheeze or history of atopy

► ***When should infants be referred?***

The vast majority of children with bronchiolitis can be managed safely at home. NICE set out specific indications for hospital assessment and admission. These need to be considered in the context of the individual and their home environment, and lower thresholds may be appropriate.

Children with existing heart or lung disease, infants <3 months, premature birth, neuromuscular disorders and immunodeficiency are associated with more severe bronchiolitis.

Factors affecting a carer's ability to look after a child with bronchiolitis should also be considered, e.g. social circumstances, confidence of carer in spotting red flags, distance to healthcare in case of deterioration.

KISS: Referral Indications [NICE 2021](#)

Consider referring children with bronchiolitis if:

- Respiratory rate >60 bpm
- Difficulty breastfeeding or inadequate oral intake (50-75% of usual volume)
- Clinical dehydration
- Persistent O2 sats <92% on air

Immediate admission (usually via 999) is indicated if:

- Apnoea (observed or reported)
- Child looks seriously unwell
- Severe respiratory distress (e.g. marked recession, grunting, RR >70bpm)
- Central cyanosis
- Persistent O2 sats on air < 90% in children aged 6 weeks or over, and < 92% in children aged under 6 weeks or children of any age with underlying conditions

Note the following are ***signs of impending respiratory failure***:

- Signs of exhaustion, recurrent apnoea, failure to maintain O2 sats despite oxygen

Lower thresholds for referral in children with underlying conditions (e.g. prematurity, existing heart, lung or neuromuscular disorders, immune deficiency etc) or if factors affecting the ability of carers to safely monitor and look after at home

► ***Management of bronchiolitis***

Treatment is supportive and parents should be counselled on maintaining fluid intake and monitoring for signs of increasing illness. Symptoms usually peak at day 3 to 5. Cough resolves in 90% by 3 weeks.

There are ***no medications that alter prognosis*** in bronchiolitis – antibiotics, inhalers, nebulisers, steroids and montelukast are not helpful and should not be prescribed. Antipyretics are only recommended in the presence of fever and distress, as per the NICE Feverish Illness in Children guidance. Children who are admitted with low O2 sats will obviously receive oxygen in secondary care, and CPAP if they have impending respiratory failure.

Traditional management such as steam inhalation or raising the head end of the cot lack research to guide recommendations.

► *How should we safety net?*

NICE is becoming more specific about safety netting advice in their guidance. ***They recommend we should provide key safety information for parents and carers to take away*** for reference when looking after children at home including:

- How to recognise developing 'red flag' symptoms
 - Worsening work of breathing
 - Fluid intake is 50-75% of normal or no wet nappy for 12 hours
 - Apnoea or cyanosis
 - Exhaustion (e.g. not responding normally, wakes only after prolonged stimulation)
- People should not smoke in the child's home because it increases the risk of more severe symptoms
- How to get immediate help from an appropriate professional if any red flags symptoms develop
- Arrangements for follow up if necessary

► *The Royal College of Paediatrics and Child Health* [RCPH Bronchiolitis Guidance 2021](#)

This guideline, with a useful pathway for primary care and community settings, provides a more nuanced 'traffic light' approach to assessment of infants with suspected bronchiolitis. See KISS summary below.

NB Learning Points & Quality Improvement Ideas

- There will be a predicted surge in cases of bronchiolitis over Autumn and Winter 2021, following the huge reduction in cases during the 'lockdown' winter of 2020/21
- Bronchiolitis usually occurs most commonly in babies under 1, peaking at 3 to 6 months, although it can occur up to the age of 2; a lack of exposure last year means that it is expected that it will be seen more often in infants aged 1 to 2, peaking at 18 months
- Bronchiolitis is a clinical diagnosis made with 1-3/7 coryza, persistent cough, tachypnoea or chest recession and wheeze or crackles on auscultation
- A key update in the NICE 2021 bronchiolitis guideline is that ***measurement of oxygen saturation using appropriate paediatric pulse oximeter*** is a key part of assessment e.g. children with a persistent oxygen saturation of <92% when breathing air be considered for admission and also management in secondary care
- The RCPCH have produced more nuanced 'traffic light guidance' for assessment
- Ensure there is appropriate safety netting information for parents and carers
 - [RCPCH Bronchiolitis, Advice for parents and carers](#)



KISS: RCPH Bronchiolitis Pathway [RCPCH 2021](#)

Suspected bronchiolitis?

- **Symptoms/signs:** snuffly nose, chesty cough, poor feeding, vomiting, pyrexia, increased work of breathing, head bobbing, cyanosis, bronchiolitis season, inspiratory crackles +/- wheeze
- **Risk factors for severe disease?** Pre-existing lung condition, immunocompromised, congenital heart disease, age <6 weeks (corrected), re-attendance, prematurity <35 weeks, neuromuscular weakness
- **Immediately life threatening signs?** Call 999, alert paed, start high flow O2 and monitor
- **Consider differential diagnosis:**
 - Temp ≥ 38.0 - think sepsis; sweaty - cardiac; unusual features of illness - other causes

Clinical findings

- **Low-risk (green)**
 - **Behaviour:** alert, normal
 - **Skin:** CRT < 2secs, moist mucous membranes, normal colour skin, lips and tongue
 - **Resp:** < 12 months < 50 breaths/min, mild resp distress - mild chest recession, no nasal flaring, no grunting, no apnoeas
 - **Sats:** 95% or over
 - **Feeding:** normal - tolerating 75% of fluid or more, occasional cough-induced vomit
- **Intermediate risk (amber)**
 - **Behaviour:** irritable, ↓ activity, ↓ response to social cues, no smile
 - **Skin:** CRT 2-3 secs, pale/mottled, pallor reported by parent/carer, cool peripheries
 - **Resp:** all ages > 60 breaths/min, ↑ work of breathing, moderate chest recession, nasal flaring may be present, no grunting, no apnoeas
 - **Sats:** 92-94% or over
 - **Feeding:** tolerating 50-75% of fluid intake over 3-4 feeds, reduced urine output
 - Any risk factors for severe disease
- **High risk (red)**
 - **Behaviour:** unable to rouse, wakes only with prolonged stimulation, no response to social cues, weak or continuous cry, appears ill to the health professional
 - **Skin:** CRT >3 secs, pale/mottled/ashen blue, cyanosis
 - **Resp:** all ages > 70 breaths/min, resp distress, severe chest recession, nasal flaring and grunting present, apnoeas
 - **Sats:** < 92%
 - **Feeding:** < 50-75% over 2-3 feeds/12 hours, appears dehydrated, significantly reduced urine output

Management

- **Green** - provide guidance and give [patient advice sheet](#), confirm carer comfortable with action
- **Amber** - seek advice from paed and/or give clear management plan agreed with parents; ensure safety netting, give [advice sheet](#) and info on where to go if child deteriorates
- **Red** - consider 999, call paed, consider O2 and any treatment required to stabilise child
- Also consider that bronchiolitis often deteriorates until day 3, which may influence advice given and referral threshold - it is particularly important in children with risk factors for severe disease

Normal values

- Resp rate - <1y=30-40bpm, 1-2y=25-35bpm
- Heart rate - <1y= 110-160bpm, 1-2y=100-150



CROUP

Croup is another childhood infection expected to 'surge' over Autumn and Winter 2019.

KISS: Croup [CKS 2019](#)

Due to parainfluenza virus, causing a seal-like barking cough and stridor.

Assessment

- **Typical presentation:**
 - Prodromal, non-specific URTI symptoms for 12-48 hours, followed by sudden-onset, seal-like barking cough, often accompanied by stridor and chest wall (intercostal) or sternal in-drawing
 - Symptoms typically worse at night and increase with agitation; hoarse voice is common
- **Differential diagnosis** (most are likely to need admission):
 - **Bacterial tracheitis** — suspect if acute deterioration with high fever, sudden onset stridor, and respiratory distress, during recovery from viral-like RTI
 - **Epiglottitis** — suspect if sudden onset high fever, dysphagia, drooling, anxiety, non-barking cough, with preferred posture sitting upright with head extended; rare due to immunisation against HiB
 - Other causes include foreign body in upper airway, retropharyngeal/peritonsillar abscess, angioneurotic oedema, allergic reaction
- **Assess severity:**
 - **Mild** – seal-like barking cough but no stridor or sternal/intercostal recession at rest
 - **Moderate** – cough with stridor and sternal recession at rest; no agitation or lethargy
 - **Severe** – cough with stridor and sternal/intercostal recession with agitation or lethargy
 - **Impending respiratory failure** – increasing upper airway obstruction, sternal/intercostal recession, asynchronous chest wall and abdominal movement, fatigue, pallor or cyanosis, decreased level of consciousness; respiratory rate > 70 breaths/minute

Management

- **Mild croup can be managed in primary care:**
 - **Treat all croup with corticosteroid therapy**
 - oral dexamethasone (0.15mg/kg single dose) - first line
 - oral prednisolone (1-2mg/kg) is an option if oral dexamethasone is not available
 - [Data](#) shows ↑ improvement at 2h lasting ≥ 24 hours, ↓ review/admission/readmission
 - [Large RCT](#) shows no difference between dex 0.15mg/kg vs pred 1mg/kg
 - Give appropriate safety-netting advice and information - [click here for a good PIL](#)
- **Admit all children with:**
 - **Moderate or severe illness, or impending respiratory failure; consider admission for children with a respiratory rate of > 60 breaths/minute**
 - **Risk factors for poor prognosis** - chronic lung disease (including bronchopulmonary dysplasia); haemodynamically significant congenital heart disease; neuromuscular disorders; immunodeficiency; age <3 months; inadequate fluid intake (50-75% of usual volume, or no wet nappy for 12 hours); adverse social circumstances or remote home
 - While awaiting admission to hospital:
 - Consider O2 if severe, dex PO/IM or inhaled budesonide, relaxing environment



CONSTIPATION IN CHILDREN

A new BMJ Practice paper ([BMJ 2021;375:e065046](#)) reviews management of constipation in children. This is welcome as the NICE guideline is almost 12 years old ([May 2010, CG99](#)).

In reality, the guideline from NICE while aged still represents the best management for childhood constipation, but the BMJ also provides some real-world, practical tips on managing this common problem.

Childhood Constipation [BMJ 2021;375:e065046](#)

The condition affects 1 in 10 children, most commonly starting during the toddler phase. It often coincides with weaning, toilet training or starting school. Around one-quarter require treatment into older childhood.

Key learning points are:

- **Early intervention with oral laxatives in addition to non-pharmacological management is crucial** to prevent chronicity, overstretched bowels with reduced motility, withholding behaviours and overflow soiling.
 - **Macrogols (paediatric preparations) are the recommended 1st line**
 - Can be mixed with any liquid the child will drink, e.g. squash, hot chocolate (!)
 - Osmotic laxatives may be preferred - lactulose is sweet, can be diluted with water and may be more palatable for some children
 - Stimulant laxatives may be required - liquid senna does not taste nice but tablets may be acceptable to children >2yo (senna tablets are licensed from 6y)
- **Parents are often not warned about the length of initial treatment** (e.g. minimum 6 weeks, usually months, occasionally years) and that **laxatives can cause side effects** (e.g. diarrhoea, abdo pain/cramps). This often leads to premature cessation of an effective therapy.
- **Careful explanation** to the child and parent/carer is very important. Visual aids may help.
- **Constipation is not the child's fault and punitive measures can make things worse.** Positive reinforcement of good behaviours (rather than success which leads to despondency and disengagement) can help to establish a regular bowel habit.
- During **disimpaction diarrhoea inevitably gets worse** before it gets better. Abdominal pain and gas may also increase - paracetamol may help manage discomfort. **Timing treatment with when school/nursery holidays may be helpful**; if not possible time of school may be required.
- **Disimpaction is successful when the child has 'watery poo'.** At this point a maintenance regime should be initiated and continued for 'at least as long as the child has been constipated'.
- **Maintenance treatment should be tapered slowly** in response to stool consistency and frequency.

KISS: Macrogol Dose for Disimpaction and Maintenance [BMJ 2021;375:e065046](#)

Age	Disimpaction regimen	Maintenance regimen
<1y	0.5-1 sachet daily	0.5-1 sachet daily
1-5y	2 sachets on day 1 4 sachets for 2 days 6 sachets for 2 days 8 sachets daily	1-6 years 1 sachet daily Max. 4 sachets daily
5-12y	4 sachets on day 1 Increase in steps of 2 sachets every 2 days Max. 12 sachets daily	6-12 years 2 sachets daily Max. 4 sachets daily
12-18y	8 sachets on day 1 Increase in steps of 4 sachets every 2 days Max. 16 sachets daily	2-6 sachets daily in divided doses for up to 2 weeks; then 2-4 sachets daily



KISS Childhood Constipation – Diagnosis and Management [BMJ 2021;375:e065046](#), [NICE 2010 CG99](#)

Diagnosing Idiopathic Constipation

Make a positive diagnosis by looking for typical findings and excluding red flags.

- **Any 2 or more of the following typical features**
 - <3 complete stools/week
 - Large hard stools or 'rabbit droppings'
 - Pain or straining on passing stool
 - Bleeding associated with hard stools
 - Previous constipation
 - Previous or current anal fissure
- **And also in children over 1 year old:**
 - Overflow soiling
 - Poor appetite or waxing and waning abdo pain that improves on passing a large stool
 - Evidence of retentive posturing (typically straight-legged, tiptoed, back arching)
- **Exclude red flags**
 - Constipation from birth or 1st few weeks of life
 - Failure/delay in passing meconium (> 48h)
 - Ribbon stools
 - Gross abdominal distention or distention with vomiting
 - Abnormalities of lumbosacral region or on lower limb MKS/neuro exam
- **Faltering growth (Amber flag)**
 - Exclusively breastfed infants have highly variable stool patterns - loose stools may occur in the first few weeks of life and then bowels may only open weekly from 6 weeks of life
- **Exclusively breastfed infants have highly variable stool patterns - loose stools may occur in the first few weeks of life and then bowels may only open weekly from 6 weeks of life**
- **Refer** if red flags or if no response to treatment within 4/52 <1y or within 3/12 >1y, or unable to wean off laxatives after 12 months treatment

Management

- **Do not use dietary or lifestyle advice alone** but good patient/parent info from [ERIC here](#)
- **Assess for and treat faecal impaction**
 - Diagnose on the basis of history, overflow soiling and/or palpable faecal mass abdominally.
 - Use the escalating regime of macrogol and advise the family about initial soiling and pain. A stimulant can be added or substituted if not effective. PRN paracetamol for discomfort.
- **Maintenance treatment** – aiming for regular passage of soft-formed stools (e.g. one daily)
 - **Macrogols first line** - start at half the dose needed for disimpaction.
 - Stimulant laxative second line (as addition or alternative to macrogol).
 - Use osmotic laxative such as docusate or lactulose if stools are hard.
 - Continue treatment for several weeks after regular bowel habit is established, or if toilet training until this is complete.
 - Use dietary and behavioural interventions once regular toileting is established.
- **Withdrawing treatment**
 - Taper gradually over months. Some children may need treatment for significantly longer.
 - No evidence that prolonged treatment leads to 'lazy bowel'.

RECURRENT COUGH IN CHILDREN

It's a common scenario each winter. Parents bring in a well-looking child, concerned about the number of infections they've suffered through in the past few months and that there is a serious underlying cause. It's also a difficult area for clinicians due to a lack of data to guide us on whether we should be concerned, and whether investigations are helpful and merited.

This *excellent BMJ paper* ([2018;362;k2698](#)) sets out the importance of identifying children who have recurrent lower respiratory tract infections, especially giving the *growing understanding that conditions such as untreated protracted bacterial bronchitis or recurrent pneumonia can be a cause of bronchiectasis* even at this young age.

► *How common is a serious disease?*

As we all know, most children with recurrent respiratory tract infection have simple viral illnesses, usually of the upper respiratory tract, and 6-10 per year is common. Acute cough lasting up to 25 days and cold symptoms 15 days is normal, much longer than most parents would think.

Thankfully serious causes are fairly uncommon: a US study showed of children admitted to hospital with pneumonia, 8% had recurrent episodes and only ~1% had an underlying illness (which was very broad, from reflux and aspiration issues through to immune deficiency or pulmonary abnormalities).

There is no consensus definition of what constitutes "recurrence" or "too many" infections, but 3 or more annual episodes of bronchitis, bronchiolitis, or pneumonia has been suggested.

► *What are the potential causes of recurrent cough?*

Identifying the cause is clearly important, as some conditions can be resolved by simple primary care treatment, while others are serious, potentially long-term and require specialist input.

KISS: Causes of Recurrent Cough

Environmental - tobacco smoke, vape fumes, air pollution

Physical - *foreign body*: specifically question parents/child about the possibility

Infection

- **Simple viral infection**
- **Protracted bacterial bronchitis (PBB)** - most common cause of moist cough for >3 week
- **Bronchiectasis** - under-recognised, has significant long-term implications and is treatable through regular chest physio and antibiotics for infections if it can be identified; associated with PBB as well as cystic fibrosis, primary ciliary dyskinesia, immunodeficiency, etc.
- **Tuberculosis** - consider if associated with fever, weight loss, fatigue, increased sweating; atypical presentations are common so keep an open mind in higher prevalence areas
- **Pertussis** - isolated cough, can occur even when immunised, may last months
- **Asthma** - may be the cause most parents are concerned about; cough-variant asthma, where the other common features of asthma are not prevalent, does occur but is rare - objective testing is important where possible

► **Remind me about protracted bacterial bronchitis...**

Protracted bacterial bronchitis (PBB) is a condition most of us have not heard of, mainly as it was only recognised as a clinical entity in 2006, but we will all be familiar with the clinical picture. PBB was usefully reviewed recently in the Annals of Thoracic Medicine ([Ann Thorac Med 2018;13:7](#)).

PBB is characterised by the presence of a chronic wet or moist cough for > 4 weeks. This separates it from the usual clinical picture of a viral URTI which may also have a prolonged cough but is almost always a dry cough after the first week or two.

PBB results from bacteria causing persistent infection of the airways enhanced by biofilm formation. A biofilm is a self-secreted matrix composed of “extracellular polymeric substances” that improve bacterial adherence to the underlying tissue, facilitate nutritional shunting and protects by decreasing antibiotic penetration. Biofilm separates this type of infection from acute respiratory infections and has been implicated in other chronic infections such as sinusitis.

In PBB children will be (by definition) clinically well, however, left untreated it is linked to the development of bronchiectasis via persistent airways inflammation (a [recent cohort study](#) found 8% of 161 children with PBB developed bronchiectasis over 2y) so it is important to identify and treat.

In the past guidelines have suggested PBB is a diagnosis of exclusion, involving CXR, sputum cultures and bronchoscopy. More recently pragmatic diagnostic criteria have been recommended which involves only clinical features without the need for investigations.

Modified Diagnostic Criteria (PBB-clinical)

[Chest 2017;151:884](#)

- Chronic wet cough (>4 weeks)
- No symptoms or signs of other causes of wet or productive cough
- Resolution of the cough after a 2-week course of an appropriate oral antibiotic (e.g. co-amoxiclav)

There is good evidence that antibiotics help this condition. A [2018 Cochrane review](#) found NNT=3 for cure and NNT=4 to prevent progression of illness in children with prolonged moist cough.

A prolonged course of antibiotics is necessary to counteract the protective effect of the biofilm. Two weeks is recommended initially and forms part of the diagnostic criteria, but guidelines also suggest a role for longer courses, especially in recurrence (realistically the remit of secondary care and after investigation). Shorter courses are often inadequate to treat the infection, leading to recurrence and resistance. Co-amoxiclav is the most commonly used antibiotic in study settings, although data is lacking on whether such a broad-spectrum antimicrobial is truly necessary.

Of course, there is a clear drive to reduce antibiotic use for upper respiratory tract infections, but this is one condition where it would appear to be “appropriate use”. This group of children with a prolonged moist cough will be a fairly small contingent compared to those with the usual short-lived viral URTIs that we typically see and certainly does not mandate blanket antibiotic prescribing.

► **How can we assess children with recurrent cough?**

The BMJ paper suggests a simple algorithm to assess children with recurrent lower respiratory infections:

► **Should we investigate children with recurrent cough?**

The paper comments that children who follow the common pattern of self-limiting, mild illnesses worse in the winter, with full recovery between episodes and who are otherwise well with a normal examination require no investigation.

It also notes that in children who have never been unwell enough to have a CXR, further investigation is unlikely to reveal much. However, it counsels where there is doubt, or for reassurance, CXR, FBC (to assess for low lymphocyte levels) and serum immunoglobulins may be merited.



Abnormalities on a CXR may indicate a range of causes, but ultimately all fall under the remit of secondary care and referral is warranted.

► *How should we manage children with recurrent cough?*

This is clearly dependent on the underlying cause. The paper recommends:

- ***“Reassurance is all that is required for otherwise healthy children”*** with a review if there is any doubt or for parental peace of mind.
- Avoid multiple courses of antibiotics in children without a specific diagnosis - short courses often result in relapse and drives concern even further
- If persistent bacterial bronchitis is suspected, 2 weeks of co-amoxiclav is appropriate - usually settles cough but recurrent cases are known, be suspicious for bronchiectasis
- If asthma is suspected but objective testing is not possible a therapeutic trial of inhaled corticosteroid (low-medium dose only) can be used - 4-8w to demonstrate a response, then cessation to show recurrence, then restarting to re-confirm response to treatment and prove the diagnosis.

KISS: Red Flags for Bronchiectasis

History

- Recurrent/chronic moist/productive cough for >4-8w; atypical “asthma” failing to respond to treatment; prior infection with predisposing organism (e.g. pertussis); recurrent/non-resolving pneumonia

Examination

- Clubbing; palpable chest secretions on coughing or forced expiration persisting over time; persistent crackles on auscultation

Investigation

- +ve sputum culture for an unusual organism, e.g. Pseudomonas; recurrent/persistent CXR changes; evidence of localised bronchial obstruction

KISS: Red Flags for Referral

- Unexplained respiratory distress from day 1 of life
- History of severe, persistent, unusual recurrent infections
- History of choking
- FHx suggestive of a genetic disorder
- Signs of chronic illness, e.g. lethargy, weight loss, or failure to thrive
- Underlying condition predisposing to infection, e.g. cerebral palsy
- Abnormal physical signs when the child is well, e/g/ clubbing, chest signs
- Wet cough for > 8 weeks continuously, even if the child seems well
- Recurrent localised crackles in the same place

KISS: Assessment of a Child with Recurrent Cough**1. Is this actually an LRTI?**

■ History and exam

• Frequency and duration of cough? Type of cough?

- **Isolated dry cough** in an otherwise well child is highly unlikely to be due to serious illness
- **Recurrent wet cough with intermittent cough-free periods** (even if just a few days) suggests repeat viral URTI and reassurance is generally all that's required
- **Daily, chronic productive-sounding cough** (children rarely expectorate) for >8 weeks (other data suggest >4w) even in an otherwise well child merits "prompt" investigation and referral
- Be suspicious of persistent wet cough in children who are unwell

• Associated symptoms? e.g. fever, SOB, fatigue, weight loss

• Risk factors? e.g. prematurity, atopy, air pollution, daycare attendance

• Often no physical signs, chest auscultation may be normal, check for clubbing

• Palpate for secretions - an interesting suggestion: the authors state this can demonstrate localised secretions requiring treatment which auscultation will miss and is often the only sign - put hands on the chest wall while the child coughs or forces expiration

■ ***If protracted bacterial bronchitis is suspected***, prescribe a 2/52 trial of co-amoxiclav then r/v; recurrent episodes still merit further assessment**2. Is it really recurrent?**

- ≥3 episodes/year, if not reassure parents

3. Is the child otherwise healthy?

- If yes consider modifiable factors (e.g. passive smoke inhalation) and arrange to follow up to check no new concerns

4. Is there suspicious of underlying disease in an "unhealthy" child?

- Include a CXR at this stage if available, refer if unavailable or concerns persist
- For red flags for bronchiectasis and other referral indicators see the end of the chapter

NB Learning Points: Recurrent Respiratory in Children

- Children with recurrent viral infections who have no symptoms between episodes and are otherwise healthy are very unlikely to have a serious illness
- Children with a wet cough for >4-8 weeks merit further consideration, which may be a trial of treatment for possible protracted bacterial bronchitis or referral to assess for other causes

ABDOMINAL MIGRAINE

When we see well children who have abdominal pain it is hard to think beyond a psychogenic cause. We probably do not consider the possibility of migraine, and yet, as this excellent BMJ paper ([2018;360:k179](#)) explains, abdominal migraine is an “important, common and under-recognised cause of recurrent abdominal pain”.

► *What is an abdominal migraine?* [BMJ 2018;360:k179](#)

Abdominal migraine is a functional disorder insofar as there is no recognised structural or biochemical abnormality. There is a strong genetic component with parents often describing similar symptoms as children, and a clear link with migraine headache, found in 70% of those with abdominal migraine but the true cause remains unclear.

UK data using older diagnostic criteria found prevalence in children of 2.4-4.2%. More recent data shows that it most commonly affects children aged 6-12yo, with up to 9% affected aged 12. Thankfully prognosis is excellent, with rates falling rapidly to 1% by age 14.

There are consensus diagnostic criteria for abdominal migraine, which the authors of the paper clearly feel are very useful but caution strictly following them may potentially miss cases (particularly with episodes shorter than 1hr). They suggest a pragmatic approach to diagnosis and an open mind as some children may not have all the features.

Definition of Abdominal Migraine

- Episodic central abdominal pain usually lasting >1hr
- Interferes with normal activity
- Occurs with ≥1 of pallor, anorexia, nausea, vomiting, photophobia, headache, OR are associated with other cyclical episodes syndromes, e.g. cyclical vomiting
- Person is well between episodes
- Normal physical and developmental exam

It would seem logical that there could be a mental health component to abdominal migraine, but we should avoid making this assumption. Data shows that levels of depression and emotional adjustment are no differences between this group and pain-free children with minor illness.

Indeed, mood change, rather than being the cause, may be a result of the migraine - it is a recognised premonitory and post-migraine symptom.

► *How do we diagnose abdominal migraine?*

As the authors state: “***the challenge is to distinguish [abdominal migraine] from organic and other functional causes of recurrent abdominal pain***”. This can be done through the identification of positive features and the exclusion of red flags. The paper sets out a useful pathway for diagnosis:

- **Assess for red flags (all the usual) with history and examination**
 - Examination should be normal other than vasomotor motor changes (e.g. pallor during an episode); development should also be normal
 - Urine dip to exclude UTI and diabetic ketoacidosis
 - Consider other neurological causes but these are rare and usually present differently (e.g. abdominal epilepsy results in brief pain, altered awareness and possibly seizure)

■ **Assess for positive features**

- Ask about vasomotor symptoms (e.g. pallor, nausea, etc.) and headache
- Check for personal or family history of migraine or other episodic disorders - cyclical vomiting and/or migraine limb pain in particular, but also paroxysmal vertigo, paroxysmal torticollis has been linked

■ **If positive features and normal examination diagnose abdominal migraine**

- With a firm diagnosis, no further investigations are required
- If a lack of positive features consider other functional conditions: functional dyspepsia, irritable bowel syndrome (abdo pain ≥ 1 /wk + change in bowel habit), functional abdominal pain syndrome) pain $>25\%$ of the time with headache, limb pain, difficulty sleeping) - refer as required

► *How can we manage abdominal migraine?*

Diagnosis and a good explanation are helpful for both the child and family - especially given that 60% of parents that shared the condition when they were children.

Management can be viewed as general, acute and preventative. There is a distinct lack of good evidence to guide us. Apart from one very small RCT on pizotifen (which had a positive result, but only had 14 participants), the rest is based on case reports and observational data in a handful of children. Expert advice is welcome then to guide us further and the BMJ suggests options that may be used in primary care, as well as those that merit more specialist input.

Either way, it is worth noting data suggests a high placebo response to any treatment (up to 66%) so this is something we can work with and be positive about.

KISS: Treatment of Abdominal Migraine in Children

General: avoid triggers (e.g. stress, missed meals, poor sleep), elimination diets lack data

Acute

- Rest in dark quiet room (almost 90% will improve with this alone)
- Simple analgesia (paracetamol 15mg/kg, ibuprofen 10mg/kg)
- Sumatriptan could be used (10mg intranasal)

Preventative

- Pizotifen 0.25mg BD as a syrup (BNF states 500mcg OD at night, increasing up to 1500mcg (1.5mg) in divided doses)
- Other treatments such as propranolol, cyproheptadine, flunarizine, sodium valproate all lack data and should be specialist initiation only

NB Learning Points: Abdominal Migraine

- Abdominal migraine affects 4% of children, and up to 9% age 12
- Simple measures such as rest are very effective, while simple analgesia can also help
- The [Migraine Trust](#) and the [American Migraine Foundation](#) both have good patient information about abdominal migraines and managing migraine in children

FALTERING GROWTH

Newborns and infants rarely have a severe illness, which can make our job of identifying those that do all the more difficult. A common feature is faltering growth, the re-brand of the term 'failure to thrive'. This area has often seemed more like art than science in general practice, but we now have clear recommendations thanks to a [recent NICE guideline](#), reviewed in the BMJ (2017;358:j4219).

Faltering Growth In Children [NICE NG75](#), [BMJ 2017;358:j4219](#)

The key to appreciating this guideline is understanding that it truly is about a team effort. An individual attempting to follow the recommendations by themselves will quickly be left incredulous. However, when each HCP understands their role it begins to make much more sense. Midwives may be the first to recognise a problem and provide initial advice. GPs have a role in assessing for severe disease and referring to specialists where doubt exists. Health visitors may spend time at the home, observing feeding patterns and family interactions, and providing one-to-one support.

This team approach is vital because faltering growth is fairly common with 5% of children affected in the first few days, weeks or months of life.

► *When should we be concerned about faltering growth?*

Faltering growth simply describes a pattern of slower weight gain than expected for the child's current age, sex and weight. The most common cause is inadequate nutritional intake and a careful feeding or eating history may reveal the driver for poor weight gain. However our role as ever is to exclude the rare but serious cause, be it an undiagnosed cleft palate or neglect (thankfully uncommon).

The pattern of weight loss differs between children in the early days of life and those after this period and so NICE consider these groups in separate pathways. The point at which NICE feel we should be concerned is summarised in the KISS below.

KISS: Thresholds for concern with faltering growth [NICE NG75](#)

Early days of life

- Up to 10% weight loss in the first few days common and considered normal
- It usually stops around day 3-4 and returns to normal by 3 weeks
- **>10% weight loss in early days of life or failure to regain birth weight before 3 weeks**

After the early days of life

- **All birth weights:** current weight <2nd centile for age
- **<9th centile:** fall across 1 or more centiles
- **9th-91st centiles:** fall across 2 or more centiles
- **>91st centile:** fall across 3+ centiles
- If there is concern weigh and measure length/height and plot with previous results on [UK growth charts](#) (use the child's red book if available) and [BMI if ≥2yo](#)
 - BMI <2nd centile suggests under-nutrition, <0.4th centile is an indication for intervention
 - If possible obtain the biological parent's heights and calculate mid-parental height - if the child is >2 centiles below this it could suggest undernutrition or a primary growth disorder - you can find this on the [2-18 years growth chart](#)

► *How should we manage significant faltering growth?*

NICE provides general advice for all health professionals, but clearly, some recommendations, e.g. observation of feeding or mealtimes, may be better undertaken by a midwife or health visitor. It is also useful for parents to keep a feeding or mealtime diary to aid assessment of nutritional intake.

If the cause appears to be nutritional it is very reasonable for us to trial management in general practice and follow up with the family as appropriate for the level of weight loss. NICE even comment there is a role for nutritional supplementation, but this can often be achieved through 'natural' high-density foods such as switching to full-fat milk from semi-skimmed or Greek yoghurt from low-fat yoghurt. This is likely to be more palatable for the child and achieve the calorie increase required without resorting to pharmaceutical nutrition options.

Where we are concerned about serious illness, significant low weight or failure to improve with primary care interventions paediatric assessment is required.

NB Learning Points and Quality Improvement Ideas: Faltering Growth

- Faltering growth affects 5% of children in the first few days, weeks or months of life
- Inadequate nutrition is the commonest cause
- 10% weight loss is acceptable in the first few days; this should be regained by 3 weeks
- In older children concern about falling centiles depends on their current centiles - smaller babies need closer attention
- Plot current weight and length/height on a growth chart, look for the trend and take into account the parental average height when considering what may be normal for a child
- Use a whole system approach, involving your midwives and health visitors

NB Quality Improvement Ideas: Faltering Growth

- Does the practice have a regular meeting with the local health visitors and midwives
 - If not is there an effective mechanism of communication to relay information and concerns about patients?
- Does the practice have a care pathway for children with faltering growth and insufficient nutrition? If not why not write a policy with your health visitors.

Patient Resources

- [NCT](#) - free, simple advice on feeding for parents
- [The Breastfeeding Network](#) - advice for breastfeeding mothers
- [Child Growth Foundation](#) - for parents of a child with underlying conditions affecting growth

KISS: Management of Faltering Growth**NICE NG75****Early days of life**■ **Clinical assessment**

- Take a detailed feeding history, examined for dehydration and underlying disease
- Consider the direct observation of feeding (by a person appropriate expertise)
 - Common contributors are ineffective suckling of breast or bottle, inadequate feeding patterns, feeding aversion, the feeding environment, parent-infant interactions and response to feeding cues, physical disorders affecting feeding
- **Only perform investigations if indicated by findings on clinical assessment** - most children do not have an underlying condition and investigations are unlikely to reveal a cause in a child who appears well with no other concerns
- **Provide feeding support**, e.g. midwife-led breastfeeding clinic
 - Supplementing with formula if breastfeeding may aid weight gain but often results in cessation of breastfeeding. Advise on expressing milk, use this in preference.
- **Refer to paediatrics** if evidence of illness marked weight loss, or failure to respond to feeding support

After the early days of life

- **Assess clinically, socially and developmentally including a detailed feeding history**
- Consider the direct observation of eating or feeding
- **Consider investigating for UTI and/or coeliac disease** (if the child has gluten)
 - It feels like an unhelpful way to phase this recommendation however it keeps our options open - it may be reasonable, in a well-child with a clear history pointing to feeding issues, to wait and assess a response first, then test later if growth does not improve
- **Refer to paediatrics if:**
 - Possible underlying disease, rapid weight loss, severe under-nutrition, slow linear growth, unexplained short stature, failure to respond to interventions, safeguarding concern
- If none of the above presents consider management in primary care-based around nutritional and feeding strategies, then reassess as appropriate to monitor progress

Management Strategies

- If necessary **advise on food choices**, taking into account what is appropriate for the child's development age, to optimise energy and nutrient density
 - Often the child is having too much energy-dense drink, e.g. milk, and is full
- If **further increases in nutritional density are required to consider short-term dietary fortification using energy-dense foods** along with a referral to a paediatric dietician
 - Supplements may be 'artificial' (e.g. PaediaSure, Pro-Cal powder) or 'natural' (e.g. full-fat milk, full fat Greek yoghurt, substitute oil for butter when cooking)
- Discuss possible **feeding and mealtime strategies**
 - E.g. encourage relaxed and enjoyable mealtimes, eat as a family, regular eating schedules, setting behaviour boundaries, avoiding coercive feeding



ENURESIS

This troubling issue has *recently been reviewed in an excellent article in the DTB (2020;58:25) and an equally good paper in the BJGP (2017;67(600)328)* with a firm focus on primary care. The fact that 'nocturnal' has been dropped from 'enuresis' internationally is not the only change. While we are used to thinking about enuresis as being 'primary' or 'secondary' types, this concept is now somewhat outdated.

The strength of these papers is the clear explanation of the underlying causes of enuresis and how this influences the success of treatments.

► *Causes of enuresis*

Enuresis is now categorised as either “**monosymptomatic**” (MNE) or “**non-monosymptomatic enuresis**” (NMNE), the latter having daytime lower urinary tract symptoms present.

NMNE is more complex with features of underlying bladder dysfunction (e.g. daytime frequency, overactive bladder, daytime incontinence) and generally *merits specialist input*.

MNE is considered appropriate for management in primary care. While often multifactorial in origin, the most common causes are:

- **Small maximum void volumes** = <65% of EBC, +/-
- **Nocturnal polyuria** = >130% EBC, +/-
- **Failure to wake in response to a full bladder**
 - EBC = expected bladder capacity = (age + 1) x 30ml

► *Identifying the cause*

A thorough history and examination assess for underlying medical conditions that may be driving enuresis, especially if the child was previously dry, excludes red flags and addresses impact on the child and family.

Having clinically excluded serious disease, the current practice employs a somewhat lucky dip approach to deciding whether first line treatment should be an alarm or medication.

However, the paper highlights that the *underlying cause of enuresis directly affects the likelihood of treatment success*.

Treatment failure in MNE is often due to initial use of the less favourable option. Identifying the underlying cause is straight-forward using a 2 step approach:

- ***Step 1 - Screening for bladder dysfunction***
 - Ask about leakage during the day, frequency (≥ 8 voids/day), urgency, etc.
 - An [excellent checklist](#) from the European enuresis guideline is a simple assessment tool and could be used by the parents as self-assessment
- ***Step 2 - Extended assessment***
 - While the paper explains this is not essential, it is advisable as it helps to identify the underlying pathophysiology of the enuresis and hence guides us as to whether medication or an alarm is the best 1st line option
 - The family complete 2 diaries: 1) a 2-day daytime bladder diary, to assess bladder capacity, 2) a nighttime urine production diary to assess nocturnal polyuria.
 - The paper recommends the [Drydawn](#) website, which has an excellent [combined diary](#), as well as providing very good patient information



► Initiating treatment

Where MNE is suspected (i.e. in the absence of daytime bladder dysfunction) treatment can be initiated in primary care. Desmopressin and bedwetting alarms are both considered first line options.

- **Small void volumes/arousal problems - bedwetting alarms** are more appropriate
 - NICE and the DTB (based on NICE) recommends alarms as 1st line if acceptable to the child & family
 - 2/3 become dry for 14 consecutive nights over 3 months, in half this persists on cessation
 - Benefit takes time so warn parents it could be a few weeks to see initial improvements
- **Nocturnal polyuria - desmopressin** would be the most appropriate 1st line
 - 70% significant response, 30% fully dry - if not fully improved at 2/52 consider changing formulation, increasing dose or bringing forward the evening dose to 1-2h before bedtime
 - 60-70% will relapse on withdrawal. This is significantly lower if a tapered withdrawal schedule is used.
- If both causes are present combination treatment can be used
- Basic advice on fluid intake and toileting should be offered (see KISS).

In the vast majority of children, treatment is successful. However, some will be resistant to therapy and the paper advises that counselling families on the possibility of failure may prevent disillusionment with treatment and dropouts.

Second-line therapies include antimuscarinics (oxybutynin is licensed for enuresis, tolterodine is often used in clinics) and TCAs (imipramine). These should only be initiated at specialist review.

International [consensus guidelines](#) note that the primary reason for treatment failure in MNE is because the child actually has NMNE. For this reason, it is recommended that if first line treatment fails children are referred to a specialist for further assessment.

NB Learning Points: Enuresis

- Enuresis in children ≥ 5 years old is considered abnormal and has negative effects on both the child and parents
- Enuresis should be considered as mono-symptomatic or non-monosymptomatic types
- A simple 2 stage approach with a simple questionnaire and combined bladder diary will identify which type of enuresis the child has, and then the likely underlying pathophysiology
- This can guide clinicians on whether an alarm or desmopressin is the most appropriate 1st line treatment

KISS: Enuresis in Children and Young People

[BJGP 2017;67\(660\):328](#), [DTB 2020;58:25](#), [NICE 2010 CG111](#)

Definitions

- **Children aged ≥ 5 yo**; <5y enuresis is considered normal and no treatment is recommended
- **Primary** (never been dry) or **2ndary enuresis** (previously dry for >6m)? (used by NICE)
- **Monosymptomatic** (MNE - no daytime Sx) or **non-monosymptomatic enuresis** (NMNE - day Sx)

Assessment

- **Clinically exclude causes of 2ndary enuresis** e.g. constipation, UTI, diabetes, neurological
 - **Urinalysis** is not routinely required but consider if daytime symptoms, recent symptoms, signs of ill health, suspected UTI or diabetes (DTB suggests check for glucose, protein and infection)
- **2 step assessment of enuresis**
 - **Screen for bladder dysfunction**, e.g. overactive bladder, [click here](#) for the questionnaire
 - **If present indicates NMNE, referral is advised**
 - Also appropriate if suspected maltreatment, learning disability, behavioural issues, etc
 - **Extended assessment**, using [2 step diary](#) to assess for nocturnal polyuria or MVV/arousal problems
 - Estimated bladder capacity = $30 \times (\text{age} + 1)$ (ml)
 - Reduced bladder capacity <65% of EBC
 - Nocturnal polyuria >130% of EBC

Management

- **Provide information on the condition, reassure families this is a common problem**
- **Advise on good fluid intake during the day and fluid restriction 2h before sleep**
 - ~1L/day up to 10kg; 1.5L/day up to 20kg; urine output ~30-40ml/kg/day
- **Lifting and waking overnight does not promote dryness long-term**
 - May be useful as short term strategy when everyone is exhausted
 - Encourage child to toilet themselves if they wake spontaneously in the night
- **Positive reward systems can be helpful**, e.g. for helpful behaviour, not just dry nights
- **If nocturnal polyuria - start desmopressin** (200mcg tablet or 120mcg sublingual)
 - s/e uncommon – headaches, abdo pain, emotional disturbances possible
 - Oral version much safer than nasal version – sodium monitoring not required
 - Advise to take at bedtime, avoid fluid 1h before and 8h after taking
 - Trial cessation every 3 months and stop if child has become dry
 - Structured withdrawal may reduce relapse rates
- **If small max voiding volume/arousal problems start bedwetting alarm**
 - 2/3 become dry; continue for 2-3 months or 14 consecutive dry nights (whichever is first)
 - Response is more gradual than medication and may be more effective by the end of treatment, with some limited evidence of greater long term effect
- **Second-line options** include anticholinergics and tricyclics but these should be specialist initiation only and, if required, referral is recommended

Patient Information

- Good resources at www.drydawn.ie and www.eric.org.uk - Eric also sells alarms



GORD IN INFANTS

In the age of modern medicine and the internet, there seems an ever-increasing need for a definitive explanation and solution to every problem. One of the in-vogue issues for infants is reflux – a huge concern for many parents, but normal babies are being over-diagnosed and over-treated.

► Gastro-oesophageal reflux disease in children and young people [NICE NG1](#)

This is a fantastic update from NICE of a common problem for the start of 2015, particularly focusing on infants with the emphasis being on reassurance and simple treatments if any at all.

Definitions are important here:

1. **Gastro-oesophageal reflux (GOR)**

- A normal physiological process that usually happens after eating in healthy infants (also children, young people and adults)
- Usually manifests as effortless regurgitation of feeds in young babies (possetting)

2. **Gastro-oesophageal reflux disease (GORD)**

- “GOR that causes symptoms (e.g. discomfort or pain) severe enough to merit medical treatment or complications”
- May involve either occult reflux (no regurgitation, “silent reflux”) or overt regurgitation
- Complications include reflux oesophagitis, recurrent aspiration pneumonia, frequent otitis media and dental erosion in those with neurodisability
- Affects various groups
 - Healthy infants; children with identifiable risk factors; adolescents as it would adults
 - Premature infants
 - Children with severe neuro disabilities, where a diagnosis can be very difficult

► Diagnosis of GORD

The key message NICE conveys in this guideline is that:

- ***Regurgitation of feeds in infants is common and normal, results from GOR and does not usually require investigation or treatment.***
 - It affects at least 40% of infants (personally I think it is much higher!)
 - Usually begins before 8 weeks of age
 - May be frequent (5% of those affected have 6+ episodes/day)
 - Decreases with time (resolves in 90% by 12 months)

Reassurance and simple advice to parents and carers in otherwise well infants is the best management.

Making an **accurate diagnosis of GORD is difficult**. When NICE reviewed the available evidence they found that **most of the symptoms commonly associated with GORD had little diagnostic value individually**. Therefore they recommend that ***in the absence of overt regurgitation, do not routinely investigate or treat for GORD if an infant or child only has one of the following features:***

- Unexplained feeding difficulties, distressed behaviour, faltering growth, chronic cough, hoarseness or a single episode of pneumonia

The **key feature on which to base management is the presence of “marked distress”**. These children are likely to have combinations of the features above. The level of parental concern is clearly important. Of course, it is also our role to identify the minority with more severe disease and/or an underlying condition. We should take a careful history and examination to exclude red flags.



KISS: Red Flags in Gastro-oesophageal Reflux in Infants and Children

[NICE NG1](#)

Gastrointestinal – NICE suggests considering acute as well as chronic causes

- **Projectile vomiting** (?pyloric stenosis), **bile-stained vomit** (green or yellow) (?obstruction)
- **Haematemesis** (?serious bleed – consider other causes, e.g. epistaxis, bleeding nipple)
- **Onset of regurgitation/vomiting after 6 months old or persisting beyond 1 year** (?Dx, e.g. ?UTI)
- **Blood in stool** (?gastroenteritis, cow's milk allergy, acute surgical condition)
- **Abdominal distension, tenderness or palpable mass** (?obstruction or acute surgical condition)
- **Chronic diarrhoea** (?CMA – see CMA section for further management)

Systemic

- E.g. signs of infection or raised intracranial pressure; infants at high risk of atopy (?CMA)

Apnoea and “apparent life threatening events” are rarely associated with GORD, but do need paediatric assessment.

Persistent back aching and/or **Sandifer's syndrome** (episodic torticollis with neck extension/rotation) are potentially useful signs (more common in children with neuro disabilities) which merit investigation.

► [Is it cow's milk allergy?](#)

A difficult question to answer. NICE comment: “*Be aware that some symptoms of a non-IgE-mediated cows' milk protein allergy can be similar to the symptoms of GORD, especially in infants with atopic symptoms, signs and/or a family history.*” So not much help.

Their guideline on food allergy suggests considering food allergy if symptoms of GORD do not respond adequately for treatment. Also, consider if chronic diarrhoea is a feature.

► [Management of GORD](#)

Infants with marked distress and frequent regurgitation can be managed conservatively in the majority of cases using a stepped care approach, with careful communication to the parents/carers. Acid-suppressing medication is only recommended in children with overt regurgitation, associated symptoms and where conservative treatment has failed. See our KISS for the full recommendations.

NB Learning Points: GORD in Infants and Children

- Regurgitation is common and normal in infants – treatment is only indicated in the presence of distressing symptoms
- Exclude red flags with history, examination and appropriate simple investigations (e.g. urine MSU)
- A trial of alginate for 1-2 weeks is appropriate if initial conservative treatment has failed
- Acid-suppressing medication should only be used when this has failed and distress remains

NB Quality Improvement: GORD in Infants and Children

- Audit children <2 years receiving PPI or H2RAs
 - Have they a documented indication? Did they trial non-pharmacological Rx first?
- Information for parents:
 - [Living with Reflux](#)
 - [NHS Choices – soothing a crying baby](#)

KISS: Management of GORD**NICE NG1**

For infants with frequent regurgitation associated with marked distress use a stepped-care approach:

Initial management:

- **Do not use positional management**
 - Data supporting prone or lateral positions is sparse, weak and outweighed by the benefits in reduction in SIDS by having children sleeping in the supine position
- **Formula-fed infants**
 - Review feeding history
 - Reduce feeding volumes only if excessive for the infant's weight, THEN
 - Offer a trial of small more frequent feeds (maintaining appropriate total daily intake) unless already using this method, THEN
 - Offer a trial of thickened formula
 - Many formula manufacturers make ready-thickened formulas or separate thickener can be added, e.g. cereal or bean gums/flour ([BNF](#))
- **Breastfed infants**
 - Undertake a breastfeeding assessment with a HCP with suitable expertise for review and advice
- **If these methods are not effective for formula or breastfed infants:**
 - Stop thickened formula if using
 - Offer alginate therapy for a trial period of 1-2 weeks
 - Gaviscon Infant is the only licensed product (birth > 2yo) – it is a powder added to formula feeds or water for breastfed babies ([BNF](#))
 - Reduction in frequency of regurgitation is the only
 - No adverse outcomes were reported in trials
 - If successful continue, but try stopping at intervals to see if the infant has recovered

Pharmacological therapy:

- Do not offer PPI or H2RAs to treat overt regurgitation occurring as an isolated symptom
- Consider a 4-week trial of PPI or H2RA for those unable to tell you about their symptoms (e.g. infants and young children, severely disabled) who have overt regurgitation with 1 or more of the following:
 - Unexplained feeding difficulties, distressed behaviour, faltering growth
- Consider a 4-week trial of a PPI or H2RA for children with persistent heartburn, retrosternal or epigastric pain
- Assess the response to the trial and consider referral if symptoms:
 - Do not resolve, or
 - Recur after stopping the treatment
- Do not offer other treatment (e.g. metoclopramide or domperidone) without specialist input

Referral (use judgment to decide on):

- Red flags unless clear cause, e.g. UTI
- If symptoms do not resolve or recur after a 4-week trial of acid-suppressing therapy
- Infants and children may have upper GI endoscopy, contrast swallow or pH studies depending on presentation



OBSTRUCTIVE SLEEP APNOEA IN CHILDREN

We have seen increasing rates of OSA in recent years, due to the obesity epidemic and improved recognition by the public and health professionals, plus new technology allowing sleep studies to be performed in the patient's home. Indeed, there is a raft of smartphone apps that can do a reasonable approximation of polysomnography.

We may not often consider OSA in children, but two very recent articles in the BMJ and BJGP highlight just how common this can be, and the detrimental effects it can have on a child.

Paediatric OSA [BJGP 2017;67:282](#), [BMJ 2017;357:j2124](#)

It is increasingly common for parents to be worried about ADHD in children but most would not link inattention and poor behaviour in childhood OSA. Excellent papers in the BJGP ([2017;67:282](#)) and BMJ ([2017;357:j2124](#)) shed light on this often-forgotten issue.

► *What are the causes?*

1-2% of children are affected by OSA. Mostly they are in preschool, but there is a spike in adolescents due to increasing levels of obesity.

The **main cause is enlarged tonsils and/or adenoids**. In otherwise healthy children this is referred to as "uncomplicated OSA".

"Complicated OSA" indicates a subgroup with underlying medical causes. The most common is now obesity, while other causes include craniofacial abnormalities (e.g. cleft palate), Down syndrome, neuromuscular disease (causing reduced tone), achondroplasia, Prader-Willi, and sickle cell.

► *How does OSA present in children?*

The presentation is different from adults. Behavioural issues, poor attention, reduced academic performance are more likely than daytime sleepiness. Snoring remains a cardinal feature.

See the KISS on the next page for the key symptoms and signs of OSA in children.

► *When should we refer?*

If the child has uncomplicated snoring then OSA is unlikely and simply waiting and reviewing in a few months is appropriate, with a referral if any concerns become apparent.

Any child with a history of regular snoring when well in conjunction with adenotonsillar hypertrophy and any features of OSA should be referred to ENT. If complicated OSA is suspected the child should be referred to paediatrics initially.

The BJGP recommends against 'watchful waiting' in primary care due to the effectiveness of treatment.

However, the BMJ paper notes that there is some early data suggesting that watchful waiting, steroid nasal sprays or oral montelukast may be effective alternatives to surgery in some children,

A child may be investigated with home pulse oximetry while they sleep - drops in saturations are indicative, although their absence does not exclude the condition and more detailed sleep studies may be required.

► *What treatments are available?*

In uncomplicated OSA, adenotonsillectomy leads to resolution of symptoms. This has been shown to improve sleep, quality of life and behavioural issues.

Where surgery fails to resolve symptoms, or if contraindicated, CPAP may be offered.

If there is a history of atopy, steroid nasal sprays may be trialled, but the referral is recommended regardless, as rhinitis may be due to adenoidal hypertrophy. In atopic children, if they have used a steroid nasal spray before surgery it should be continued postoperatively to prevent re-accumulation of tonsillar and adenoidal tissue.

Where obesity is the underlying cause, adenotonsillectomy is also effective. Healthy lifestyle/weight management input is also recommended, with referral as appropriate.

KISS: Features of Paediatric OSA

History

- Snoring +/- apnoea episodes during sleep; cyanotic episodes
 - Parental concern about the child's breathing while asleep is a key factor
- A child sleeps with neck arched backwards (opens up the airway)
- Symptoms exacerbated by URTI
- Does speech have a nasal quality? (Implies nasal obstruction)
- Secondary enuresis
- Daytime symptoms - vital to ask as parents may not attribute these to nighttime Sx
 - Behavioural problems, e.g. inattention, hyperactivity, irritability (25% present with ADHD features); poor academic performance
- Consider the past medical history - atopy? muscle weakness? developmental anomalies?

Examination

- Mouth-breathing; nasal speech; stertor (snoring sound when awake)
- Inspect the tonsils - large tonsils are a sensitive but not specific feature
- Inspect nasal cavity (rhinitis, polyps) and ears (glue ear, may indicate adenoid hypertrophy)
- Check height, weight, BMI - plot on a centile chart
- Are there any craniofacial or syndromic abnormalities?

NB Learning Points: OSA

- Paediatric OSA affects 1-2% of preschool children and is mostly a result of adenotonsillar hypertrophy
- Common features are snoring, pauses in breathing at night and behavioural problems
- A range of tools can help with initial assessment but a sleep study is required to confirm the diagnosis

INFECTIVE CONJUNCTIVITIS

Conjunctivitis isn't the sort of condition where you would expect there to be much controversy, but goodness me we had a bit of a Hoo-Ha earlier in 2021 with chloramphenicol eye drops in the under 2s! Thankfully this seems to have been resolved now, but if you missed it all, here is a summary and why on earth everyone was banging on about boron for a few months...

[Our blog](#) gives more detail on the background to this issue, but in summary, there have been concerns that boron may impair fertility if given above certain levels and it is used as a buffering agent in chloramphenicol eye drops. Back in 2017, this was [addressed by the EMA](#) which made **recommendations on thresholds of boron that were acceptable in medicines**, and that for children under 2 years old the maximum exposure should be 1mg/day. The upshot is that concerns were raised that in younger children under 2, use of chloramphenicol eye drops would reach that threshold, leading many manufacturers (and indeed medicines management teams) to state **chloramphenicol eye drops should not be used in the under 2s**. This issue and the poor communication around it was highlighted in the [DTB recently](#) and led to a fairly firm rebuttal from the **Royal College of Ophthalmologists stating** *"Chloramphenicol eye preparations have been widely used in children of all ages for many years with no documented adverse effects on fertility. To our knowledge, there is no new scientific data from human studies to support this change to the product license"*.

This could have had major implications for us, as the alternative options for under 2s with conjunctivitis were not great, to say the least - Fusidic acid has a much narrower spectrum of activity and may not give suitable cover (and is horribly expensive at £35/tube, making it black listed on many formularies). Quinolone and cephalosporin products obviously have real risks of driving resistance and are specifically not recommended by the RCOphth for bacterial conjunctivitis as they should be reserved for severe eye disease e.g. keratitis. So that basically left us with chloramphenicol eye ointment (which does not contain boron), but trying to apply eye ointment to a wriggling toddler is likely to be well-nigh impossible!

Thankfully sense has prevailed with a very recent communication from the [MHRA July 2021](#) stating:

- Following a review of the available toxicological data and a calculation of daily exposure to boron from a typical dosing regimen, we have concluded that the balance between the benefits and risks of chloramphenicol eye drops containing borax or boric acid remains positive for children aged 0 to 2 years.
- **Chloramphenicol eye drops can be safely administered to children aged 0 to 2 years where antibiotic eye drop treatment is indicated.**
- A typical regimen of **one drop, applied typically 3 to 4 times a day, to both eyes**, would result in a daily exposure well below the safety limit for children aged 0 to 2 years.

Although this has ended up being a complete storm in a teacup, it may have had one useful outcome - which is to **question whether antibiotic eye drops are needed at all, especially in children** - as we know the majority of cases are viral, and even if bacterial the [NICE/PHE guidance](#) states evidence that **65% will resolve by day 5 and 74% by day 7 without any antibiotic treatment**. So if nothing else this is a good reminder not to reach for the antibiotic drops for most...

KISS: Infective Conjunctivitis[CKS 2022 NICE/PHE antibiotic prescribing guidance](#)**Assessment:**

- Red-eye, pain and photophobia absent, may be some discomfort with gritty sensation if infective, may be itch if allergic.
- Normal acuity, purulent or watery (viral) or mucoid (allergic) discharge, eyelid swelling with conjunctival oedema may be present.
- It is very difficult to distinguish between viral and bacterial conjunctivitis clinically.
- **Consider red flags:**
 - Reduced visual acuity.
 - Marked eye pain, headache or photophobia.
 - Red sticky eye in a neonate (within 30 days of birth).
 - History of trauma (mechanical, chemical or ultraviolet) or possible foreign body.
 - Copious rapidly progressive discharge — may indicate gonococcal infection.
 - Infection with a herpes virus.
 - Contact lens wearer (see below).

Management:

- **Self-care advice:**
 - Reassure it is a self-limiting illness and usually settles in 1-2 weeks.
 - Lubricant eye drops (available over the counter) may reduce discomfort.
 - Clean eye regularly with cotton wool/water.
 - Wash hands, avoid sharing pillows and towels to reduce the spread of infection.
- There is **no reason to exclude children from school or childcare settings** unless there is an outbreak at which point the HPA should be involved.
- **Antibiotics should be avoided in the majority. Even if bacterial ~75% will resolve by day 7.**
- However, if not resolved by 5-7 days or severe symptoms consider ocular antibiotics.
- **Prescribing ocular antibiotics:**
 - Only use ocular antibiotics for severe disease and even then consider delayed script to use in 3 days if symptoms not resolving.
 - **First line chloramphenicol:**
 - 0.5% drops — apply 1 drop 2 hourly for 2 days then 4 times daily for 5 days.
 - (NB recent [MHRA review July 2021](#) has concluded that **chloramphenicol eye drops still safe to use in children <2** despite the concerns over boron, but use standard lower dose 1 drop 3-4 times per day.)
 - 1% ointment — apply four times daily for 2 days, then twice daily for 5 days.
 - **Fusidic acid 1% eye drops** — can be used second line but has less gram -ve cover - apply twice daily for 7 days.
- **Conjunctivitis associated with contact lens wear:**
 - If topical fluorescein does not identify any corneal staining (if facilities/skills available):
 - Advise them to immediately stop contact lens use.
 - Advise regular bathing/cleaning of the eyelids with cotton wool soaked in sterile saline or boiled and cooled water to remove any discharge.
 - Contact lenses should be kept out until all symptoms of the infection have gone.
 - **However, have a low threshold for referral to ophthalmology if there is any suspicion of corneal involvement as this is a potentially sight-threatening condition.**



BLEPHARITIS

So common and annoying for patients, we have new guidance to guide us...

KISS: Blepharitis

Based on [NICE-PHE Antibiotic Guidance](#), [CKS 2019](#), [BMJ 2012](#)

Background:

- Incredibly common recurrent condition, most common in middle-aged and older adults but can occur at any age.
- The symptoms (inflamed, sore, itchy, scratchy, burning, gritty eyes) are often much worse than the signs suggest.
- Useful to divide into two broad types, which can co-exist:
 - **Anterior blepharitis** at outer margins of lids where lashes emerge:
 - Usually staphylococcal or seborrhoeic
 - May co-exist with seborrhoeic dermatitis
 - **Posterior blepharitis**, at the inner margins of the lid touching the eye:
 - Usually due to meibomian gland dysfunction
 - Associated with dry eyes, chalazion and acne rosacea

Management:

- **Lid hygiene** is essential for all patients at least twice daily:
 - Warm compress twice daily for 10 to 15 minutes, gentle massage of lids is especially important for posterior.
 - Eyelid cleaning is especially important for anterior blepharitis.
 - For patient info on how to do this see [NHS Patient Info](#),
 - NB 'baby shampoo' is no longer recommended in NHS advice.
- **Second line:**
 - If lid hygiene alone is ineffective after 2 weeks, offer topical chloramphenicol ointment BD for 6 weeks.
- **Third line:**
 - Oral oxytetracycline 250mg or 500mg bd for 4 weeks, or oral doxycycline 50mg or 100mg daily for 4 weeks.
- **Fourth line:**
 - Oral oxytetracycline or doxycycline for 8 weeks.
- **Recurrence prevention:**
 - On-going regimen of lid hygiene.
 - Avoid eye makeup and contact lenses (which may interfere with meibomian gland dysfunction and increase the risk of secondary keratitis).
 - Treat dry eyes with lubricant drops.
 - Treat associated seborrhoeic dermatitis and acne rosacea.
 - Consider referral for surgical removal or persistent chalazia.

CORNEAL ULCERS AND ACUTE RED EYE

I'm sure many of us find examining the eye in General Practice a challenge. We can all remember pictures being shown to us in the lovely colour of the various ulcers we may see but identifying them in a 10-minute consultation is not easy. So it was handy that the BJGP kicked off 2018 with a short review article on corneal ulcers in general practice ([Br J Gen Pract 2018; 68 \(666\): 49-50](#)) to help guide us on what to look out for, which ties in nicely with the assessment of the acute red eye and 'red flags'.

[BJGP Jan 2018](#)

First a quick reminder on some of the terminology

- **Keratitis** = inflammation of the cornea vs. **Keratopathy** = non-inflammatory disease of the cornea
- **Entropion** = inversion of eyelid vs. **Ectropion** = eversion of eyelid

Points to consider in the history

- **Onset** - acute onset typical of trauma; chronic mild or moderate discomfort worsening as the day goes on or with tasks e.g. reading/computer work is typical of dry eyes or blepharitis
- **Pain** - higher pain levels more likely with microbial keratitis or mechanical abrasion
- **Progression** - acute onset and rapid resolution typical of mechanical abrasion; progressive worsening typical in microbial keratitis
- **Risk factors:**
 - **Always ask about contact lens use** - any presentation of red-eye in a contact lens user needs an urgent referral for suspected microbial keratitis
 - Cold sores and atopy are risk factors for herpes simplex keratitis
 - Grinding, hammering, and chiselling are associated with corneal foreign bodies

Examination

- If high pain scores consider topical local anaesthetic drops if available (stops blepharospasm)
- Assess visual acuity with Snellen chart - a new reduction in vision is a red flag
- Check for entropion and ectropion - entropion can cause corneal abrasion whereas ectropion can cause dry eye and exposure keratopathy
- Examine with an ophthalmoscope especially looking for fluffy white lesions (represents infiltration)
- Stain with fluorescein and look for yellow-green uptake using blue light on ophthalmoscope; **importantly use the smallest amount of dye possible** - a full drop of 1% fluorescein is too much and will flood the eye and not fluoresce until the tears have diluted it

Typical features and management

- **Microbial keratitis** - fluffy white lesion with irregular border; usually central with significant fluorescein uptake; **requires urgent referral**; do NOT start topical antibiotics in the community
- **Herpes simplex keratitis** - irregular lesion with dendriform ('tree-like') pattern on fluorescein uptake; **requires urgent referral**
- **Corneal abrasion** - usually very painful; uniform fluorescein uptake with sharp border; can be managed with topical 1% chloramphenicol
- **Dry eye/exposure keratopathy** - mild discomfort; pinpoint fluorescein uptake; treat with lubrication; refer only if symptoms persist (see the separate chapter on dry eye for further detail)



GISS: Red Eye (Serious Causes and Red Flags)

[CKS Oct 2021](#); [BMJ 2017;358:j3614](#); [Br J Gen Pract 2018; 68 \(666\): 49-50](#)

Assessment:

- History - visual changes; pain; foreign body sensation; photophobia; contact lens wearer.
- Examination - visual acuity; discharge; pattern of redness; pupil reaction; fluorescein stain.
- Documentation - especially important to document -ve findings if planning to manage red eye in primary care.

Features that may suggest a serious cause of red eye (i.e. potential 'red flags'):

- Reduced visual acuity (VA).
- Deep pain within the eye.
- Contact lens use (**note all cases of red-eye in contact lens wearers need consideration of urgent specialist assessment** - the commonest cause is bacterial keratitis, often caused by *Pseudomonas*, which generally does not respond to usual first line topical antibiotics e.g. chloramphenicol; so if concern about bacterial keratitis or ulcer on fluorescein staining (or unable to stain) do not start antibiotic treatment in primary care and refer immediately).
- Photophobia.
- High-velocity injuries (e.g. injuries occurring while hammering/chiselling), or involving glass.
- Chemical eye injury.
- Ciliary injection (injection of limbal vessels causing redness mainly around the cornea).
- Fluorescein staining indicating ulcer or bacterial keratitis (keratitis usually causes significant fluorescein uptake with fluffy white lesions with irregular borders).
- Unequal or misshapen pupils, or abnormal pupillary reactions.
- Pain on pupillary constriction.
- Conjunctivitis in an infant in the first 28 days of life.

Conditions requiring urgent (on the day) referral:

Condition	Typical features in history	Typical features on examination
Acute glaucoma	Unilateral pain; headache; lights seen surrounded by haloes; nausea/vomit	Reduced VA; tender hard eye; ciliary injection; fixed mid-dilated pupil
Corneal ulcer or contact lens red eye	Photophobia; blurred vision; pain; may start with FB sensation	Severe conjunctival injection; discharge; ulceration on fluorescein stain
Corneal foreign body (FB)	Symptoms after wind blowing into eye or use of hammering/grinding tools; FB sensation; pain; photophobia	May see FB on examination but VA and pupil reflexes often normal
Anterior uveitis (Iritis) - within 24hrs	Pain; photophobia; watering; may have history of spondyloarthritis	Ciliary injection; constricted or distorted pupil; VA may be normal
Scleritis - within 24 hrs	Severe pain	Reduced VA; abnormal pupil reaction

- Other conditions requiring urgent (on the day) referral:
 - Penetrating eye injuries or high-velocity foreign body (see above).
 - Chemical injuries.
 - Neonatal conjunctivitis (within first 28 days).

DRY EYE SYNDROME

KISS: Dry Eye Syndrome

[BMJ 2016;353:i2333](#); [CKS 2017](#)

Aetiology

- Aqueous insufficiency (decreased aqueous secretion from lacrimal glands)
- Evaporative - caused by meibomian gland dysfunction or external causes (allergy, topical medications, contact lens use)

Clinical features

- Ocular discomfort, light sensitivity, watering eyes; typically worse with prolonged visual tasks and exposure to wind or air conditioning
- In severe cases, inflammation may lead to complications e.g. corneal damage but this is rare and usually associated with other underlying conditions e.g. Sjogren's syndrome

Referral

- **Urgent** - if suspected serious cause e.g. keratitis, iritis, ulcer (also see KISS: red-eye)
- **Routine** (urgency dependent on clinical judgement) if:
 - Suspected underlying condition e.g. Sjogren's syndrome
 - Abnormal lid anatomy or function
 - Symptoms not controlled after 12-16 weeks (consider optometrist initially if available)

Management

- **Review drugs/medications:** antihistamines, B-blockers, oestrogens, TCAs, SSRIs, topical eye medication (may have a preservative in)
- **Treat other underlying conditions:** allergic conjunctivitis (see separate chapter), rosacea, Sjogren's syndrome, blepharitis ([click here for good patient information](#))
- **Lifestyle measures:**
 - Warm compress, lid hygiene and massage ([info in blepharitis leaflet above](#))
 - Reduce contact lens use
 - Review environmental triggers:
 - Computer use (lower screen below eye level, increase blink frequency, take regular breaks)
 - Dry humidity (consider humidifier)
 - Smoking
- **Tear supplements:**
 - Mild/moderate disease - artificial tears (over-the-counter treatment may be sufficient for many)
 - More severe disease - preservative-free artificial tears plus ocular ointments at night
 - Use **drops in the daytime** - start with less viscous formulations (e.g. hypromellose) but need regular application; more viscous formulations (e.g. carbomers, polyvinyl alcohol) need less frequent application but more likely to cause blurring and stinging
 - **Gels and ointments best used at night** (can cause blurring of vision)
 - No evidence to suggest any one product is superior so start with the lowest-cost preparations
 - **Preservative-free formulations** are more expensive - consider if more severe disease (need drops >4 x/day), soft/hybrid contact lens use, intolerant of preservative
- **Patient information/education** - ensure realistic expectations; [click here for patient info](#)

ALLERGIC EYE DISEASE

Allergic eye disease is common, affecting 10-20% of people globally. It rarely causes complications, but it can cause a significant negative impact on the quality of life and productivity. Knowing which drops to use, when and for how long can be confusing for both patient and GP alike, so the recent clinical update in the BMJ in November 2017 ([BMJ 2017;359:j4706](#)) was very welcome! We summarise all this in the KISS below, but **first a few highlights/detail on the primary care management of this tricky disease:**

► Treatment options for acute symptoms (alone or in combination):

- **Topical antihistamines** (e.g. azelastine) - well tolerated and only need BD application so good for kids at school/concordance in general; insufficient evidence to suggest one agent better over another so switching is recommended if treatment failures or side effects.
- 2nd generation **oral antihistamines** (e.g. cetirizine, loratidine).
- **Topical vasoconstrictors** - short-acting so require frequent use; can result in rebound symptoms on cessation; should not be used long term or in those with CVD or hypertension.

► Treatment options for chronic symptoms:

- **Mast cell stabilisers** (e.g. sodium cromoglycate, nedocromil sodium) - can cause a burning sensation on application and need QDS application, but generally well tolerated and a Cochrane review in 2015 ([Cochrane Database Syst Rev. 2015 Jun 1;\(6\):CD009566](#)) concluded they were effective in both SAC and PAC; note however they take several weeks to become effective so are not good for acute symptoms
- Use **combinations** of topical antihistamines, oral antihistamines and mast stabilisers if single agents are ineffective; also, consider switching to alternative agents in the same class.

► What is prognosis like and how can we/our patients improve outcomes?

- **Managing expectations is key** - our patients need to know that the aim of treatment is to improve symptoms to resume normal activities; it is not to eliminate symptoms.
- Seasonal allergic conjunctivitis (SAC) and perennial allergic conjunctivitis (PAC) have a chronic course with recurrent exacerbations, but they do tend to improve over time.
- We need to warn patients that **treatments take several weeks to suppress symptoms.**
- Outcomes are generally dependent on adherence to treatment.
- If patients have predictable flares, **start treatment continuously for several weeks before** as well as during flare periods.
- If flares are unpredictable or symptoms occur throughout the year, consider long-term treatment.
- [Click here for a good patient information leaflet](#)

NB Learning Points: Allergic Eye Disease

- Red flags for acute referral include pain +/- photophobia, blurring or loss of vision, red-eye in a contact lens wearer, evidence of corneal involvement, recent eye surgery.
- Use topical +/- oral antihistamines for acute symptoms.
- Do not use mast stabilisers for acute symptoms as they take several weeks to become effective but there is evidence for their use in chronic symptoms.
- Manage patient expectations - the aim of treatment is to improve symptoms to resume normal activities, not to eliminate symptoms.
- If patients have predictable flares start treatment several weeks before their flare period.
- [Click here for good patient information leaflet](#)

KISS: Allergic Eye Disease [BMJ 2017;359:j4706](#)**Assessment:**

- **History and examination:**
 - Itching is the predominant symptom; also ask about discharge, duration of symptoms and exacerbating factors; is it seasonal? how much is it affecting the quality of life?
 - Check visual acuity (VA); look for corneal and limbal (around the iris) irregularities; consider fluorescein stain if pain, loss of vision or photophobia
- **Red flags (require an acute referral):**
 - Pain +/- photophobia; blurring or loss of vision; red-eye in contact lens wearer; corneal involvement (ulcer, opacity or spots, limbal changes); recent eye surgery
- **Differential diagnoses:**
 - Infective conjunctivitis - eyes tend to be affected sequentially and discharge tends to be more mucous; URTI symptoms usually present
 - Other differentials include serious causes of red eye (see separate chapter/KISS on 'corneal ulcers and acute red eye') and dry eye disease (see the chapter on 'dry eyes')

Causes:

- **Seasonal allergic conjunctivitis (SAC)** - bilateral conjunctival redness with watery/mildly mucous discharge; associated with childhood atopy; most frequent in spring and summer, triggered by pollens; often associated with nasal symptoms (rhinoconjunctivitis)
- **Perennial allergic conjunctivitis (PAC)** - presents as for SAC, but doesn't follow seasonal pattern; triggered by environmental allergens e.g. house dust mites, animals, fungal spores
- Rare causes include **vernal and allergic keratoconjunctivitis** - severe forms of allergic eye disease; present with severe symptoms and often red flags e.g. keratitis, corneal erosions; need to be under secondary care - usually need topical steroids +/- immune suppressants

Management:

- **Preventative measures:**
 - Cold compress; avoid rubbing eyes; refrigerated preservative-free lubricants (need to use >4-6 times/day); ?allergen avoidance (evidence for effectiveness poor)
- **Acute symptoms (alone or in combination):**
 - **Topical antihistamines (AH)** (e.g. azelastine) - well tolerated; only need BD application; switch to an alternative if treatment failures or side effects
 - 2nd generation **oral AH** (e.g. cetirizine, loratadine)
 - **Topical vasoconstrictors** - short-acting so require frequent use; can result in rebound symptoms; don't use long term or in those with CVD or hypertension
- **Chronic symptoms:**
 - **Mast cell stabilisers** (e.g. sodium cromoglycate, nedocromil sodium) - need QDS application; can cause burning sensation; take several weeks to become effective
 - **Combinations** of topical AH, oral AH and mast stabilisers if single agents ineffective
- **Patient information:** [Click here for a good patient information leaflet](#)
 - **Manage expectations** - the aim of treatment is to improve symptoms to resume normal activities not to eliminate symptoms; treatments take several weeks to suppress symptoms; outcomes are generally dependent on adherence to treatment
 - If patients have **predictable flares**, start treatment continuously for several weeks before as well as during flare periods; if flares are **unpredictable** or symptoms occur throughout the year, consider long-term treatment

DENTAL CONDITIONS IN GENERAL PRACTICE

A paper and qualitative study [BMJ Open 2015;5:e008551](#) confirms what we know: GPs regularly see dental problems, often because patients cannot access urgent dental care. We are not trained to do this, not covered medico-legally to do so and most patients end up receiving a prescription for antibiotics despite the fact that they are largely ineffective. Problems such as acute dental abscess require dental surgical intervention. Deferring dental intervention can result in increased morbidity as well as increased antibiotic resistance due to unnecessary prescribing. The paper concludes 'the prescription of antibiotics by GPs may contribute to both patient morbidity from untreated disease and emerging antimicrobial resistance'.

So, how should we manage acute dental pain? We have excellent recent papers [BJGP2016;66;236](#) and [BMJ2015;350:h1300](#) to guide us:

1 Management of acute dental infections [BJGP2016;66;236](#) and [BMJ2015;350:h1300](#)

What causes toothache? And how should it be treated in primary care?

We need to exclude non-dental causes (e.g. TMJ pain, sinusitis, parotids, trigeminal neuralgia, giant cell arteritis etc) but acute toothache is usually due to **pulpitis**. This is inflammation of the pulp, the soft fibrous tissue within the tooth which contains the neurovascular structures. Characteristics of pulpitis include:

- Severe pain in mouth and jaw, stimulated by hot and cold
- Tooth pain when biting
- Pain may be localised or radiate, including to the ear
- There is no infection of the surrounding tissue or associated swelling
- **Pulpitis does not respond to antibiotics.** It requires analgesia, pending surgical management by a dentist.

Acute dental abscess needs local surgical treatment to allow drainage and so early referral to a dentist is advised. **Antibiotics should not be prescribed even if it is not possible to start dental treatment immediately.** Analgesia should be prescribed. Antibiotics should only ever be prescribed in patients with significant immune compromise or if there are signs of local or systemic spread.

What 'red flags' should we worry about?

Severe dental infection can spread into surrounding soft tissues and cause life threatening complications. These 'red flag' symptoms and signs warrant immediate hospital referral:

Red Flags (warranting immediate hospital referral)

- Signs of spreading infection
 - Difficulty swallowing, breathing or speaking
 - Difficulty opening the mouth (trismus, with mouth opening of less than 2 finger breadths)
 - Swelling in the floor of the mouth (the raised tongue sign) or drooling
- Signs of sepsis (check vital signs)
- Rapid progression

So, if we are not to prescribe antibiotics what is our role in acute dental pain?

The BMJ paper emphasizes what the role of the doctor is when patients present with us with acute dental pain.

Our role is:

- To exclude other causes of orofacial pain e.g. TMJ pain, giant cell arteritis etc.
- To identify and immediately refer to hospital those with red flag symptoms & signs
- To advise other patients to see a dentist for evaluation and surgical management, and to prescribe appropriate analgesia in the interim
- We should avoid potentially diverting those who need to see a dentist by inappropriately prescribing antibiotics, which are likely to be ineffectual and may delay presentation to a dentist and worsen outcomes. Furthermore, our medical indemnity will not cover us for this as it is outside the scope of our practice
- *Antibiotics should only be prescribed if there is localised or facial swelling but no possibility of access to dental care in the next few hours*

2 NHS Scotland guidelines

[Acute Dental Problems, NHS Scotland Guidelines](#)

This excellent, interactive online guidance has been produced for NHS Scotland. ***The guidance reinforces that most cases of acute dental pain are due to pulpitis and that antibiotics are not indicated.*** The correct management is to provide analgesia and to seek dental care.

Most patients presenting to their GP will have one of the 6 symptoms below – for each of these there are simple algorithms or ‘click-through guidance’ on management

- Pain
- Swelling
- Bleeding
- Trauma
- Ulceration
- Altered sensation or appearance

The bottom line: antibiotics are likely to do more harm than good in acute dental pain. Our role is to exclude signs of serious infection, provide adequate analgesia and signpost appropriate dental care.

ORAL HEALTH

KISS: Acute Dental Pain

Based on [BJGP2016;66;236](#) & [BMJ2015;350:h1300](#)

Presentation	Likely cause	Management
Short, sharp pain Exacerbated by cold stimuli	Reversible pulpitis	Dental review, antibiotics not needed
Dull, aching, severe & persistent pain No swelling Exacerbated by thermal stimuli	Irreversible pulpitis	Urgent dental review Antibiotics are ineffective
Tender tooth with inflammation of surrounding tissue	Pulp necrosis follows untreated pulpitis. The death of the pulp often gives temporary relief of the pain, but the offending tooth will be tender to touch and pressure	Urgent dental review Tooth may temporarily respond to antibiotics but GP prescription not advised
Severe tenderness to touch, swelling and tooth mobility	An apical abscess, often due to untreated periodontitis	Needs surgical drainage by dentist but may respond in short term to antibiotics if a dental review in the next few hours is not possible. If prescribed, low dose amoxicillin is first choice.

NB Quality Improvement Ideas: Acute Dental Pain

- Audit of patients presenting with acute dental pain and the prescription of antibiotics in the practice
 - You should be able to demonstrate a clear change in practice here and complete an audit cycle!
- Put some patient information, using the links below, on your website with details of how to access urgent dental care and when they need to see a GP
- Resources
 - [NHS Scotland Interactive Guidance](#)
- Patient information
 - [NHS Choices: Toothache](#)
 - [NHS Choices: Dental Abscess](#) ('there isn't a lot your GP can do')
 - [How to access urgent NHS dental care](#) (including out-of-hours)

BURNING MOUTH SYNDROME, GEOGRAPHIC TONGUE & ORAL CANDIDIASIS

Miserable for patients, often a heartsink condition for doctors, the strange condition of **burning mouth syndrome (BMS)** has recently been the subject of a new systematic review (1a) [Clinical Evidence 2016;01;1301](#)

This is welcome, as these common yet 'medically unexplained' syndromes are often neglected in the literature. However, sadly the evidence base is poor.

What is burning mouth syndrome BMS?

It has previously been called various names including glossodynia, stomatodynia, sore tongue etc but BMS is the now preferred term. It is simply ***unexplained discomfort or pain in the mouth with no known medical or dental cause***. It is often localised to the tongue and/or lips but it can involve all of the oral mucosa. Most, but not all, patients have a subjective sense of dryness associated with it.

It is common, affecting 15% of all adults and up to 1/3 of postmenopausal women. Most people have some form of other chronic morbidity or are on medication and there is a high psychological co-morbidity. There is an association with generalised anxiety disorder, depression and adverse life events.

It is a diagnosis of exclusion once other causes have been ruled out. The most accepted definition is classified by the International Headache Society.

IHS Definition: Burning mouth Syndrome

An intraoral burning sensation for which no medical or dental cause can be found. Pain may be confined to the tongue and subjective dryness of the mouth, paraesthesia and altered taste may be associated symptoms.

Diagnostic criteria:

A: Pain in the mouth present daily and persisting for most of the day

B: Oral mucosa is normal in appearance

C: Local and systemic disease have been excluded

What causes BMS?

Whilst by definition it is idiopathic, recent research suggests that with sensitivity testing there are peripheral and central neurophysiological changes in BMS and that it may be a ***neuropathic pain syndrome***. Tongue biopsies show changes in nerve fibres and functional MRI also shows changes similar to those seen in neuropathic pain.

What is the prognosis?

There are no good quality prospective cohort studies but anecdotal reports suggest that approximately 50% have spontaneous remission over 6 or 7 years.

How do I make the diagnosis?

It is a clinical diagnosis of exclusion in the presence of an oral mucosa which is normal in appearance, once other causes have been excluded (see KISS summary below)

What does the evidence say about available treatments?

Unfortunately, there is very little evidence to guide us, and the most that there is have been small studies with limited follow-up. However, they conclude that:

- **Cognitive Behavioural Therapy is likely to be beneficial**
 - Two studies have been done in patients with BMS which suggest benefit in reducing pain intensity at 6 months
 - Other studies on unexplained oro-facial pain and CBT has been shown to help in chronic pain, especially with improving mood and coping strategies
- **Tricyclics**
 - Although TCAs are of course widely recommended for unexplained syndromes, chronic and neuropathic pain there are no RCTs evaluating them in BMS
 - We have a lack of evidence rather than evidence of lack, so TCAs may be helpful as in other types of neuropathic pain
- **SSRIs and SNRIs**
 - Similarly, there are no placebo controlled RCTs so we have a lack of evidence
 - If patients have associated depression or generalised anxiety, clearly they should be tried
- **Benzydamine mouthwash**
 - One placebo controlled trial, no better than placebo
- **Clonazepam**
 - Topical or oral clonazepam, two studies suggest it may be of benefit compared to placebo
 - Because of concerns around dependency clonazepam use 'should be limited'

What about geographic and fissured tongue? [NEJM2016;374:670](#)

Geographic tongue (benign migratory glossitis) is clinically distinct from BMS, but it may be symptomatic.

- Highly common, geographic tongue affects 2% of the population
- It is a benign condition associated with disturbance of the filiform papillae of the tongue
- Patients present with migrant, oval red areas surrounded by a well-defined white rim
- So-called fissured tongue (NB which is what it says on the tin...) often accompanies it
- It is usually asymptomatic, but symptoms of pain or burning can be triggered by eating spicy or acidic foods

What about oral candidiasis? [Clinical Evidence 2013;11:1304](#)

As we all know, this is very common in primary care and it is an important differential in BMS. Candida carriage on the tongue is present in 65% of people. Overt opportunistic infection causes painful red or white lesions of the oropharynx which can affect taste and also speech. ***It commonly causes a sore mouth with a burning tongue and altered taste.*** It may appear as white curd-like plaques on the tongue and buccal mucosa, but may also cause smooth red patches or white hyperplastic patches.

As well as obvious cases of immunosuppression (such as drugs, chemotherapy and HIV) in primary care it is commonly associated with haematinic deficiency, denture wear, use of corticosteroid inhalers, antibiotic use, anticholinergics (and other drugs causing dry mouth) and smoking. In recurrent infection trying to eliminate these risk factors is key.

For treatment, the evidence suggests that miconazole gel is more effective than nystatin drops (at least in children) and in people with more severe and recurrent infection oral fluconazole for treatment and weekly prophylaxis if needed is effective (NB bear in mind important interaction between oral miconazole and warfarin, [Drug Safety Update 2016](#)).

KISS: Burning Mouth Syndrome

[Clinical Evidence 2016;01;1301](#) & [World J Gastro 2013;19:665](#)

Diagnosis:

- Oral mucosa normal in appearance + daily pain in the mouth
- **Careful examination and simple tests to exclude oral, systemic or dental causes**
 - Oral (e.g. geographic tongue, recurrent aphthous ulcers)
 - Pre-malignant lesions e.g. erythroplakia (unexplained red patches) & erythroleukoplakia (unexplained red and white patches) which represent carcinoma in situ, consider dental opinion and if persist for >3 weeks urgent referral (see cancer guidelines section)
 - Dermatological (e.g. oral lichen planus)
 - Gastroenterological (e.g. acid reflux)
 - Infections (e.g. oral candida, HSV)
 - Allergies (e.g. oral allergy syndrome)
 - Vitamin (e.g. Vitamin B12) or hormone (e.g. thyroid) deficiency
 - Dental cause (e.g. ill-fitting dentures)
 - Drug side effects (especially if they cause dry mouth, such as anti-cholinergics)
 - **Suggested bloods:** FBC, ferritin, B12, folate, glucose OR HbA1c, TSH and autoantibodies (NB consider HIV test if appropriate)
- More common in postmenopausal women (but we do not know if HRT helps)
- Physical and psychological co-morbidities are common

Management:

- **Reassurance and explanation**
 - BMS maybe unpleasant but *it is not harmful*, it may go away and treatments may help to ease the discomfort
- Advise full dental check-up to exclude dental cause
- Lifestyle guidance
 - Frequent sipping of water, sugar-free gum, sucking on ice
 - Avoid irritants e.g. hot, spicy foods; acidic fruits and juices; alcohol; mouthwashes
 - Stress reduction e.g. advise patients follow the [Mindfulness for Health](#) programme
- Optimise psychological co-morbidities
- Refer for CBT if available
- If drug treatment is needed, consider
 - Low dose tricyclic
 - SSRI if associated depression or anxiety

Patient information:

[Patient Information, BMJ Best Practice](#)

[Patient Information, British Dental Health](#)

DEMENTIA

Dementia remains (and will continue to remain) a Hot Topic for us all in primary care. It will surprise no one that the numbers of patients we are seeing in General Practice continue to rise substantially; in 2017 the number of people with a formal diagnosis of dementia on GP registers had risen to >450,000, a **rise of >45% over the past 8-10 years**. In [June 2018 NICE](#) published comprehensive updated guidance on the assessment and management of people living with dementia [NICE NG97, 2018](#). It was a chunky document and much of it focuses on what we do best in General Practice - giving holistic care to patients and their families, and making sure the patient is at the centre of all decision-making. More recently we have had a useful review in the [DTB May 2020](#) reminding us of the importance of reviewing anticholinergic medications both when assessing patients with suspected dementia, but also after diagnosis, as their drugs are an important iatrogenic contributor to non-cognitive symptoms of dementia. And whilst on the subject of non-cognitive symptoms of dementia we have had a recent update article ([BMJ 2022;376:e069187](#)) highlighting the importance of, and evidence behind, non-pharmacological management of this distressing phenomenon and also evidence reviewed in [Drug and Therapeutics Bulletin 2022;60:68](#) that mirtazapine does not improve agitation, and may cause harm.

► **Non-cognitive symptoms of dementia** [BMJ 2022;376:e069187](#) [DTB2022;60:68](#)

The management of non-cognitive symptoms of dementia (also known as neuropsychiatric symptoms in dementia or BPSD - behavioural & psychological symptoms of dementia) is a major problem for patients, carers and GPs alike and was only briefly dealt with in the NICE guideline so a recent BMJ review article ([BMJ 2022;376:e069187](#)) was welcome. We have summarised the key points in the KISS below, but a few key messages we should be aware of:

- Neuropsychiatric symptoms of dementia are common, are difficult to treat and cause distress for carers.
- First assess for (and treat if present) causes of delirium.
- There are a **range of non-drug options available**, and although the evidence base is inevitably uncertain given the nature of the interventions studied, the likelihood is that these interventions are as effective (if not more so) than drug therapies. **Examples include massage and touch therapy, occupational therapy, animal therapy, cognitive stimulation, and reminiscence therapy.**
- Medications should only be used in limited situations where severe distress or danger to the person, or others, is present.
- There are **well-known risks of antipsychotics** in this situation (falls, stroke, fractures and death) and prescribing of these drugs is declining, but there has been a concomitant increase in prescribing of other drugs (antidepressants and anticonvulsants) and the safety of these is also being questioned.
- **If antipsychotics are needed aim to attempt de-prescribing after 3 months** - [Cochrane 2018](#) review evidence (albeit low quality) suggests that discontinuation of antipsychotics in people with BPSD and dementia after >3 months did not lead to worsening symptoms.

Whilst on the topic of psychotropic drug prescribing in BPSD, Mirtazapine has been increasingly prescribed in recent years for agitated behaviour in patients with dementia. **A recent well-designed, double-blind placebo-controlled UK trial SYMBAD (Study of mirtazapine for agitated symptoms in dementia)** [Lancet 2021](#) found **no benefit of mirtazapine compared to placebo, and a potentially higher mortality with the use of mirtazapine**. This study was reviewed in the [Drug and Therapeutics Bulletin 2022;60:68](#):

- 204 patients with dementia, average age 82, and agitation that had not responded to non-pharmacological management were randomised to either placebo or mirtazapine initially 15mg daily escalating and aiming for a dose of 45mg daily over 4 weeks.
- Agitation levels were assessed using a scoring system and there was a decrease in scores in both groups at both 6 and 12 weeks but there was **no difference between the 2 arms**.
- However, there was an **increase in adverse events (192 vs 225), and mortality (7 vs 1) in the mirtazapine group vs placebo group**.
- A study with a clear conclusion: the **data do NOT support the use of mirtazapine as a treatment for agitation in dementia**.

► *Prescribing anticholinergic medicines to people initiating dementia medicines* [DTB May 2020](#)

This [DTB review](#) focused on an Australian cohort study and has some important reminders for us. It reviewed >4000 people initiating dementia medications (mostly donepezil) and looked at the proportion given anticholinergic medications in the 6 months before, and the 6 months after they had been initiated on their dementia medication.

- ~1/3 had been dispensed an anticholinergic either before or after the dementia drug had been started, and **worryingly of those given medication in almost 60%, it was an antipsychotic**, which as we know has significant risks in this group.
- **Almost 10% were given an anticholinergic medication after the dementia medication was initiated** with risperidone and prochlorperazine being the most commonly prescribed. This has a 'double-whammy' effect of not only rendering the dementia medication less effective but can also contribute to non-cognitive symptoms of dementia (see below).

As the commentary discusses this study highlights ongoing high anticholinergic medication use in people at risk of, or with dementia; although an Australian study it is quite possible we are falling into the same trap here in the UK. **These drugs are associated with cognitive, functional and motor decline, increased healthcare utilisation and higher mortality.** It was particularly worrying that antipsychotics were still being prescribed so regularly, although other drugs e.g. amitriptyline and oxybutynin still featured commonly. It is a salient reminder to review medications when assessing people with cognitive decline, as well as regularly reviewing for anticholinergic medication use in those diagnosed with dementia.

We have also had recent research [BMJ2018;361:k1315](#) and [JAMA 2019](#) showing a 'robust association' between anticholinergic drugs and future risk of dementia. The risk was greatest with exposure to antidepressants, urological and anti-Parkinson's drugs with anticholinergic side effects. An associated editorial [BMJ2018;361:k1722](#) states '*anticholinergics, in general, should be avoided in older adults*'.

► *What about case finding for dementia?*

This has been a thorny issue and there has been a lot of debate over the past few years about case finding for dementia but [NICE NG97](#) specifically recommended we should NOT undertake case finding for dementia unless part of a clinical trial (that is 'actively assessing people who are at risk for a particular disease, before they present with symptoms and before there is clinical suspicion of the condition').

► *Risks during hospital admission for people with dementia* [NICE NG97 2018](#)

There was an important section towards the end of the NICE guidance specifically addressing the issue of hospital admission. As GPs we are often under pressure to admit people with dementia for intercurrent illness by family or carers, sometimes on the premise that it will be 'safer' for them to be in hospital rather than at home or in a community setting, when we know that is often not the case. So this small section is very important in helping to give us support when we are making those very difficult decisions about whether, or not, to admit a person with dementia to hospital:

- Be aware of the **increased risk of delirium** in people with dementia who are admitted to hospital.
- When considering hospital admission take into account the **value of keeping them in their own environment**.
- If considering admitting a person with severe dementia balance current medical needs with the additional **harms they may face in hospital** e.g. disorientation and delirium, longer length of stay, increased morbidity on discharge and increased mortality.

KISS: Assessment and Management of Dementia
[NICE NG97 2018 BMJ 2018;361:k2438](#)

Assessment:

- **History** - from both the patient and someone who knows them well.
- **Cognitive testing:**
 - NICE have suggested a number of cognitive tests to consider using in primary care, although interestingly the GPCOG test ([click here](#)) which has been widely used was not mentioned.
 - NICE point out that a **normal score on cognitive testing does not rule out dementia**.
 - We can use **short cognitive tests** rather than the longer versions such as MMSE e.g. 6-CIT - ([click here](#)), Mini-Cog ([click here](#)) the 10-point cognitive screener, the 6-item screener, the Memory Impairment Screen and [Test Your Memory TYM Test](#).
- **Rule out reversible causes:**
 - Do appropriate **blood and urine tests** (not specified by NICE what we should do but previous [guidance from the BMJ](#) suggests we should do FBC/ESR/U&E/thyroid function/B12 and folate and consider other tests e.g. MSU if clinically appropriate).
 - **Could this be delirium?** Often not easy. NICE say we should not use standard cognitive testing to distinguish delirium from dementia, and they make the pragmatic suggestion that if the diagnosis is unclear we should treat for delirium first, before re-assessing for dementia.
 - **Review drugs** that may cause/contribute to cognitive impairment, **especially anticholinergics**: Consider using the [Anticholinergic Cognitive Burden Scale](#) to review drugs with anticholinergic properties; the most common ones prescribed with high burden are anticholinergics used for overactive bladder and the tricyclic antidepressants.
- **Referral:**
 - Once reversible causes have been investigated refer to a specialist dementia diagnostic service.
 - If the person has suspected **rapidly progressive dementia refer to neurology** for consideration of other conditions e.g. CJD.

Management:

- Actively encourage and enable people with dementia to give their opinions and views.
- Make sure patients and families have access to relevant **information and support services** - e.g. **Alzheimer's Society UK** is a fantastic resource for patients and families - [click here](#)
- All people with a diagnosis of dementia need to **inform the DVLA** - [click here](#)
- **Advanced care planning:** Do they wish to make advanced decisions regarding future care or refusing treatment? Have they got lasting powers of attorney in place?
- Consider **interventions to promote cognition, independence and well-being** e.g.:
 - Cognitive rehabilitation or occupational therapy to support functional ability.
 - Offer a range of activities to promote well-being that are tailored to the patient's preference.
- **Do NOT offer** the following: Ginseng, Vitamin E supplements, herbal formulations, acupuncture.
- **Incontinence** - many of the drugs used for overactive bladder have a significant anticholinergic burden and should be avoided, but the beta-agonist mirabegron (with appropriate BP monitoring [MHRA Guidance](#)) is an option [NICE TA290](#)
- Assess for **sensory problems** including hearing and sight - improving these can improve quality of life as well as reduce falls: Encourage people with dementia to have eye tests every 2 years.
- **Pharmacological management of cognitive symptoms** ([click here](#) for a visual summary):
 - Initiation of pharmacological treatment for AD should be done on the advice of a specialist but once that decision has been made the first prescription can be done in primary care.
 - Memantine can be initiated in primary care without specialist advice in people with an established diagnosis of AD.
 - See KISS below for a summary of drugs and indications.

KISS Summary of Dementia Drugs

Based on [NICE NG 97 June 2018](#) & [DTB2014;52;69](#)

Drug & indications	Dosage regimen	Common side effects	Important interactions
Donepezil: All stages of AD All stages of Lewy body (LB) dementia PD dementia Vascular dementia - only if comorbid AD, LB or PD dementia	5mg daily, increased after 1 month if needed to a max of 10mg daily	Serious SEs are rare: GI (nausea, vomiting, diarrhoea, anorexia) SEs common and dose-related. Also: Headaches, insomnia bradycardia (caution with AV block; monthly monitoring at onset, stop if HR<50) tremor, agitation, dreams, hallucinations; caution with asthma, COPD, urine retention and peptic ulcer	Anticholinergic and rate-limiting drugs P450 inhibitors and inducers
Galantamine: All stages of AD Mild/mod LB dementia only if donepezil or rivastigmine not tolerated PD dementia	4mg bd for 4 weeks, then 8mg bd for 4 weeks, then 8-12mg bd maintenance dose extended-release: 8mg daily, then 16mg daily after 4 weeks, maintenance 16-24mg daily	Ditto	Ditto
Rivastigmine: All stages of AD All stages of Lewy body (LB) dementia PD dementia Vascular dementia - only if comorbid AD, LB or PD dementia	1.5mg bd, increase by 1.5mg at 2 weekly intervals to a maximum of 6mg bd patch: 4.6mg/24 hours increase after 4 weeks to 9.5mg/24 hours; can increase to 13.3mg/24 hours after 6 months if meaningful decline	Ditto GI side effects are less with the patch	Ditto, but no problem with hepatic enzyme blockers or inducers (so fine with phenytoin, carbamazepine etc.)
Memantine: Mod/severe AD as mono- or dual therapy with AChE inhibitors LB & PD dementia if AChE inhibitors C/I or not tolerated Vascular dementia - only if comorbid AD, LB or PD dementia	5mg daily, increase by 5mg every week to a max of 20mg daily If AD diagnosis established can start in GP without specialist advice	Fewer and less serious side-effects than the AChEIs. Constipation, headache, somnolence, dizziness most common s/e	May enhance levodopa

KISS: Behavioural and Psychological Symptoms of Dementia (BPSD)

[DTB 2014 BMJ 2015](#), [Oxford BPSD guidance NICE NG97 2018 BMJ 2022;376:e069187](#)

Background:

- **75% of patients with dementia experience BPSD** including:
 - Psychotic symptoms e.g. delusions, hallucinations.
 - Behavioural disturbances.
 - Mood changes.

Assessment:

- **Presentations may be varied:**
 - Agitation/aggression e.g. hitting kicking, restlessness, screaming.
 - Depression e.g. sadness, slow movements, early morning waking.
 - Delusions e.g. false belief someone is trying to harm them or steal from them.
 - Hallucinations e.g. hearing, feeling, seeing things or people that aren't real.
 - Anxiety e.g. SOB, worry, fear of something about to happen.
 - Apathy e.g. loss of interest in activities or ADLs.
 - Disinhibition e.g. impulsiveness, saying or doing inappropriate things.
 - Motor disturbance e.g. pacing, restlessness, wandering, repetitive activities.
 - Night behaviours e.g. night waking, excessive daytime napping.
 - Changes in eating e.g. changes in food preference, weight gain or loss.
- **Consider could this be Lewy body or Parkinson's dementia?**
 - Characterised by a long history of vivid visual hallucinations and fluctuating cognition.
 - Get specialist advice as this is part of dementia and can respond well to Acetyl Choline esterase inhibitors.
- **Assess contributing factors:**
 - Physical, environmental and iatrogenic causes must be investigated and treated first:
 - Adopt a person-centred approach, **applying the PAIN approach**:
 - P = physical problems e.g. infection, constipation, pain.
 - A = activity related e.g. washing, dressing (need to modify).
 - I = iatrogenic e.g. drug side-effects.
 - N = noise. Consider and calm the patient's environment.
- **Exclude physical causes of delirium:**
 - Usually short history <1 week of increased confusion, hallucination, delusions etc.
 - Common issues include:
 - Constipation.
 - Untreated pain (one study has shown a systematic approach to better pain management can reduce agitation in dementia).
 - Infection, especially pulmonary and UTI.
 - Consider alcohol, drug withdrawal or side-effect.
- **Review medications.** BPSD are worsened by:
 - Drugs with anticholinergic properties e.g. tricyclics, sedating antihistamines, anticholinergic bladder drugs etc.
 - Benzodiazepines.
 - Some analgesics e.g. tramadol.

KISS: (BPSD) Continued

[BMJ 2015](#) [Oxford BPSD guidance](#) [NICE NG97 2018](#) [BMJ 2022;376:e069187](#)

Management:

■ **Non-pharmacological management:**

- These options should be **tried first wherever possible**.
- There is level 1a evidence from systematic reviews of the efficacy of behavioural and psychological approaches, including:
 - Training care staff to understand and manage BPSD, ideally in MDT format.
 - Ensuring the patient's environment is calm, safe and predictable.
 - Cognitive stimulation - structured therapy aimed at promoting cognitive function e.g. orientation, reminiscence, art therapy and games.
 - Occupational therapy.
 - Social interaction.
 - Exercise including aerobic, resistance or balance training.
 - Behavioural therapies e.g. reminiscence therapy when patients are helped to re-live past.
 - Experiences lead to improvements in mood, behaviour and well-being.
 - Animal therapy.
 - Music therapy and sensory stimulation including massage and touch therapy have also been shown to help.
- If mild/moderate depression present in patients with mild/moderate dementia consider psychological therapies.

■ **Pharmacological management:**

- **Depression** - NICE advise that antidepressants should not be routinely offered to people with mild/moderate depression and mild/moderate dementia unless indicated for a pre-existing mental health problem.
 - Note we have recent evidence from the [SYMBAD trial Lancet 2021](#) that **mirtazapine does not improve agitation in dementia** and may cause increased harm including mortality.
- **Insomnia/anxiolytics:**
 - Benzodiazepines should be avoided because their use is associated with further cognitive decline, falls and fractures.
 - Long-term use of sedating antihistamines should be avoided due to the anticholinergic effect.
 - NICE advise melatonin should NOT be used for the management of insomnia.
- **Antipsychotics:**
 - Should **generally be avoided due to the significant risk of harm** - they are associated with an increased risk of stroke, falling, fractures and death.
 - However, they can be considered if there is a risk of harm to self or others, or if there is agitation, hallucinations or delusions causing severe distress.
 - They should be used at the lowest effective dose for the shortest possible time and reviewed every 6 weeks to check if still needed.
 - **Attempt to de-prescribe by about 3 months** - we have [Cochrane 2018](#) review evidence (albeit low quality) that discontinuation of antipsychotics in people with BPSD and dementia after >3 months did not lead to worsening symptoms.
 - The exception would be those with chronic psychotic illnesses e.g. schizophrenia - withdrawal of antipsychotics should only be done by older persons mental health specialists.
 - They should NOT be used in patients with dementia with Lewy Bodies or Parkinson's dementia; they can worsen motor features and can cause severe antipsychotic sensitivity reactions.

NB Quality Improvement Ideas: Dementia

- Audit of patients with Alzheimer's dementia:
 - Have they had a drug review and stopped meds which adversely affect cognitive function, **especially those with high anticholinergic burden?**
 - Have they been offered a cognitive-enhancing drug?
 - Have they had an eye test in the last 2 years and has their hearing been assessed?
 - How many patients are taking anti-psychotics? These should all be reviewed with a view to stopping if possible due to the increased risk of worsening cognitive symptoms as well as an increased risk of stroke and death.

NB Learning points: Dementia

- Do not actively 'case find' or screen for dementia but assess clinically, supported by short cognitive tests and suspected cases.
- Use **short cognitive screening tests** (rather than the longer tests e.g. MMSE) as part of our assessment for dementia in primary care:
 - Examples include the 6-CIT ([click here](#)) and mini-cog ([click here](#))
- We have had a recent DTB review **highlighting the risks of anticholinergic drugs** in people either with or at risk of dementia:
 - Review medications, both in people you are assessing for possible dementia and in those with a confirmed diagnosis, especially those drugs with high anticholinergic burden - [Click here](#) for info.
- NICE have updated their guidance on dementia drugs (see KISS above):
 - The decision to initiate these drugs should only be made by specialists with expertise in diagnosing and treating dementia, but the first prescriptions can be done in primary care.
 - However, for people with established Alzheimer's dementia who are already taking AChE inhibitors, memantine can be initiated in primary care without specialist advice.
- For behavioural and psychological symptoms of dementia (BPSD):
 - Non-pharmacological options should be used first line.
 - Use the [algorithm](#) produced by Oxford Health Trust.
 - Do not use mirtazapine.
 - Antipsychotics should only be used if there is a risk of harm to self or others, or if there is agitation, hallucinations or delusions causing severe distress.
 - If antipsychotics are used, aim to de-prescribe after 3 months.

FRAILTY AND SARCOPENIA

A recent paper in the Lancet [Lancet 2019;394:1376](#) describes **frailty** as '*the new frontier of medicine*', with growing awareness of the importance of recognising the syndrome as a major predictor of adverse outcomes. A fascinating recent BJGP paper [BJGP2020;70;200](#) also discusses **sarcopenia**, loss of muscle bulk and strength, which is distinct from frailty but a principal contributor to the vulnerability that defines it.

With an ageing, multimorbid population there is an increasing **recognition of frailty as a medical syndrome in its own right**. *It is frailty, not age, that best predicts adverse outcomes such as dependency, admissions, medical interventions and death*. There is a strong body of opinion in the literature that we should stop thinking of vulnerable older people as having collections of single diseases and instead focus on the global issue of frailty and to consider frailty as a long-term condition in its own right to be managed in a pro-active way. This idea is now strongly influencing health policy and reform, with an increasing drive for us to recognise and 'diagnose' frailty and to introduce frailty screening. In England, identification of frailty became a contractual requirement back in 2017. *But is this a worthwhile endeavour and what, if anything, works?*

New research published in 2019 [BJGP2019;69:e61](#) suggests that *interventions in primary care can be effective to help delay or reverse frailty*. We have also had new reviews of the evidence published in the last year [Lancet 2019;394:1365](#) and [Lancet 2019;394:1376](#).

A *significant potential benefit of the identification of frailty* was discussed in the Lancet review [Lancet 2019;394:1365](#): *identification of frailty in clinical practice has the potential to provide more holistic, person-centred care based on patients' wishes and aimed at optimising function, preserving independence and protecting patients from unnecessary medical treatments and potential harm*.

► *Delaying and reversing frailty: a systematic review of primary care interventions* [BJGP2019;69:e61](#)

As this paper points out, although frailty screening is widely recommended in primary care (and indeed was made a contractual requirement for GPs in England back in 2017) there is lack of clear guidance on what should be done once frailty is identified. There is no consistent approach as to how frailty is managed in primary care once it is identified. This new research looking at the effectiveness of primary care interventions is therefore very important to help inform practice.

- It was a systematic review including data from 46 studies and 16,000 patients
- A broad range of frailty interventions were included e.g. physical exercise, nutrition, medication including hormone supplementation, home visits and counselling
- **Results:**
 - A significant improvement in frailty status was demonstrated in 71% of the studies
 - Interventions with both strength training and protein supplementation were consistently the most effective in terms of relative effectiveness and ease of implementation
 - Typical exercise regimen would be 25 minutes at home 4 times per week, comprising exercises for leg and arm strengthening and also for balance and coordination
 - Nutrition or protein supplementation regimens included dietary emphasis on milk, eggs, tuna, chicken or if preferred 2 x 200ml of nutritional supplement formula daily
 - Other effective interventions included mild intensity exercises, exercise such as Tai Chi, education and health promotion
 - There was some evidence for comprehensive geriatric assessments and home visits
 - **Conclusion:** *a number of interventions have evidence of efficacy, but analysis of the evidence suggests that a combination of strength training and protein supplementation is the most effective*

This study usefully informs us that a number of interventions are worthwhile, especially home-based exercises with strength and balance training and dietary protein supplementation. However, the outcomes measured are improvements in frailty scores, and the paper does not report reductions in more meaningful outcomes such as hospital admission, loss of independence etc.

► *Sarcopenia: why it matters in General Practice* [BJGP2020;70;200](#)

This excellent debate and analysis paper re-introduces us to the issue of **sarcopenia**, which with all the talk of frailty in recent years had rather fallen out of the medical lexicon. *However, loss of muscle strength, bulk and declining physical performance is a key contributor to the syndrome of frailty and most interestingly the area where interventions to improve outcomes have shown the greatest benefit* **Key points:**

- Sarcopenia was recognised as a disease in ICD back in 2016, and yet is a diagnosis rarely made or coded or clinically acknowledged in primary care
- Sarcopenia contributes to frailty, but is distinct from it: sarcopenia is considered a disease which may be primary (age-related), or secondary (e.g. low physical activity, chronic inflammatory disease, malnutrition etc) whereas frailty is a syndrome
- Sarcopenia is distinct from cachexia, a catabolic state typically associated with cancer and organ failure
- The criteria for the diagnosis of sarcopenia are low muscle strength, low muscle quantity or quality and low physical performance
- Involuntary weight loss is common with sarcopenia, as with frailty; however, sarcopenia can co-exist with obesity
- *Sarcopenia matters in clinical practice because it is highly predictive of adverse health outcomes including falls, loss of independence and increased mortality*
- **Interventions:**
 - There is growing evidence on the effectiveness of resistance training, exercises to improve muscle strength and mass
 - The evidence relating to nutritional interventions is less clear, although a combination of diet and increased dietary protein is usually recommended
- **Conclusion:** *sarcopenia is a key contributor to the syndrome of frailty and highly predictive of adverse health outcomes including falls and loss of independence; identification is clinically important, as interventions focussed on resistance training and increasing dietary protein have been shown to improve outcomes*

KISS: Frailty & Sarcopenia Background Points

[BJGP2020;70;200](#) and [BJGP2019;69;e61](#) and [Lancet 2019;394:1365](#)

- **Definition:** frailty is a state of physiological vulnerability with diminished capacity to handle stressors increasing the risk of illness, falls, dependence, disability and death
 - A key clinical implication of frailty is that it results in poor functional or homeostatic reserve so that an apparently small 'insult' (e.g. a new drug, a UTI, a biochemical imbalance etc.) can result in a large and disproportionate change in health and social state (e.g. by precipitating an admission, a fall, a decline from independence to dependency etc.). An acute event which may be inconsequential in a non-frail person can present with a steep and prolonged functional and cognitive decline in a patient with frailty.
- There is international consensus that recognises frailty as medical syndrome highly predictive of hospital admissions, loss of independence and death; it is recognised as an emerging global health burden with major implications for clinical practice and public health
- **Sarcopenia** is classified as a disease by the ICD; the criteria for the diagnosis are low muscle strength, low muscle quantity or quality and low physical performance; it strongly contributes to the vulnerability characteristic of the syndrome of frailty, and is similarly associated with adverse health outcomes
- The **prevalence** of frailty is estimated at 11% in adults aged over 65, and > 50% aged over 80
- Frailty can occur at any age, and it is biological age and not chronological age which is the strongest predictor of frailty
- A variety of **diagnostic tools** are used. A commonly used tool is Fried's frailty phenotype with ≥3 criteria from:
 - Exhaustion: self-reported
 - Unexplained weight loss: self-reported or measured > 5% in one year
 - Slowness: slow gait speed
 - Weakness: weak grip strength
 - Low physical activity
- **Common presentations of frailty** include delirium, loss of mobility or functional independence, falls or a non-specific failure to thrive or cope. In primary care frailty typically presents with
 - **Non-specific** fatigue, weight loss and frequent infections
 - **Falls.** Balance and gait impairment are major features of frailty
 - **Delirium.** Acute confusion characterised by rapid onset of fluctuating confusion
 - **Fluctuating disability** with 'good days and bad days' is characteristic
- **Frailty screening** is widely recommended, and a contractual requirement for GPs in England
- It is important to remember that the syndrome of frailty refers to physical frailty; care with words and careful explanation is needed, as the word 'frail' implies a weakness which our emotionally strong, fiercely independent and proud older patients with physical frailty absolutely do not have [BJGP2017;67;216](#)
- **Interventions:** a number of interventions have evidence of efficacy, but analysis of the evidence suggests that a combination of strength training and protein supplementation is the most effective

NB Quality Improvement Ideas: Frailty

- Look out for, and use the [electronic Frailty Index](#).
- Build a population of patients coded as frail and audit them:
 - Have they been referred for multidisciplinary/geriatric assessment?
 - Have they had a medicines optimisation review? Use the [STOPP START Toolkit](#) when prescribing
 - Have they had a falls risk assessment?
- See [Frailty: 10 Top Tips for Primary Care](#)
- Read Atul Gawande's brilliant book [Being Mortal](#) for further inspiration! And constantly ask your frail older patients: what is really important to you?

NB Learning points & QI ideas: Frailty & Sarcopenia

- **Frailty** is an internationally recognised clinical syndrome which predicts a broad range of adverse health outcomes
- **Sarcopenia** is classified as a disease by the ICD; the criteria for the diagnosis are low muscle strength, low muscle quantity or quality and low physical performance; it strongly contributes to the vulnerability characteristic of the syndrome of frailty, and is similarly associated with adverse health outcomes
- The recognition of frailty has the potential to enable us to deliver more holistic, patient-centred care based on patients' wishes and function rather than on single diseases
- There is a new paradigm for older people and frailty: we need to stop thinking of the 'frail elderly' (a passive label) and start thinking of an 'older person living with frailty', an active diagnosis of long-term condition where outcomes may be improved with active management
- Frail patients are vulnerable to major adverse consequences (e.g. admissions, falls, loss of independence, delirium etc.) from relatively minor 'insults' e.g. UTI or starting a new drug
- The diagnosis is based on patients having three or more of: unintentional weight loss, weakness, fatigue, slow gait speed and inactivity. There is a new [electronic Frailty Index](#) already integrated into GP systems which prompts a possible diagnosis
- There is emerging evidence for primary care interventions to delay or reverse frailty, including strength and balance training and protein supplementation
- Care with language...Whilst our patients may be physically frail, they frequently show incredible emotional strength, power and resilience. They are anything but 'weak'...

KISS: Management of Frailty

Based on [BJGP2019;69:e61](#) & [BJGP2017;67;235](#) & [Fit For Frailty 2015](#)

- Carry out a full and comprehensive **holistic review** of the medical, functional, psychological and social needs of the patient (NB with nursing and social care colleagues, ideally with an overall care coordinator)
- Ensure that **reversible medical conditions** are considered and addressed
- Identify **personal goals**
- Encourage **muscle strength training** (refer for supervised classes if available) and advise an increase in **dietary protein**
- **Protect the patient from potential stressor** events e.g. prompt treatment of UTIs, careful consideration when prescribing etc
- Refer to **geriatric medicine** for comprehensive geriatric assessment (CGA)
 - Including physical and occupational therapy and mental health input when needed
- Conduct **personalised medication reviews** and optimise medication
 - E.g. using [STOPP START Toolkit](#)
- **Anticipatory care planning**
 - Generate a personalised care and support plan which documents
 - Treatment goals
 - Management plans
 - Plans for urgent care
 - End of life plans when appropriate
 - Ensure that robust systems are in place for sharing (e.g. with out of hours, emergency services and secondary care) and regular review of care plans

Common problems in frailty which can be addressed to improve outcomes include:

- Falls and poor mobility (the most important intervention [BMJ2017;358;j4478](#))
- Cognitive impairment
- Continence
- Weight loss and malnutrition
- Polypharmacy
- Low mood and depression
- Alcohol and smoking
- Vision problems
- Social isolation and loneliness
- **Additional advice**
 - Look after your feet
 - Keep warm, get ready for winter
 - Make your home safe
 - Vaccinations
- **Patient resources:**
 - [NHS Choices](#) including [home-based exercises for frail older people](#)
 - Great patient-friendly resources and tips from [Age UK](#)
 - [Frailty: what it means and keeping well over winter](#)
 - [Patient advice on maximising nutrition from food](#)

MULTIMORBIDITY

Multimorbidity is one of the most important Hot Topics of our time and improving the care of people with multimorbidity one of the greatest challenges we face. Multimorbidity is a major challenge for elderly people but it is of course not restricted to the elderly; many people aged under 65 have to live with multimorbidity and its consequences. Back in 2016/2017 we had major new guideline from NICE [NICE: Multimorbidity, NG56 2016](#) discussed in [BJGP2017;67;658](#), a new report from the RCGP [Responding to the needs of patients with multimorbidity 2016](#) and [NICE 2017, KTT18 Multimorbidity & Polypharmacy](#). Although over 5 years old now, the principles of these guidelines are as relevant now (and probably more so) than ever.

Multimorbidity: Background points

[BJGP2017;67;658](#), [NICE 2017, KTT18](#), [RCGP2016](#), [NICE 2016](#) and [BMJ2016;354;i4843](#)

- Multimorbidity is defined as having two or more long-term conditions (LTCs).
- **Multimorbidity is the 'norm' in UK primary care:**
 - 1 in 6 of all our patients have ≥ 2 LTCs.
 - 65% of people over 65 and 80% of those over 85 have ≥ 2 LTCs.
 - The number of patients with multimorbidity is increasing with the ageing population.
 - Multimorbidity is not just a problem for the elderly & occurs earlier in deprived areas.
- **Multimorbidity is associated with:**
 - **Disease burden:** decreased quality of life with increased mortality, functional decline, polypharmacy, adverse drug events and high use of unplanned healthcare.
 - **Treatment burden:** of tests, appointments & medication.
 - Care that is disease-centred, fragmented and poorly coordinated.
 - Increased healthcare usage, including unplanned admissions.
 - **Psychological distress:**
 - Anxiety and depression are common.
 - 23% of people with one LTC report depression, compared to 40% with 5 or more.
- **Single disease guidelines are often not applicable** in patients with multimorbidity. Aggressive management of risk factors is often a major treatment burden, not evidence-based and inappropriate.
- The most common pair of conditions are osteoarthritis and a cardiometabolic condition, such as hypertension or diabetes. A common triad is:
 - A cardiometabolic condition
 - A painful condition e.g. osteoarthritis
 - A mental health condition e.g. anxiety, depression
- Patients with multimorbidity have great need but are most likely to receive poor and fragmented care.

► The NICE Multimorbidity Guideline [NICE: NG56 2016](#)

'It's our responsibility to deliver person-centred care, not disease-focused treatment', Prof Mark Baker, NICE

This could be argued to be the most important guideline that NICE have produced for GPs, being the first to take an explicit 'patient first' rather than 'disease first' approach. It was summarised in [BJGP2017;67;658](#) & [BMJ2016;354;i4843](#). It aims to *optimise care for adults with multimorbidity by tailoring care to the individual patient* and has the potential to significantly change the way long-term conditions (LTCs) are managed in the NHS. It represents a major shift in national guidelines, explicitly recognising the deficiencies in the evidence base emphasising *a patient-centred approach by firmly placing clinical judgement and patient preference at the centre of the decision-making process*.

The aim of the guideline is '*to improve quality of life by reducing treatment burden, adverse events and unplanned or uncoordinated care*'. It is more general in scope, and shorter on specific recommendations than many guidelines, but is simple and pragmatic and represents a fundamental change in approach from care that is disease-focused to care that is person-centred which many GPs will welcome.

Definition:

The guideline takes a very broad definition. *Multimorbidity is defined as the presence of two or more long-term health conditions*, which can include:

- Defined long-term physical and mental health conditions e.g. diabetes and schizophrenia.
- Learning disability + another long-term physical or mental health condition.
- Symptom complexes e.g. frailty and chronic pain.
- Sensory impairment e.g. sight or hearing loss.
- Alcohol or substance misuse.

General principles:

We need to be aware that the management of risk factors for future disease can be a major treatment burden for people with multimorbidity and that the evidence base in patients with multimorbidity is often lacking. The potential risks and benefits of individual treatments in people with multimorbidity and their preferences for treatment needs to be carefully considered.

An approach that takes multimorbidity into account should be considered if:

- The person requests it.
- They find it difficult to manage day-to-day activities.
- They have both long-term physical and mental health conditions.
- They have frailty or falls.
- They frequently have unplanned or emergency care.
- They are prescribed multiple medications.

How do we identify patients who may benefit from an approach to care that takes account of multimorbidity?

We can identify patients who may benefit from this approach either:

- Opportunistically.
- Proactively using an electronic tool e.g. the electronic Frailty index [eFI](#)
- Searching for people with polypharmacy:
 - Adopt a 'multimorbidity approach' for adults of any age on 15 or more prescribed medications (these include topical treatments and inhalers).
 - Consider adopting this approach if prescribed 10 to 14 regular medications, or fewer than 10 but increased risk of adverse events.
- Consider assessing people with multimorbidity for frailty.

And how should we assess for Frailty?

(NB it is now internationally accepted that Frailty is a medical syndrome, highly predictive of adverse outcomes. See the Frailty chapter for a more detailed discussion.)

- Do not assess when the person is acutely unwell.
- Assessment of gait speed:
 - Informal e.g. time taken to walk from the waiting room.
 - Formal e.g. time taken to walk 4m (>5s suggests Frailty) or the [Timed Up and Go Test](#) (>12 s indicates frailty).
 - [PRISMA-7](#) Questionnaire.
 - Self-reported ('how would you rate your health status on a scale of 0 to 10?', scores of 6 or less suggest frailty).

Principles of the 'Multimorbidity Approach':

This is an approach to care that takes multimorbidity into account, involves a personalised assessment focussed on the patients' needs and priorities and develops an individualised management plan with the aim of improving quality of life by reducing treatment burden, adverse events and unplanned care.

When adopting the Multimorbidity Approach we need to **adopt a patient-centred approach to focus on:**

- How their conditions and treatments interact and impact on their quality of life.
- Their individual needs, preferences, priorities and goals (NB 'what is important to you?').
- Improving quality of life by reducing treatment burden and adverse events.
- Improving coordination of care across services.

Practical steps include:

- Discussing the purpose of a trimmed-down, multimorbidity approach.
- Establishing disease and treatment burden, including impact on mental health.
- Establishing individual patients' goals, values and priorities and tailor care to this.
- Review medications taking into account evidence of likely benefits and harms.
- Agree on an individualised management plan, making it clear the goals and plans for care, who is responsible for coordination of care and timing of follow-up and access to urgent care.
- Stopping medicines at highest risk of adverse events e.g. falls, GI bleeding, acute kidney injury
- Be alert to the possibility of depression, anxiety and chronic pain.

Reviewing medications:

- Assess the relative benefits and side effects of different treatments and making individualised decisions based on patients' wishes.
- For treatments intended to relieve symptoms (e.g. pain) ask if the treatments are actually helping. If they are unsure, consider reducing or stopping.
- A key part of the guideline is that it includes a [database of treatment effects](#) to find information on the relative evidence base of different treatments, to help prioritise which ones to stop.
- Consider using the [STOPP/START](#) tool in older people. Discuss with people with frailty or limited life expectancy whether they wish to continue treatments aiming to improve prognosis, which may offer them limited benefit.
- Tell a person who has been taking bisphosphonates for at least 3 years that there is no consistent evidence of further benefit from continuing beyond 3 years or harms from stopping after 3 years.

[NICE, Database of Treatment Effects 2016](#)

- This is a spreadsheet providing estimates of benefit of drugs recommended in single disease guidelines, compared to placebo for specific outcomes and is designed to help inform discussion on the relative benefits of different drugs.
- It looks complex at first sight, reading the instruction sheet first is recommended!
- This evidence typically comes from 'single disease' studies and are not necessarily appropriate for people with multimorbidity, but help to give an estimate of relative benefits by comparing orders of magnitude of efficacy.
- It will display benefits in terms of reductions in absolute risk or NNT.
- **Examples:**
 - Anticoagulant in AF: NNT = 40 over 1 year to prevent a stroke.
 - ACEi in CKD: NNT = 51 to prevent progression of CKD.
 - Depression & antidepressants: NNT = 4 over 1 year to prevent relapse of depression.
 - Statins, primary prevention: NNT = 170 over 1 year to prevent a non-fatal MI, 595 over 1 year to prevent one death.
 - Hypertension drug therapy: NNT = 570 over one year to prevent one stroke, 840 to prevent one MI and 1050 to prevent one death.

The management plan:

- After discussion of disease and treatment burden, and the patients' individual goals and priorities, an individualised management plan should be established.
- ***Responsibility should be assigned to a named professional to coordinate care.***
- The plan should cover aspects of care, such as medications and who is involved in their care.

Resources for patients:

[Healthtalk Online: Living with multiple health problems](#)

[NICE patient information on multimorbidity](#)

The RCGP 2016 Report [Responding to the needs of patients with multimorbidity 2016](#) discusses the barriers we face in adopting an effective approach to multimorbidity. The major one of course is...TIME!! Longer consultations are needed as well as integrated, multidisciplinary care. For example, there is clear potential for the role of a practice-based pharmacist in providing medication reviews and their needs to be greater integration with social care.

KISS: Multimorbidity

Based on [NICE: Multimorbidity, NG56 2016](#), [BJGP2017;67;658](#) & [BMJ2015;350:h176](#)

'Person-centred care, not disease-focused treatment'

- **The aim of the NICE guideline** is 'to improve quality of life by reducing treatment burden, adverse events and unplanned or uncoordinated care'.
- **Multimorbidity is defined as the presence of 2 or more long-term conditions (LTCs)**
 - These can be any physical or mental health LTC, including:
 - Frailty, Chronic Pain, Alcohol or substance misuse, visual or hearing loss etc
 - Multimorbidity is a problem for people of all ages, not just the elderly
- **Actively identify patients with multimorbidity**
 - Opportunistically
 - Proactively using an electronic tool e.g. [QAdmissions](#), [PEONY](#) or [eFI](#))
 - Patients with polypharmacy (≥ 10 meds)
 - Patients with Frailty
- **Adopt a 'Multimorbidity Approach'**
 - An approach to care that takes multimorbidity into account and involves a personalised assessment of the patient's needs and wishes, with the aim of reducing treatment burden, improving function and maximising quality of life
 - Discuss the approach with the patient and take a shared care approach
 - **Continuity of care:** assign a named doctor & care-co-ordinator
 - **Target**
 - **What matters most to the patient:** 'what is bothering you most?', 'What would you like to focus on today?'
 - **Function and not disease**
 - Would patient benefit from physio, OT, podiatry, audiology etc?
 - **Shared risk factors** common across different LTCs e.g. smoking, obesity etc.
 - **Optimise medicines:** Medicine's management and Polypharmacy
 - Structured drug reviews incorporating de-prescribing.
 - Assess relative benefits and side effects of different treatments & make individualised decisions on de-prescribing
 - Use the [database of treatment effects](#) to find information on the relative evidence base of different treatments, to help prioritise which ones to stop.
 - Consider using the [STOPP/START](#) tool in older people
 - Discuss with people with frailty or limited life expectancy whether they wish to continue Rx aiming to improve prognosis, which may offer them limited benefit
 - Stop medicines at highest risk of adverse events e.g. falls, GI bleeding etc
 - **A management plan** including medications, patients' wishes and priorities, professionals involved in care & named care co-ordinator
- **Extended consultations**
 - Schedule longer 'extended care consultations' (NB hopefully providing fewer but longer and better consultations for these patients)
 - Consider running specific multimorbidity clinics for complex patients

KISS: Improving Care of People with Multimorbidity

Practical tips from the [NICE multimorbidity guideline](#):

- Use clinical judgement and an electronic tool, such as eFI or QAdmissions to help **identify people** with multimorbidity
- Look for, and diagnose, **Frailty** on the basis of gait speed and self-reported health status
- Searching for **polypharmacy**, and consider a multimorbidity approach in all people on 10 or more regular medications (including topicals and inhalers)
- Focus on **patients' priorities** (e.g. 'what is important to you?' 'What do you want to achieve with your treatment?')
- Establish an individualised **management plan**
- Have fewer, longer appointments with planned, regular review
- When considering **stopping medication** use the [database of treatment effects](#) to find information on the relative evidence base of different treatments and consider. Consider using the [STOPP/START](#) tool in older people
- Consider **stopping bisphosphonates** after 3 years
- Stop medicines at highest risk of adverse events e.g. falls, GI bleeding, acute kidney injury
- Assess all patients with physical multimorbidity for **mental health problems**
- Target function and not disease, and consider functional status of people with multimorbidity e.g. would they benefit from physiotherapy, OT, podiatry etc?

NB Practice Points: Multimorbidity

- This new guideline introduces a fundamental change in approach, as we move from disease-orientated care to person-centred care in people with more than one long-term condition. It has the potential to reduce overtreatment, improve safety and patient outcomes.
- Make a point of actively identifying patients with multimorbidity.
- Discuss with them adopting a multimorbidity approach in line with their needs, values and wishes, focused on reducing their treatment burden and improving their functional status rather than on disease-specific outcomes.
- Identify what matters most to the patient and establish a management plan.
- Optimise medication, using the [database of treatment effects](#) & the [STOPP/START](#) tool.
- Maximise continuity, consider introducing multimorbidity-specific clinics and fewer but longer extended consultations.

NB Quality Improvement Ideas

- Actively search for patients with multimorbidity using QAdmissions or eFI.
- Make an active diagnosis of Frailty using simple clinical criteria.
- Perform structured medication reviews and aim to reduce treatment burden in patients on 10 or more meds.

FALLS PREVENTION

'The single most serious threat older people face is falling' Atul Gawande

Ask an elderly person what is the biggest risk to their health and they are likely to say cancer, dementia or vascular disease. They are wrong, their biggest risk is from falling over. Ask them what the most important thing to them is and they are likely to say maintaining their independence. Falls are the main reason why older people lose independence. ***It is becoming increasingly clear that one of, if not the single most useful thing we can do to help preserve the quality of life of our frail, older people is to help to prevent them from falling over.*** The physical, psychological and social consequences of falling are immense and ***falls are preventable.*** Interventions are effective and can reduce the subsequent risk of falling significantly.

We had a 'state of the art review' on falls prevention in older people [BMJ2016;353:i1419](#), and to guide us further we have had guidelines from the UK [NICE 2013 CG161](#) and the US [CDC 2013 Fall Guidance STEADI](#). Although a few years old now, these guidelines are even more relevant now than when originally published - an ageing population, which in some cases will have de-conditioned due to lockdown measures and concerns over COVID means falls prevention will remain an ongoing 'Hot Topic'.

Falls: Background

[BMJ2016;353:i1419](#)

- 30% of community-dwelling people aged over 65 fall each year, and 50% aged over 80 fall.
- After a first fall, 66% fall again within a year.
- 68% of falls result in injury:
 - 24% need medical care, 35% result in functional decline.
- The human cost of falling is huge: distress, pain, injury, hospitalisation, loss of independence and mortality:
 - 95% of hip fractures are caused by falls.
 - After a hip fracture 40% end up in institutional care, 20% never walk again and 30% die within a year.

1 What are the main risk factors for falling?

NICE [NICE 2013 CG161](#) did an extensive evidence review and have highlighted the risk factors in the box below. ***Note that the single biggest risk factor is a history of a previous fall***, followed by gait and mobility problems. In their guidance, they, therefore, emphasize that:

- Older people should be routinely asked by health professionals whether they have fallen in the last year and asked about the frequency, context and characteristics of the fall.
- All patients who have had a fall in the last year or have abnormal gait or balance should be offered a multifactorial falls risk assessment.
- Older people reporting a fall or considered at risk, check for balance and gait deficits using a test of balance and gait:
 - The [Timed Up And Go Test](#) TUGT is recommended as a simple, baseline pragmatic test (a new technology has been developed using sensors and software, to give a more accurate assessment of risk, called the Quantitative TUG Test [QTUG](#) which is being evaluated).
- Whilst single risk factors are important, it is the combination of multiple risk factors which greatly increases risk. 70% of falls have multiple, interacting causes.

Risk Factors for Falling

Based on [BMJ2016;353:i1419](#), [NICE 2013 CG161](#) and [US STEADI Guideline](#)

Whilst single risk factors are important, it is the combination of multiple individual & environmental risk factors which most increases risk. 70% of falls have multiple, interacting causes.

The following factors are the **MOST predictive** of falling in the community:

- Previous falls history.
- Fear of falling.
- Gait and balance problems.
- Impaired mobility, especially muscle weakness and foot problems.
- Postural hypotension.
- Visual impairment.
- Cognitive impairment.
- Urinary incontinence.
- Home hazards.

Important **additional risk factors** include:

- Medication
 - Especially polypharmacy, psychotropic drugs, anticholinergics and antihypertensives/diuretics causing postural hypotension.
- Low BMI (consider malnutrition as a possible contributing cause).
- Generalised pain.
- High alcohol intake.
- Chronic diseases e.g. arthritis, diabetes, Parkinson's, stroke etc.

The [US STEADI Guideline](#) has produced a useful [Fall Risk Checklist](#)

2 *What interventions help to prevent falls?*

Balance and exercise training is effective and is a key component of intervention recommended by [NICE CG161 2013](#) and NICE quality standards [QS876 2017](#).

- An RCT from 2015 showed that amongst women with reduced mobility and at falling risk aged 75 to 85, and living at home, a two-year **balance and exercise training programme significantly reduces the number of injurious falls** [BMJ2015;351:h3830](#)
- This is supported by a more recent Cochrane meta-analysis [CD012424 2019](#) which confirms that **exercise programmes significantly reduce falls**:
 - This covered >23,000 community-dwelling older people over 108 RCTs.
 - There was **high certainty evidence that exercise programmes (most with balance, functional exercise +/- resistance exercises) significantly reduce falls by 23%** (RR 0.77, CI = 0.71-0.83).
 - Based on an illustrative risk of 850 falls in 1000 people over 1 yr this equates to 195 fewer falls.
 - Tai Chi probably also reduces falls risk.
 - Other outcomes were less certain but with **lower certainty exercise programmes also seem to reduce fall-related fracture risks** (RR 0.73, CI = 0.56-0.95), and possibly also falls requiring medical attention.
 - **Conclusion: Exercise programmes that involve predominantly balance and functional exercise significantly reduce falls risk and probably reduce fall-related fractures in older adults.**

KISS: Fall Risk Assessment

Based on [BMJ2016;353:i1419](#), [BGS 2016 Falls Risk Assessment](#) & the [STEADI](#) toolkit

- **Be proactive:** ask all patients over 65 if they have fallen in the last year, and adopt a proactive approach to secondary prevention 'falls are the biggest risk older people face'.
- **Patient has fallen:**
 - Refer all patients to a specialist falls service for multifactorial assessment and intervention, including an exercise & balanced training programme.
 - Identify risk factors, using the [Fall Risk Checklist](#).
 - Physical examination including:
 - Gait, strength and balance e.g. [Timed up and Go Test](#).
 - CVS check including for postural hypotension (20mmHg drop in SBP on standing is significant).
 - Neurological exam e.g. for Parkinson's, ataxia etc.
 - Cognitive screening.
 - Eyes and feet.
 - Malnutrition screen if recent weight loss, or muscle wasting.
 - Lab tests: As you feel clinically appropriate.
 - Medication review and medicine's optimisation (see multi-morbidity section):
 - De-prescribe all unnecessary medications, but the most important drugs are
 - Any psychotropic drug, especially benzodiazepines.
 - Any drug with anticholinergic side effects.
 - Any drug causing postural hypotension.
 - Use the RCP 'traffic light' guide [Medicines & Falls Checklist](#).
 - Encourage a proactive approach to fall prevention and also home exercises.
 - Consider vitamin D supplementation 800iu daily in patients at risk.
 - Follow up within 30 days and review.

Resources for patients:

- Useful advice for patients on fall prevention from:
 - [NHS Choices](#) including [home-based exercises](#) for frail older people.
 - Great patient-friendly resources and tips from [Age UK](#).
 - Great resources from [STEADI](#) on how to stay independent.
- ***Five things that your patients and elderly relatives can do to prevent falls:***
 - Begin an exercise programme to improve your leg strength and balance.
 - Go to Tai Chi classes.
 - Ask your GP to review your medications.
 - Get an annual eye check-up.
 - Make your home safer: remove clutter and tripping hazards, install railings and grab bars, have good lighting, especially on stairs.

NB Learning Points and Quality Improvement Ideas: Falls

- Falls are the biggest risk older people face and can be prevented.
- Take a pro-active approach to asking all older people if they have fallen in the last year, and an active approach to secondary prevention.
- We have high quality evidence that exercise programmes that incorporate functional exercise and balance components significantly reduce falls risk, and probably fall related fractures as well.

END OF LIFE CARE

Providing high-quality end of life care to our dying patients and support to their families is one of the most important aspects of our work in primary care and is simultaneously challenging and deeply rewarding work.

End of life care is always a Hot Topic wherever you work, and trying to improve the experience of dying for our terminally ill patients and their families will always be a constant goal for us. Although now no longer a component of the QoF Quality Improvement Domain for those in England, the principles from that project remain relevant and important. We should continue to focus on the wider aspects of care for patients who are expected to die within months, in addition, to support for their carers, such as those outlined below in the recently published [Daffodil Quality Standards](#) and the [NHS England End of Life Care Programme](#).

Improving end of life care is part of the [NHS Long Term Plan](#). Central to the plan are the [Ambitions for Palliative & End of Life Care](#) national framework report for 2015-2020, produced by the [National Palliative and End of Life Partnership](#), which includes NHSE, RCGP, RCN, RCP, the CQC, the GMC, PHE and a number of charitable partnerships including MacMillan and Marie-Curie. This chapter deals with these broader aspects of providing personalised care.

► *Palliative Care Guidelines* [Scottish Palliative Care Guidelines 2019](#)

For help and guidance with *symptom control*, the excellent [Scottish Palliative Care Guidelines 2019](#) were updated in 2019. Easy to navigate, these guidelines are a superb resource for evidence-based advice on symptom control, including difficult to manage symptoms such as breathlessness, hiccough and delirium. They also provide useful guidance on 'just in case' [anticipatory prescribing](#) for patients in the last days or weeks of life for the four medications that may be required for symptom control (opioids for pain relief e.g. morphine, anxiolytic e.g. midazolam, anti-secretory e.g. hyoscine and anti-emetic e.g. levomepromazine). They also provide guidance on hydration, and whether or not patients should be considered for hydration with [subcutaneous fluids](#).

► *Advance care planning* [BJGP2020;70:e28](#)

Identifying people who would benefit from a palliative care approach at an early stage is crucial to optimise outcomes and is recommended in all palliative care and end of life guidelines, see below. A common theme of end of life care has been that it has not been planned with the patient, taking their preferences into account, and that crucial information has not been shared between different healthcare teams. Advanced care planning and sharing of this information is seen as essential to provide personalised and well co-ordinated care.

To facilitate the process of developing and sharing advanced care plans, *electronic palliative care coordination systems* are now being developed and widely introduced across the UK.

Recent research from Scotland has looked at the efficacy of an electronic care coordination system known as the *Key Information Summary (KIS)* is a national electronic care coordination system which is being rolled across Scotland. The KIS is a care plan generated in primary care and enables the health professional to record essential information such as resuscitation status, preferred place of care and death, treatment escalation plans, carer details and anticipatory medicines. The KIS is then shared across different emergency and out-of-hours settings.

- The study examined data from 18 General Practices across Scotland
- Of those with advanced progressive illness, 69% had a KIS
- People with cancer were much more likely to have a KIS (80%) than those with organ failure (47%)
- Most patients who died with frailty or dementia had a KIS started > 1 year before death
- There was significant variation in how completely the KIS was completed, but 95% of the KISs were considered either useful or very useful
- The presence of a KIS was positively associated with dying in a community setting (Odds ratio 3.7)
- **The high uptake of the KIS is described as a significant achievement, however, to improve further more information is needed on informal carers, regular updating and more KISs for patients with organ failure**
- **Conclusion:** *the Scottish care co-ordination electronic KIS is having significant uptake and having a positive impact on care*

Similar regional systems throughout the UK are being developed (e.g. [Coordinate My Care](#) personalised urgent care plan in London; patients are helped to enter their preferences into a secure online system which can then be accessed by all the healthcare professionals who may be involved in that patient's care e.g. GP, Out of Hours, 111, paramedics, A and E etc)

So advance care planning is seen as an essential component for improving end of life care. ***But, what constitutes good advance care planning?*** We have guidance from [Gold Standards Framework Advance Care Planning Guidance](#)

KISS: Advanced Care Plans

[Gold Standards Framework Advance Care Planning Guidance 2019](#)

- Advance care planning is the key to means of improving care near the end of life and enabling the better provision of care in line with the persons' needs and wishes.

5 simple steps to advance care planning:

- **Think**
 - Think about the future, what is important to you and what you want to happen or not to happen when you become unwell.
- **Talk**
 - With family, friends or ask someone to be your advocate or proxy spokesperson.
- **Record**
 - Write down your wishes and thoughts, or ask your proxy to do on your behalf.
- **Discuss**
 - Discuss your plans with your doctor, nurse and carers.
- **Share**
 - Share this information with others who need to know and review it regularly.

► Talking about death and dying [RED-MAP framework 2020](#)

So, identifying people coming towards the end of their life and making advanced care plans is one thing, but how do we actually discuss death and dying with people when they are at that stage? It is a subject that is becoming less taboo but can still feel difficult to discuss. The excellent RED-MAP framework, developed in Scotland and recently updated during the COVID-19 outbreak, helps guide us:

KISS: Talking about death and dying

[RED-MAP 2020](#)

R = READY:

- Build a relationship, outline reason for discussion, check who should be involved.
- *We need to talk about your treatment and care. Who else do we need to speak to?*
- *We are doing our best to care for you, but we are worried about your condition.*

E= EXPECT:

- Find out what the person and family know and expect.
- *Can I ask what you know about your health problems and how you are now?*

D = DIAGNOSIS:

- Explain what you know about their underlying condition in short chunks with pauses to check for responses; acknowledge and share uncertainty; keep terms simple and clear; kindness matters.
- *You are less well because...*
- *We hope you will improve with...but I am worried about how you are...*
- *I am sorry to tell you (persons name) is very ill...(s)he could die with this illness...*

M = MATTERS:

- Pause to let them/family take in information; Find out what matters to them.
- *Can we talk about what's important for you now and what we can do to help?*

A = ACTIONS:

- Talk about realistic options for treatment, care and support; be honest and clear about what can help and what won't work.
- *Intensive care and ventilation do not help everyone. Can we talk about that?*
- *Has anyone talked about cardiopulmonary resuscitation or CPR? CPR is a treatment that can restart the heart, but it does not work when a person is in very poor health or dying.*
- *Whatever happens, we will continue to care for you and your family.*
- *We give treatments for any symptoms like breathlessness, pain or distress.*

P = PLAN:

- Use available forms/online systems to record and share care plans and DNACPR decisions.
- *We record and share plans so everyone knows about them.*

Phrases to avoid, which can make people feel abandoned or deprived of treatment and care:

- There is 'nothing more' we can do.
- 'Ceiling' of treatment or care or 'withdrawing' treatment.
- Treatment is 'futile'.

► *Daffodil 'Core Quality Standards' for End of Life Care* [BJGP2018;68:114](#)

In the UK, new GP Core Standards for Advanced Serious Illness and End of Life Care have been produced called [Daffodil standards](#), introduced in an excellent recent BJGP editorial [BJGP2018;68:114](#). *The BJGP editorial points out how in the near future care of the dying will be even more demanding than we have envisaged.* It is estimated that the numbers of people needing palliative care are estimated to rise by 25% by 2040. The main driver will be dementia as well as cancer.

The RCGP have joined with Marie Curie to focus on end of life care as a clinical priority and has produced new core standards called [Daffodil Standards](#) which were *launched in 2019*. They outline eight core domains in which practices can aim to ensure compassionate, safe and effective care and are evidence-based.

Key to their success will be the development of health and social partnerships. As the authors sagely illustrate, conventional palliative care interventions are often pathology and morbidity driven whereas in 'real-life' psychosocial determinants (such as loneliness, social isolation, depression, anxiety and inability to cope) often create the greatest problems. Personalised care and support planning will be key, but the standards also aim to de-professionalise many aspects of palliative care as we increasingly develop and work in partnership with social services, the voluntary sector and also the communities in which we work.

The Daffodil standards are recognised by the CQC as a robust framework that provides evidence of the quality of care.

KISS: Daffodil Quality Standards for End of Life Care

[Daffodil Standards 2019](#) and [BJGP2018;68:114](#)

- UK General Practice Core Standards for Advanced Serious Illness and End of Life Care.
- Each standard consists of a set of quality statements, evidence-based tools & quality improvement steps.
- Key to success will be in developing health and social partnerships to effectively assist in the psychosocial aspects of care.
- By adopting the standards, GPs commit to making improvements in ≥ 3 of 8 core areas each year, with the aim of 8/8 after 3 years.
- The 8 core areas are:
 - Professional and competent staff.
 - Early identification.
 - Carer support, before and after death.
 - Seamless, planned & coordinated care.
 - Assessment of unique needs of the patient.
 - Quality care during the last days of life.
 - Care after death.
 - General Practices being hubs with Compassionate Communities.

Back in 2019/20 The England GP QoF contract changed by retiring some indicators and introducing new quality improvement domains, one of which was End of Life Care. This has now been changed to newer Quality Improvement areas but the principles of the programme remain relevant, and serve as a useful benchmark for auditing and improving end of life care in primary care.



KISS: End of Life Quality Improvement Domain, GP Contract England 2019

[GP Contract England 2019/20](#) & [Quality & Outcomes Framework 2019](#)

Although now retired from the England GP contract, the 2019/20 Quality Improvement Domain on End of Life Care recommendations remain a useful framework for benchmarking our end of life care provision, and can serve as guidance for ongoing quality improvement:

- GP practices can demonstrate **continuous quality improvement activity** focused on end of life care
 - Based on [Daffodil standards](#) or suggestions below
 - For each activity, practices should identify and agree with their own objectives which are SMART (specific, measurable, achievable, relevant and time-bound)
- GP practices have **participated in network activity to share and discuss learning activity**; this would usually include participating in a minimum of two network peer review meetings
- **Key steps in quality improvement activity involves:**
 - Identifying patients in need of end of life care
 - Early identification and support of people with advanced progressive disease who may die in the next 12 months
 - Identification and support for family/caregivers
 - Assessing their needs and preferences
 - Pro-actively planning their care
 - Well planned and coordinated care that is responsive to their changing needs
- **Practices will need to**
 - Evaluate their current quality of end of life care and identify areas for improvement
 - This will usually involve a **retrospective audit of deaths**
 - Identify **quality improvement activities** and set improvement goals for performance, for example:
 - An audit standard that the practice was successfully anticipating 60% of deaths and identifying them on a 'supportive care register'
 - Increasing the proportion of people who died from serious advanced illness who were offered personalised care and support plan discussions
 - A reliable system in place to monitor and enable improvement
 - An increase in the proportion of people who died where a family member, next of kin or carer has been identified and offered support
 - Implement an improvement plan
 - Participate in a minimum of two GP network peer review meetings
 - Complete a specific QI monitoring template in relation to this module

NB Learning Points and Quality Improvement Ideas

- Recent guidelines and quality standards such as the [Daffodil Standards](#), strongly support an improved, personalised, holistic approach to end of life care involving health & social care.
- Early identification of dying patients, sensitive communication, a supportive care register, **personalised advance care plans** and a personalised holistic approach to care are crucial.
- Electronic advance care plans and 'palliative care coordination systems' are being rolled out across the UK, such as the KIS in Scotland and [Coordinate My Care](#) in London.
- Further excellent resources for helping to manage end of life care can be found in the Scottish [REDMAP framework](#), which includes **helpful guidance on the difficult subject of talking about death and dying**.
- For symptom control and guidance on anticipatory prescribing, see the excellent [Scottish Palliative Care Guidelines 2019](#)

ANAPHYLAXIS

We have recently had newly updated guidelines from the [Resus Council May 2021](#) and [NICE CG 134 2020](#)

► *Background*

- Anaphylaxis is a severe, life-threatening, generalised or systemic hypersensitivity reaction that is characterised by rapidly developing airway and/or breathing and/or circulation problems usually associated with skin and mucosal changes.
- Triggers - broad range, and many cases idiopathic; most common community triggers are foods (esp. nuts), drugs (esp. penicillins, cephalosporins, NSAIDs, aspirin) and venom including stings.

► *Controversies in diagnosis and management*

BMJ journals paper - [Myths, facts and controversies in the diagnosis and management of anaphylaxis](#) presents a number of key evidence-based learning points:

- Absence of skin changes or an identified trigger does NOT exclude the diagnosis of anaphylaxis.
- **IM Adrenaline and IV fluids are first line treatment for anaphylaxis**
 - It is easy to think we should deliver adrenaline, antihistamine and glucocorticoid simultaneously to treat anaphylaxis successfully. This is NOT the case.
 - Deliver IM adrenaline every 5 minutes with IV fluids until the patient has responded clinically.
 - Only once the patient has improved consider additional treatment with an antihistamine.
- Antihistamines alone cannot treat anaphylaxis. They help skin changes only.
- Growing evidence that glucocorticoids do NOT reduce the incidence of biphasic reaction and potentially have no place in the treatment of anaphylaxis.

New [American anaphylaxis guidelines](#) from January 2020 add further weight to these concepts. They state:

- The only effective treatment for anaphylaxis is prompt IM Adrenaline.
- No evidence supports the use of antihistamines or glucocorticoids in the treatment of anaphylaxis.
- Antihistamines may help skin changes such as flushing or urticaria only.

This evidence has been incorporated into the new [Resus UK guidelines May 2021](#) with some key changes:

- **Greater emphasis on IM adrenaline** and to repeat after 5 minutes if ABC problems persist.
- Increased emphasis on **avoiding sudden changes of posture** - maintain supine position (or semi-recumbent if difficulty with breathing).
- **IV fluid bolus** after 2nd adrenaline/5 minutes if ABC problems persist.
- **Corticosteroids are now NO LONGER advised.**
- **Antihistamines only for persisting skin symptoms** and ONLY once resus complete (i.e. do not use to treat ABC/underlying anaphylaxis).

► *Assessment*

- Anaphylaxis is a clinical diagnosis and is likely when all 3 of the following criteria are met (but not all criteria needed to make diagnosis):
 - Sudden onset and rapid progression of symptoms (typically over minutes).
 - Life-threatening problems with:
 - Airway (e.g. swelling, hoarse voice, stridor) and/or
 - Breathing (e.g. SOB, wheeze, cyanosis) and/or
 - Circulation (e.g. pale, clammy, tachycardic, hypotensive)
 - Skin and/or mucosal changes (flushing, urticaria, angioedema) can be subtle or absent in up to 20% of reactions. There can also be GI symptoms (e.g. vomiting, abdominal pain, incontinence).
- Patients will look and feel unwell, are usually anxious and may experience a “sense of impending doom”.

► Management

- Call ambulance and call for help.
- Lie patient flat and with legs raised if hypotensive. Those with airway or breathing difficulties will be more comfortable sitting upright/semi-recumbent. Avoid changes in posture. Remove trigger if present.
- Give IM adrenaline (doses below) ASAP (anterolateral aspect middle third of thigh):
 - Repeat dose after 5 minutes if inadequate response.
 - Do NOT give IV adrenaline in primary care.
 - Continue to give IM adrenaline every 5 minutes until adequate response.

DRUG	ADULT AND CHILD >12	6-12 YRS	6 MONTHS - 6 YRS	<6 MONTHS
Adrenaline 1:1000 (1mg/ml) IM - Repeat every 5 minutes until adequate response	500mcg (0.5ml)	300mcg (0.3ml)	150mcg (0.15ml)	100-150mcg (0.1-0.15ml)

- Establish airway and give high flow oxygen.
- Monitor pulse, BP and oxygen saturations.
- IV fluid challenge (if possible) if inadequate response after 5 minutes:
 - 500-1000ml in adult; 10ml/kg in child.
- Do NOT give corticosteroids.
- Bronchodilators can be considered in addition if wheeze present but should NOT be used as an alternative to adrenaline in the presence of ongoing respiratory problems.
- Antihistamines do NOT form part of acute anaphylaxis treatment and should **only be given one** resuscitation is complete if persisting skin symptoms (e.g. urticaria, angioedema):
 - Oral non-sedating antihistamines are now preferred cetirizine:

< 2 years	2-6 years	6-11 years	12+/adults
250mcg/kg	2.5-5mg	5-10mg	10-20mg

- Admit all patients receiving adrenaline due to the risk of a biphasic reaction (a life-threatening recurrence of symptoms after initial presentation without re-exposure to trigger).

FOOD ALLERGY IN CHILDREN

There has been a startling increase in the prevalence of food allergy. Peanuts are the perfect example: in just 10 years the rate of peanut allergy in children from Western countries has doubled to 3%. Unsurprisingly there is intense academic interest in why allergies develop and how to prevent them.

1 Hydrolysed formula to prevent allergy [BMJ 2016;352:i974](#)

There is a wide consensus that for babies breastfeeding is best. However, amazingly, European/UK, US and Australian allergy guidelines recommend if *infants at high risk of allergy require supplemental milk in the first 4 months of life "a documented hypoallergenic formula is recommended"*, not just cow's milk formula.

This is based on the conclusion of a 2006 Cochrane review ([CD003664](#)) which found "limited evidence that prolonged feeding with a hydrolysed formula compared to a cow's milk formula reduces infant and childhood allergy and infant cow's milk allergy". There was no evidence of benefit compared to breastfeeding.

This very recent systematic review published in the BMJ ([2016;352:i974](#)) examined current data involving hydrolysed formula and allergy. It found:

- *No consistent evidence that partially or extensively hydrolysed formula reduces the risk of allergic or autoimmune outcomes including eczema compared to cow's milk formula*
- There was evidence of conflict of interest, high or unclear bias risk and publication bias

The linked editorial ([BMJ 2016;352:1143](#)) compliments *the strength of this new review* and its rigorous approach to assessing the quality of the available data. It agrees with calls from the authors of the paper for changes to current infant feeding guidelines in light of this evidence. This spurious finding should not be used to undermine the message about breastfeeding, and where that remains not an option, standard cow's milk formula can be used without the need to medicalise and complicate feeds.

2 Prevention of Food Allergy In Infants [Lancet 2016;389:276](#), [NEJM 2015;372:803](#)

There is a growing body of data suggesting that certain common food allergies may be prevented with early introduction of the possible allergen into the infant's diet.

The latest data comes from an RCT published in The Lancet ([2016;389:276](#)) which recruited infants aged 4-5 months old with eczema in Japan. The infants were randomised to receive heated egg powder from 6 months or placebo, alongside aggressive treatment of their eczema.

- *8% of the egg powder group developed egg allergy vs 38% in the placebo group*
- Adverse events after ingestion of the egg were the same in both groups (15% vs 18%)

This follows a fantastic piece of UK research on early introduction of peanuts. It comes from the simple observation that children of Jewish origin in the UK were 10 times as likely to have peanut allergy as their Israeli counterparts, who are given peanuts as part of their diet from 7 months rather than 1 year of age.

- 640 infants (age 4-11 months) at high risk of developing peanut allergy (with a history of severe eczema, egg allergy or both) were randomised to either *consume or avoid peanuts until 60 months* of age
 - Peanut consumption consisted of at least 6g peanut protein per week in at least 3 separate meals; ideally using Bamba, a popular Israeli peanut-based snack
 - They were further stratified into two cohorts by pre-existing sensitivity to peanut extract determined by an initial skin-prick test
 - Around 10% of the initial recruitment group were excluded from the trial due to marked reaction (defined as a wheal >4mm)

- **Primary outcome: proportion of children with a peanut allergy at 60 months of age**

- Determined by follow-up skin prick test and oral food challenge
- Serum-specific IgE, IgG and IgG4 antibodies were also measured

Results:

- **Negative skin prick cohort** (N = 530)
 - 13.7% in the avoidance group vs 1.9% in the consumption group had a peanut allergy
- **Positive skin prick cohort** (N = 98)
 - 35.3% in the avoidance group vs 10.6% in the consumption group had a peanut allergy
- There were no significant differences in the incidence of serious adverse events

Conclusions: The early introduction of peanuts and eggs to the infant diet leads to a marked decrease in the frequency of development of allergy in a higher-risk population. UK guidance has already changed and peanut products can now be recommended from 6 months of age.

3 Desensitisation in peanut allergy [Lancet online 2014](#)

What about children with an established peanut allergy? There has been the theory that desensitization can be induced using oral immunotherapy (OIT) - exposure to a causative allergen in very small doses at which level they do not cause a reaction and is then are gradually increased over time.

This UK-based study, run by allergy specialists, recruited children aged 7-16 years old with immediate hypersensitivity reaction to peanuts and came in two parts:

- **Stage 1** - children were randomised to powdered peanut or control group
 - 24/39 (62%) receiving OIT became desensitised, 84% could tolerate 800mg of powdered peanut (the equivalent of 5 peanuts)
 - 0/46 in the control group were desensitised.
- **Stage 2** - the control group then received OIT for 26 weeks.
- **Results:**
 - After the second phase, 91% could tolerate 800mg of peanut, 54% tolerated 1400mg
 - Side effects were mild
 - Most were gastrointestinal (around 1/3 had some nausea, 1/3 vomited at some point)
 - IM adrenaline was used on one occasion only (so after 0.01% of total doses given).

The authors concluded that “**oral immunotherapy successfully induced desensitisation in most children with a peanut allergy with a clinically meaningful increase in peanut threshold**”. This clearly has very positive implications for those who are very restricted by their allergies. The safety profile is good, but this is research in its early stages and the paper cautions the technique should not be used outside of the specialist setting at present.

KISS: Referral for Food Allergy in Children and Young People

BMJ 2014;348:g3695, NICE 2011 CG116

Suspected food allergy? Is it:

- **Immediate/IgE mediated (Type 1) hypersensitivity**
 - Presents early with classical symptoms of urticaria, wheeze and vomiting
- **Delayed/non-IgE hypersensitivity**
 - Presents later with persistent symptoms that may involve different organ systems.
- Distinguish from food intolerances (e.g. lactose intolerance)

IgE mediated	Non-IgE mediated
Skin	
Pruritis	Pruritis
Erythema	Erythema
Acute urticarial or angioedema	Atopic eczema
Gastrointestinal	
Nausea	GORD
Colicky abdo pain	Loose or frequent stools
Vomiting	Blood and/or mucous in stools
Diarrhoea	Abdo pain or infantile colic
Oral pruritis	Constipation
	Food refusal or aversion
	Perianal redness
	Pallor, tiredness, faltering growth
Respiratory	
Upper: itching, sneezing, rhinorrhoea, congestion	
Lower RT: cough, tightness, wheeze, SOB	

Investigations

- IgE mediated allergy – offer skin prick tests and/or blood tests for specific IgE antibodies to the suspected foods and co-allergens
- Non-IgE mediated allergy – try eliminating the suspected allergens for 2-6 weeks, then reintroduce
 - Seek advice from a dietician with about nutrition

Referral is recommended in the following situations:

- Any acute systemic reaction (e.g. anaphylaxis) or severe delayed reactions
- Confirmed IgE-mediated allergy and concurrent asthma
- Where the diagnosis is uncertain
- Faltering growth with ≥ 1 GI symptom
- Possible multiple food allergies (particularly egg with a second major food allergy)
- No response to an elimination diet
- Tests are –ve but there is strong clinical suspicion of IgE-mediated allergy
- Significant atopic eczema where multiple or cross-reactive food allergies are suspected
- Persisting parental suspicion of food allergy despite lack of supporting history
- Persistent egg allergy beyond the age of 6-8 years
 - The BMJ article suggests secondary care referral for IgE-mediated allergy may not be required if symptoms are mild (such as limited to the skin), there is no coexisting asthma and the diagnosis is clear (with supporting investigations)

KISS: Cows' Milk Allergy

[BMJ 2013;347:f5424](#)

Diagnosis

- IgE mediated? – reactions usually within 20mins of ingestion of cows' milk, OR
- Non-IgE mediated? – more chronic picture, temporal association less likely.
- Investigations:
 - Skin prick or specific IgE blood testing in suspected IgE mediated CMA
 - Trial elimination diet for 4 weeks with reintroduction for non-IgE mediated CMA

Management

- Educate patients on avoidance of cows' milk and treatment of reactions if exposed
 - Antihistamines first line in milder reactions
 - Adrenaline autoinjector if:
 - CVS or respiratory reactions to any food or co-existing asthma
 - Consider in those with reactions to small amounts of food, e.g. airborne or skin exposure, those in remote areas or reactions in teenagers.
- Replace cows' milk formula
 - In infants
 - 1st line is extensively hydrolysed formula
 - 2nd line is an amino acid-based formula (or 1st line if severe reactions, faltering growth, sx while breastfed)
 - In >6 months
 - Also, consider soya milk (IgE mediated CMA only), oat milk, rice milk >5yo
 - Consider if additional calcium supplementation is required.
 - In exclusively breastfed infants the mother should exclude cows' milk and take Ca/Vit D
- Continue elimination diet for a minimum of 5 months and until 1 year of age.
 - Consider reintroduction of food after this non-IgE mediated CMA.
 - If child remains intolerant, continue elimination diet and retry in 6 months.
 - Repeat specific IgE or skin prick testing in IgE-mediated CMA until negative.

Referral to secondary care/allergy specialist should be considered if systemic reactions, severe GI/resp symptoms, comorbid disease, especially asthma, restricted diet or parental concern.

NB Learning Points: Food Allergy in Children

- Early introduction of peanuts to the diet substantially reduces the risk of developed allergy
- Food allergy can manifest as IgE and non-IgE hypersensitivity... consider the latter in any child with chronic symptoms, particularly chronic GI symptoms, significant GORD or troublesome atopic eczema.
- NICE eczema guidance suggests we should consider food allergy in all children with moderate or severe eczema that is not adequately controlled.
- Children with a history of anaphylaxis should have adrenaline autoinjectors – they should always be prescribed and carried in pairs and used at the first sign of a severe reaction

Quality Improvement Ideas: Food Allergy in Children

- Can you commission an allergy testing service in primary care for your locality?
- Audit children prescribed adrenaline autoinjectors
 - Are they being prescribed in pairs?
 - Is there documented advice about carrying them?

ATOPY AND ALLERGIC RHINITIS

At least 20% of the UK population has some allergic disease and the rates continue to increase year on year. There is also increased public interest in testing for underlying triggers which poses a problem for GPs who have limited time and limited access to allergy tests. A new study in the BJGP explores if a simple questionnaire can exclude atopy as effectively as skin prick testing.

► Screening for Atopy [BJGP 2017;67\(657\):e293](#)

The important premise for this study, recently published in the BJGP ([2017;67\(657\):e293](#)), is that many of the general population believe they have an allergy, and, while it is common, most do not. Five times as many people (10-12% of the population) believe they have a food allergy or intolerance as can be proven on testing (1-2%).

Testing itself has limited access and is hard to interpret due to high rates of false positives. Could patients be successfully screened using simple allergy-based questions in primary care?

■ Method

- An 11-question screening tool was developed from existing data and expert opinion
- 143 adult participants were recruited in Scotland through advertising in GP surgeries, the local university and local asthma clinics
- Participants completed the questionnaire and then underwent skin prick testing for common aeroallergens (house dust mites, cat, dog, mixed grasses)
 - The questionnaire was considered positive if there was at least 1 positive response, and negative if all questions were answered with negative responses

■ Results

- 81 participants (56.6%) were diagnosed as atopic on skin prick testing
- Comparison with questionnaire answers gave a:
 - Sensitivity 96.3%, specificity 33.9%
 - Positive predictive value 65.6%, negative predictive value 87.5%
- Out of the 11 questions, a combination of 4 gave a reasonably high NPV of 87.5%
 - Do you, or have you ever had, hay fever?
 - Do your allergy symptoms vary when you go from place to place (e.g. on holiday)?
 - Do any of your parents or siblings have, or have ever had, hay fever?
 - Is there a specific trigger that always sets off your allergy symptoms?

The authors ***conclude that this simple set of questions is predictive of non-atopic status***. This could help guide general practitioners when further allergy investigations may be appropriate.

Further testing in a larger study population is necessary to improve accuracy further before we can be really confident about a negative result. The negative predictive value is based on a total of 24 participants. Nevertheless, this is a promising start to managing this growing group of patients without resorting to investigations.

KESS: Allergic Rhinitis in Children

[BMJ 2014;348:g4153](#)

15% of children are affected. IgE mediated allergic reaction of nasal mucosa. Classified as:

- **Intermittent** - <4 days per week or < 4 consecutive weeks
- **Persistent** - ≥ 4 days week and >4 consecutive weeks
- **Severity: mild:** impaired sleep, ADLS/work/school, troublesome sx; **moderate/severe:** if any of the former present

Diagnosis: clinical

- **History:** ask about symptoms, triggers, severity and response to treatment; associated atopy
- **Examination:** look in nose for swollen mucosa (may appear slightly grey) and any anomalies such as septal deviation, hypertrophied turbinates, polyps; assess cranial nerves to exclude tumours
- **Swabs are not recommended as rarely helpful**
- **Allergy testing:** not usually recommended (most children will be positive) unless resistant to treatment or has concurrent asthma

Referral

- If red flags present – unilateral sx, bloody nasal d/c, pain, visual/cranial nerve disturbance
- If poorly controlled sx, troubling side effects, uncertain Dx, chronic rhinosinusitis, polyps

Treatment

- If mild and not troubling advice and conservative therapy may be sufficient
- Allergen avoidance is hard and often ineffective
- Nasal saline irrigation: inexpensive, effective and well-tolerated – can use from 1 month of age
- **Pharmacological therapy**
 - **Mild intermittent** – consider antihistamine, nasal steroids or leukotriene receptor antagonist
 - **Mod/severe intermittent or mild persistent** - use any of the above three
 - If effective continue for 1 month then consider stopping and monitor
 - If not effective consider stepping up treatment, e.g. combination therapy
 - **Mod/severe persistent** – again use any of the above three
 - Review at 2-4 weeks
 - If effective continue treatment for >1 month, stepping down if controlled
 - If failure to improve, check compliance, review diagnosis, then add or increase nasal steroid dose
 - If ongoing nasal blockage add a decongestant or short-course oral steroid
 - Failure to improve – refer
 - Steroid nasal spray highest efficacy.
 - Avoid 1st gen antihistamines (e.g. Piriton) – less effective, slower acting, longer sedation
 - For ocular symptoms, oral or topical antihistamines can be effective
- [Allergy UK](#) has lots of information and resources for patients.
- [NHS Choices](#) also has decent information reflecting current practice.

KISS: Hay Fever/Seasonal Allergic Rhinitis & Conjunctivitis in AdultsBased on [NICE-CKS 2018](#), [BMJ Best Practice 2018](#), [DTB 2013](#), [Cochrane](#)■ **Management**• **Self-help and allergen avoidance**

- Useful PIL on self-help from [Allergy UK](#) and NHS Choices
- Nasal and eye saline irrigation decreases symptoms and improves the efficacy of nasal treatments

• **Intranasal steroids**

- Are more effective than oral antihistamines for nasal symptoms and have similar efficacy for ocular symptoms
- If significant obstruction use drops first to get control [PIL on correct use](#)

• **Antihistamines**

- Rapid onset of action, so more suitable for prn use
- Intranasal antihistamine (azelastine) seem more effective than oral

• ***If symptoms uncontrolled with self-help, intranasal steroid and oral anti-histamine***

- Check the correct technique for sprays and drops
- Advise saline irrigation
- If nasal congestion, consider a short-term decongestant for up to 7 days to boost the efficacy of intranasal steroid
- If persistent watery rhinorrhoea, add in *nasal anticholinergic ipratropium*
- If persistent nasal itching/sneezing, add in *antihistamine spray* (azelastine)
 - A combination spray of nasal steroid and antihistamine (fluticasone and azelastine) is available
- If associated asthma symptoms add in *oral montelukast*

• ***Eye symptoms***

- If intranasal steroids and oral antihistamines are ineffective, consider topical antihistamine drops e.g. azelastine, olopatadine, ketotifen etc.
- Mast cell stabilisers e.g. cromoglicate, nedocromil are an option but require regular dosing for several weeks

• ***Short-term oral steroids?***

- If symptoms are uncontrolled by standard treatment and advice and seriously affecting quality of life, then short courses of steroids are indicated especially if needed to cover important life events
- Adults: prednisolone 0.5 mg/kg for 5 to 10 days
- ***Persistent troubling symptoms?*** Refer to an allergy specialist for testing and for consideration of immunotherapy



PENICILLIN ALLERGY

This recent, very interesting paper in the DTB ([2017;55:3:33](#)), subsequently republished in the BMJ ([2017;358:j3402](#)) discusses the role of testing patients with possible penicillin allergy to help guide us on whether this important group of antibiotics is a viable treatment option.

► How common is penicillin allergy?

10% of the population are documented as having an allergy to penicillin. Most of these patients do not have a true allergy, however, studies fail to agree on what percentage actually do - figures range from <10% to around 20%. The rest will have had a simple 'adverse reaction' (e.g. vomiting) or a coincidental reaction, be it idiopathic or relating to the underlying illness the antibiotics were meant to be treating. The ***primary reason for allergy to develop is repeated exposure to penicillin-based antibiotics***. The greater the exposure, the greater the risk. It remains unclear if there is a familial element, although it has been suggested people with atopic allergies may be at increased risk. Data shows why having an accurate diagnosis is important. Beyond simply our exasperation at prescribing another course of Cipro, patients with ***suspected penicillin allergy spend more time in hospital, have poorer clinical outcomes and higher rates of C. diff, MRSA and vancomycin-resistant enterococcus infections***. Furthermore, data suggests that ***correct diagnosis reduces broad-spectrum antibiotic use, duration of hospital admission, mortality and treatment costs***.

► Who should avoid penicillins?

Patients with **immediate hypersensitivity reactions** (anaphylaxis, urticaria or rash immediately after taking the drug), **severe non-immediate reactions**, or a **+ve skin test should avoid penicillins**. ***But what about all those patients who develop a 'bit of a rash' around the end of treatment?***

The DTB quotes the BNF which states: "patients with a history of minor rash (e.g. non-confluent, non-pruritic rash restricted to a small area of the body) or a rash that occurs more than 72 hours after penicillin administration are probably not allergic to penicillin. In these individuals, penicillin should not be withheld unnecessarily for serious infections; the possibility of an allergic reaction should, however, be borne in mind. Other beta-lactam antibiotics (including cephalosporins) can be used in these patients".

► Who should be tested?

What about the rest? Due to the risk of serious reactions, testing for penicillin allergy must be undertaken by a specialist, so clearly 10% of the UK population cannot be referred. The **British Society for Allergy and Clinical Immunology (BSACI) recommend investigating penicillin allergy in patients with the following:**

- 'Multiple antibacterial allergy' label
- Personal Hx of beta-lactam hypersensitivity in people requiring frequent antibiotics
 - e.g. bronchiectasis, CF, immunodeficiencies, hypo/asplenia... diabetes...
- Beta-lactam hypersensitivity and requirement for treatment with a specific beta-lactam
- Hx of anaphylaxis during general anaesthetic when penicillin was given alongside other drugs

► How are patients tested?

There is no single test for penicillin allergy, so 3 steps are used to exclude true penicillin allergy:

■ Detailed clinical history

- Poor differentiator but clearly information needs to be gathered on why allergy is suspected, potential causative medications and the nature/timing of any reactions
 - Reactions may be immediate (IgE) or non-immediate (T-cell mediated) reactions
 - IgE: onset usually <1h, Sx range from urticaria > anaphylaxis
 - T-cell: onset usually days, most commonly maculopapular and urticarial rashes
 - Severe reactions, e.g. Stevens-Johnson syndrome, are thankfully less common

■ Skin Testing - 1st line investigation

- ***A positive test confirms penicillin allergy***
- Helpful in both IgE and T-cell mediated allergy; better shortly after the reaction
- Skin prick, intradermal and patch testing may be used
- Should only be performed in specialist centres with resuscitation equipment
- Much less useful in patients with non-immediate reactions - oral provocation is better in this situation

■ Oral Provocation Testing - 2nd line investigation

- ***Negative skin tests are not sufficient to exclude allergy***, oral provocation is required
- Not recommended if +ve skin test or in people with life-threatening delayed reactions, asthma or receiving beta-blockers
- Again, should only be undertaken under specialist supervision

► What do we need to advise patients with penicillin allergy?

A history of allergy should be recorded in the patients' medical records and in correspondence with secondary care. In suspected or confirmed allergy, patients should receive relevant information about the allergy and medications that should be avoided. The DTB advises that patients carry this information at all times and recommends using a MedicAlert product or equivalent.

Adrenaline autoinjectors are not routinely prescribed given that exposure in the community is unlikely.

► Can we prescribe cephalosporins to patients with penicillin allergy?

Patients with penicillin allergy should also avoid cephalosporins, at least in primary care. While the DTB reports that cross-reactivity here appears to have been over-estimated, safety cannot be guaranteed.

The main issue appears to be 1st ceph (and possibly certain 2nd-generation), which share similar chemical side chain structures. 10% of patients will get cross-reactivity. In 3rd-generation ceph it has been reported as only 2-3%, and a meta-analysis even suggests rates were 'negligible'.

In reality, this still limits our ability to prescribe. Commonly used oral ceph such as cefalexin and cefaclor should be avoided, and later generation ceph are unlikely to be prescribed in primary care in any case, and even then only used in a supervised environment (hospital) where other antibiotics may be ineffective.

NB Learning Points: Penicillin Allergy

- The majority of patients with reported penicillin allergy do not have a true allergy
- Testing should be limited to patients with multiple antibiotic allergy labels, and those who require frequent courses of antibiotics, e.g. patients with immunodeficiency
- Skin testing and oral provocation are required to exclude allergy
- Cephalosporins should also be avoided in primary care for patients with possible allergy

NB Quality Improvement Ideas: Penicillin Allergy

- Audit your practice population for patients with multiple antibiotic allergies
- Would they benefit from referral to an allergy specialist to identify safe antibiotics for future infections?



REM SLEEP BEHAVIOUR DISORDER

A recent paper in the [BJGP BJGP 2023;73:40](#) has explored this fascinating diagnosis. Awareness of this condition is low amongst patients and doctors, but early identification in primary care is important because of a strong link with neurodegenerative disease and the potential for treatment.

KISS: REM Sleep Behaviour Disorder

Based on [BJGP BJGP 2023;73:40](#)

■ Background

- REM (rapid eye movement) sleep behaviour disorder (RBD) is characterised by vivid dreams and 'dream enactment behaviour' including kicking, shouting, punching or falling out of bed
- Patients present reporting vivid and violent dreams, and may have unexplained injuries; frequently a bed partner will report that they 'act out their dreams' with kicking and punching during sleep
- During normal REM phase of sleep we dream vividly, the eyes move around rapidly but all other skeletal muscle tone is lost and we have atonia; this is managed via the brainstem, and if this pathway is damaged atonia is lost and people can physically act out their dreams
- It is estimated to occur in up to 2% of the population, and is most common in men aged over 50
- There is a strong link between RBD and neurodegenerative disease which is a major reason for the importance of early identification in primary care

■ RBD and neurodegenerative disease

- RBD affects 50% of people with Parkinson's disease (PD), 80% of people with dementia with Lewy bodies (DLB) and almost 100% of people with multi-system atrophy and can be the first presenting symptom in these syndromes and an early sign of neurodegenerative disease
- RBD can occur in otherwise healthy people, when it is known as idiopathic or isolated RBD (iRBD); this isolated form is associated with a 80% to 90% risk of development to an neurodegenerative disease such as PD or DLB within 10 years

■ Management of suspected RBD

- GPs have a *'crucial role in identifying the symptoms of RBD'*
- Symptoms may be difficult to distinguish from other sleep disorders such as night terrors (talking or shouting during sleep) and periodic limb movement disorder (vigorous limb movements during sleep) so specialist confirmation of the diagnosis is necessary
- If you suspect a possible case, ask the single initial screening question *'Have you been told, or suspected yourself, that you seem to act out your dreams whilst asleep?'*
 - This screening question has been validated with a diagnostic accuracy of 94% sensitivity and 87% specificity compared to the diagnostic gold standard of video polysomnography, and a positive response should trigger a referral
 - Referral to neurology or sleep specialist depending on local pathway for diagnostic confirmation (via overnight sleep study for video polysomnography) and further assessment, including for early signs of neurodegenerative disease and possible entry into trials for prevention
- *Idiopathic or isolated RBD may be triggered by prolonged use of antidepressants including SSRI, SNRI and TCA*
 - Consider dose reduction or withdrawal of antidepressants in suspected cases of RBD
- Behavioural advice, includes alcohol avoidance and sleep protection using pillow barriers etc
- Look for early signs of neurodegenerative disease in suspected cases e.g. Parkinson's symptoms or signs, or cognitive impairment and visual hallucinations for LBD
- **Medication**
 - Clonazepam (given at a dose range of 0.125 to 3mg) at night improves symptoms in approx 2/3 of patients, but caution in elderly due to drowsiness, confusion and falls risk
 - Melatonin (dose range 2mg to 6mg) safer option and effective in approx 1/3 of cases

MALNUTRITION

A recent BMJ editorial [BMJ2020;371:m4593](#) describes malnutrition as the new 'silent pandemic'. The longstanding problem of malnutrition has been compounded by the impacts of the 2020 pandemic, and as we are now in the midst of a huge cost-of-living crisis the problems are likely to increase significantly. A recent report [NIHR/BAPEN Malnutrition Report 2021](#) estimates that malnourished adults account for 30% of hospital admissions, 35% of care home admissions and 10% of those presenting to GPs. It is estimated to affect 5% of the population, 10% of those aged over 65. **Hospital admissions for malnutrition have doubled in the last decade** [NHS Data 2021](#). It is frequently under-recognised and under-treated, yet of course has a major impact on all bodily symptoms increasing the risk of disease, admission, dependency, mortality and significantly slowing recovery times.

To help us to recognise and manage the problem an updated guideline [Managing Adult Malnutrition in the Community 2021](#) has been published, and endorsed by the RCGP, the RCN and the British Dieticians Association. It is recommended that clinicians should routinely check for malnutrition, using a validated screening tool such as the MUST tool below. Multiple studies have indicated that at least 25g to 30g of high-quality protein is necessary at each meal to optimally build or maintain muscle mass in older people and those who are unwell. Protein is expensive...

KISS: Clinical Consequences of Malnutrition

[Managing Adult Malnutrition in the Community 2021](#)

- Feeling TATT, weak, frequent illness and slow recovery
- Reduced muscle strength, sarcopenia and frailty
- Increased risk of falls and injury from falls
- Slower recovery from illness and surgery
- Poorer clinical outcomes e.g. increased mortality
- Impaired immune function, wound healing and psycho-social function including anxiety, depression and cognitive function
- Anaemia and biochemical consequences e.g. hypoalbuminaemia

NB Learning Points & Quality Improvement Ideas

- Malnutrition has been described as the new 'silent pandemic', and is likely to increase significantly as a consequence of the cost of living crisis
- Be aware of the potential impact of malnutrition on all aspects of health and recovery, and if suspected check using the MUST screening tool
- See KISS below for diagnosis and management

KISS: Malnutrition in the Community

Based on [Managing Adult Malnutrition in the Community 2021](#)

- *Malnutrition is defined as a deficiency of energy, protein and other nutrients that causes adverse effects on the body, the way it functions and clinical outcomes; it can be disease-related or caused by social factors*
- Screen for malnutrition opportunistically (e.g. new nursing home resident) or upon clinical concern (e.g. on appearance, unplanned weight loss, poor wound healing, falls, recurrent infections etc)
- Assess at-risk patients using the **MUST screening tool** [click here](#)
 - LOW-RISK ► routine care
 - MEDIUM-RISK ► observe and monitor, dietary advice to maximise intake
 - HIGH-RISK ► dietician support, food first + oral nutritional supplement (ONS)
- **Diagnose malnutrition if:**
 - BMI < 18.5
 - BMI < 20 and unintentional weight loss of >5% in last 3-6/12
 - Any BMI, and unintentional weight loss of > 10% in last 3 to 6/12
 - A clinical diagnosis of malnutrition can be made if weight/BMI is not available
 - Consider: *are there any safeguarding issues?*
- **Management**
 - Check for dysphagia...
 - Refer to community dietician/specialist nutrition nurse; set goals and monitor progress
 - **'Food first'** approach at all times to maximise oral intake
 - Provide clear written information to patient and carers [click here](#)
 - Advise high-calorie/high-protein foods (e.g. cheese, butter, cream, full-fat milk etc)
 - Small, frequent meals and snacks
 - Advise fortifying diets and nutritional 'calorie boosters' [click here](#)
 - Consider adding a multivitamin and mineral supplement
 - **Prescribe *additional* ONS** in those at high risk, or eat little or nothing for 5 days, or poor absorption due to illness
 - Prescribe 1 to 3 ONS per day with clear instructions (e.g. 'take 3 times a day between regular meals'), to be used in addition to the above dietary advice
 - Start with a mixed flavours selection
 - Assess after 4 to 6 weeks to monitor use and compliance
 - If long-term need (eg COPD, cancer, frail elderly) then continue 2 ONS per day
 - Prescribe 1 starter pack then 60 preferred ONS per month
 - check compliance at 6 weeks, and monitor every 12 weeks & stop once the target weight is reached and stable

What additional guidance is out there to help us? [Malnutrition Pathway](#)

RELATIVE ENERGY DEFICIENCY

A new Clinical Practice paper in the [BJGP June 2022](#) highlights a common condition with wide-ranging consequences for affected patients but limited awareness among clinicians - RED-S.

Key points from this paper on RED-S, relative energy deficiency in sport, include:

- This is a condition that **affects men and women** - while previously acknowledged in the 'female athlete triad' RED-S is not limited by gender and similar hormonal/metabolic disturbance is seen in males
- This is a condition which **can affect anyone undertaking sports at any level**, not just high-level athletes (but arguably can be applied to any person with prolonged low energy availability)
- The widely held notion that **certain physiological consequences of exercise, such as oligo-/amenorrhoea, are normal in athletes is incorrect** - in fact they are an indication of RED-S
- A number of important consequences such as **bone density loss and impaired growth may be irreversible if diagnosis of the condition is delayed**
- While disordered eating is more common in male and female athletes than in the general population RED-S is not a condition specifically driven by eating disorders, rather any cause of low energy availability
- This is a **condition which is extremely treatable** with recognition and guidance

► *What causes RED-S?*

RED-S results from low energy availability - a long-term imbalance between energy availability and demands. This may be intentional or unintentional. Sustained periods of negative balance drives physiological changes in the body in an attempt to save energy, resulting in disruption of the hormonal axis and subsequent multi-system dysfunction.

By definition, RED-S may be present in patients undertaking sport, but in reality, the same physiological processes may affect any person with long-term low energy deficiency (think child with failure to thrive, people with eating disorders, etc.). However, it is also true that **people undertaking sports have specific issues predisposing them to developing RED-S such as the inherent increase in energy expenditure, and the environments they exist in**, with pressure around eating/performance/body aesthetics, plus an increased risk of eating disorders (20% higher prevalence than in non-athletes).

► *What physiological changes occur?*

Prolonged relative energy deficiency results in a number of changes to normal body function.

The hypothalamic-pituitary axis may be disrupted resulting in reduced thyroid hormones, growth hormones, sex hormones (FSH & LH initially and subsequently testosterone & oestradiol), and gut hormones such as blood glucose and insulin. This causes a wide range of issues including lower metabolic rate, altered body composition (favouring fat over muscle), amenorrhoea, reduced bone density, reduced energy levels, and dyslipidaemia. Many of these will combine to cause reduced athletic performance.

Cortisol production increases, compounding these existing issues and impairing immunity, so people with RED-S often find they have frequent illness. Anaemia may also be present due to gut hormone changes limiting iron absorption.

Common clinical features include:

- Fatigue, poor athletic performance, recurrent injuries and minor illness
- Sleep disturbance, GI disturbance, lack of concentration, irritability, ↓ mood, ↑ anxiety
- Hormonal disturbance
 - Amenorrhoea/oligomenorrhoea in females, loss of morning erections in males
 - Low libido
 - Delayed puberty and impaired growth in younger people (falling off their centiles)
 - Loss of bone density, fractures including stress fractures
- Symptoms may be vague, and especially easy to miss in males.

Weight loss is not always present and RED-S can occur in people with a normal BMI as the compensatory mechanisms which drive the complications allow weight to be maintained.

► How can we investigate RED-S?

Investigations aim to identify RED-S through characteristic changes in hormone levels, identify where the hormonal axis has been disrupted, and assess for alternative causes. Important differentials include PCOS, prolactinoma, pregnancy, ovarian insufficiency and primary male hypogonadism.

KISS: Investigating Possible RED-S BJGP 2022

All athletes	FBC, TFTs, ferritin, vit B12, vit D, 9am cortisol, prolactin Coeliac screen/faecal calprotectin if ongoing GI disturbance DEXA scan if athletes at high risk of poor bone health, e.g. amenorrhoea, 2+ stress fractures
Female	FSH, LH, oestradiol, testosterone, pregnancy test
Male	Testosterone, LH

Tips on interpretation:

- In RED-S most hormone levels are at the low end of normal, or below normal in severe cases
 - The exception is cortisol which may be raised due to high-stress levels
- Females:
 - Always exclude pregnancy (be guided by history in paediatric cases)
 - Testosterone levels help separate the functional hypothalamic amenorrhoea of RED-S and PCOS - in the former levels are usually lower, the latter levels may be raised
 - Beware that COCP use may result in falsely normal hormone levels
- Males
 - Low testosterone points to hypogonadism.
 - LH then distinguishes type: raised suggests primary hypogonadism (testicular pathology), reduced suggests secondary hypogonadism (pituitary-hypothalamic origin including RED-S)

Severity of RED-S can be made using the [RED-S CAT risk assessment tool](#), a validated tool which can also guide on the risk of sport participation and return to play.

For example, anorexia nervosa is high risk - sport participation 'may pose serious jeopardy to his/her health and may distract the athlete from devoting the attention needed for treatment and recovery' and no competition or training is recommended plus the use of a written 'treatment' contract is advised. A proforma is included in the tool.

► *How can we treat RED-S?*

Education is crucial. Patients with unintentional low energy availability may simply need advice on how to maintain adequate nutrition. Those with intentional LEA may need discussion around the negative effects that RED-S may have on athletic performance as well as driving other complications.

Management revolves around finding a strategy for adequate fueling in the individual and should include carbohydrate intake pre- and post-exercise as even in-day deficiency may contribute to RED-S.

Medications are often used to treat the complications from RED-S but are inappropriate. For example, the COCP to 'correct' amenorrhoea, testosterone to correct deficiency or thyroxine for borderline levels are all inappropriate treatments. They fail to correct the underlying drivers of hormone dysfunction - increasing energy levels is the necessary treatment.

Referral may be necessary - firstly if any red flags are present (the authors would include absent or irregular periods, loss of morning erections, delayed puberty, stunted growth, eating disorder, headaches (?pituitary tumour), markedly raised prolactin levels (tumour)), but also dietician input may be helpful, and where improvements are not seen (or behavioural change is not made) referral to endocrinology is appropriate. If eating disorders are present then the paper recommends a wider multi-disciplinary approach including psychiatry and sport medicine specialist.

NB Learning Points & Quality Improvement Ideas: Relative Energy Deficiency in Sport

- RED-S is due to persistent low energy available and results in wide-ranging physiological changes in the body including hormonal dysfunction, tiredness, poor performance, bone density loss and anaemia
- It is a diagnosis of exclusion and simple blood tests including hormone profile is appropriate
- Treatment is to increase relative energy levels, usually through increased energy consumption
- In severe cases participation in sport should be avoided and consideration of referral to specialists such as endocrinology, sports medicine physicians and/or psychiatry
- Patient education is vital. There are excellent online resources:
 - [Health 4 Performance](#) in association with the British Association of Sport and Exercise Medicine
 - [RED-S.com](#)

MYALGIC ENCEPHALOMYELITIS/CHRONIC FATIGUE SYNDROME

In October 2021 NICE published their new guideline on myalgic encephalomyelitis or encephalopathy (ME)/chronic fatigue syndrome (CFS) [NICE NG206 2021](#). This is an important update for a long-neglected condition, which makes significant changes to practice. ME/CFS is a long-term, distressing and disabling condition which has a significant negative impact on the quality of life for patients so the new guidance is welcome to help us to recognise, understand and manage the condition better. As is often the case, when the guideline was published the immediate response from a front-line clinician was commonly 'but where are the resources to deliver this?'. It is a measure of how seriously NICE are taking this long-neglected condition and the impact that it will have on NHS resources that they have produced an unprecedented [NICE 2022 ME/CFS implementation statement](#), stating it is 'critically important that specialist services for people with ME/CFS are available and commissioned in line with the guideline'.

The guideline introduces significant changes from the previous 2007 version, which recommended that CBT and graded exercise therapy (GET) should be offered to people with ME/CFS. ***The new guidance is clear that patients with suspected ME/CFS should NOT be advised to exercise or use more energy than they have, to manage their daily activity according to their limits, not to 'push through' their symptoms and to rest and convalesce as needed.*** Any exercise programme should be overseen by a ME/CFS specialist team and CBT should only be offered to help people to manage their symptoms if they wish it. See the KISS summary below.

Another important change from the 2007 guideline is with the diagnostic criteria and that ***the diagnosis can now be made after just 3 months*** and makes useful suggestions for focussed investigations in suspected cases. Post-exertional malaise is a key diagnostic pointer.

The NICE guidance overtly states that it is important for clinicians to recognise that people with CFS/ME may have experienced prejudice and disbelief and feel stigmatised by people, including health professionals, and therefore may have lost trust in them. ***Acknowledging the reality of living with ME/CFS is vital to build trust and an effective therapeutic relationship.***

The guideline recommends ***early and prompt referral to multi-disciplinary specialist ME/CFS teams*** to confirm the diagnosis and to guide management. This team should ideally include clinicians from a range of specialities (e.g. rheumatology, rehab medicine, endocrinology, immunology, infectious diseases etc) as well as therapists and psychologists with special training in CFS/ME. As with many other current NICE guidelines, this seems a very 'aspirational' recommendation at best at odds with what GPs have available and what commissioners can afford to provide. The implementation statement is clear however that these services need to be commissioned, and ***'commissioners should consider options to support primary care in implementing the recommendations'***.

There is no doubt that primary care teams will struggle to get the timely, multi-disciplinary specialist support that their patient needs and the guideline recommends, but in the meantime, the guidance is useful for us in primary care to help improve our understanding of the condition and its management. Even though in many areas of the NHS we currently lack the multi-disciplinary access to specialist help, GPs will play a key role in early recognition and diagnosis and offering patients ongoing regular support and clinical review.

An important area which the guideline does NOT address is the potential overlap of cases with long covid. We discuss long covid separately in the Covid-19 chapter, but it is *reported that up to 25% of people after Covid-19 may suffer from prolonged debilitating fatigue which meets diagnostic criteria for CFS/ME [BMJ 2021;374:n1995](#)*. Not all patients with long covid will have ME/CFS but those who do will need recognition and support, and it is likely that the prevalence of CFS/ME will increase significantly in the next few years. Research and evidence-based guidance is desperately needed here.

KISS: ME/CFS

Based on [NICE 2021, CG206](#)■ **Background**

- The prevalence of ME/CFS in community-based studies is relatively low; a meta-analysis found the pooled prevalence for self-reported CFS was 3% and following clinical assessment 1% [CKS 2021](#)

■ **General principles**

- Be aware that ME/CFS is a complex, multi-system, fluctuating condition in which symptoms can change unpredictably in nature and severity and affects everyone differently with varying degrees of impact
- *Acknowledge to the person the reality of living with ME/CFS* and how symptoms affect them; take time to build a supportive empathetic relationship and use a person-centred approach
- Patients should be offered regular monitoring and review in primary care

■ **Diagnosis**

- *Timely and accurate diagnosis* is important so that patients can get appropriate care
- *Suspect ME/CFS* if the person has had *all of the persistent symptoms below* for a minimum of 6 weeks in adults and 4 weeks in children and young people, their symptoms are not explained by another condition and the symptoms have a significant impact on function
- *Diagnose CFS/ME* if the person has had *all of the persistent symptoms below* for a minimum of 3 months in adults, children and young people, their symptoms are not explained by another condition and the symptoms have a significant impact on function
- **Symptoms for suspecting or diagnosing CFS/ME (all should be present)**
 - *Debilitating fatigue that is worsened by activity, not caused by excessive physical, cognitive or emotional exertion and is not significantly relieved by rest*
 - *Post-exertional malaise after activity in which the worsening of symptoms is often delayed by hours or days, is disproportionate to the activity and has a prolonged recovery time*
 - *Unrefreshing sleep or sleep disturbance, which may include feeling exhausted, flu-like and stiff on waking or hypersomnia*
 - *Cognitive difficulties ('brain fog') which may include problems finding words or numbers, difficulty in speaking, problems with STM and difficulty concentrating or multi-tasking*
- **If the above symptoms are all present, and ME/CFS is suspected**
 - Arrange full clinical medical assessment including physical examination and assessment of the impact on psychological and social well-being
 - Investigations to exclude other diagnoses, for example (but not limited to)
 - Urinalysis, FBC/U&E/LFT/TFT/CRP or ESR/Calcium + phosphate, HBA1c, ferritin, coeliac serology, creatinine kinase
 - Use clinical judgement as to whether additional tests may be needed e.g. Vit D, B12, serology if specific infection suspected and 9 am cortisol if adrenal insufficiency suspected
- **Additional symptoms**
 - Be aware that additional symptoms may be associated with ME/CFS
 - Orthostatic intolerance and autonomic dysfunction, temperature hypersensitivity, flu-like symptoms, intolerance to alcohol or medications, heightened sensory sensitivity to light, sound, touch and smell, and pain including pain on touch and myalgia, headaches, eye pain, abdominal pain etc
- **Advice for people with suspected ME/CFS**
 - When ME/CFS is suspected, give people personalised advice about managing their symptoms and advise them *not to use more energy than they perceive that they have*, they should manage their daily activity within their limits and not 'push through', to rest and convalesce as needed and to maintain a healthy balanced diet

KISS: ME/CFS continuedBased on [NICE 2021, CG206](#)

- **Severity classification**

- **Mild ME/CFS**

- Able to self-care and do light domestic tasks; most will be working or in education but have likely stopped leisure pursuits and often reduce hours or take time off to rest and recover

- **Moderate to severe ME/CFS**

- Reduced mobility, restricted activities of daily living, have usually stopped work or education; often have disturbed sleep at night and may need to rest in the day

- **Very severe ME/CFS**

- In bed all day and dependent on care, need help with activities of daily living; often very sensitive to sensory stimuli

- **Referral**

- *Refer patients with persistent symptoms at 3 months to a ME/CFS specialist team* (to paediatrician in children and young people) to confirm the diagnosis and to develop a personalised care and support management plan. Every child and young person should be referred to a paediatrician

- **Management**

- Patients should be offered personalised, holistic care by specialist CFS/ME multi-disciplinary teams based around a personalised care and support plan agreed with the patient
 - The care and support plan will include guidance on self-management, including self-directed energy management within the limits of the patient and with goals set by them
 - Patients should receive personalised guidance on symptom management including sleep and rest, physical functioning and mobility, orthostatic intolerance and pain
 - This should include personalised guidance on energy management tailored to patient-set targets, to reduce the risk of post-exertional malaise or worsening of symptoms by exerting their limits
 - Patients should only be offered a personalised **physical activity** or exercise programme overseen by a physiotherapist if they feel ready to progress to this and if they would like to incorporate physical exercise into managing their ME/CFS
 - **Cognitive behavioural therapy (CBT)** should only be offered to patients if after discussing it they would like to use it to help them to manage their symptoms and improve their quality of life
 - **Primary care advice and management**
 - *Do not advise people with ME/CFS to undertake exercise that is not part of a programme overseen by a ME/CFS specialist team*
 - Be aware of flare-ups (a temporary worsening of symptoms which lasts a few days) or a relapse (sustained and marked exacerbation of symptoms that lasts more than a few days)
 - Help to identify possible triggers, advise to reduce activity levels and to monitor symptoms and do not return to previous levels of activity until the flare-up has resolved
 - Offer adults with CFS/ME a review of their care and support plan in primary care at least once a year, and children and young people once every 6 months
 - Assess for any new symptoms, and investigate as clinically appropriate

NEWS: NATIONAL EARLY WARNING SCORE

The **National Early Warning Score** or NEWS has been in widespread use in secondary care for almost a decade now but has only started to be used in primary care recently. GPs routinely record structured sets of observations in ill patients but do not necessarily calculate a formal NEWS score. However, they are now often asked for a patient's NEWS score before an ambulance or hospital transfer. This has been increasingly the case since the start of the COVID-19 pandemic, despite concerns that this scoring tool has not been researched and validated in primary care. So, some recent primary care research [BJGP2020;70:e374](#) and an accompanying editorial [BJGP2020;70:272](#) is very welcome.

► Just remind me what the NEWS score is?

The [National Early Warning Score NEWS](#) was first produced in 2012 by the Royal College of Physicians and updated in 2012 to make it the [NEWS2 score](#). It was based on research showing that failure to recognise or act on signs that a patient is clinically deteriorating (especially with sepsis) was leading to avoidable deaths. Recognising and responding to patient deterioration with appropriate escalation of care is the key aim of the score.

So, what is it? It is a physiological scoring system based on patients observations used in secondary care to predict clinical deterioration, whether patients' care needs escalation and the urgency needed of that escalation. It is based on respiration rate, oxygen saturation, systolic BP, pulse rate, temperature and new onset of altered mental state (such as confusion, disorientation or agitation). Recognising the importance of new-onset of confusion or delirium was the addition to the original NEWS score in 2017 to become NEWS2.

The adoption of NEWS2 has been part of the drive in recent years to identify and manage sepsis earlier and has been seen as vital to standardise how patients who are deteriorating are identified and responded to, and to streamline efficient communication with a 'common language' across the NHS. See figure below for how the score is calculated.

The NEWS2 Scoring System

Physiological Parameter	Score						
	3	2	1	0	1	2	3
Respiration Rate	≤8		9-11	12-20		21-24	≥25
SpO2 Scale 1 (%)	≤91	92-93	94-95	≥96			
SpO2 Scale 2 (%)	≤83	84-85	86-87	88-92 ≥ 93 on air	93-94 on oxygen	95-96 on oxygen	≥97 on oxygen
Air or oxygen?		Oxygen		Air			
Systolic blood pressure (mmHg)	≤90	91-100	101-110	111-219			≥220
Pulse (per minute)	≤40		41-50	51-90	91-110	111-130	≥131
Consciousness				Alert			CVPU
Temperature (°C)	≤35.0		35.1-36.0	36.1-38.0	38.1-39.0	≥39.1	

CVPU is the consciousness/cognitive state (confusion, voice, pain, unresponsive), and scores 3 if there is new onset of confusion, disorientation and/or agitation. This may be subtle but any new confusion, disorientation or agitation would score 3.

An aggregate score of ≥ 5 is the key threshold for an urgent response and sepsis screening, and a score of ≥ 7 of an emergency critical care response.

The key thing with the NEWS2 score is that it was developed for secondary care use, which is where it has been researched and validated. *So why is it now being used in primary care?*

► *Why has the NEWS2 score now started being used in primary care?*

In [2018 NHS England made use of the score mandatory](#) in all acute hospital *and* ambulance trusts. They said that paramedics should use the score for *'pre-hospital patients who are ill and at risk of deteriorating'* i.e. in the community. Use of the score in pre-hospital settings has also been [encouraged by NHS Wales](#) since 2019. Since then GPs have started being asked by ambulance trusts what the score is before transferring a patient to hospital and anecdotally it has been in much wider use since the onset of the COVID-19 pandemic.

The [NICE COVID-19 guideline](#) has a specific recommendation that 'NEWS2 may be useful', but that a face-to-face consultation should not be arranged solely to measure it given that it includes BP and O2 saturation measurement.

So, use of the score has started to be used in primary care mostly for communication with paramedics and also secondary care. It is easy to see why. Primary care is inherently complex and it is tempting to have a simple, objective measure to make complex clinical decision-making easier, to iron out variations in care and to create a 'common language' which is understood by paramedics and secondary care.

However, concerns have been raised that its use has not been researched or validated in primary care. A [recent evidence review](#) by the Centre of Evidence-Based Medicine in April 2020 looking at NEWS2 in the context of assessing possible COVID-19 patients in primary care concludes that *'NEWS2 has not been validated in COVID-19 patients nor in primary care'*, and that enthusiasm for use in primary care may be *'premature'*. Since then new primary care-based research and a BJGP editorial has been published.

► *Use of the NEWS in primary care, new research in 2020*

Against this background, two welcome new studies have been published looking at the use of NEWS in acutely unwell patients in primary care in June 2020.

- **Implementation of the NEWS score in patients with suspicion of sepsis**, [BJGP 2020;70:e381](#)
 - In 2015 the West of England Academic Health Science Network recommended calculation of the NEWS score for acutely unwell patients at referral into secondary care
 - This was an observational study, comparing outcomes in patients with suspected sepsis compared to the rest of England from 2015 to 2019
 - **Results:** In the West of England cohort there was no increase in suspected sepsis admissions with the use of the score, but there was a decrease in mortality rates in suspected sepsis compared to the rest of England
 - **Conclusion:** this is the first study showing that use of the NEWS score in pre-hospital care is associated with improved outcomes in suspected sepsis
 - *'This supports the recommendation that NEWS should be used in pre-hospital care for acutely unwell patients and, to the author's knowledge, for the first time demonstrates that this leads to an improvement in outcomes.'*
- **Association between the NEWS score in primary care and clinical outcomes** [BJGP2020;70:e374](#)
 - This was an observational study which prospectively collected NEWS values for 13,000 GP referrals into acute care
 - NEWS values were then examined to assess associations with process measures and clinical outcomes
 - **Results:** Higher NEWS values were associated with faster transfer of patients by ambulance, faster medical review in secondary care, higher rates of mortality, ITU admission and length of hospital stay
 - **Conclusion:** A higher NEWS value calculated at GP referral into secondary care is associated with faster transfer, quicker medical review in hospital and poorer clinical outcomes
 - *'The findings should increase GPs' trust in using NEWS2 to communicate patient acuity to ambulance and secondary care staff...it should reassure GPs that when used in conjunction with clinical judgement calculating NEWS2 in primary care is worthwhile in terms of improving clinical processes of assessment and treatment of their patients'*

► *So, should we be using NEWS2 routinely in primary care?*

An excellent editorial [BJGP2020;70;272](#) was published in June 2020 was published in response to these two pieces of new research and discussed the wider implications. It acknowledges the positive outcomes of the research but suggests there are other things to consider before widespread adoption.

- The NEWS2 score may help augment decision making and identify patients for faster transfer and hospital admission, but we cannot say if it will help with triage of patients at an early stage of illness
- It is not clear if high scores should over-ride clinical judgement and prompt admission, or indeed if low scores may lead to false reassurance in the face of gut feeling and clinical uncertainty (NB a low score cannot reliably exclude sepsis or serious pathology)
- It is unclear if a formal NEWS2 score in primary care provides safer or better care than a structured set of recorded observations
- Although a NEWS2 score of ≥ 5 is the threshold for urgent action from ambulance staff and in hospital, we lack guidance on 'thresholds for action' linked to the score in primary care (NB and a low score cannot exclude
- There are potential medico-legal implications, *'that a high documented NEWS2 score will provide a clear nudge to admission that may prove irresistible to the retrospective gaze of a clinical negligence lawyer'*

In conclusion, if the clinician calculates and uses the NEWS score after the decision to refer, this new research shows that it can help to prioritise resources, speed up transfer time and focus clinical attention. However, if it is to be used to help or augment decision-making, there is uncertainty and a risk that it may interfere with rather than augment clinical decision-making as *'no score can communicate the gut feeling of an experienced clinician'*.

NB Learning Points & Quality Improvement Ideas

- Since 2018 use of the NEWS2 score has been made mandatory for use in acute hospital trusts and ambulance trusts in England for pre-hospital patients and its use is being encouraged in other countries within the UK NHS
- Using the score allows for a 'common language' to be used across the NHS, understood by paramedics and secondary care
- GPs are increasingly being asked for a patient NEWS2 score in acutely ill patients prior to hospital transfer, leading to disquiet of many in the GP community due to a lack of primary care research and validation
- In 2020 two large new observational studies from primary care have published. These show that use of the score in the community is associated with improved outcomes in patients with suspected sepsis, and a higher NEWS2 score calculated at the point of referral leads to faster ambulance transfer and quicker medical assessment in secondary care
- An accompanying editorial stresses that the use of the score should not replace experienced clinical judgement and clinical decision-making, however, this research shows that it can help to prioritise resources, speed up transfer time and focus clinical attention
- If you use the score, if it is elevated it needs action but it cannot exclude sepsis or serious pathology if the score is normal...
- **Quality Improvement Idea**
 - Start calculating the score prior to hospital transfer once you have made the clinical decision that admission is necessary
 - A NEWS2 score of ≥ 5 is the key threshold for an urgent response
 - However, do not be falsely reassured if the score is < 5 if your 'gut' clinical feel is that this patient needs hospital care

THE ENVIRONMENT AND GP

There is a growing sense of urgency, not least amongst the global youth, that significant and rapid change is required to limit the increasingly evident damage to the planet. While most of us acknowledge climate change, pollution and the loss of biodiversity are hugely important issues, as medics, we might not make the association with our patient's health and the illness it can drive.

In 2020 the [NHS acknowledged that climate change represents the greatest threat to long-term health of the nation](#) and committed to being carbon net-zero by 2040, with an 80% reduction by the end of the 2020s. 4% of the UK carbon emissions are from healthcare. The [RCGP has declared a climate emergency](#).

But what is the role of general practice as a whole and individual practices? While thinking about how we can adapt our practices to become more environmentally friendly can be daunting there are good reasons for ourselves and our patients to get involved. Becoming a 'greener practice':

1. **Helps improve the health of our patients now, as well as in the future**
2. **Reduces patient reliance on medicine and improves self-care, reducing practice demand**
3. **Save money for practices**

Our KISS summary has more details about how to reduce a practice's carbon footprint, but here are some easy things we can all do which can make a big impact:

NB: Environmental Easy Wins for Practices

These simple ideas have the potential to save your practice money, reduce the number of consultations and improve the health of your patients now and in the future.

- **Switch to a renewable energy provider** - this may even be cheaper than your current rates
- **Inhalers - prescribe dry powder instead of metered-dose where clinically appropriate** - 4% of all NHS carbon emissions are due to MDIs and up to 1/4 of a practice's footprint. See the Asthma section for optimising mild asthma and reducing salbutamol MDI use.
- **Use social prescribing** - many health problems result from inactivity and isolation; getting patients linked into local groups to get people outdoors, active and together shifts the onus away from medicine, improves health and reduces treatment burden for patients and work burden for us
- **Think about waste** - this can be as simple of putting the right rubbish in the right bin; recycle in the practice - you do it at home, you can do it at work as well; go paper-lite (or -less), and when you do need to print set it to print on both sides of the page, you've just halved your paper consumption.

1 Air Pollution

It has taken 8 years since the tragic death of Ella Adoo-Kissi-Debrah in 2013 to acknowledge the polluted environment around her was a significant factor in her death. This 9-year-old girl with severe asthma had 27 hospital admissions including several episodes of respiratory arrest in the 3 years before she died - a terrifying ordeal for a small child and her family.

In 2020, after years of campaigning by the family to re-open and review Ella's case, the [Coroner for Inner South London concluded that her death was due to acute respiratory failure, asthma, and "air pollution exposure"](#).

While many of us naturally assume air pollution can drive ill health and even death, this is the first time (in the UK, and possibly the world) that air pollution has been formally recognised as causing death.

It has taken a horrific, shocking case for the public and policymakers to pay attention to this problem but evidence has been mounting for years that air pollution has a significant effect on not just the respiratory system but the whole body and drives a wide range of diseases.

- Fine particulates, which have a diameter of <2.5 microns (PM2.5), are principally caused by transport and combustion sources (including biofuels burned in the home) and enter the body with inhalation where they spread not only to the respiratory and cardiovascular systems but the whole body including the brain. They result in local and systemic inflammation and are associated with a variety of diseases.
- The WHO recommends PM2.5 levels should average <10 µg/m³ over 24h. Most cities in the UK routinely breach these levels. Globally up to 7 million deaths annually are associated with particulate pollution (WHO 2018).
- Here is a tiny selection of papers highlighting the negative effects on health of fine particulates:
 - **Short-term exposure to fine particulate matter and hospital admissions** [BMJ 2019;367:l6258](#)
 - This paper examined 214 disease groups for temporal association with short-term exposure to fine particulates in people aged 65yo and over in the US. It found:
 - Positive associations between risk of hospital admission, fine particulate levels and CVD, respiratory disease, Parkinson's, diabetes, phlebitis, thrombophlebitis, and thromboembolism
 - Each 1 µg/m³ rise in PM2.5 levels drives increase in hospital admissions, bed days and costs.
 - **Air pollution and family determinants of asthma and wheeze in children** [BMJ 2020;370:m2791](#)
 - This Danish case-controlled study of 1-15yos found PM2.5 to be associated with an increased risk of developing asthma or persistent wheeze.
 - Hazard ratios increased 1.05 each 5 µg/m³ PM2.5 increase.
 - **Air quality and out-patient visits during the Beijing Olympics** [SoTE 2010;408:1226](#)
 - This paper showed rates of out-patient asthma visits at a local hospital during the Beijing Olympics, where there were significant restrictions on transport and industrial emissions, were reduced from 12 to 7 per day average compared to two months prior where no restrictions were in place and air quality was significantly worse.
 - **Reduction in fine particulate air pollution and mortality: extended follow up** [AJRCC 2005](#)
 - This seminal paper examining population health in 6 US cities and levels of air pollution followed mortality rates over 25 years and found a reduction in PM2.5 exposure leads to a decrease in mortality and that the effects were in part reversible; HR 0.73 per 10 µg/m³ reduction.

It is reasonable to ask how can practices reduce local air pollution. There is clearly a limit to the direct effects we can have but we can lead by example, try to reduce our use of traditional cars to get to work and encourage patients to use 'active travel' - walking or cycling. As general practice sees the whole population we are in a unique position to positively influence behaviour which can help the individuals health through increased activity, the local area through less particulate pollutants, and the world through reduced emissions.

2 Climate Change and Health

While currently, all thoughts are on the SARS-CoV-2 pandemic in the medium to long term climate change represents a much larger threat to the global population and, ultimately, our own patients.

The [Lancet 2020 Countdown Report](#) on health and climate change highlight some of the key effects of rising global temperatures on human health, both directly and indirectly:

- Weather-related issues
 - Rising mortality due to heat driven by increased heat extremes such as prolonged heatwaves
 - The last decade has been the warmest on record in human history which
 - Surprisingly Europe has experienced the greatest increases in heat-related mortality, driven by the high proportion of vulnerable populations, e.g. the elderly and those with multi-morbidity, and >100k excess deaths linked to heat
 - Wildfire increase - Australia, the USA and Africa saw particularly large increases
 - Flood and drought - Europe saw unusually high durations of drought
 - Extreme weather events - global increases in variability and intensity

- Infectious diseases
 - Warmer temperatures allow vectors associated with hot areas to spread further
 - Dengue, malaria and Vibrio food infections have all increased in frequency and area, although mosquito-borne infections remain uncommon in Europe for now.
 - In the UK the Lyme season has extended into the winter months over the past a decade
- **Food Security and undernutrition**, e.g. increasing challenge for crop growing with global yields down
- **Migration due to displacement**, e.g. from rising sea levels & famine
 - There is a clear link between these two, affecting the most vulnerable populations on the planet
 - Wealthy countries can proactively help with these issues - ultimately without support, it will impact on them whether they like it or not.

3 Improving the Practice Carbon Footprint

In the KISS below we briefly discuss how practices can start to consider their carbon footprint and make positive changes to reduce the impact.

In the NB Learning Points box, there are links to some of the growing number of resources that can help us understand our impact and how to drive change.

For the large number of you who are interested in this area, if you haven't already consider signing up for the excellent and free [Green Impact for Health Toolkit](#). This collaboration between the RCGP and National Union of Students sets out a framework for practices wishing to become more environmentally friendly and sustainable. It has a broad range of ideas and information which practices can use to consider what positive changes they can make. There is also a series of awards a practice can work towards as they become more confident with these changes.

NB Learning Points & Quality Improvement Ideas: The Environment and GP

Many thanks to Dr Matthew Sawyer, GP and director of [SEE Sustainability](#), for his advice on practice carbon footprints and strategies to help reduce environmental impact.

- Reducing health's carbon footprint is a key goal in the NHS Longer-Term Plan and detailed in [Greener NHS](#)
- The [BMA has a report](#) on how GP practices can help and what help we will need
- The RCGP has declared a climate emergency - this [webpage has links](#) to various resources

Improving Carbon Literacy

- You can use these calculators to work out your [individual](#) and [practice](#) carbon footprint
- The [BBC has a carbon calculator for food](#) which is really clear to use and easy to understand
- [SEE Sustainability](#) has free educational resources and more in-depth courses

Making Changes in Practice

- The [Green Impact for Health Toolkit](#) is free and the most comprehensive resource available - sign up!
- [Greener Practice](#) was set up by GPs to help share ideas and information about climate change and how to help - it can link you in with other local GPs who are also interested
- [Greeninhaler.org](#) is a comprehensive website for everything you (and patients) need to know about the environmental impact of inhalers and how to minimise that impact
- [Complete the Cycle](#) is a pharmacy inhaler recycling scheme set up by GSK - see which local pharmacies can recycle inhalers and encourage patients to use the service
- [Parkrun](#) - one of the simplest ways people can get involved and active, free and UK-wide, practices can sign up as well

Patient Information

- For information on climate change [watch this BBC documentary](#) - be warned, it's quite depressing
- [Greener Practice](#) has patient resources on a variety of topics including too much treatment, etc.

Quality Improvement Ideas:

- Lots of QI which can help both the environment and patients
- Review salbutamol use. If high then asthma control needs to be optimised using ICS or ICS/LABA. When prescribing use a DPI version rather than MDI.
- Review patients with poorly controlled hypertension. Discuss the role of physical activity and weight. Use social prescribing to help find a local group that patients can join and ask if a referral to a weight-loss service would be welcomed.
- Review patients with multimorbidity and multiple medications. Can they be rationalised and deprescribed?

KISS: General Practice, Carbon Footprint and Sustainability■ **Carbon footprint**

- The amount of carbon dioxide released into the atmosphere as a result of the activities of a particular individual, organization or community
- Measured in tonnes of CO₂
- Actually refers to "CO₂ equivalent" to include other greenhouse gases, e.g. methane
- Mean CO₂ per person in the UK is 12.7 tonnes/year
- To put that in perspective, 1 tonne of CO₂ is equivalent to driving 2500 miles in an av. petrol car

■ **Carbon literacy**

- Awareness of the carbon costs & impacts of everyday activities & the motivation to ↓ emissions

The carbon footprint of a practice can be considered in 3 parts:

1. Operational costs from day-to-day running
2. Medication and treatment costs
3. Investigation costs

It may be possible to improve each of these areas through behaviour (B) or technological (T) change.

■ **Operational costs:**

- **Energy use** (40-45%) - installing energy efficiency measures pays for itself in 1-3 years
 - Switch to a renewable energy provider
 - B: using less, turning off, reducing heat loss by shutting doors; T: light sensors, LED lights, see if there are local solar panel scheme for business
- **Travel, both patients and staff** (25%, greater in rural practices, lower in urban)
 - **Patient travel**
 - B: high emissions esp. in rural areas, great opportunity to encourage 'active travel' (e.g. walking, cycling), highlight personal health and community benefits
 - T: use of remote consulting can reduce unnecessary physical attendance
 - **Staff travel** - lead by example, both for staff and patients
 - B: encourage local staff to walk/cycle, cycle to work schemes, changing facilities, storage
 - T: encourage electric vehicles (car/e-bike), have charging points to reduce range anxiety
- **Professional services** (20%)
 - B: recycling, not reimbursed but cost of a provider is low; T: remote meetings can work for all
- **Medical and office equipment & consumables** (10-15%)
 - B: go paper-lite with electronic prescribing, emailing Fit notes; T: reusable equipment
 - Opportunity to ask providers for more sustainable products and services

■ **Medication and treatment costs:**

- **Green 'social' prescribing:** an intrinsic part of the NHS Long Term Plan
 - A huge range of options such as walking/cycling, green gyms, gardening, conservation, spending time in outdoor spaces, group activities to reduce isolation
 - Potential to reduce medical issues, e.g. BP, diabetes, pain, and so reduce associated prescribing, consultations, travel - this really is a win-win.
- **Reduce unnecessary prescribing**
 - We often overtreat patients, driving iatrogenic morbidity. Deprescribe where possible. Use evidence-based tools such as [STOPP START](#) to reduce. Ensure patients are using it correctly to prevent waste and maximise the benefits of their treatment.
 - **Switch inhalers where clinically appropriate**, from metered-dose to dry powder, recycle
 - MDI propellants have between 1000 - 3000 greater greenhouse warming potential than CO₂
 - MDIs continue to leak propellant long after the medicine is used up, pharmacies can recycle
- **Investigations** - every action has a cost, ask yourself, do they really need this test?
 - B: reduce unnecessary investigations, e.g. rpt cholesterol
 - T: home monitoring reduces need for travel, saves consultations and improves pt outcomes

LEARNING DISABILITY

Before COVID-19 came along and upended our world as we knew it, two very important areas were about to come across our primary care radar - namely the early diagnosis of cancer, and also improving care for people with learning disabilities. This was in part due to the two new [QI components of QOF 2020/21](#) covering these two areas, as they were seen as areas where primary care could make a significant impact in improved care. The springboard for the LD component was the [CIPOLD report from 2013](#), which highlighted systemic failures in the care of people with learning disabilities. The COVID-19 pandemic caused significant disruption to the implementation of QoF, although the LD component was included in the QI component of the [2021/22 contract](#).

Even though the QoF requirement for Learning Disabilities quality improvement has been and gone, the importance of improving LD service provision has been highlighted by [new legislation in July 2022](#) requiring training in LD and autism. This has been updated in [CQC requirements November 2022](#). The new requirement does not specify a training package or course for staff, however, in conjunction with HEE the government has developed and trialled a training programme, which it is calling the [Oliver McGowan Mandatory Training on Learning Disability and Autism](#), as its preferred training programme.

However, in whichever part of the UK you work, improving care for our patients with learning disabilities should remain a priority. Why?

- A learning disability (LD) is a *'significantly reduced ability to understand complex information or learn new skills; a reduced ability to cope independently; and a condition which started before adulthood with a lasting effect'*.
- There is significant health inequality for people with LD:
 - Average age of death for people with LD is 23 years younger (male) and 27 years younger (female); commonest causes of death are pneumonia, aspiration pneumonia, sepsis, dementia, CVD, epilepsy.
 - The [CIPOLD report 2013](#) concluded that 42% of deaths for people with LD are premature, with the commonest reasons being delays to diagnosis and treatment, problems with identifying needs and providing appropriate care.
- People with LD are more likely to have co-morbidities - most commonly obesity, asthma, constipation, dysphagia, GORD, epilepsy, dementia, mental health problems.
- People with LD are less likely to take up national screening programmes and have flu vaccinations.
- There is considered to be significant over-prescribing of psychotropic drugs in people with LD, especially antipsychotics, which increases the risk of harm; [data from PHE 2015](#) based on GP records showed <50% of people with LD being prescribed antipsychotics had a licensed indication recorded.
- Currently only ~25% of the expected 1.1 million people with LD are recorded on the LD QOF register.

Whilst the QOF QI component for LD has been and gone clinically it is an area that should be a priority for us all, and the principles, aims, learning points and QI ideas are something that we should all consider implementing. A summary of the principles of the QI domain are in the KSS below and include lots of useful resources for us to work with.

Overview of QI module - Main Aims:

- Improve the accuracy of the GP register by increasing the identification and coding of people with LD.
- Increase uptake of annual health checks.
- Optimisation of medications, with a focus on appropriate prescribing of antipsychotics.
- Recording the need for reasonable adjustments required.
- Consider wider use of community support with local LD services and social prescribers.

KISS: Learning Disability QI component of QOF 2020/21**What do practices need to do? 5 steps****1. *Identify areas for improvement by assessing the current quality of care:***

- Improve accuracy and increase prevalence of LD QOF register - NHSE has produced a useful document on improving identification of people with LD including a list of codes that do indicate a LD, codes that may indicate a LD and an 'inclusion tool' to help us identify who would benefit from being added to the register - [click here](#).
- Complete a whole practice training needs analysis in LD awareness.
- Reflect on practice's overall approach to caring for people with LD e.g. rate of annual health check completion, cancer screening and vaccination rates, long term condition management.
- Evaluate how reasonable adjustments are identified, highlighted and implemented.
- Consider involving the local community LD team, where possible, to provide support and advice on identifying areas for improvement.

2. *Create an improvement plan:*

- Identify area(s) for improvement (see QI box below for examples to consider) based on the diagnostic phase above, which should focus on:
 - Increasing number of people on LD register - for NHSE document to help [click here](#).
 - Increasing uptake of annual health checks aiming for national target of 75% of those on LD register (aged ≥ 14 years old); there is [meta-analysis evidence](#) that health checks for people with LD are effective in identifying previously unrecognised health needs, including life-threatening conditions. There is further guidance and support on health checks from NICE - see KISS below ([NG 54 2016](#)) and the RCGP ([LD toolkit on annual health checks](#)).
 - One important aspect of the health check is a medication review - general practice has a key role in optimising prescribing to reduce harm, particularly the over-prescribing of psychotropic medications; see [further information](#) on the STOMP initiative which has lots of useful information including support and help for both health professionals and families/carers in managing challenging behaviour without medication from the [Challenging Behaviour Foundation](#).
 - Increasing numbers of people with LD of all ages having annual flu immunisation - the commonest cause of death in people with LD is respiratory tract infection.
 - Improving provision and understanding of Reasonable Adjustments and use of the Flag - for further information on the Reasonable Adjustment Flag [click here](#).
- Practices should choose their own QI activities and set their own targets based on their baseline audits and search results; these should be challenging but realistic and validated by network peers as part of the initial network review meeting.
- Multiple small tests of change are recommended.
- Practices should aim to ensure improvement is continuous and QI becomes routine.

3. *Implement the plan:*

- It is recommended that the plan and activities involve the whole practice team and (where practicable) people outside the practice e.g. LD services, social prescribers, network pharmacists, patient/family/carers support groups.

4. *GP network peer review meetings:*

- A key objective is to enable shared learning across the network, especially around wider system issues that impact on quality of care that require collective responses.
- Contractors should participate in a minimum of 2 network peer review discussions (usually through the PCN) and a record of attendance maintained.
- Network to decide timing of meetings, but it is recommended that the first meeting takes place early to decide on QI activities, with the second towards the end to share outcomes/learning.

5. *Reporting and verification:*

- Complete the QI monitoring template - self-declare activity described has been completed and attendance at a minimum of 2 network review meetings.
- Commissioners may require written evidence that QI activity has been undertaken and of attendance at the peer review meetings.

KISS: Annual Health Checks in People with Learning Disability

Based on NICE NG54, 2016

- GPs should offer an annual health check to all adults with a LD and all children and young people with LD who are not having an annual general check with a paediatrician.
 - Health checks should be based on a [standardised template](#).
- **Communication:**
 - Speak to the person with LD directly rather than talking about or over them.
 - Use clear, straightforward, unambiguous language.
 - Make adjustments, communicate at an appropriate pace and use different methods/formats.
 - Regularly check understanding.
- **Involve a family member** or carer who knows the patient well in the review.
- **Include the following in the annual health check:**
 - A mental health review, including known or suspected mental health problems and how they may be linked to physical problems.
 - A full physical health review (see example [standardised template](#)).
 - An agreed shared care plan for managing any problems.
 - During annual health checks for adults with Down's syndrome ask them and their family/carers about any changes that might suggest the need for a dementia assessment e.g. change in behaviour, loss of skills or a need for more prompting.
- **Consider a mental health problem** if there any change in behaviour e.g. loss of skills, social withdrawal, irritability, avoidance, agitation, loss of interest in activities they used to enjoy
- *If you suspect serious mental illness or dementia refer to a psychiatrist specialising in assessing and treating mental health problems in LD.*

NB Learning Points & Quality Improvement Ideas: Learning Disability

- [The CIPOLD Report 2013](#) is shocking. It concludes that ~42% of people with LD die prematurely through failure of care, particularly due to poor assessment and delayed diagnosis.
- New legislation in the [Health and care Act July 2022](#) requires mandatory training in LD.
- There is strong evidence that a process of systematic health checks in primary care is effective at detecting significant undiagnosed health problems in adults with LD. Include an assessment of mental health as part of your annual check - behaviour change may be a pointer to this. The other key component of the health check is to undertake a medication review - in particular to identify over-prescribing of harmful antipsychotics.
- Be aware of the concept of '*diagnostic overshadowing*'. Are you missing a health problem by attributing a behaviour or symptom directly to their LD?
- Further resources:
 - [Health Information for people with LD](#)
 - [Resources for Health Professionals from Mencap](#)
- **Quality Improvement Ideas: Learning Disability**
 - Are you completing annual health checks for all your LD patients? Use the KISS above to help guide what to include in those annual checks.
 - Aim to increase the proportion receiving the flu vaccination within 12 months.
 - Aim that every death of a person on the LD register is considered a significant event with documented reflective analysis on what lessons can be learned including identification of good practice and areas for improvement.
 - Training in LD is now a statutory requirement - is this being implemented in your practice? Consider the [Oliver McGowan Mandatory Training on Learning Disability](#) supported by HEE.

OVERWEIGHT / OBESITY

The COVID pandemic has given us much to think about and has affected our population in multiple ways. But one result has been the recognition that obesity has been a [major risk factors for hospitalisation and death](#). This is on the back of the well-known complications of cardiovascular and respiratory disease, and the increasing recognition that obesity is a major risk factor for many cancers. Even before the pandemic, we have had data ([Lancet 2016;388:2492](#)) showing that people welcome an opportunistic conversation about their weight by their GP (only 1 in 500 thought it was not appropriate). This will be reassuring for GPs in England, with the [enhanced service for weight management being rolled on again into 2022/23](#) (see below KISS).

There is growing data on the success of low-energy diets for weight loss (see below) but not everyone can manage these challenging interventions. There have been a dearth of pharmacological options to provide an alternative, but with the introduction of GLP-1 analogues, this looks to be changing.

Finally, we also have the **NICE guideline on obesity (CG189)** which was **updated in September 2022**. There are changes to recommendations on how we assess and talk to people with overweight and obesity - factors we know that are crucial if we are to support people in losing weight and improving their health.

1 **NICE guideline on obesity updated September 2022** [CG189](#)

The NICE guideline on obesity had some significant changes in its recent update. The recommendations on management are largely unchanged, reflecting the uncertainty/heterogeneity around diet interventions, and the fact that although evidence is growing for pharmacological interventions, the cost-effectiveness is still under scrutiny. However, the **big changes are around how we assess and talk to people about their weight**, which is welcome. It has become perfectly clear that previous pushes to simply identify people with obesity, measure their BMI and refer them to weight loss programmes in a 'tick box' manner is not helpful, and is likely to be counterproductive. **So what's new?**

► *Assessment and classification of obesity*

It is now well recognised that we need to be careful about using BMI as the sole arbiter of defining people's weight, as it does not take into account a person's central adiposity which is the key driver for complications.

- For those with **BMI <35** measure **waist-to-height ratio** (see box below) as well as BMI as this **better predicts central adiposity and health risks**.
- Other groups we need to be careful with when interpreting BMI:
 - Those with high muscle mass - BMI poorly reflects central adiposity.
 - Elderly - higher BMI may have a protective effect during ageing.

KISS: How do I measure the waist-to-height ratio?

Measure

- Find the bottom of the ribs and the top of the hips.
- Wrap a tape measure around the waist midway between these points (this will be just above the belly button) and breathe out naturally before taking the measurement.

Calculate

- Measure waist circumference and height in the same units (either both in cm, or both in inches). If you know your height in feet and inches, convert it to inches (for example, 5 feet 7 inches is 67 inches).
- Divide waist measurement by height measurement. For example:
 - 38 inches divided by 67 inches = waist-to-height ratio of 0.57 or
 - 96.5 cm divided by 170 cm = waist-to-height ratio of 0.57.

Classification now takes into account both BMI and waist-to-height ratio:

- **By BMI:**
 - Healthy weight = BMI 18.5-24.9
 - Overweight = BMI 25-29.9
 - Obesity class 1 = BMI 30-34.9
 - Obesity class 2 = BMI 35-39.9
 - Obesity class 3 = BMI ≥ 40
 - For those with a South Asian, Chinese, other Asian, Middle Eastern, Black African or African-Caribbean family background reduce BMI thresholds by 2.5. (e.g. obesity class 1 = BMI 27.5-32.4)
- But as above use BMI with caution, especially if BMI < 35 and use **waist-to-height ratio as a better indicator of health risks:**
 - Healthy = ratio 0.4-0.49
 - Increased central adiposity = ratio 0.5-0.59
 - High central adiposity = ratio > 0.6
- **For children** NICE recommend we continue to use age-adjusted BMI charts, however for the majority of children (those with BMI < 91 st centile) we should use waist-to-height ratio as a better predictor of health, with the same thresholds as for adults above.
- We should encourage people to monitor their waist-to-height ratio at home with the **simple message that they should aim to keep their waist to half their height** (i.e. a ratio of < 0.5).

► *Discussing weight*

This is probably the most important change to the guideline. NICE identify that we obviously need to make sure we discuss the long-term risks of being overweight but are **explicit in the recognition that this must be done in a sensitive and positive way**. This is crucial *'because the stigma associated with obesity can affect people's mental and physical health. This can lead to further weight gain and make them less likely to engage with healthcare professionals.'*

Language matters. How we discuss people's weight is crucial to help engagement. If we get this wrong, as discussed by NICE, it can generate worsening self-esteem due to body image and a cycle of worsening behaviours and sustained or worsening weight gain.

So how do we talk about weight? NICE give one specific piece of advice and direct us to other resources to help with this:

- **1.2.13 Ask the person's permission before talking about the degree of overweight, obesity and central adiposity, and discuss it in a sensitive manner.**
- Further help can be found in the Obesity UK Language Matters support document (see KISS below)

Regarding the **management of obesity**, the guideline remains unchanged, with the principles of a holistic approach to weight loss key. This may involve behavioural interventions, physical activity, diets (no specific recommendations apart from that if very low-calorie diets are considered they should be done as part of a multicomponent weight management strategy), appropriate referral to tier 3 management services and surgical interventions (primary care management/monitoring after bariatric surgery is summarised in a KISS below).

Regarding **medications**, there is nothing new from NICE. Orlistat is the only medication mentioned as an option alongside diet, exercise and behavioural approaches - the evidence base is questionable and side effects are high. There are emerging options for the pharmacological management of weight loss, notably the GLP1 agonists (discussed in more detail below) but as it currently stands NICE do not feel there is enough evidence or cost-effectiveness data to support use generally. As discussed below liraglutide is recommended in specific circumstances in tier 3 weight loss services for those with obesity and pre-diabetes. [Semaglutide is in the process of being appraised by NICE](#) for use in both adults and children aged 12-17.

KISS: Discussing weight**Language Matters: Obesity UK****General Principles:**

- Seek permission.
- Use language that is:
 - Not judgemental.
 - Is person-centred (AKA person first i.e. a person with obesity, not an obese person).
 - Collaborative and engaging.
- Recognise that some words or phrases are unacceptable to individuals, whatever the intention - be flexible to how we discuss weight and listen to the person's own words and phrases.
- Don't blame and don't generalise.
- Be positive - focus less on the weight loss per se/risks associated and more on the achieved and potential gains to health and quality of life.
- Stick to the evidence.

How do we do this in practice? Some example phrases to consider:

- **Starting the conversation** - 'would you mind if we spoke about your weight? Where do you think you're at?'
- **Refer back to topics already mentioned** - 'And as you said, your weight's crept up a bit..you said you'd like to lose some weight because you're feeling breathless...
- **Acknowledge positive actions** - 'It's fantastic that you've taken up swimming. Don't worry that your weight hasn't come down yet, the benefits to your health go beyond weight loss'.
- **Focus on potential gains** - 'if you were to lose weight, how might life be different' or 'you mentioned that you'd like to be able to kick a ball around with your daughter; if we can lose a small amount of weight you'll find that much easier'.
- Help your patients **set realistic goals**, which for some may be not gaining further weight - 'what are your targets? How can we achieve this together?'

2 Pharmacological Options for Weight Loss

There are currently three licensed pharmacological options for weight loss in the UK: orlistat, liraglutide (Saxenda) and naltrexone-bupropion (Mysimba).

► GLP-1 Analogues

This class of medication is an analogue of the incretin hormone, which is secreted by the gut in response to food. It causes insulin secretion (in the presence of glucose), reduces plasma glucagon levels, delays gastric emptying and suppresses appetite. The latter two are thought to be why these drugs have a role in weight loss. The class currently is the area of greatest interest, research and, no doubt, lobbying and marketing in drug-mediated weight loss.

Liraglutide has been established in this area for several years, as the brand *Saxenda*. It has a different dose and indications for use than when used in diabetes management (max. dose 3mg vs 1.8mg daily). The seminal paper was published in The Lancet in 2017 ([Lancet 2017;389:1399](#)), part of the SCALES programme - a group of trials looking at the role of liraglutide on obese patients, "pre-diabetics" and diabetics. The data was reviewed in the [DTB 2017;55:78](#).

- **RCT of >3700 obese patients without diabetes (BMI ≥ 30 or ≥ 27 + hypertension or dyslipidaemia), all given diet and exercise counselling then randomised to liraglutide 3mg S/C daily or placebo injections for 56 weeks**
- **Results:** the active group has a clinically as well as statistically significant level of weight loss
 - Weight loss - 8% (8.4kg) liraglutide vs 2.6% (2.8kg) placebo
 - % achieving $\geq 5\%$ loss of body weight - 63.2% vs 27.1% (NNT = 3)
 - % achieving $\geq 10\%$ loss of body weight - 33.1% vs 10.6% (NNT = 5)
 - 72% vs 64% completed the 56 weeks, considered fairly average for obesity trials
 - 10% on liraglutide dropped out due to side effects vs 4% on placebo GI side effects were very common (nausea 40% vs 14%, vomiting 16% vs 4%). Pancreatitis occurred in 7 trial participants on liraglutide vs 1 on placebo (2.4 cases/1000pts/yr vs 0.6).

At the end of the study, those still on liraglutide were randomised to continue or be switched to placebo and their weight was monitored for a further 12 weeks. Those continuing the active treatment gained 0.7% of their body weight back compared to 2.9% switching to placebo. The initial placebo group's weight increased by 0.3%.

Direct comparison with orlistat is lacking but the DTB suggests data indicates liraglutide 3mg will result in greater average weight loss.

So, ***liraglutide used in this manner appears to be effective at leading to weight loss*** (feeling nauseated for 56 weeks probably helps) and while adverse effects will be problematic ***the main issue is cost. ~£180 for 28 days is hard to stomach.***

NICE now recommends liraglutide for managing weight loss in obesity and overweight ([TA664, Dec 2020](#)), albeit with some significant restrictions in place.

NICE TA664: Liraglutide for Weight Loss

- Liraglutide is recommended alongside a reduced-calorie diet and increased physical activity in adults, only if:
 - BMI ≥ 35 (or ≥ 32.5 in minority ethnic groups) AND
 - Non-diabetic hyperglycaemia (HbA1c 6.0-6.4%/42-47mmol/mol or fasting glucose 5.5-6.9mmol/L) AND
 - High CVD risk based on the presence of risk factors, e.g. hypertension, dyslipidaemia AND
 - It is prescribed in secondary care by a specialist multidisciplinary tier 3 weight management service AND
 - The company provides as part of a commercial agreement (i.e. they give the NHS a decent deal)

Interestingly the company proposal to NICE was for 2 years of treatment in those who benefitted (i.e. $>5\%$ loss of total body weight) but the committee thought this is too stingy given that obesity/overweight is a long-term condition and ongoing treatment is likely to be wanted and required in this group, so treatment could be for an indefinite duration.

Does liraglutide help maintain weight loss?

A recent paper in the *NEJM* ([2021;384:1719](#)) examined the role of liraglutide in maintaining weight loss in patients with obesity (BMI >32) after an 8-week low-calorie diet, and compared it to liraglutide + exercise, exercise alone or placebo alone. Active interventions were continued for 1 year, N = 195.

- Mean weight loss after 8-week low-calorie diet = 13kg
- At 1 year, mean weight loss was:
 - Exercise alone -4.1kg
 - Liraglutide alone -6.8kg
 - Liraglutide + exercise -9.6kg
 - The latter two groups showed a statistically significant difference in weight
 - Only the combined groups saw improvements in HbA1c, insulin sensitivity and cardiovascular fitness.
- **Conclusion:** a combination of liraglutide and exercise improved healthy weight loss maintenance more than either treatment alone.

Is this a class effect?

Recently, positive data has been published on once-weekly semaglutide with the [STEP 1](#) and [STEP 2](#) trials (Semaglutide Treatment Effect in People With Obesity programme), the former in patients with obesity without diabetes, the latter with T2DM, and another study showing positive results for tirzepatide.

STEP 1 ([NEJM 2021;384:989](#))

- RCT semaglutide 2.4mg S/C weekly vs placebo for 68weeks, in addition to lifestyle intervention
- 1961 adults, BMI ≥ 30 (or ≥ 27 + ≥ 1 weight-related condition), no diabetes
- Primary endpoints: % change in body weight, weight reduction of at least 5%
- **Results:**
 - % change body weight: 14.9% semaglutide v 2.4% placebo (this was mean 15.3kg v 2.6kg)
 - Loss of 5% body weight: 86% v 32%
 - Loss of 10% body weight: 69% v 12%
 - Loss of 15% body weight: 51% v 5%
 - Greater improvement with semaglutide in cardiometabolic risk factors and physical functioning
 - Nausea and diarrhoea were common with semaglutide, reported as typically transient, mild-moderate severity, and subsiding over time.
 - Discontinuation rates were 4.5% semaglutide vs 0.8% placebo
- **Conclusion:** 2.4mg semaglutide in addition to lifestyle intervention provides sustained clinically significant reduction in body weight.

STEP 2 ([Lancet 2021;397:971](#))

A very similar RCT, this time with patients with T2DM and BMI ≥ 27 . This also included semaglutide 1.0mg (which is the licensed dose for diabetes management) as well as 2.4mg and placebo.

- Mean % body weight loss was 9.6% with semaglutide 2.4mg vs 7% 1.0mg vs 3.4% with placebo
- Loss of 5% body weight occurred in 69% 2.4mg vs 57% 1.0mg vs 29% placebo
- Adverse events were common in all groups, but highest in the higher dose of semaglutide (88% 2.4mg vs 82% 1.0mg vs 77% placebo)

The authors concluded 2.4mg semaglutide resulted in a clinically meaningful decrease in body weight compared to placebo. Another [NEJM paper published in June 2022](#) reported on tirzepatide - a novel glucose-dependent insulinotropic polypeptide and glucagon-like peptide-1 receptor agonist - at 3 different doses in >2500 people with BMI > 30 (or >27 & 1+ weight-related complication) over a 6-year duration. Higher doses resulted in a 21% reduction in mean body weight vs 3% with placebo. >90% had greater than 5% body weight reduction and 57% had >20% (vs 3% in the placebo group). There was improvement in all measured cardiometabolic markers. Side effects were fairly common on treatment, reported as mild to moderate, but only 6-7% on higher doses discontinued the therapy. Most side effects occurred during dose escalation. With these 3 options, the GLP-1 analogues represent a new and exciting option for medium-term weight management for many people, the pay-off being a high potential for low-risk side effects, particularly during the early stages of treatment. The next step is developing services equipped to deal with the potential demand and for clinicians to broaden the conversation when we discuss weight loss with our patients.

Is there a role for GLP-1 analogues in weight loss in children?

We know that the earlier a person controls their weight the better and that most weight gain starts in childhood. But would we want to use medication to break the weight gain cycle in kids? Regardless, researchers have already conducted trials in this area.

A recent paper in the NEJM ([2020;382:2117](#)) examined the utility of liraglutide in adolescents with obesity, aged 12-18. This compared liraglutide 3.0mg to placebo in addition to lifestyle measures for 56 weeks in a RCT of ~300 children, with a further 26 weeks follow-up after cessation of treatment. Percentage body weight is not a good marker in growing children, so a change in BMI from baseline was the primary endpoint. 43% of the liraglutide group achieved $\geq 5\%$ reduction in BMI compared to 19% on placebo. After discontinuation, there was a marginally greater increase in BMI in the liraglutide group. Minor adverse events were, as with adults, common. Severe AE was reported as 2.4% in the active group v 4% in placebo, not a significant difference but a signal that these treatments are not without risk.

► **Naltrexone-Bupropion** [NICE 2017 TA494](#)

This new obesity treatment, Mysimba, is marketed as an adjunct to a reduced-calorie diet and increased physical activity for weight management in adults. BMI ≥ 30 , or ≥ 27 if weight-related comorbidities. It is a twice-daily oral medication costing £73.00 per 28 days at full dose. **NICE last year recommended against its use on the NHS.**

- They assessed data from 4 linked clinical RCTs that compared Mysimba to placebo, with weight checks at 28 or 56 weeks.
- Mean % reduction in weight in the treatment group was 5.1-9.7% vs 1.3-5.5% on placebo.
- Perhaps surprisingly effects on people with T2DM were less than in studies of other treatments.

There is no long-term data to understand how the medication benefits people, weight gain following treatments and the role of re-treatment. There are also some nagging safety concerns after a large, separate trial examining the medication's cardiovascular safety (LIGHT study) was stopped early due to a number of irregularities in the study. While the data they had gained did not appear to show an increase in CVD risk, the initial benefits early data had suggested were much less favourable vs placebo further to the trial.

③ **Dietary modifications for weight loss - DROPLET Trial** [BMJ 2018;362:k3760](#)

The Doctor Referral of Overweight People to Low Energy total diet replacement Treatment (DROPLET) study ([BMJ 2018;362:k3760](#)) investigated how GP referral to a commercially-provided low-energy "total diet replacement" (TDR) programme (the Cambridge Weight Plan) compared with usual care.

- 278 adult patients with BMI >30 were recruited and randomised to either:
 - **Total diet replacement 12-week programme** - 8w TDR (810kcal/day) then 4w food re-introduction and regular behavioural support vs **Usual care** - series of appt with a practice nurse for behavioural weight management advice.
- **Primary outcome:** weight change from baseline at 12 months
- **Secondary outcomes:** biomarkers for cardiovascular and metabolic risk
- **Results**
 - **Weight change: mean -10.7kg vs -3.1kg TDR v usual care**
 - Weight loss was greatest at 6 months
 - 45% achieved a weight loss of $>10\%$ at 1 year
 - Biomarkers in the TDR group improved for HbA1c, dBP, and QRisk2 ($\downarrow \sim 1\%$)
 - Triglycerides were reduced by 0.3 but cholesterol levels did not change
 - **Reported adverse events: 51% vs 30% but were mild (mod>worse 11.% vs 12%)**
 - Constipation (1in7), fatigue (1in12), headache (1in17), dizziness (1in22)

The authors concluded that *TDR combined with behavioural support seems to be “accepted, well-tolerated, and leads to greater weight loss with larger improvements in CV risk than currently available weight loss programmes offered in primary care”.*

And there's the rub - this is commercially available so we can be very positive about its benefits for patients wishing to give it a try, but NHS weight-loss services, by and large, are unable to provide it.

The [associated editorial by the chief investigator](#) bemoans a system failing this group of patients despite an increasingly substantial evidence base demonstrating benefit. Part of this is funding evidence-based interventions such as TDR, part is a change in culture amongst healthcare professionals to start the conversation, a conversation our patients really do want to have.

One question that still remains to be answered is the longevity of such interventions. Given that the maximum weight loss in the intervention group was seen at 6 months, and then decreased from 15kg to 10kg, is this a trend that will continue, or can that (significant) improvement be maintained? Longer-term data is still lacking and would be very welcome for patients, clinicians and commissioners.

KISS: NHS England Weight Management Enhanced Service 2022/23

- **Obesity defined** as BMI ≥ 30 or ≥ 27.5 for Black, Asian and other minority ethnic groups.
- **Component 1 - developing a supportive environment**
 - Ensure necessary training of all staff involved in referral and signposting conversations.
 - Develop and implement a protocol for identification and support of patients living with obesity.
 - Review and maintain the practice obesity register - as a minimum recording needs to be at a level compared to either march 2020 or march 2021 (whichever was highest).
- **Component 2 - patient support and referral**
 - For individuals on the QOF Obesity Register make an individual assessment of patient readiness to engage with weight management service and record the outcome of this, including a BMI reading.
 - If ready, refer to a weight management service including:
 - NHS Digital Weight Management services for those with hypertension and/or diabetes.
 - Local authority tier 2 weight management services.
 - Diabetes prevention programme for those with non-diabetic hyperglycaemia.
 - NHS low calorie diet programmes (if available) for those with T2DM and BMI >27 , or BMI >25 for Black, Asian and other ethnic minority groups.
 - Tier 3 or 4 services.

NB Learning Points and Quality Improvement Ideas: Obesity

- Patients welcome a discussion about their weight, but language matters as highlighted in the updated NICE guideline 2022 - Obesity UK has helpful guidance as to how to best frame discussions with patients about their weight.
- NICE also recommend a shift from just using BMI to assess weight to also assessing waist-to-height ratio, which is a better predictor of central adiposity and thus health risk.
- We should give patients the simple message to aim to keep waist half that of height (i.e. ratio <0.5).
- Total diet replacement programmes are highly effective for weight loss (at least for up to 1 year).
- **Quality Improvement Projects:**
 - Review the information about Language Matters from Obesity UK above - how can you and your colleagues change and improve your discussions for people with obesity?
 - **Practice quality improvement project for people who have had bariatric surgery :**
 - Create a database of patients who have had bariatric surgery.
 - Ensure annual review with appropriate investigations.
 - Are they taking suitable multivitamin OTC preparations plus other recommended supplements.

KISS: Post-Bariatric Surgery Monitoring and Management**BOMSS 2020****Postoperative care and monitoring:**

- Follow-up care should remain with the bariatric surgery centre for the first 2 years.
- Thereafter people should have a **minimum annual review as part of a shared care agreement.**
- **Blood test monitoring** - most need to be done at 3, 6 and 12 months then annually:
 - U&E/renal function
 - LFTs
 - FBC and ferritin
 - B12 and folate
 - Vitamin D and calcium
 - PTH (if not done pre-operatively)
 - Fat-soluble vitamins A, E and K after malabsorptive surgery (NB check Vit E if unexplained anaemia or neuropathy)
 - Trace minerals zinc, copper, selenium after SG (sleeve gastrectomy), RYGB (roux-en-Y gastric bypass) or BPD/DS (duodenal switch)
 - HbA1c if pre-op diabetes
 - Lipids if pre-op dyslipidaemia

Post-operative supplementation:

- **A complete multivitamin and mineral supplement is recommended, which should contain thiamine, iron, zinc, copper, selenium, folic acid (400-800mcg/day)**
 - For SG, RYGB and BPD/DS this should include a minimum of 2mg copper and 15mg zinc
- **In addition to the multivitamin take:**
 - **Iron** - standard OD dose (sulphate 200mg, fumarate 210mg), but may need increasing for menstruating women.
 - **Vitamin B12** - discussion on whether this should be oral or IM (BOMSS recommend IM injections every 3 months).
 - **?Vitamin D** - levels >75 nmol/L considered adequate and many multivitamins have Vit D but after malabsorption surgery 2000-4000 iu/day may be needed.
 - **Fat-soluble vitamins** - Vitamins A (10,000 iu), E (100 iu) and K (300mcg) daily after malabsorptive surgery.

Post-operative problems:

- **Unexplained fatigue or anaemia** - check serum zinc, copper and selenium.
- **Prolonged vomiting/dysphagia/poor oral intake:**
 - Refer back to the bariatric surgery centre.
 - Are at risk of thiamine deficiency - give additional thiamine 200-300mg daily plus Vit B co-strong 1-2 tablets TDS - given these immediately whilst awaiting review.
- **Neurological symptoms** - treat for thiamine deficiency (as above) immediately and check B12, folate, copper and Vit E and refer to a neurologist.
- **Vit A deficiency**/night vision disturbances - given 10,000-25,000 iu oral daily for 1-2 weeks, recheck levels in 3 months.
- **Vit E deficiency** - give 100-400 iu oral daily, recheck levels in 3 months.
- **Vit K deficiency** - give 1-2mg oral daily, recheck levels in 3 months.
- **Zinc or copper deficiency** - given 2 forceval daily and recheck levels in 3 months; if severe deficiency refer for specialist advice.



SMOKING

The most important hot topic in smoking is electronic cigarettes. While the medical community debates around the relative safety of e-cigarettes the public have already voted with their fingers and taken a gamble on their health. In 2018, 6.3% of the UK population were current vapers, a 70% increase from 2014 (see [Office of National Statistics](#)). Over half report, they started as an aid to stopping smoking.

But in the past year, there has been growing concerns around the safety of e-cigarettes following a wave of deaths in the USA. This has prompted bans to various degrees in many countries around the world, although there is currently no suggestion for this in the UK.

1 **E-Cigarettes versus Nicotine Replacement Therapy** [NEJM 2019;380:629](#), [BMJ 2019;365:l2036](#)

This new UK RCT which recruited adult smokers from smoking cessation clinics in London, Leicester and East Sussex, published in the [NEJM in Jan 2019](#), is one of the first to compare e-cigarettes to nicotine-replacement therapy.

- 886 participants were randomised to either an NRT of their choice for 3/12 or an e-cigarette starter pack with a recommendation to buy further e-liquids of their choice
- All participants had weekly behavioural support for at least 4 weeks
- **Primary outcome:** abstinence at 1 year - those lost to f/up were considered non-abstinent
- **Results**
 - **1-year abstinence rate: 18% of the e-cig group vs 9.9% of NRT group** (RR 1.83)
 - 80% of successful quitters still used e-cigs at 1 year vs 9% of those using NRT
 - Throat and mouth irritation was more common with e-cigs (65% v 51%), nausea more so with NRT (38% v 31%)
 - The e-cig group reported greater declines in cough; no difference was seen with wheeze or SOB
- **Conclusion: e-cigarettes are more effective for smoking cessation than NRT** when both were used with behavioural support.

The *linked editorial* ([NEJM 2019;380:678](#)) explores some of the issues this data raises:

- While almost double the number of people successfully quit conventional cigarettes, the majority of e-cig users continue vaping, whereas the majority of NRT who successfully quit stopping all nicotine products so the effectiveness must be balanced against the short and long-term safety (see below).
- It points out that varenicline and bupropion for smoking cessation demonstrate comparative rates of smoking cessation, without longer-term exposure.
- Beyond simply safety, there are societal considerations - adult use normalises ECs and therefore may help drive use in younger people.

The paper was also reviewed by the *National Institute of Health Research and published in the BMJ* ([2019;365:l2036](#)). It notes that this research "*signals a potential shift towards provision of e-cigarettes as part of NHS stop smoking services*". It comments that the two methods also have a significant cost discrepancy - e-cig starter pack = £30.25 v 3 months of NRT = £120 - and that the former "might be a good economic choice", as well as being more clinically effective.

The NB authors feel the paper may not have fully considered the longer cost implications for the patient of e-cigarettes and the potential exposure of younger people when parents make the switch, but of course, this is very difficult to weigh against the greater numbers that stop conventional smoking and the incredible harm we know continuing smoking causes.

2 Are E-Cigarettes Safe?

In 2015 [Public Health England published a paper](#) stating that e-cigarettes were 95% less harmful than traditional cigarettes. This was met with criticism by some due to a lack of solid data on which to base the key figure.

Safety claims are hard to swallow amid reports of 2600 cases of e-cigarette lung injury and 60 deaths seen in the USA during 2019.

► *E-cigarette or vaping-associated lung injury (EVALI)*

This type of injury now has an official term: **e-cigarette or vaping-associated lung injury or EVALI**. The [MHRA published a report in January 2020](#) instructing clinicians to "**have a high index of suspicion in patients presenting with respiratory symptoms where there is a history of e-cigarette use or vaping in the past 30 days**" and report them via the Yellow Card Scheme. It encourages us to ask about vaping routinely as we would traditional cigarette smoking.

► *What caused the lung injury cases in America?*

This is an important question because according to a [UK panel of experts on vaping](#) there has been only one 'victim' (it's unclear if that means 'death') in the UK in the past 10 years, although in Jan 2020 the BBC reported on 2 UK deaths and 20 injuries due to vaping. Nevertheless, something specific and fairly unique appears to be happening in the USA.

The CDC report that 86% of cases were linked to use of tetrahydrocannabinol (THC) in vaping devices. Vitamin E is an additive in THC vaping products and it is speculated that this may have been the driver although other chemicals may also have contributed. While the exact pathology remains unclear EVALI is linked with lipoid pneumonia, chemical pneumonitis, giant cell interstitial pneumonitis and hypersensitivity pneumonitis.

Regardless, the UK has higher and more consistent safety regulations for e-cigarettes, in particular, no vitamin additives are permitted, and so these causes are unlikely to be seen on this side of the Atlantic.

► *Is there evidence of benefit on health?*

A recent paper published in the Annals of Internal Medicine ([2017;166:390](#)) compared the urine and saliva levels of a range of tobacco-associated carcinogens and toxins in cigarette smokers, e-cigarette users and nicotine replacement therapy users. It found that all measured levels were significantly lower in people using e-cigarettes or NRT, suggesting that they are substantially safer.

There is conflicting data on the effects within the body. A [2019 study of 114 smokers in Dundee](#) found that within 1 month of switching to e-cigarettes their arterial dilatation have improved, equivalent to improving their cardiac relative risk by 13%. The flip side is a [new longitudinal study from the USA](#) found e-cigarette use associated with an increased risk of lung disease (OR 1.3), although this was lower than conventional cigarette use (OR 2.6) and combined use (OR 3.3). Slightly worryingly the study found that 90% of participants were in fact 'dual users'.

And e-cigarettes are a heterogeneous bunch. As a NEJM review ([2016;375:1372](#)) notes, e-cigarettes each have their own specific chemicals, which when 'vaped' can produce potentially dangerous compounds. A [2018 Thorax](#) paper showed that while e-cigarette liquid may appear non-toxic when vaporised the resulting chemicals can induce an inflammatory state within the lungs, and while not necessarily carcinogenic the authors speculate this could drive the development of COPD.

So, while it remains likely they are safer than traditional cigarettes, they cannot be assumed to be risk-free, and long-term safety data is required (but will be a long time coming). A NEJM editorial ([2019;380:678](#)) sensibly suggests that given no appropriate duration of e-cig treatment is known, we should recommend patients use the lowest effective dose to suppress cravings and have a clear timeline for complete cessation.

► *Do e-cigarettes encourage younger people to smoke?*

A key concern is that e-cigarettes will act as a gateway to traditional tobacco smoking. E-cigarettes should specifically not be marketed at younger people, but there have been concerns that this market is a key area for companies and that ultimately they will target this group one way or another. "Appealing" flavours is one such method, and have driven such concern in the US (see [NEJM ed 2019;380:679](#)) that the FDA is banning the sale of most flavoured vape liquid.

A new paper in Paediatrics ([2017;139:2](#)) examined smoking habits in US adolescents between 2004 and 2014. Overall the rate of smoking fell from 15.8% to 6.4%. However, when the authors reviewed data on combined cigarette and e-cigarette use they found a small increase between 2011 and 2014, from 11.4% to 12.2%.

Troublingly, it found that e-cigarettes were been taken up in adolescent groups that would be low risk for traditional tobacco smoking, and also that users of only e-cigarettes were more likely to be smoking properly a year later.

This highlights the need for tight legislation on the marketing of e-cigarettes. While the message that they are safer than normal cigarettes is important for traditional and current smokers, allowing them to be marketed as a lifestyle product, particularly to younger groups, may cause harm in the long run.

NB Learning Points: Smoking

- E-cigarettes have been shown to be a useful aid to smoking cessation, however, 80% of patients continue to vape, unlike the minority who continue on NRT after stopping conventional cigarettes. Bupropion and varenicline have similar (or better) efficacy without continuing use.
- Safety concerns remain regarding e-cigs and the figure of 95% safer than conventional cigarettes has been called into question, although it remains likely that overall they are safer than conventional cigarettes.
- EVALI - e-cigarette and vaping-associated lung injury - is now a recognised complication of vaping.
- We should ask about vaping routinely in cases of respiratory illness and if vaping is suspected as a potential cause it should be reported via the Yellow Card Scheme.
- For a comprehensive review of what e-cigs are and how they work to see the [BMJ 2018;360:j5543](#).
- Brief interventions in the consultation are effective and may be the impetus for a patient to stop smoking - [NCSCT](#) will provide training to healthcare professionals to help with the discussion

KISS: Smoking Cessation [BMJ 2014;348:f7535](#)

Use the “Very Brief Advice” approach to identify smokers – takes 30 seconds, [click here for training](#)

- ASK – establish smoking status: “I see from your records you’re a smoker – how’s that going?”
- ADVISE – recommend a combination of support and drug treatment as the most effective approach
- ASSIST – if the patient wishes refer to practice nurse or local smoking cessation clinic, or prescribe and arrange follow-up appointments with yourself

Assessment of nicotine dependence – 3 simple questions:

- Minutes after waking to 1st cigarette – <30mins suggests dependence, <5mins severe
- Number of cigarettes per day – 10+ indicative
- Cravings or withdrawal symptoms on previous quit attempts

Counselling and/or behavioural therapy

- Individual and group programmes and telephone, text and internet interventions very effective.

Pharmacological therapy – encourage the use of support services and follow up as required.

- Nicotine replacement therapy
 - Available in long-acting (patches) and short-acting (gum, inhalator, mouth or nasal spray, lozenge, micro-tablet)
 - Can be used in all groups, including pregnant women and adolescents.
 - Pregnant women should be cautioned that nicotine may not be risk-free for the foetus, though trials have not reported adverse events, but is considered safer than cigarettes.
 - Use for at least 8 weeks – [starting prior to quitting date may improve quit rates](#).
 - Use combinations of NRT if withdrawal symptoms are not controlled.
 - Side effects are uncommon and mild
 - Nausea, headache and dizziness improve with time.
 - Patches can cause skin irritation and disturb sleep
 - Oral NRT may cause sore mouth, hiccups and heartburn.
- Varenicline (Champix)
 - Can be used in adult smokers including those with chronic illness.
 - Contraindicated in pregnancy and childhood.
 - Start treatment 1-2 weeks prior to quitting date.
 - 500mcg OD for 3/7, then 500mcg BD for 4/7, then 1mg BD for 11 weeks
 - Start packs are available which make titrate easy for patients and us
 - A subsequent 12 weeks can be used after the initial course which may reduce relapse rates further.
 - Side effects include nausea (30%, usually mild).
 - Reports of depression, agitation, behaviour changes, suicidal ideation have not been substantiated in trials, in both patients with and without psychiatric disorders.
- Bupropion (Zyban)
 - Can be used in adult smokers including those with chronic illness.
 - Contraindicated in pregnancy, patients with a history of current or past seizures, concurrent use of monoamine oxidase inhibitors.
 - Caution with medications that lower seizure threshold.
 - Start treatment 1-2 weeks prior to quitting date.
 - Initially 150mg OD for 6 days then 150mg BD for 7-9 weeks
 - Maximum dose 150mg in patients at risk of seizures.
 - Can be combined with NRT if withdrawal symptoms are not controlled.

ALCOHOL

We are all aware of reported benefits in the reduction of cardiovascular disease ([NEJM 1988;319:267](#)) with up to moderate drinking of alcohol, but more recently concerns have been raised regarding its effects elsewhere in the body, particularly an increase in the risk of cancer, which appears dose-dependent above low > moderate consumption ([BMJ 2015;351:h4238](#)). This data is what prompted a reduction in the UK recommended weekly intake for men from 21u to 14u in the past year.

Two recent papers, published in the BMJ ([BMJ 2018;362:k2927](#) and [BMJ 2017;357:j2353](#)), gained significant media attention. Both explore the effect of alcohol on cognition and memory. As medicine has become more successful at preventing and treating illness, our focus perhaps needs to change from the effects of alcohol on our hearts in our 60s and consider our minds in our 80s.

1 Alcohol consumption and cognitive decline [BMJ 2017;357:j2353](#), [BMJ 2018;362:k2927](#)

These new papers use data from the Whitehall II cohort study, where 10,308 civil servants were recruited in 1985 and had socio-demographic, health and lifestyle variables measured every 5 years for 30 years through self-reported questionnaires.

The first paper ([BMJ2017;357:j2535](#)) was the first to challenge the widespread notion among the public (and medics) of the health benefits of alcohol consumption.

- In the Whitehall II cohort study 550 men and women (mean age 43 at trial entry, mostly white, middle-class males), non-alcohol dependent on CAGE screening were randomly selected from the original cohort, to match the original characteristics of the study population
- Participants had ***weekly alcohol intake recorded, cognitive testing*** plus an ***MRI brain*** at the end of the study to assess for physical alterations (changes in grey/white matter and hippocampal atrophy - a marker for Alzheimer's disease)
- **Results**
 - Higher alcohol consumption was associated with an increased risk of hippocampal atrophy and white matter changes plus a faster decline in lexical fluency
 - Very heavy drinking (>30u/wk) had an odds ratio of 5.8 (95% CI 1.8-18.6)
 - Moderate drinking (7-14u/wk) had an odds ratio of 3.4 (1.4-8.1)
 - Light drinking had no protective effect
 - No association was found with cognitive performance, semantic fluency or word recall
- **Conclusion:** *there is a dose-dependent association between alcohol consumption over 30 years and hippocampal atrophy plus lexical fluency*

The second paper ([BMJ 2018;362:k2927](#)) looked more directly at the risk of dementia from alcohol consumption.

- 9,087 participants from the Whitehall II cohort aged 35-55 at study inception (1985-88) were checked for a diagnosis of dementia from hospital, UK mental health database and mortality register records and compared to self-reported alcohol consumption on ≥2 questionnaires.
- **Results**
 - 397 cases of dementia were recorded over 23 years, mean age at diagnosis = 76y
 - Abstinence in midlife was associated with a higher risk of dementia (HR 1.47) compared to those consuming 1-14units per week
 - This excess risk appeared to be linked partly to cardiometabolic disease as subgroup analysis showed abstainers without cardiometabolic disease to have less risk of dementia than those with (HR 1.33 vs 1.47)
 - Drinking >14u/wk was associated with a higher risk of dementia - 17% increase per 7u increase above 14u/wk

- **Conclusion:** *The risk of dementia was increased in people who abstained from alcohol in midlife or consumed >14u/wk*

As the linked editorial ([BMJ2018;362:k3164](#)) neatly puts it: *“it’s complicated”*. The first study shows radiological changes in the brain linked in a dose-dependent fashion, the second study shows that consumption >14u/wk is linked to an increased risk of dementia.

This provides some clarity on possible “safe levels” of alcohol to consume and it appears to be in line (surprisingly) with current government recommendations.

The difficulty is for abstainers. While this data suggests no alcohol could also be harmful to cognition over time there are of course other health considerations such as the increase in the risk of liver disease and cancer. Getting data to clarify the relative merits would be hard, ethically dubious and take many, many years.

A second linked editorial ([BMJ 2017;357:j2645](#)) summarises its thoughts on the issue: *“with increasing longevity, maintenance of brain health into older age is the key priority of our time”*.

2 **Vitamin B supplements in alcohol dependency** [BJGP 2017;66\(656\):134](#)

This interesting article in the BJGP ([2017;66\(656\):134](#)) discusses when vitamin B supplementation is warranted. NICE guidance only recommends thiamine for the prevention of Wernicke-Korsakoff syndrome (as it results from thiamine deficiency!). In some health regions, GPs are being encouraged to stop vitamin B co-strong supplements, but as the article explains if you are deficient in one, you are likely to be deficient in all the B vitamins (including folate which is B9).

Vitamin B deficiencies may cause a variety of issues including Wernicke-Korsakoff syndrome (B1/thiamine), dementia (B2), anaemia (B2, B9/folate, B12), dermatitis and glossitis (B2, B3, B6), enteropathy (B6), cardiomyopathy (B6, B12) as well as neuropathic/psychiatric issues seen with low B12 and folate.

Most B vitamins cannot be tested for but a low serum folate implies a deficiency in others.

The paper highlights those at risk of deficiency:

- Older patients with a long history of regular excessive drinking and dependence
- Patients with generally poor nutrition, losing weight or low body weight, and diarrhoea
- Evidence of deficiency in 1 B vitamin - e.g. peripheral neuropathy, persistent skin rash, glossitis, mucous membrane ulcers (B2, B6, niacin/B3) - or low serum folate levels

The treatment is **oral vitamin B compound strong tablets (2 tablets BD or 1 QDS) and thiamine** (the dose is insufficient in vit B co-strong). There are no reported side effects from treatment. They are recommended for problem drinkers if:

- **Symptoms/signs of vit B deficiency as above**
- **Evidence of poor nutrition and malabsorption**, e.g. low BMI, weight loss >5% in past 6/12
- **Has co-morbidities likely to increase malabsorption**, e.g. coeliac, Crohn’s, chronic disease
- **Before assisted withdrawal, when body demand increases**

Proactive, prophylactic treatment with a simple, safe treatment may prevent significant morbidity or worse in this group.

3 Brief Interventions on Smoking and Excessive Alcohol Consumption [BJGP 2016;66:e1](#)

This great study in the BJGP ([2016;66:e1](#)) explains that while brief interventions for excessive alcohol consumption have been shown to be effective and cost-effective, unlike smoking advice they are rarely used in primary care.

- Data was taken from household surveys from the Smoking and Alcohol Toolkit surveys, which tracks information about a new group of 1800 UK adults each month
- Of 1775 smokers, 50% recalled receiving a brief intervention regarding smoking in the past year
- Of 1100 people drinking excessively (AUDIT score 8+), 6.5% remembered receiving advice
 - Those given advice were more likely to have higher AUDIT scores and be male

The authors discussed why it appears that so few excessive drinkers are given advice about reducing alcohol intake. QOF is likely to play a significant role, with direct financial incentives to discuss smoking. Also, it may be that smokers are perceived as more willing to engage in discussion, plus the pathway for helping reduce or stopping smoking is clearer and easily accessible within the practice.

NB Learning Points and Quality Improvement Ideas: Alcohol

- Consider opportunistic screening in higher-risk groups with either the **FAST** or **AUDIT-C** questionnaire, going on to complete the remainder of the **AUDIT** questions if indicated.
- **Offer brief interventions** for those scoring 8-19 on AUDIT, and follow up with reassessment.
- For **people with comorbid depression or anxiety disorders**, alcohol misuse should be treated first. Assess and treat if depression or anxiety persists after several weeks of abstinence.
- **Young people** with problem drinking should be assessed fully and if needed referred to child and adolescent mental health services for treatment, e.g. individual and/or family therapy.
- The NHS '[Change 4 life](#)' website has an interactive drinks checker and useful patient information
- **Quality Improvement Idea:**
 - **Audit patients with alcohol dependence:**
 - Have they been offered a referral to an alcohol specialist service?
 - Has relapse prevention medication been discussed/prescribed? Acamprosate is increasingly used, has a good evidence base and is a relatively safe drug (see KISS below on alcohol dependence)
 - Are they being given thiamine and vitamin B supplementation as per KISS below?

KISS: Alcohol - Background and Assessment**NICE CG115, BMJ 2015;350:h715 CKS 2022****Background:**

- Recommended limit - 14 units/week for men and women, spread over at least 3 days.
- However, it is increasingly clear there is no 'safe level' of alcohol use; recent data ([BMJ 2017;357:j2353](#); [BMJ 2018;362:k2927](#)) has shown a dose-dependent association between alcohol consumption and hippocampal atrophy plus lexical fluency as well as a significantly increased risk of dementia for those regularly drinking >14 units/week.
- The ONS defines binge drinking as >8 U/day for men, 6 U/day for women.
- Harmful drinking is defined as a pattern of alcohol consumption causing health problems directly related to alcohol, which may be physical, psychological or both.
- Alcohol dependence is characterised by craving, tolerance and a preoccupation with alcohol and continued drinking in spite of harmful consequences.
- Alcohol units:
 - 1 unit of alcohol = 10ml (8g) of pure ethanol = total volume of drink (ml) x ABV divided by 1000.
 - 1 unit of alcohol = half-pint of low (3%) alcohol beer/cider, small measure (25ml) of spirits (40%).
 - Examples of common drink measures ([click here](#) for unit calculator):
 - Standard glass (175ml) of 13% wine = 2.3 units
 - Small bottle (330ml) of 5% lager = 1.7 units
 - 1 pint of 4.5% cider/lager = 2.6 units

Screening:

- Use the AUDIT-C tool for initial screening ([click here for the tool](#)); for other tools click [here](#)
- If score is ≥5 (or 3+ and <18 or >65y) continue to the full AUDIT questionnaire ([click here](#))
- AUDIT score ≤7 = provide positive feedback and reiterate sensible drinking limits
- AUDIT score 8 – 19
 - Patient in the hazardous or harmful drinking groups – offer brief intervention, 5-10mins (see below)
 - Use motivational interviewing techniques for this can be useful, e.g. change talk
- Consider referral for specialist assessment if a patient asks for such help, major alcohol-related harm, AUDIT score >20 or alcohol dependence syndrome.
- Assess psychiatric status – if co-morbid depression/anxiety treat alcohol issues first

Category of drinking	Definition	AUDIT score
Low risk	No safe level but drinks	≤7
Hazardous (increasing risk)	Regularly drinks: ■ >3u/day women, >4u/day men	8-15
Harmful (high risk)	Regularly drinks: ■ >6u/day or ■ 35u/week women or 50u/week men	16-19
Dependence	3+ features such as compulsion to drink, tolerance, withdrawal symptoms, self-neglect or harm due to alcohol use	≥20+

KISS: Management of Harmful Drinking**CKS 2022 BMJ 2015;350:h715****Structured Brief Advice:**

- The aim is to deliver this in 5-10 minutes.
- Use **FRAMES principles**:
 - Feedback - on the person's risk of having alcohol problems
 - Responsibility - change is the person's responsibility
 - Advice - provision of clear advice when requested
 - Menu - what are the options for change?
 - Empathy - an approach that is warm, reflective, and understanding
 - Self-efficacy - optimism about the person's ability to change their own behaviour
- **Implementing structured brief advice**:
 - Important to convey the concept of **increasing risk with increasing consumption** - suggest aiming to cut down alcohol rather than stopping is likely to be more achievable and will reduce risks (unless concern about alcohol dependence in which case abstinence should be the goal - see below).
 - Be clear on the **risks of harmful drinking** e.g. increased risks of hypertension, liver disease, cancer, mental health problems, cognitive problems and dementia.
 - But also **emphasise the benefits of cutting down** e.g. improvements to health and well-being, better sleep, weight reduction, saving money etc.
 - Start with an **open-ended 'transitional' statement** e.g. 'How important is it for you to change your drinking?'
 - **Consider a 'readiness ruler'** - ask patients to rate on a scale of 1-10 how ready they are to change - what are the barriers? What would make the number on the 'readiness ruler' go up?
 - Be **firm that it is the person's responsibility to make a change, but be empathetic to their situation** e.g. 'it can be very difficult to make these changes when everyone around you is drinking heavily'.
 - Recognise and avoid high-risk situations.
 - Recognise and address cues and triggers for drinking.
 - Set goals and establish drinking targets.
 - Give practical advice on how to reduce alcohol consumption (see below).
 - Direct person to further avenues of support e.g. [DrinkAware](#), [AlcoholChange](#).

Practical Advice on Reducing Alcohol (click here for more advice from [DrinkAware](#))

- Measure your drinks e.g. with a unit measuring cap.
- Remove temptation - don't automatically re-stock, move alcohol out of view, don't have alcohol chilling in the fridge 'ready to go'.
- Manage what you buy - decide on a weekly limit and stick to it.
- Alternate alcoholic drinks with water if you are drinking.
- Delay your drinking - only start drinking when you are eating, not before.
- Take drink-free days and increase these over time.
- Try non-alcoholic drinks - non-alcoholic beers/wines, mocktails - then have these ready to drink.
- Change routines and look for alternative activities (this can help address cues/triggers) - try a non-alcoholic drink and tasty snack instead of a beer while watching telly, reading or explore other interests/hobbies, and try new exercise regimes.
- Build new habits by addressing the cues, routines and rewards.
- Remember building new habits is a marathon, not a sprint and will take time.
- Celebrate your success with (non-alcoholic!) rewards.

KISS: Alcohol Dependence [CKS 2022](#) [NICE CG115](#) [BMJ 2015;350:h715](#)

Assess severity:

- **Recognise potential alcohol dependence** - craving, tolerance and a preoccupation with alcohol and continued drinking in spite of harmful consequences, or AUDIT score ≥ 20 .
- **Recognise alcohol withdrawal:**
 - Mild - hypertension and tachycardia, anorexia, anxiety, emotional lability, insomnia, irritability, diaphoresis, headache, and fine tremor.
 - Moderate - worsening mild symptoms plus agitation and coarse tremor.
 - Severe/delirium tremens - worsening moderate symptoms plus confusion/delirium, generalised tonic-clonic seizures (this may be the first manifestation of alcohol withdrawal for some people), auditory, visual, or tactile hallucinations, hyperthermia subsequent to psychomotor agitation.
 - NB - Mild to moderate withdrawal may start as early as 4 to 6 hours after the last drink, and peak at 24 to 36 hours. Severe withdrawal symptoms occur after 24 hours and usually peak on day 2. Delirium tremens generally occur after 3 days of abstinence or decreased drinking.
- Recognise features of **Wernicke's encephalopathy** - Confusion, loss of coordination/ataxia, eye paralysis/ophthalmoplegia, nystagmus, memory disturbance, hypothermia, hypotension, and coma.

Management:

- Patients with features of Wernicke's encephalopathy need urgent admission for parenteral thiamine.
- Admission may also be needed for those at high risk of alcohol withdrawal seizures or delirium tremens; consider lower threshold if frail, cognitive impairment, LD, multiple co-morbidities.
- For those not needing admission offer **referral to specialist alcohol support services**.
- **Abstinence is the preferred option for dependence**, however, do NOT advise sudden unsupervised reductions in alcohol - direct to local alcohol support services for advice on alcohol reduction.
- Consider alcohol focussed psychological therapies.
- Advise people with alcohol dependence they are **legally obliged** to [notify the DVLA](#); clinicians may need to break confidence and notify the DVLA if they continue to drive when not fit to do so.
- **Vitamin replacement/supplementation:**
 - **Thiamine** if malnourished +/- decompensated liver disease +/- medically assisted withdrawal planned - if mild deficiency suspected 50-100mg daily if more severe deficiency suspected 200-300mg daily in divided doses.
 - **Vitamin B?** Not addressed by NICE and often not in local guidance as a result but [BJGP 2017; 67 \(656\)](#) recommends **Vit B co-strong** 2 tablets BD (or 1 QDS) if symptoms/signs of Vit B deficiency or evidence of poor nutrition (e.g. low BMI, weight loss $>5\%$ in past 6/12), or co-morbidities likely to increase malabsorption (e.g. coeliac, Crohn's, chronic disease), or before assisted withdrawal.
- **Medically assisted withdrawal:**
 - Assisted withdrawal programmes should involve drug management and psychological support and should be **specialist-led** (i.e. we should NOT be doing this in routine General Practice) - without the correct support successful withdrawal is unlikely and there are considerable clinical risks.
 - Fixed-dose regimes of long-acting benzodiazepines (e.g. chlordiazepoxide, diazepam) usually used.
- **Relapse prevention:**
 - **Mutual aid facilitation** should be recommended e.g. [AA](#) or [Smartrecovery](#) (general addictions).
 - **Drugs: Acamprosate** is increasingly used in shared care guidelines as it is a relatively safe drug with few side effects and is safe if the person starts drinking again (but should be stopped if drinking persists for $>4-6$ weeks); usually prescribed for 6 months (or longer).
 - [Systematic review](#) evidence suggest NNT of only 12 to prevent return to any drinking.
 - dosing as 333mg tablets - 2 TDS or if $<60\text{kg}$ 2 mane/1 midday/1 nocte.
 - Routine blood tests not needed but avoid if creatinine >120 .

EXERCISE

Exercise is relevant for just about everything else in this book!. It is an effective intervention for just about everything!! It really is the closest thing we have to a 'panacea'. *Exercise is listed as evidence-based prevention or treatment in 39 UK national guidelines* ([The Lancet 2012](#))

The evidence base is enormous but, for example, a recent large 2014 cohort study, n=55k with 15 years of follow-up, [J Am Coll Cardiol 2014;64:472](#) (11a) confirms that people who run for leisure have an associated reduction in their cardiovascular mortality by 55% and their all-cause mortality by 30% even if they just run a modest amount (an hour a week). A protective effect was seen regardless of running duration, leading many media reports to pick up on the benefits of even small amounts of daily exercise. The most important determining factor was persistently and consistently running over time.

► Exercise and Motivational Interviewing

But we are so busy, how can we make a difference?

Whilst the health benefits of exercise are obvious to us time is limited and there are many challenges and barriers to us providing effective health promotion. There is however **evidence that brief advice and interventions in primary care DO work** and this is now in NICE guidance. Many different methods have been studied, *but a combination of brief motivational interviewing, guided motivation and written physical activity* on prescription appears to work best (1a). [The Lancet 2012](#)

[NICE 2013 Guidance on Brief Interventions](#)

KISS: Brief Interventions in Primary Care:

Based on [The Lancet 2012](#) and [NICE brief interventions for physical activity](#)

- A brief intervention is a 3 step process
 - Identify inactive adults
 - Give them appropriate advice, backed up with written information and resources
 - Offer follow-up to check the progress
- Stress the health gains from activity e.g. with obesity, move the focus away from weight loss and towards the physical and mental health gains from activity
- Take into account the individuals preference and needs and agree on attainable goals
- Adopting a collaborative guiding rather than directing style, eliciting the patient's own motivation to change using 'change talk'.
- For those who like models, four steps in motivational interviewing are:
 - **Engage** with the patient, establish an agreed **focus** for the conversation, **evoke** their motivation to change and **plan** change
- Provide written information about the benefits of exercise and the local opportunities to be active
- Encourage exercise on referral, if available
- Follow up at 3 to 6 months

See [Motivational Interviewing from Motivate2Move](#) link for more information on how including example consultation videos

There are some great resources and tools out there to help us. *It would be a great use of your CPD time for the year ahead to familiarise yourself with some of these and to change practice to take a more proactive approach to encourage exercise.* You just can't tell people often enough.....!!

Will smartphones have a role to play in behaviour change?► **The SMART MOVE Study** [1b] [BJGP 2014;64:e384](#)

This small but well-designed RCT (n=139, based in primary care) from Galway, **evaluated the effectiveness of a smartphone app to increase physical activity in primary care**. 90% of smartphone users are in possession of their phone 24 hours a day, making them potentially very powerful tools for a lifestyle change. Other studies have shown that pedometers generally are associated with boosting activity, but they require the purchase and carriage of additional equipment and people generally give up on using them long term. The potential advantage of the smartphone is that people always carry them and no additional equipment is needed.

All participants were given information and advice about exercise, and half were randomised to use a simple smartphone app (The Accupedo-Pro Pedometer app available at www.accupedo.com; this counts your steps accurately if you walk with your phone in pocket, bag or belt; the app is available for iPhone and Android, and there is a free version or the pro version used in this study costs £2.50; the app records steps, gives feedback on step count and calories burnt, a graphic visual display and goal setting etc; other very similar apps are widely available)

Result: *Compared to the information-only group, the group that used the app increased physical activity over the 8 weeks of the study (an improved daily 'step' count of 25%, or 1,000 steps per day from a baseline of 4,000 which is 'clinically meaningful') and this was maintained over the 8 week period.* The control group had an initial improvement in exercise, which was not maintained.

Conclusion: simple, inexpensive or free smartphone apps can boost activity in primary care.

Quality Improvement Ideas: Exercise

- Exercise really is the closest thing we have to a panacea, so put this in your PDP and spend some time on this in the next year making your lifestyle change advice more effective
- Brief interventions are effective in primary care at improving exercise rates – don't give up!
 - Ask, advise stressing health gains, utilise resources such as exercise on referral schemes, back up with written advice and offer follow-up
- Motivational interviewing is effective at improving uptake. Consider a course or training, or check out the links below.

[NB Hot Topics Blog Helping GPs promote Exercise](#)

[Motivate 2 Move educational package for GPs](#) ('The busy doctor's guide to health promotion' from NHS Wales, highly recommended! Includes a brief video on motivational interviewing and how this can be done quickly in consultations, with 2 consultations in 6 minutes!)

[Moving Medicine](#)[Physical Activity Guideline Factsheets for different age groups](#)[NHS Tips on Getting Active](#)

NOVEL PSYCHOACTIVE SUBSTANCES

'Legal highs' have hit the headlines in both the mainstream and medical media. The term does them a disservice - firstly, trading them is illegal (although currently, possession is not); secondly, the term implies a degree of safety, which is far from accurate. Indeed these drugs have the potential to be more dangerous and more addictive than traditional illicit substances - as demonstrated by the current 'epidemic' of Spice throughout the UK.

With at least 8% of younger people trying legal highs, these timely, informative papers in the BMJ ([2017;356:i6848](#) and [2017;356:i6814](#)) shed some very needed light on the subject.

► **What are novel psychoactive substances?** [BMJ 2017;356:i6848](#)

Novel psychoactive substances, or NPS, are chemical compounds that mimic existing recreational drugs. Even by 2015, there were >500 different substances being monitored and hundreds more will have been created since. They are broadly categorised into four groups: stimulants cannabinoids, hallucinogens and depressants.

■ Stimulants

- Mephedrone is the most commonly available
- Structurally similar to MDMA, cocaine and amphetamines
- May be swallowed, inhaled, injected or used PR depending on the variant
- **Effects:** increased empathy and emotion (serotonergic drugs), euphoria, mania (dopaminergic drugs)
- **Risks:** agitation, anxiety, psychotic symptoms, hypervigilance, cardiovascular toxicity (e.g. arrhythmias), hyperthermia. Seizures, delirium and respiratory failure may occur. Serotonin syndrome may occur, especially with concomitant use of SSRIs, etc.

■ Cannabinoids

- Commonly known as 'spice'
- Synthetic cannabinoid receptor agonists; usually smoked, but 'vape' versions exist
- **Effects:** feeling 'stoned', may be stimulating and sedating, anxiogenic and anxiolytic
- **Risks:** less predictable than smoking cannabis. Anxiety, paranoia, marked agitation, psychotic symptoms. May cause a hangover. Also seen: confusion, cognitive impairment, slurred speech, excessive sweating, hypertension, tachycardia, renal failure, pulmonary damage, MI, seizures, and stroke.
- Higher potential for addiction and withdrawal effects than cannabis

■ Hallucinogenics

- May be 'dissociatives' or 'psychedelics'
- Dissociatives are generally more harmful. Early types include ketamine and PCP, but methoxetamine is now common. This has more intense and longer-lasting effects.
- **Risks:** data mostly from older compounds such as ketamine. Deaths are mostly accidental, linked to impulsivity and carelessness, although direct toxicity has been reported. Other features: aggression, psychosis, catatonia, respiratory failure. May cause longer-term neurocognitive and negative mood effects, abdo symptoms, arrhythmia, blackouts and severe ulcerative cystitis and renal damage.
- Psychedelics produce psychedelic effects, rather than true hallucinations. They are generally lower risk than other NSPs, although some have stimulants properties which can lead to stimulant side effects

■ Depressants

- These may be benzodiazepines or opioids, with similar effects and side effects to their established relatives. Commonly they may be designed with a much longer duration of action, which may make overdose more likely.

► **How can we manage people using NPS?** [BMJ 2017;356:i6814](#)

Just like with traditional drug use, identifying users can be very difficult. As ever, there are opportunities to explore possible drugs use, and we should explore the possibility when patients present with low mood, anxiety, palpitations, etc. especially in younger people. Direct questioning may be appropriate.

The second BMJ paper recommends where someone reports the use of NPS a sensitive, non-judgemental approach is vital with the offer of help if the patient wants it. It advises we avoid pejorative words like 'addict' and adopt a supportive motivational approach.

KISS: Assessing Patients using Novel Psychoactive Substances

- Take a thorough history, mental state exam, and physical examination
 - BP, pulse, temperature and level of consciousness of particularly important
- Discuss drug use
 - Establish type(s) used, e.g. stimulant, cannabinoid, etc.
 - Establish a method of use, e.g. oral, snorting, IV, etc.
 - Establish patterns of use - quantity, frequency, other drugs (prescription, OTC, illicit) and alcohol use
 - Establish harmful effects - consider acute and chronic effects
 - Physical symptoms/signs, mental state, social functioning, sexual health
- Consider safeguarding issues, risk of exploitation, etc.
- Consider social and environmental issues which might drive drug use or prevent cessation
 - Risk factors for longer-term use include age <18y, early use (<15yo), poly-drug use, antisocial behaviour, being affected by other's drug use or domestic violence, being an in need child or on a protection plan

Simple recreational users may not require or want further input. The key is identifying those with problematic use (is it having a detrimental effect on their physical or mental health, socially, at work, etc.) who would benefit from continuing professional input. In this group referral to the relevant specialist service (e.g. substance misuse team) would be appropriate. Also, consider the wider harm that may be caused - are there children at home? Is a safeguarding referral required?

The paper recommends using motivational interviewing as a helpful technique in promoting change. This is more effective than simply encouraging cessation. The FRAMES approach is a well-established model in drug services which is simple and quick to use.

KISS: FRAMES Motivational Interviewing Model

- **Feedback** - discuss potential adverse outcomes, listen to their responses
- **Responsibility** - emphasise it is up to the individual to decide if they wish to change their behaviour
- **Advice** - Straightforward advice on changing drug use
- **Menu** - provide therapeutic options and facilitate their decision making
- **Empathy** - have a non-judgemental and warm clinical approach
- **Self-efficacy** - project optimism that the person can positively change their life

NB Learning Points: Novel Psychoactive Substances

- NPS are increasingly commonly used and may be more potent, dangerous and addictive than the traditional drugs they attempt to emulate
- NPS fall into 4 categories: stimulants, cannabinoids, hallucinogens and depressants
- If a patient divulges they use NPS an open, non-judgemental approach is required, and remember to assess their physical and mental health as well as discussing drug use
- Use the FRAMES approach when helping people using NPS to make decisions about changing their behaviour
- There are a number of useful websites with information on NSPs for health professionals:
 - [UK National Poisons Information Service](#)
 - [NEPTUNE](#) (novel psychoactive treatment UK Network)
 - [NPS resource pack from GOV.UK](#)
- And for the public
 - FRANK - confidential drugs advice - www.talktofrank.com
 - [EROWID](#) - non-profit advice website



BEING A GP IN 2023

Much as we all love General Practice and helping our patients, we cannot sugar-coat the fact that we are practising in tough times. Years of underinvestment in the NHS in general and primary care in particular, unprecedented patient demand and a crisis in secondary and social care has led to a critical workload and workforce problem in primary care. GPs and their teams are working harder than ever and yet despite this, we have faced unfounded and unfair criticism in the media. We are providing a critical service for people, saving and improving lives every day.

Working in primary care has always been challenging but it is now more than ever with our huge workload. However, as we enter into 2023 and beyond there are emerging reasons to be cheerful! We are seeing a welcome return to face-to-face consulting, longer 15-minute consultations being rolled out, a broader skill mix and range of professionals in primary care, plans for better direct access to diagnostic tests and a record number of new GP trainees entering the profession. Furthermore, politicians and policymakers will *have* to see that to provide an efficient, effective and cost-effective health care it is primary care where the investment is most needed. A strong system of primary care and generalism is needed now more than ever.

Our view is that being a GP is a wonderful job. The variety is endless, it is constantly intellectually demanding and rewarding, we never stop learning and every day we make a positive difference in people's lives. It is both a privilege and an honour. However, it is also bloody hard work and at times frankly exhausting! It is all too easy to become worn down and to suffer from burnout. So, do look out for yourself and your colleagues and if you are struggling check out the information and key resources on our 'Burnout' page.

Thank you for doing our course. We really hope that you have found it a good educational experience and that the book and associated material proves to be a useful resource to help you and your patients. **And good luck** back in your consulting room tomorrow. You have no idea what you'll be dealing with, but at least you know you'll be making a positive difference in people's lives.

All the very best, and we hope to see you again in a year's time!

Simon Curtis, Neal Tucker, Siobhan Becker, Kate Digby, Zoe Norris, Rob Walker, Stephanie de Giorgio, Duncan Hill, Sarah Davies, Mark Hadden, Will Duffin, Nick Jones, Andy Ward, Philippa Davies and Rachel Brettell

March 2023



NB 'Takeaways' (Ideas to change practice/to audit/to discuss with colleagues)	Handout Part and Page



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GLOSSARY

This simple glossary has been included by request, especially for doctors new to the UK and our complicated language of acronyms!

BNF	British National Formulary
CBT	Cognitive behavioural therapy
CCG	Clinical Commissioning Groups (GP-led organisations responsible for the local organisation and delivery of healthcare)
CHM	Commission on Human Medicines (formerly the Committee on Safety of Medicines)
DTB	Drugs and Therapeutics Bulletin
FDA	Food and Drug Administration (US drug regulatory body)
GMS	General Medical Services
KISS	'Keep it simple' summary
MeRec	Prescribing bulletins produced by the NPC
NICE	National Institute for Health and Clinical Excellence
NPC	National Prescribing Centre (www.npc.co.uk)
OTC	'Over the counter'
QOF	Quality and Outcomes Framework (of the 'GMS' GP Contract)
Rx	Treatment
SIGN	Scottish intercollegiate guidelines network

STATISTICAL ABBREVIATIONS

Download our Keep it Simple Guide to Evidence Based Medicine for a simple description and explanation of these terms (and other research terminology and simple statistics)

AR	absolute risk
ARR	absolute risk reduction
CI	confidence interval
HR	hazard ration
NNT	number needed to treat
NNH	numbers needed to harm
NNS	numbers needed to screen
NNI	numbers needed to invite in screening studies
OR	odds ratio
RCT	randomised control trial
RR	relative risk, or Risk Ratio
RRR	relative risk reduction